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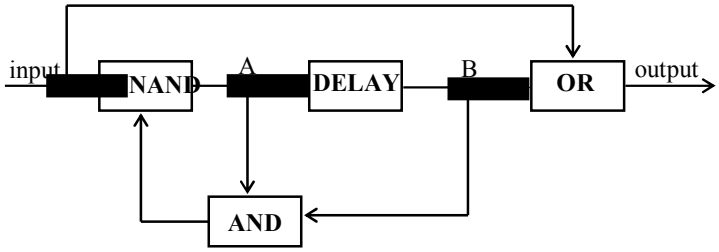
Theory of Automata

Lecture N0. 23
Reading Material

Introduction to Computer Theory Chapter 8

Summary
Mealy machines in terms of sequential circuit

Example
Consider the following sequential circuit



The following four types of boxes are used in this circuit
NAND box (NOT AND): For the given input, it provides the complement of Boolean AND output.
DELAY box (Flip Flop box): It delays the transmission of signal along the wire by one step (clock pulse).
OR box: For the given input, it provides the Boolean OR output.
AND box: For the given input, it provides the Boolean AND output.
The current in the wire is indicated by 1 and 0 indicates the absence of the current.
There are two points A and B w.r.t. to which following four states of the machine are identified according to the presence and absence of current at these points *i.e.*
$\mathbf{q_0}$**(A=0, B=0)** ≡ (0,0)
$\mathbf{q_1}$ **(A=0, B=1)** ≡ (0,1)
$\mathbf{q_2}$ **(A=1, B=0)** ≡ (1,0)
$\mathbf{q_3}$ **(A=1, B=1)** ≡ (1,1)
The operation of the circuit is such that the machine changes its state after reading 0 or 1. The transitions are determined using the following relations
new B = old A
new A = (input) **NAND** (old A **AND** old B)
output = (input) **OR** (old B)
It is to be noted that old A and old B indicate the presence or absence of current at A and B before inputting any letter. Similarly new A and new B indicate the presence or absence of current after reading certain letter.
At various discrete pulses of a time clock, input is received by the machine and the corresponding output string is generated.
The transition at the state q_0 after reading the letter 0, can be determined, along with the corresponding output character as under

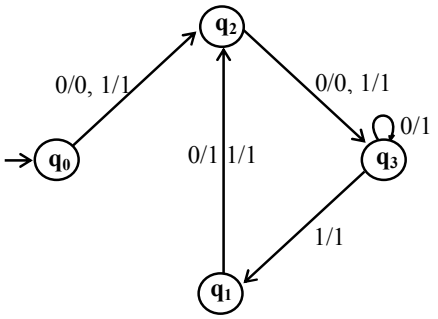
new B = old A = 0
new A = (input) **NAND** (old A **AND** old B)
= 0 **NAND** (0 **AND** 0) = 0 **NAND** 0
= 1
output = (input) **OR** (old B) = 0 **OR** 0 = 0

Thus after reading 0 at q_0 new B is 0 and new A is 1 i.e. machine will be at state (1,0) ≡ q_2 and during this process it's output character will be 0.
The remaining action of this sequential circuit can be determined as shown by the following suggested transition table of the corresponding Mealy machine

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Old state	Inputting 0		Inputting 1	
	State	Output	State	Output
$\mathbf{q_0} \equiv (0,0)$	$(1,0) \equiv \mathbf{q_2}$	0	$(1,0) \equiv \mathbf{q_2}$	1
$\mathbf{q_1} \equiv (0,1)$	$(1,0) \equiv \mathbf{q_2}$	1	$(1,0) \equiv \mathbf{q_2}$	1
$\mathbf{q_2} \equiv (1,0)$	$(1,1) \equiv \mathbf{q_3}$	0	$(1,1) \equiv \mathbf{q_3}$	1
$\mathbf{q_3} \equiv (1,1)$	$(1,1) \equiv \mathbf{q_3}$	1	$(0,1) \equiv \mathbf{q_1}$	1

The corresponding transition diagram may be as follows



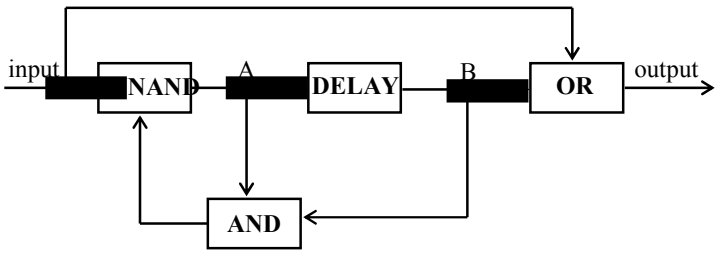
Note: It may be noted that if the string 00 is read at any state, it results in ending in state q_3.

Running the string 01101110 on the previous machine, the output string can be determined by the following table

Input		0	1	1	0	1	1	1	0
States	$\mathbf{q_0}$	$\mathbf{q_2}$	$\mathbf{q_3}$	$\mathbf{q_1}$	$\mathbf{q_2}$	$\mathbf{q_3}$	$\mathbf{q_1}$	$\mathbf{q_2}$	$\mathbf{q_3}$
output		0	1	1	1	1	1	1	0

Following is a note regarding the sequential circuit under consideration

Note



It is to be noted that in this sequential circuit, delay box plays an important role in introducing four states of the machine.

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Theory of Automata

Lecture N0. 24
Reading Material

Introduction to Computer Theory Chapter 9

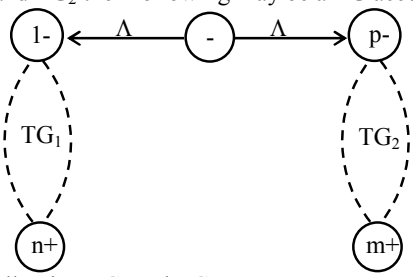
Summary
Regular languages, complement of a language, theorem, proof, example, intersection of two regular languages

Regular languages
As already been discussed earlier that any language that can be expressed by a RE is said to be regular language, so if L_1 and L_2 are regular languages then $L_1 + L_2$, L_1L_2 and L_1^* are also regular languages. This fact can be proved by the following two methods

By Regular Expressions
As discussed earlier that if r_1, r_2 are regular expressions, corresponding to the languages L_1 and L_2 then the languages $L_1 + L_2$, L_1L_2 and L_1^* generated by $r_1 + r_2$, r_1r_2 and r_1^*, are also regular languages.

By TGs
If L_1 and L_2 are regular languages then L_1 and L_2 can also be expressed by some REs as well and hence using Kleene's theorem, L_1 and L_2 can also be expressed by some TGs. Following are the methods showing that there exist TGs corresponding to $L_1 + L_2$, L_1L_2 and L_1^*

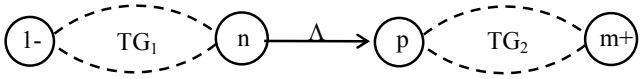
If L_1 and L_2 are expressed by TG_1 and TG_2 then following may be a TG accepting $L_1 + L_2$



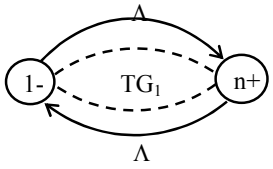
If L_1 and L_2 are expressed by the following TG_1 and TG_2



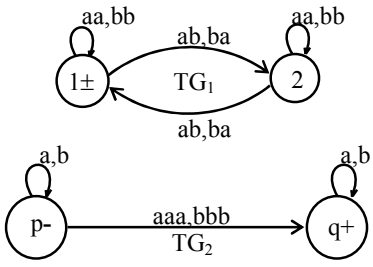
then following may be a TG accepting L_1L_2



also a TG accepting L_1^* may be as under

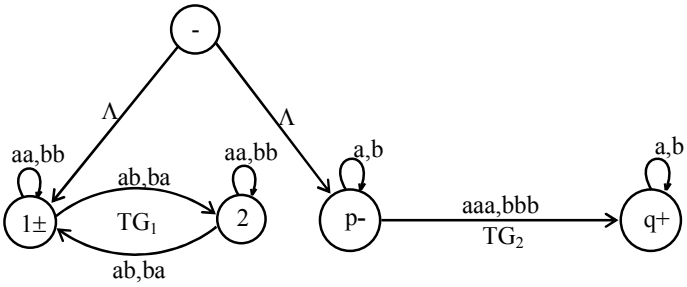


Example
Consider the following TGs

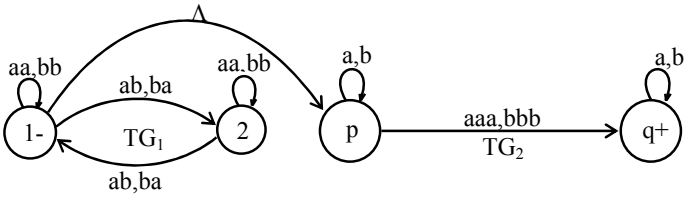


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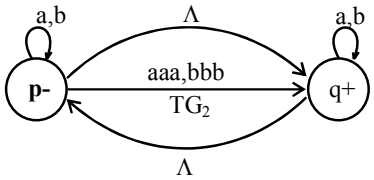
Following may be a TG accepting L_1+L_2



also a TG accepting L_1L_2 may be



and a TG accepting L_2*



Complement of a language

Let L be a language defined over an alphabet Σ, then the language of strings, defined over Σ, **not belonging to L,** is called **Complement of the language L**, denoted by L^c or L'.

Note

To describe the complement of a language, it is very important to describe the alphabet of that language over which the language is defined.

For a certain language L, the complement of L^c is the given language L *i.e.* $(L^c)^c = L$

Theorem

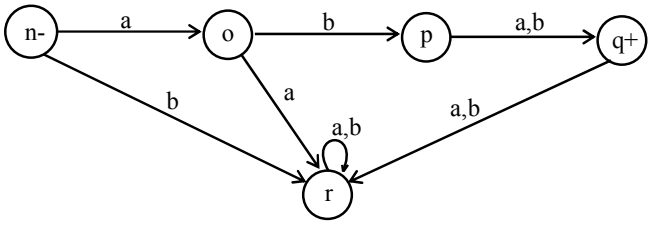
If L is a regular language then, L^c is also a regular language.

Proof

Since L is a regular language, so by Kleene's theorem, there exists an FA, say F, accepting the language L. Converting each of the final states of F to non-final states and old non-final states of F to final states, FA thus obtained will reject every string belonging to L and will accept every string, defined over Σ, not belonging to L. Which shows that the new FA accepts the language L^c. Hence using Kleene's theorem L^c can be expressed by some RE. Thus L^c is regular.

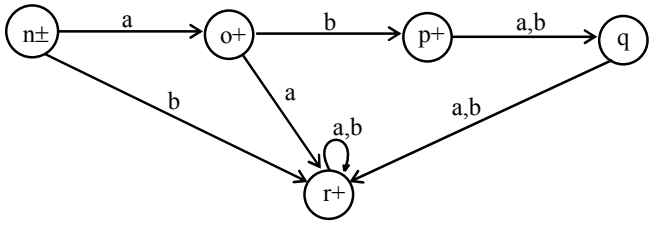
Example

Let L be the language over the alphabet Σ = {a, b}, consisting of only two words aba and abb, then the FA accepting L may be



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Converting final states to non-final states and old non-final states to final states, then FA accepting L^c may be



Theorem

Statement

If L_1 and L_2 are two regular languages, then $L_1 \cap L_2$ is also regular.

Proof

Using De-Morgan's law for sets

$(L_1^c \cup L_2^c)^c = (L_1^c)^c \cap (L_2^c)^c = L_1 \cap L_2$

Since L_1 and L_2 are regular languages, so are L_1^c and L_2^c. L_1^c and L_2^c being regular provide that $L_1^c \cup L_2^c$ is also regular language and so $(L_1^c \cup L_2^c)^c = L_1 \cap L_2$, being complement of regular language is regular language.

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Theory of Automata

Lecture N0. 25
Reading Material

Introduction to Computer Theory Chapter 9,10

Summary
Intersection of two regular languages, examples, non regular language, example

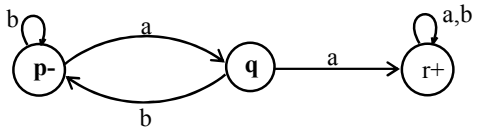
Theorem
Statement
If L_1 and L_2 are two regular languages, then $L_1 \cap L_2$ is also regular.

Proof
Using De-Morgan's law for sets
$(L_1^c \cup L_2^c)^c = (L_1^c)^c \cap (L_2^c)^c = L_1 \cap L_2$
Since L_1 and L_2 are regular languages, so are L_1^c and L_2^c. L_1^c and L_2^c being regular provide that $L_1^c \cup L_2^c$ is also regular language and so $(L_1^c \cup L_2^c)^c = L_1 \cap L_2$, being complement of regular language is regular language.
Following is a remark

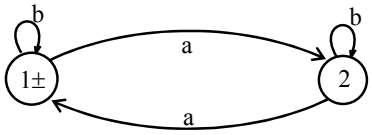
Remark
If L_1 and L_2 are regular languages, then these can be expressed by the corresponding FAs. Finding regular expressions defining the language $L_1 \cap L_2$ is not so easy and building corresponding FA is rather harder. Following are example of finding an FA accepting the intersection of two regular languages

Example
Consider two regular languages L_1 and L_2, defined over the alphabet $\Sigma = \{a, b\}$, where
L_1 = language of words with **double a's.**
L_2 = language of words containing **even number of a's.**
FAs accepting languages L_1 and L_2 may be as follows

FA_1

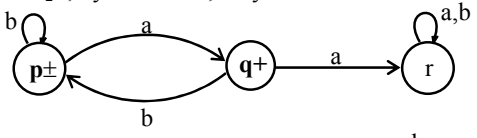


FA_2

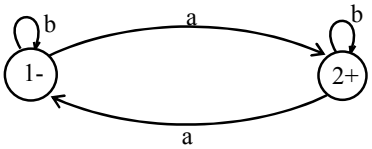


Their corresponding REs may be
$r_1 = (a+b)^*aa(a+b)^*$
$r_2 = (b+ab^*a)^*$
Now FAs accepting L_1^c and L_2^c , by definition, may be

FA_1^c



FA_2^c

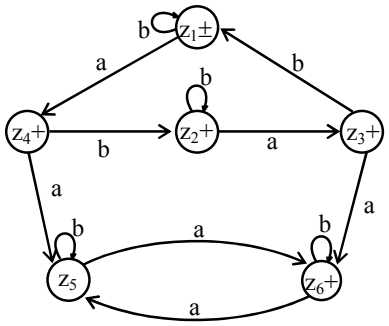


Now FA accepting $L_1^c \cup L_2^c$, using the method described earlier, may be as follows

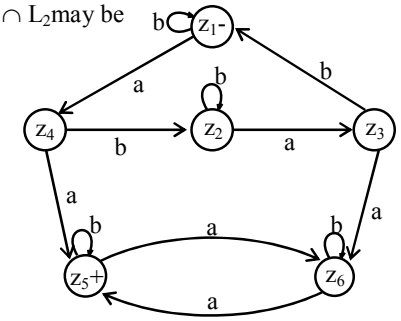
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Old States	New States after reading	
	a	b
$z_1\pm\equiv(p,1)$	$(q,2)\equiv z_4$	$(p,1)\equiv z_1$
$z_2+\equiv(p,2)$	$(q,1)\equiv z_3$	$(p,2)\equiv z_2$
$z_3+\equiv(q,1)$	$(r,2)\equiv z_6$	$(p,1)\equiv z_1$
$z_4+\equiv(q,2)$	$(r,1)\equiv z_5$	$(p,2)\equiv z_2$
$z_5\equiv(r,1)$	$(r,2)\equiv z_6$	$(r,1)\equiv z_5$
$z_6+\equiv(r,2)$	$(r,1)\equiv z_5$	$(r,2)\equiv z_6$

Here all the possible combinations of states of FA_1^c and $FA_2^c$are considered

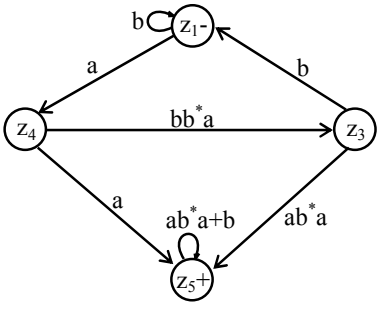


An FA that accepts the language $(L_1^c \cup L_2^c)^c = L_1 \cap L_2$may be

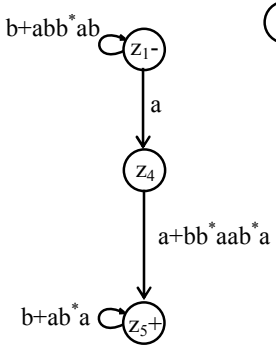


Corresponding RE can be determined as follows

The regular expression defining the language $L_1 \cap L_2$ can be obtained, converting and reducing the previous FA into a GTG as after eliminating states z_2 and z_6

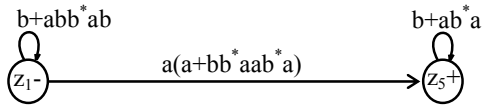


after eliminating state z_3

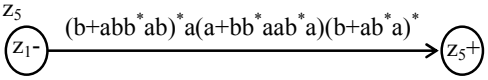


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z_4 can obviously be eliminated as follows



eliminating the loops at z_1 and z_5



Thus the required RE may be $(b+abb^*ab)^*a(a+bb^*aab^*a)(b+ab^*a)^*$

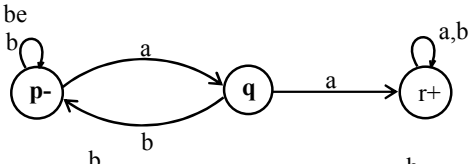
FA corresponding to intersection of two regular languages (short method)

Let FA_3 be an FA accepting $L_1 \cap L_2$, then the initial state of FA_3 must correspond to the initial state of FA_1 and the initial state of FA_2.

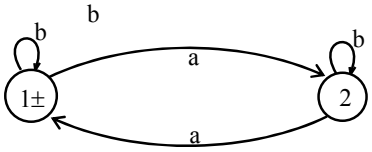
Since the language corresponding to $L_1 \cap L_2$ is the intersection of corresponding languages L_1 and L_2, consists of the strings belonging to both $L_1$and L_2, therefore a final state of FA_3 must correspond to a final state of FA_1 and FA_2. Following is an example regarding short method of finding an FA corresponding to the intersection of two regular languages.

Example

Let $r_1 = (a+b)^*aa(a+b)^*$ and FA_1 be



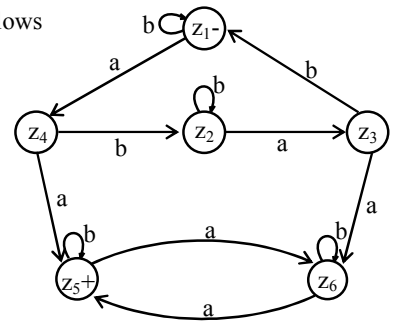
also $r_2 = (b+ab^*a)^*$ and FA_2 be



An FA corresponding to $L_1 \cap L_2$ can be determined as follows

Old States	New States after reading	
	a	b
z_1-≡(p,1)	(q,2)≡z_4	(p,1)≡z_1
z_2≡(p,2)	(q,1)≡z_3	(p,2)≡ z_2
z_3≡(q,1)	(r,2)≡z_6	(p,1)≡ z_1
z_4≡(q,2)	(r,1)≡ z_5	(p,2)≡ z_2
z_5+≡(r,1)	(r,2)≡ z_6	(r,1)≡ z_5
z_6≡(r,2)	(r,1)≡ z_5	(r,2)≡ z_6

The corresponding transition diagram may be as follows



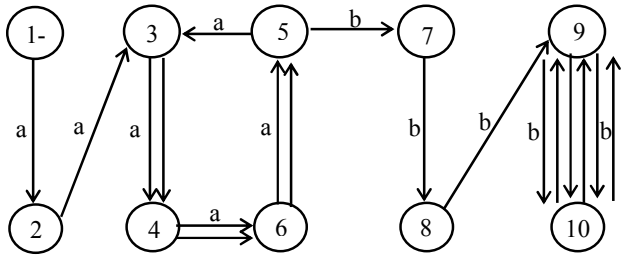
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Nonregular languages

The language that cannot be expressed by any regular expression is called a **Nonregular language**.
The languages **PALINDROME** and **PRIME** are the examples of nonregular languages.
Note: It is to be noted that a nonregular language, by Kleene's theorem, can't be accepted by any FA or TG.

Example

Consider the language L = {Λ, ab, aabb, aaabbb, …} *i.e.* $\{a^n b^n : n=0,1,2,3,\ldots\}$
Suppose, it is required to prove that this language is nonregular. Let, contrary, L be a regular language then by Kleene's theorem it must be accepted by an FA, say, F. Since every FA has finite number of states then the language L (being infinite) accepted by F must have words of length more than the number of states. Which shows that, F must contain a circuit.
For the sake of convenience suppose that F has 10 states. Consider the word $a^9 b^9$ from the language L and let the path traced by this word be shown as under



But, looping the circuit generated by the states 3,4,6,5,3 with a-edges once more, F also accepts the word $a^{9+4} b^9$, while $a^{13}b^9$ is not a word in L. It may also be observed that, because of the circuit discussed above, F also accepts the words $a^9(a^4)^m b^9$, m = 1,2,3, …
Moreover, there is another circuit generated by the states 9,10,9. Including the possibility of looping this circuit, F accepts the words $a^9(a^4)^m b^9(b^2)^n$ where m,n=0,1,2,3,…(m and n not being 0 simultaneously).Which shows that F accepts words that are not belonging to L.

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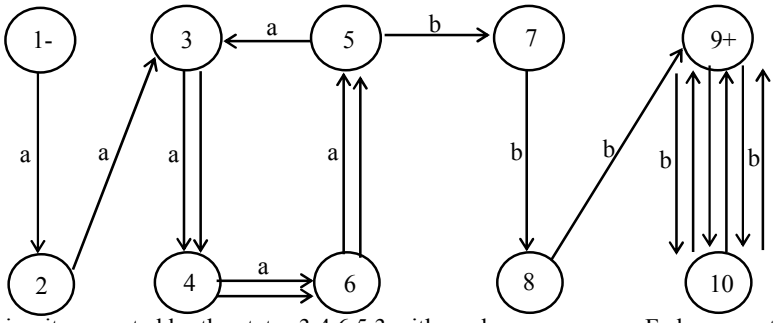
Theory of Automata

Lecture N0. 26
Reading Material

Introduction to Computer Theory Chapter 10

Summary
Example of nonregular language, pumping lemma version I, proof, examples

Example
Consider the language L = {Λ, ab, aabb, aaabbb, …} *i.e.* $\{a^n b^n : n=0,1,2,3,\ldots\}$
Suppose, it is required to prove that this language is nonregular. Let, contrary, L be a regular language then by Kleene's theorem it must be accepted by an FA, say, F. Since every FA has finite number of states then the language L (being infinite) accepted by F must have words of length more than the number of states. Which shows that, F must contain a circuit.
For the sake of convenience suppose that F has 10 states. Consider the word $a^9 b^9$ from the language L and let the path traced by this word be shown as under



But, looping the circuit generated by the states 3,4,6,5,3 with a-edges once more, F also accepts the word $a^{9+4} b^9$, while $a^{13}b^9$ is not a word in L. It may also be observed that, because of the circuit discussed above, F also accepts the words $a^9(a^4)^m b^9$, m = 1,2,3, …
Moreover, there is another circuit generated by the states 9,10,9. Including the possibility of looping this circuit, F accepts the words $a^9(a^4)^m b^9(b^2)^n$ where m,n=0,1,2,3,…(m and n not being 0 simultaneously).Which shows that F accepts words that are not belonging to L.
Similarly for finding FAs accepting other words from L, they will also accept the words which do not belong to L.
Thus there is no FA which accepts the language L. which shows, by Kleene's theorem, that the language L can't be expressed by any regular expression. It may be noted that apparently $a^n b^n$ seems to be a regular expression of L, but in fact it is not. The observations made from this example, generalize the theorem (also called the Pumping lemma) regarding the infinite regular language as follows

Pumping Lemma
Statement
Let L be any infinite regular language (that has infinite many words), defined over an alphabet ∑ then there exist three strings x, y and z belonging to Σ^* (where y is not the null string) such that all the strings of the form xy^nz for n=1,2,3, … are the words in L.

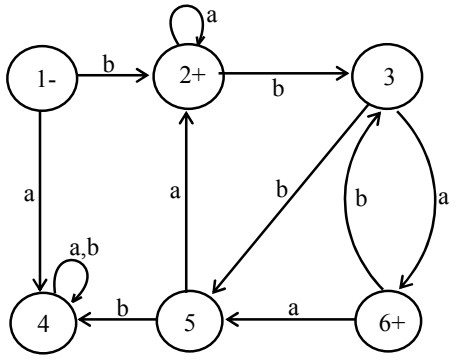
Proof
If L is a regular language, then according to Kleene's theorem, there exists an FA, say, F that accepts this language. Now F, by definition, must have finite no of states while the language has infinitely many words, which shows that there is no restriction on the length of words in L, because if there were such restriction then the language would have finite many words.
Let w be a word in the language L, so that the length of word is greater than the number of states in F. In this case the path generated by the word w, is such that it cannot visit a new state for each letter *i.e.* there is a circuit in this path.
The word w, in this case, may be divided into three parts
The substring which generates the path from initial state to the state which is revisited first while reading the word w. This part can be called x and x can be a null string.

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The substring which generates the circuit starting from the state which was lead by x. This part can be called as y which cannot be null string.
The substring which is the remaining part of the word after y, call this part as z. It may be noted that this part may be null string as the word may end after y or z part may itself be a circuit.
Thus the word may be written as w = xyz where x,y and z are the strings, also y can't be a null string.
Now this is obvious that, looping the circuit successively, the words xyyz, xyyyz, xyyyz, … will also be accepted by this FA *i.e.* xy^nz, n=1,2,3, … will be words in L.

Remark: In the above theorem, it is not affected if the z-part has circuit. To prove the theorem it is only to find a circuit and then looping that circuit, is all that is needed. While looping the circuit the volume of the string y (or z) is pumped, so the theorem is also called the Pumping lemma. Following are the examples

Example
Consider the following 5 states FA, say, F which accepts an infinite language



Let the word w = bbbababa, belonging to the language L, so that the length of word is greater than 6 (the number of states in F).
In this case the path generated by this word is such that it cannot visit a new state for each letter *i.e.* there is a circuit in this path.
The word w, in this case, may be divided into three parts.
The substring which generates the path from initial state to the state which is revisited first while reading the word w. This can be called as part x and this may be null string.
The substring which generates the circuit starting from the start state which was lead by x, this part can be called as y and this cannot be null string.
The substring which is the remaining part of the word after y, this part can be called as z. It may be noted that this part may be null string as the word may end after y or z-part may itself be a circuit.
Thus the word w may be written as w = xyz, where x,y,z are strings belonging to Σ^* and y cannot be null string.
The state 2 is such that it is revisited first while reading the word w. So the word w can be decomposed, according to pumping lemma, as w = xyz = (b)(bba)(baba)
If y-part of w is continuously pumped, the resulting strings will be accepted by F and hence will be words in the language accepted by F. Thus, by pumping lemma, the language accepted by F is regular.

Remark: If the pumping lemma is applied directly on the language L = $\{a^n b^n : n=0,1,2,3,\ldots\}$, it can be observed that for the word w = (aaa)(aaaabbbb)(bbb)
where x = aaa, y = aaaabbbb and z = bbb
xyyz will contain as many number of a's as there are b's but this string will not belong to L because the substring ab can occur at the most once in the words of L, while the string xyyz contains the substring ab twice. On the other hand if y-part consisting of only a's or b's, then xyyz will contain number of a's different from number of b's. This shows that pumping lemma does not hold and hence the language is not regular.

Example
Consider the language **EQUAL**, of strings, defined over Σ={a,b}, with number of a's equal to number of b's, *i.e.* **EQUAL** = {Λ ,ab,aabb,abab,baba,abba,…}
From the definition of **EQUAL,** it is clear that $\{a^n b^n\} = a^* b^* \cup$ **EQUAL**
Obviously $a^* b^*$ defines a regular language while $\{a^n b^n\}$ has been proved nonregular.
Using the theorem that intersection of two regular languages is, regular; it can be proved that the **EQUAL** is not regular. Because if it is considered regular then the language $\{a^n b^n\}$ will, being intersection of regular languages, be regular language, which is impossible.

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Following are the remarks regarding these examples

Remarks

In the previous examples, languages are proved to be regular or nonregular using pumping lemma. In fact to prove a certain language to be regular, it is not needed to use the full force of pumping lemma *i.e.* for a word with length greater than the number of states of the machine, decomposing the word into xyz and for a language to be regular it is sufficient that xyyz is in L. The condition that xy^nz is in L for n>2, provides that the language is infinite.

Consider the language PALINDROME and a word w = aba belonging to PALINDROME. Decomposing w = xyz where x=a, y=b, z=a. It can be observed that the strings of the form xy^nz for n=1,2,3, …, belong to PALINDROME. Which shows that the pumping lemma holds for the language PALINDROME (which is non regular language). To overcome this drawback of pumping lemma, a revised version of pumping lemma is to be introduced.

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Theory of Automata

Lecture N0. 27
Reading Material

Introduction to Computer Theory Chapter 10

Summary
Pumping lemma version II, proof, examples, Myhill Nerode theorem, examples

Pumping Lemma version II
Statement
Let L be an infinite language accepted by a finite automaton with N states, then for all words w in L that have langth more than N, there are strings x,y and z (y being non-null string) and length(x) + length(y) ≤ N s.t. w = xyz and all strings of the form xy^nz are in L for n = 1,2,3, …

Proof
The lemma can be proved, considering the following examples

Example
Consider the language PALINDROME which is obviously infinite language. It has already been shown that the PALINDROME satisfies pumping lemma version I (previous version). To check whether the new version of pumping lemma still holds in case of the PALINDROME, let the PALINDROME be a regular language and be accepted by an FA of 78 states. Consider the word $w = a^{85}ba^{85}$.
Decompose w as xyz, where x,y and z are all strings belonging to Σ^* while y is non-null string, s.t. length(x) + length(y) ≤ 78, which shows that the substring xy is consisting of a's and xyyz will become $a^{\text{more than } 85}ba^{85}$ which is not in PALINDROME. Hence pumping lemma version II is not satisfied for the language PALINDROME. Thus pumping lemma version II can't be satisfied by any non regular language. Following is another example in this regard

Example
Consider the language **PRIME**, of strings defined over Σ = {a}, as $\{a^p : p \text{ is prime}\}$, *i.e.*
PRIME = {aa, aaa, aaaaa, aaaaaaa, …}
To prove this language to be nonregular, suppose contrary, *i.e.* **PRIME** is a regular language, then there exists an FA accepts the language PRIME. Let the number of states of this machine be 345 and choose a word w from PRIME with length more than 345, say, 347 *i.e.* the word $w = a^{347}$
Since this language is supposed to be regular, therefore according to pumping lemma xy^nz, for n = 1,2,3,… are all in PRIME.
Consider n=348, then $xy^nz = xy^{348}z = xy^{347}yz$. Since x,y and z consist of a's, so the order of x, y, z does not matter *i.e.* $xy^{347}yz = xyzy^{347} = a^{347}y^{347}$, y being non-null string and consisting of a's it can be written $y = a^m$, m=1,2,3,…,345.
Thus $xy^{348}z = a^{347}(a^m)^{347} = a^{347(m+1)}$
Now the number 347(m+1) will not remain PRIME for m = 1,2,3, …, 345. Which shows that the string $xy^{348}z$ is not in PRIME. Hence pumping lemma version II is not satisfied by the language PRIME. Thus PRIME is not regular.

Strings belonging to same class
Consider a regular language L, defined over an alphabet ∑. If, two strings x and y, defined over ∑, are run over an FA accepting the language L, then x and y are said to belong to the same class if they end in the same state, no matter that state is final or not.

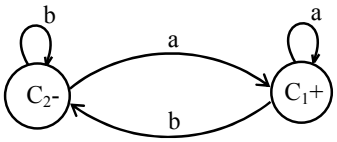
Note: It is to be noted that this concept of strings x and y can be compared with indistinguishable strings w.r.t. L (discussed earlier). Equivalently, the strings x and y are said to belong to same class if for all strings z, either xz and yz belong to L or xz and yz don't belong to L.

Myhill Nerode theorem
Statement
For a language L, defined over an alphabet∑,
L partitions Σ^* into distinct classes.

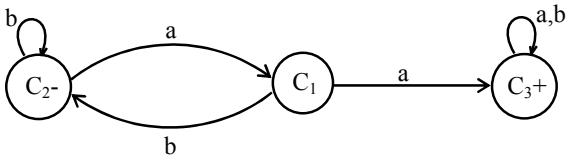
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If L is regular then, L generates finite number of classes.
If L generates finite number of classes then L is regular.
The proof is obvious from the following examples

Example
Consider the language L of strings, defined over ∑ = {a,b}, **ending in a**.
It can be observed that L partitions $\sum^*$ into the following two classes
C_1 = set of all strings ending in a.
C_2 = set of all strings not ending in a.
Since there are finite many classes generated by L, so L is regular and hence following is an FA, built with the help of C_1 and C_2, accepting L.



Example
Consider the language L of strings, defined over ∑ = {a,b}, **containing double a.** It can be observed that L partitions $\sum^*$ into the following three classes
C_1 = set of all strings without aa but ending in a.
C_2 = set of Λ and all strings without aa but ending in b.
C_3 = set of all strings containing aa.
Since there are finite many classes generated by L, so L is regular and hence following is an FA, built with the help of C_1, C_2 and C_3, accepting L.



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Theory of Automata

Lecture N0. 28
Reading Material

Introduction to Computer Theory Chapter 10

Summary

Examples of Myhill Nerode theorem, Quotient of a language, examples, Pseudo theorem: Quotient of a language is regular, prefixes of a language, example

Example

Consider the language L which is EVEN-EVEN, defined over Σ = {a,b}. It can be observed that L partitions Σ* into the following four classes

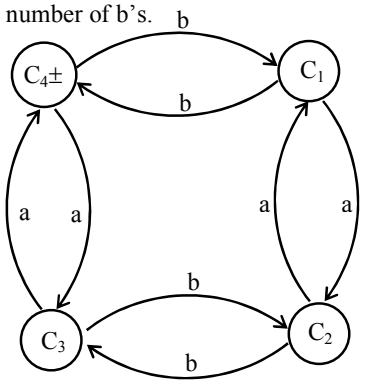
C_1 = set of all strings with even number of a's and odd number of b's.

C_2 = set of all strings with odd number of a's and odd number of b's.

C_3 = set of all strings with odd number of a's and even number of b's.

C_4 = set of all strings with even number of a's and even number of b's.

Since there are finite many classes generated by L, so L is regular and hence following is an FA, built with the help of C_1, C_2, C_3 and C_4, accepting L.



Example

Consider the language L = {w ∈ {a,b}*: length(w) ≥ 2, w ends in either ab or ba}.

It can be observed that L partitions Σ* into the following seven classes

C_1 = set containing only null string.

C_2 = set containing only letter a.

C_3 = set containing only letter b.

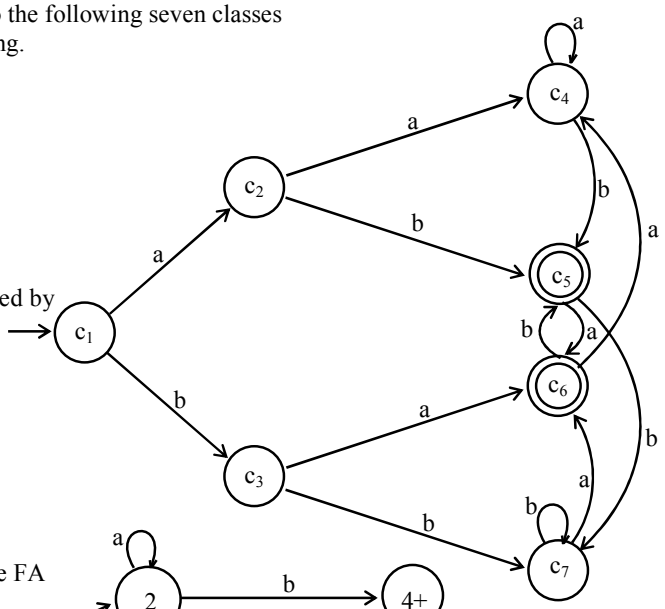
C_4 = set of strings ending in aa.

C_5 = set of strings ending in ab.

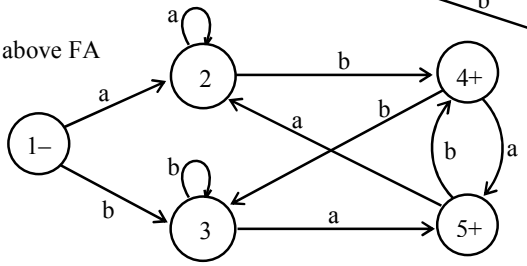
C_6 = set of strings ending in ba.

C_7 = set of strings ending in bb.

Since there are finite many classes generated by L, so L is regular and hence FA shown aside, is built with the help of C_1, C_2, C_3 , C_4, C_5 , C_6 and C_7, accepting L



Following is an FA equivalent to the above FA



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Note It can be noted, from the above two FAs accepting the same language, that if the language L, partitions Σ^* into n distinct classes, then L may partition Σ^* into finite many distinct classes other than n.

Quotient of a language into another

Remark The theorem has been proved to show under what conditions a language is regular. It has also been proved that the product of two regular languages is regular.

The question arises that whether there exists a theorem showing that quotient of regular languages is regular. There is a problem in defining the quotient of two regular languages. There is an approach in defining the quotient of regular languages i.e. the language Q is said to be **quotient of two regular languages** P and R, denoted by Q=R/P if PQ=R.

It is to be noted that this definition does not determine a unique language e.g. for P=Q=R expressed by a* then PQ=R and so Q=R/P i.e. a*=a* / a*. But for Q={∧}, P=R expressed by a*, PQ=R is still true which shows that Q={∧}=R/P expressed by a* / a*

Similarly, for the same P and R, Q may be taken as {∧},{a},{aaaa},{aaaaaaaa}, … Thus there exist infinite many choices for defining the quotient language in this case of one-letter alphabet.

Pseudo theorem

Statement

For three languages P,Q and R, while PQ=R the language Q must be regular if both P and R are regular.

(Note: It is to be noted that since this theorem is not true, so the theorem is called pseudo theorem.)

Disproof

The theorem can be disproved by contradiction i.e. supposing that Q is regular.

Let P=a*, Q be the product of $\{a^nb^n:n=0,1,2,\ldots\}$ and b* then PQ=a*$\{a^nb^n\}$b*=a*b*=R which shows that R is regular.

To disproof this theorem, it is sufficient to prove that Q is not regualr. By definition, the words in Q are of the form a^xb^y where $x \leq y$. Let Q be regualr and hence there exists an FA that accepts Q. Suppose the number of states in this machine be N. Now the word a^Nb^N is also in Q and must be accepted by this FA.

Since the number of states in this machine is N, there must be a circuit in this machine to run the substring a^N. Thus while accepting the word a^Nb^N, the machine looping the circuit once again, can accept the word $a^{\text{more than N}}b^N$, which is not in Q. Hence it is impossible to find any FA that accepts exactly the language Q. Thus Q is not regular and hence the theorem is disproved.

Prefixes of a language in another language

If two languages R and Q are given, then the language **the prefixes of Q in R** denoted by **Pref(Q in R)** is the set of strings of letters that, when concatenated to the front of some word in Q to produce some word in R *i.e.* Pref(Q in R) = the set of all strings p such that there exists words q in Q and w in R such that pq = w. Following are the examples in this regard

Example

Let Q = {aa,abaaabb,bbaaaaa,bbbbbbbbbb} and R = {b,bbbb,bbbaaa,bbbaaaaa}

It can be observed that aa and bbaaaaa occur at the ending parts of some words of R, hence these words help in defining the language pref(Q in R). Thus pref(Q in R) = {b,bbba,bbbaaa}

Note*:* The language of prefixes may be consisting of word L, while there is also a possibility that this language may not contain any string (even not the null string).

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Theory of Automata

Lecture N0. 29
Reading Material

Introduction to Computer Theory Chapter 10, 11

Summary
Example of prefixes of a language, Theorem: pref(Q in R) is regular, proof, example, Decidablity, deciding whether two languages are regular or not ?, method 1, example, method 2, example

Example
Let Q and R be expressed by ab*a and (ba)* respectively i.e. Q={aa,aba,abba, …} and R={∧, ba, baba, bababa, …}. aba is the only word in Q which can make a word in R, because the words in R don't contain the double letter. Thus pref(Q in R) = {b, bab, babab, …}, which can be expressed by b(ab)* or (ba)*b.

Remark
It may be noted that the language R cannot be factorized with the help of language Pref(Q in R) *i.e.* Pref(Q in R)Q is not equal to R in general. However the following theorem shows that the language pref(Q in R) is regular if R is regular, no matter whether the language Q is regular or not.

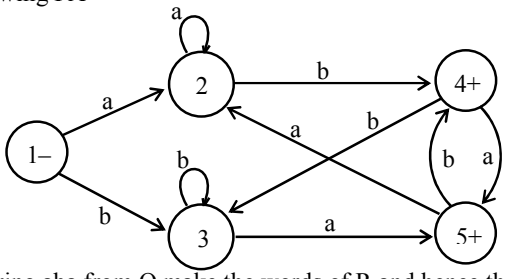
Theorem
Statement
If R is regular language and Q is any language (regular/ nonregular), then Pref(Q in R) is regular.

Proof
Since R is regular there exists an FA that accepts this language. Choose a state, say, s of this FA and see whether this state can trace out a path ending up in a final state while running words from Q. If this state traces out a path ending up in a final state for any of the words of Q, mark this state with certain colour.
Repeat this process for remaining states of the machine. If at least one state of this machine is marked then it can be shown that the language Pref(Q in R) is non-empty. Now build a new FA with some marked states by considering the initial state that of original FA and final states which are marked. The machine, thus obtained accepts exactly the language Pref(Q in R). Thus Pref(Q in R) being accepted by an FA is regular.

Remark
There is a problem in deciding whether a state of FA should be marked or not when the language Q is infinite. This proof just gives non constructive method to prove that Pref(Q in R) is regular.

Example

Consider the languages Q={aba, abb} and R = {w ∈ {a,b}*: length(w) ≥ 2, w ends in either ab or ba}, where R may be accepted by the following FA



It can be observed that the string aba from Q make the words of R and hence the states 1,2,3,4 and 5 can easily be marked. Thus from the given FA, making the states 1, 2 and 3 to be final as well, the resulting FA will accept the language pref(Q in R). Moreover it can be observed that pref(Q in R) can be expressed by (a+b)*, which is the RE corresponding to the resulting FA as well.

Decidability
Effectively solvable problem

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A problem is said to be effectively solvable if there exists an algorithm that provides the solution in finite number of steps *e.g.* finding solution for quadratic equation is effectively solvable problem, because the quadratic formula provides an algorithm that determines the solution in a finite number of arithmetic operations, (four multiplications, two subtractions, one square root and one division).

Decision procedure

If an effectively solvable problem has answer in yes or no, then this solution is called decision procedure.

Decidable problem

A problem that has decision procedure is called decidable problem *e.g.* the following problems
The two regular expressions define the same language.
The two FAs are equivalent.

Determining whether the two languages are equivalent or not ?

If L_1 and L_2 are two regular languages, then they can be expressed by FAs. As shown earlier, L_1^c , L_2^c , $L_1 \cup L_2$, $L_1 \cap L_2$ are regular languages and the methods have already been developed to build their corresponding FAs. It can be observed that $(L_1 \cap L_2^c) \cup (L_1^c \cap L_2)$ is regular language that accepts the words which are in L_1 but not in L_2 or else in L_2 but not in L_1 . The corresponding FA cannot accept any word which is in both L_1 and L_2 i.e. if L_1 and L_2 are equivalent, then this FA accepts not even null string. Following are the methods that determine whether a given FA accepts any words

Method 1

For FA corresponding to $(L_1 \cap L_2^c) \cup (L_1^c \cap L_2)$, the regular expression can be determined that defines the language accepted by this FA. From that regular expression one can determine whether this regular expression defines any word or not? Following are the steps to be followed
Remove all *s from the regular expression.
Separate the right part of + and the plus itself.

The regular expression thus obtained if contains atleast one word then the language is not empty otherwise the language is empty.

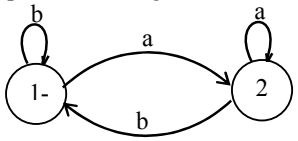
Example

For $(a+\wedge)(a^*b+ba^*)(a^*+\wedge)^*$ to be the regular expression of $(L_1 \cap L_2^c) \cup (L_1^c \cap L_2)$, it is required to find whether this language accepts any string or not?
After removing all *s the RE will be $(a+\wedge)(ab+ba)(a+\wedge)$
After separating the right part from + and the + itself the RE will be aaba
As this language contains atleast aaba as its word, so this language is not empty.

Remark

Sometimes, while determining regular expression for a given FA, it is impossible to write its regular expression *e.g.*

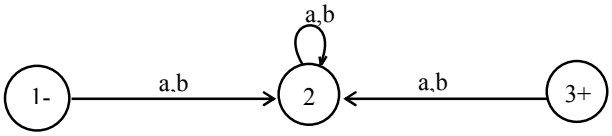
FA_1



FA_2



FA_3



Method 2

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To examine whether a certain FA accepts any words, it is required to seek the paths from initial to final state. But in large FA with thousnads of states and millions of directed edges, without an effective procedure it is impossible to find a path from initial to final state. Following are the steps of this procedure

Mark the initial state.

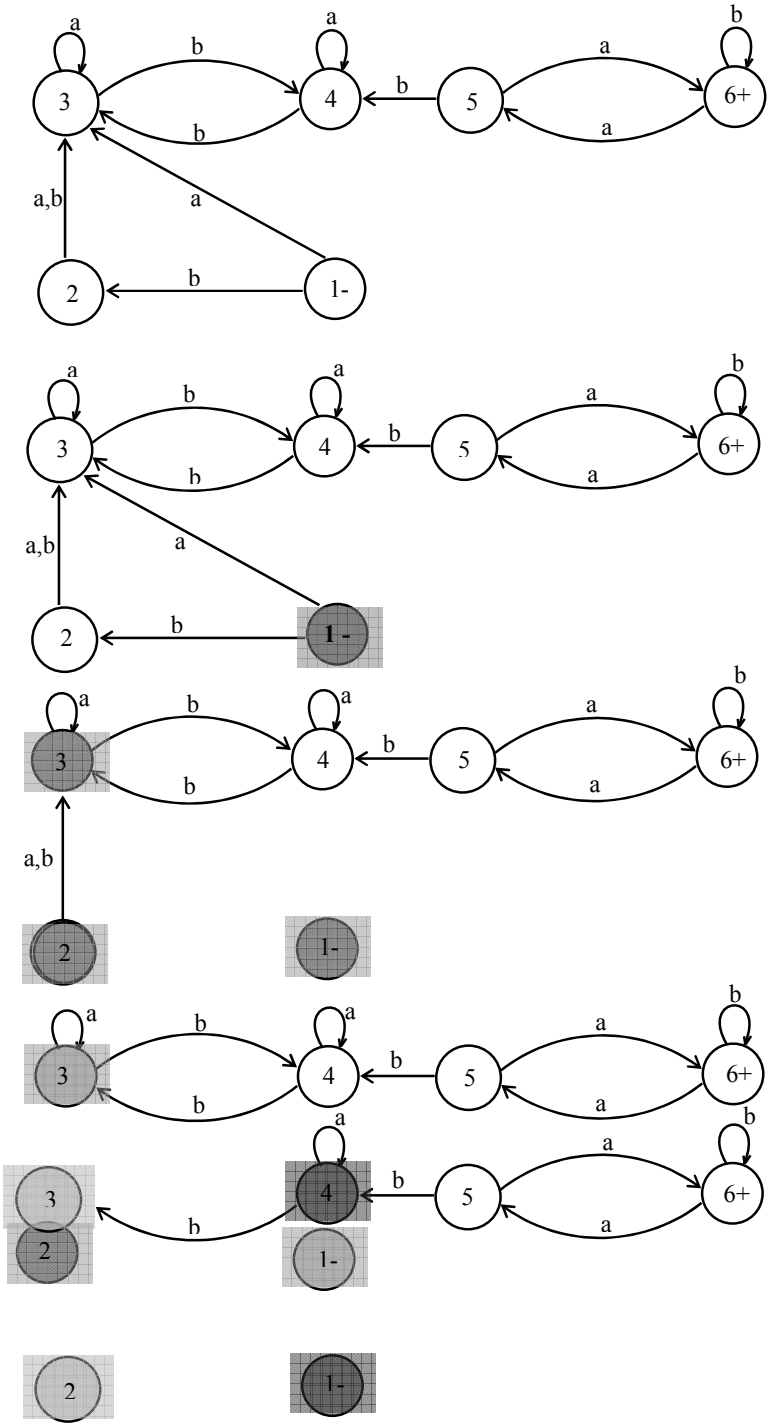
For every marked state follow each edge that leads out of it and marked the concatenating states and delete these edges.

Repeat step 2 until no new state is marked.

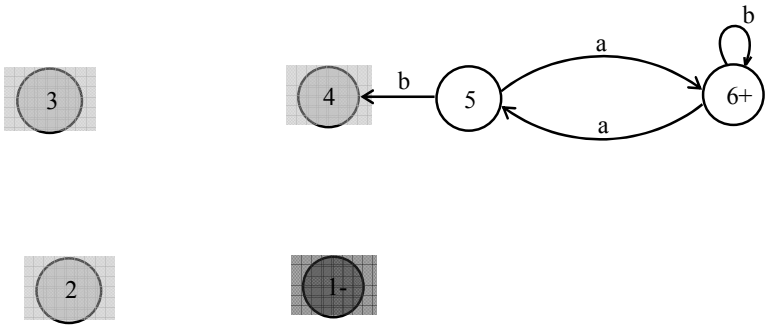
If any of the final states are marked then the FA accepts some word, otherwise not.

Example

Suppose it is required to test the FA, which is given below, whether it accepts any string or not? Applying method 2 as



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This FA accepts no string as after applying method 2, the final state is not marked.

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Theory of Automata

Lecture N0. 30
Reading Material

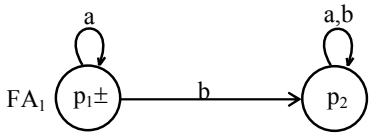
Introduction to Computer Theory Chapter 11,12

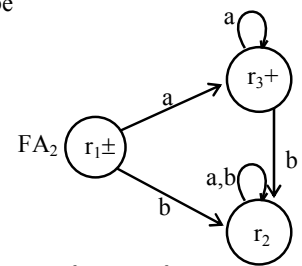
Summary

Deciding whether two languages are equivalent or not, example, deciding whether an FA accept any string or not, method 3, examples, finiteness of a language

Example

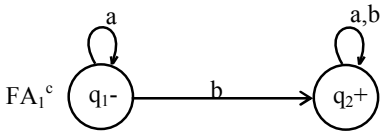
Consider two languages L_1 and L_2, expressed by the REs $r_1=a^*$ and $r_2=\wedge+aa^*$ respectively. To determine whether L_1 and L_2 are equivalent, let the corresponding FAs be

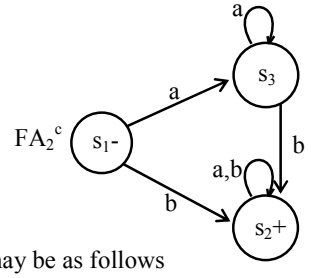




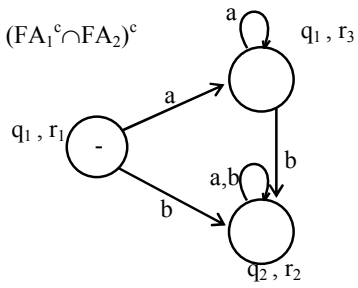
As discussed earlier, the FA corresponding to the language $(L_1 \cap L_2^c) \cup (L_1^c \cap L_2)$ helps in this regard *i.e.* if this FA accepts any word then L_1 and L_2 are not equivalent other wise L_1 and L_2 are equivalent.

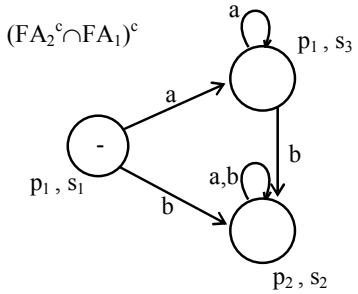
Following are the FAs corresponding to L_1^c and L_2^c





FAs corresponding to $(FA_1^c \cap FA_2)^c$ and $(FA_2^c \cap FA_1)^c$ may be as follows



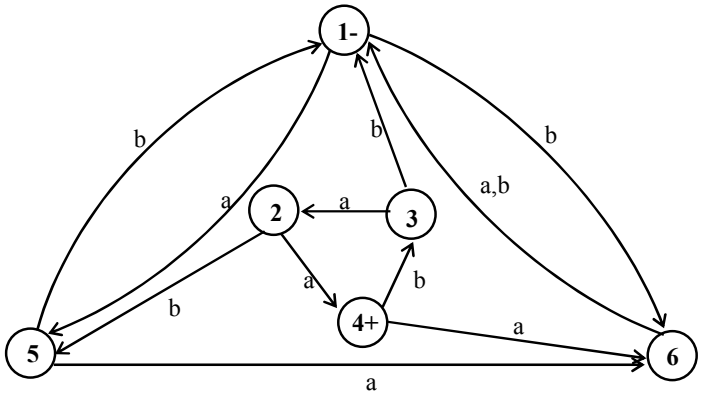


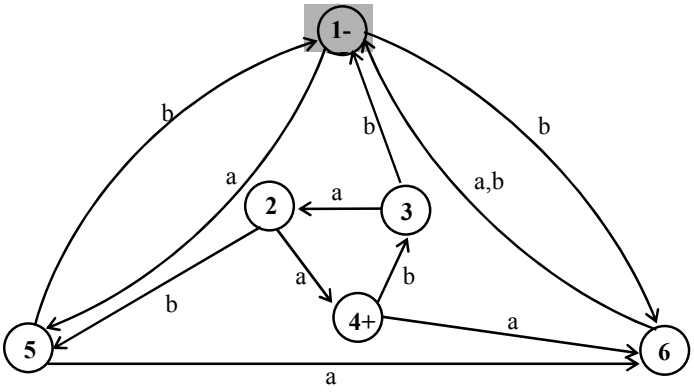
Both the FAs have no final state, so these FAs accept nothing. This implies that their union will not also accept any string. Hence FA corresponding to the language $(L_1 \cap L_2^c) \cup (L_1^c \cap L_2)$ accepts nothing. Thus both the languages are equivalent.

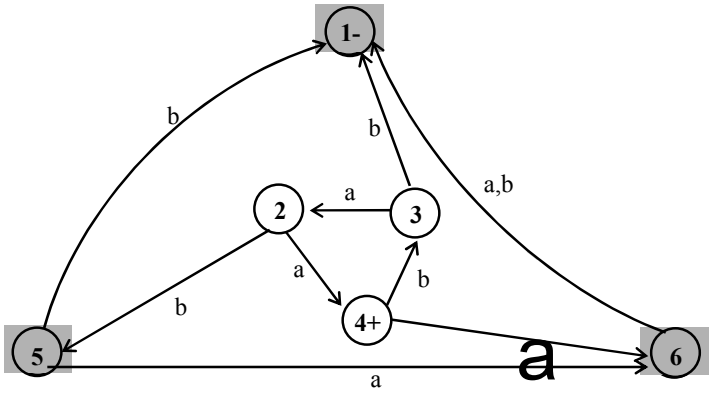
Example

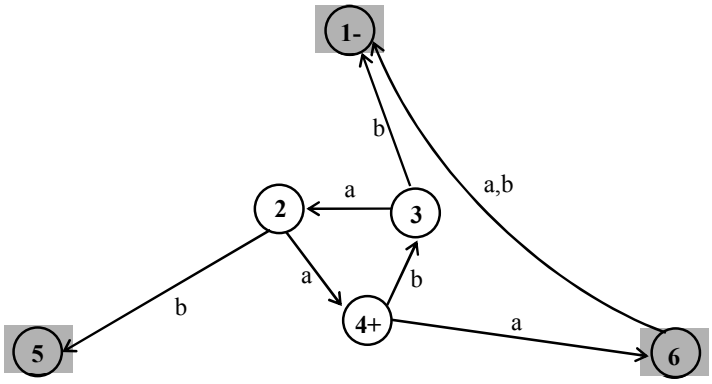
Following is an FA, for which it is required to decide whether it accepts any string or not? Using steps discussed in method 2, the following FA can be checked whether it accepts any word or not.

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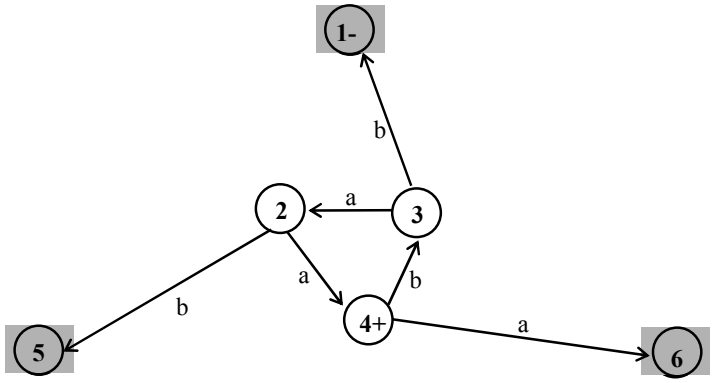








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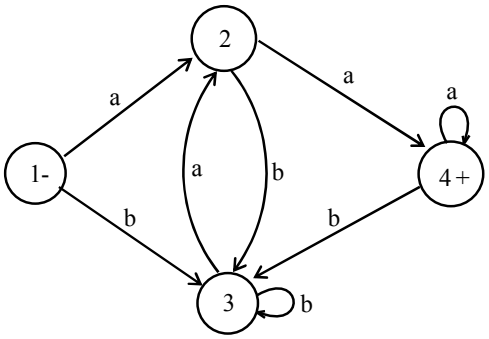
As no final state of the FA is marked, so the given FA accepts no word.

Method 3

If the FA has N states, then test the words of length less than N. If no word is accepted by this FA, then it will accept no word.

Note: It is to be noted that all the methods discussed above, to determine whether an FA accepts certain word, are effective procedures.

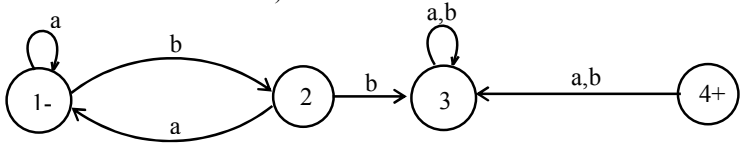
Example: To determine whether the following FA accepts certain word, using method 3, all the strings of length less than 4 (*i.e.* less than the number of states of the FA) are sufficient to be tested



i.e. ∧, a, b, aa, ab, ba, bb, aaa, aab, aba, abb, baa, bab, bba, bbb are required to be tested.
It can be observed that the strings aa, baa, aaa are accepted by this FA, hence the language accepted by this FA is not empty.

Example

Consider the following FA. To determine whether this FA accepts some word, all the strings of length less than 4 (*i.e.* less than the number of states of this FA) are to be tested



It can be observed that none of the strings Λ, a, b, aa, ab, ba, bb, aaa, aab, aba, abb, baa, bab, bba, bbb, is accepted by this FA. Thus the given FA cannot accept any word.

Observation

To find whether a regular expression defines an infinite language or not? The following possibilities are required to be checked.

If a regular expression contains * then it may define an infinite language, with the exception $\wedge^*$ as $\wedge^* = \wedge$ *e.g.* $(\wedge+a\wedge^*)(\wedge^*+\wedge)^*$ defines finite language. While $(\wedge^*+a\wedge^*)^*$ $(\wedge^*+\wedge)^*$ defines an infinite language.

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There are so many ways to decide whether an FA accepts a finite language or an infinite. Following is a theorem in this regard

Theorem

Let F be an FA having N states
If F accepts a word w s.t. $N \leq \text{length}(w) < 2N$
then F accepts infinite language.
If F accepts an infinite language then there are some words w s.t. $N \leq \text{length}(w) < 2N$
The first part can be proved, using the pumping lemma version II.

Remark

There is an effective procedure to decide whether an FA accepts finite or infinite language. For a machine with N number of states, the total number of strings to be tested, defined over an alphabet of m letters, is $m^N + m^{N+1} + m^{N+2} + \ldots + m^{2N-1}$. After testing all these strings on the machine, if any is accepted then the machine accepts infinite language other wise not. *e.g.* for machine of three states and alphabet of two letters, $2^3 + 2^4 + 2^5 = 56$ strings are to be tested.

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Theory of Automata

Lecture N0. 31
Reading Material

Introduction to Computer Theory Chapter 12

Summary

Context Free Grammar, Terminals, non-terminals, productions, CFG, context Free language, examples.

Context Free Grammar (CFG)

The earliest computers accepted no instructions other then their own assembly language. Every procedure, no matter how complicated , had to be encoded in the set of instructions, LOAD, STORE, ADD the contents of two registers and so on. The major problem was to display mathematical formulas as follows

$$S = \sqrt{\frac{(8-0)2+(7-10)2+(11-10)2}{2}}$$

or

$$A = \frac{\frac{1}{2}+9}{4+\frac{8}{21}+\frac{5}{3+\frac{1}{2}}}$$

So, it was necessary to develop a way of writing such expressions in one line of standard typewriter symbols, so that in this way a high level language could be invented. Before the invention of computers, no one would ever have dreamed of writing such complicated formula in parentheses *e.g.* the right side of formula can be written as

((1/2)+9)/(4+(8/21)+(5/(3+(1/2))))

The high level language is converted into assembly language codes by a program called compiler.
The compiler that takes the user's programs as its inputs and prints out an equivalent program written in assembly language.
Like spoken languages, high level languages for computer have also, certain grammar. But in case of computers, the grammatical rules, don't involve the meaning of the words.
It can be noted that the grammatical rules which involve the meaning of words are called **Semantics**, while those don't involve the meaning of the words are called **Syntactics**.
e.g. in English language, it can not be written " Buildings sing ", while in computer language one number is as good as another.
e.g. X = B + 10, X = B + 999

Remark
In general, the rules of computer language grammar, are all syntactic and not semantic. A law of grammar is in reality a suggestion for possible substitutions.

CFG terminologies

Terminals: The symbols that can't be replaced by anything are called terminals.
Non-Terminals: The symbols that must be replaced by other things are called non-terminals.
Productions: The grammatical rules are often called productions.

CFG

CFG is a collection of the followings
An alphabet Σ of letters called terminals from which the strings are formed, that will be the words of the language.
A set of symbols called non-terminals, one of which is S, stands for "start here".
A finite set of productions of the form
non-terminal → finite string of terminals and /or non-terminals.

Note
The terminals are designated by small letters, while the non-terminals are designated by capital letters.
There is at least one production that has the non-terminal S as its left side.

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Context Free Language (CFL)

The language generated by CFG is called Context Free Language (CFL).

Example

$\Sigma = \{a\}$

productions:

$S \rightarrow aS$

$S \rightarrow \wedge$

Applying production (1) six times and then production (2) once, the word aaaaaa is generated as

$S \Rightarrow aS$

$\Rightarrow aaS$

$\Rightarrow aaaS$

$\Rightarrow aaaaS$

$\Rightarrow aaaaaS$

$\Rightarrow aaaaaa\Lambda$

$= aaaaaa$

It can be observed that prod (2) generates Λ, a can be generated applying prod. (1) once and then prod. (2), aa can be generated applying prod. (1) twice and then prod. (2) and so on. This shows that the grammar defines the language expressed by a*.

Example

$\Sigma = \{a\}$

productions:

$S \rightarrow SS$

$S \rightarrow a$

$S \rightarrow \Lambda$

This grammar also defines the language expressed by a*.

Note: It is to be noted that Λ is not considered to be terminal. It has a special status. If for a certain non-terminal N, there may be a production $N \rightarrow \Lambda$. This simply means that N can be deleted when it comes in the working string.

Example

$\Sigma = \{a,b\}$

productions:

$S \rightarrow X$

$S \rightarrow Y$

$X \rightarrow \Lambda$

$Y \rightarrow aY$

$Y \rightarrow bY$

$Y \rightarrow a$

$Y \rightarrow b$

All words of this language are of either X-type or of Y-type. *i.e.* while generating a word the first production used is $S \rightarrow X$ or $S \rightarrow Y$. The words of X-type give only Λ, while the words of Y-type are words of finite strings of a's or b's or both *i.e.* $(a+b)^+$. Thus the language defined is expressed by $(a+b)^*$.

Example

$\Sigma = \{a,b\}$

productions:

$S \rightarrow aS$

$S \rightarrow bS$

$S \rightarrow a$

$S \rightarrow b$

$S \rightarrow \Lambda$

This grammar also defines the language expressed by $(a+b)^*$.

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Example

$\Sigma = \{a,b\}$

productions:

$S \rightarrow XaaX$

$X \rightarrow aX$

$X \rightarrow bX$

$X \rightarrow \Lambda$

This grammar defines the language expressed by (a+b)*aa(a+b)*.

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Theory of Automata

Lecture N0. 32
Reading Material

Introduction to Computer Theory Chapter 12

Summary
Examples of CFL, EVEN-EVEN, EQUAL, Language of strings containing bbb, PALINDROME, $\{a^n b^n\}$, Language of strings beginning and ending in different letters, Parsing tree, example

Example
$\Sigma = \{a,b\}$
productions:
$S \rightarrow SS$
$S \rightarrow XS$
$S \rightarrow \Lambda$
$S \rightarrow YSY$
$X \rightarrow aa$
$X \rightarrow bb$
$Y \rightarrow ab$
$Y \rightarrow ba$
This grammar generates EVEN-EVEN language.

Example
$\Sigma = \{a,b\}$
productions:
$S \rightarrow aB$
$S \rightarrow bA$
$A \rightarrow a$
$A \rightarrow aS$
$A \rightarrow bAA$
$B \rightarrow b$
$B \rightarrow bS$
$B \rightarrow aBB$
This grammar generates the language EQUAL(The language of strings, with number of a's equal to number of b's).

Note
It is to be noted that if the same non-terminal have more than one productions, it can be written in single line *e.g.* $S \rightarrow aS$, $S \rightarrow bS$, $S \rightarrow \Lambda$ can be written as $S \rightarrow aS|bS|\Lambda$
It may also be noted that the productions $S \rightarrow SS|\Lambda$ always defines the language which is closed w.r.t. concatenation *i.e.*the language expressed by RE of type r*. It may also be noted that the production $S \rightarrow SS$ defines the language expressed by r^+.

Example
$\Sigma = \{a,b\}$
productions:
$S \rightarrow YXY$
$Y \rightarrow aY|bY|\Lambda$
$X \rightarrow bbb$
It can be observed that, using prod.2, Y generates Λ. Y generates a. Y generates b. Y also generates all the combinations of a and b. thus Y generates the strings generated by (a+b). It may also be observed that the above

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CFG generates the language expressed by (a+b)*bbb(a+b)*. Following are four words generated by the given CFG

$S \Rightarrow YXY$
$\Rightarrow aYbbb\Lambda$
$\Rightarrow abYbbb$
$\Rightarrow ab\Lambda bbb$
$= abbbb$

$S \Rightarrow YXY$
$\Rightarrow bYbbbaY$
$\Rightarrow b\Lambda bbbabY$
$\Rightarrow bbbbabbY$
$\Rightarrow bbbbabbaY$
$\Rightarrow bbbbabba\Lambda$
$= bbbbabba$

$S \Rightarrow YXY$
$\Rightarrow \Lambda bbbaY$
$\Rightarrow bbbabY$
$\Rightarrow bbbabaY$
$\Rightarrow bbbaba\Lambda$
$= bbbaba$

$S \Rightarrow YXY$
$\Rightarrow bYbbbaY$
$\Rightarrow b\Lambda bbba\Lambda$
$= bbbba$

Example
Consider the following CFG
$S \rightarrow SS|XaXaX|\Lambda$
$X \rightarrow bX|\Lambda$
It can be observed that, using prod.2, X generates Λ. X generates any number of b's. Thus X generates the strings generated by b*. It may also be observed that the above CFG generates the language expressed by (b*ab*ab*)*.

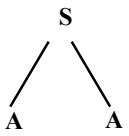
Example
Consider the following CFG
$\Sigma = \{a,b\}$
productions:
$S \rightarrow aSa|bSb|a|b|\Lambda$
The above CFG generates the language PALINDROME. It may be noted that the CFG
$S \rightarrow aSa|bSb|a|b$ generates the language NON-NULLPALINDROME.

Example
Consider the following CFG
$\Sigma = \{a,b\}$
productions:
$S \rightarrow aSb|ab|\Lambda$
It can be observed that the CFG generates the language $\{a^n b^n: n = 0,1,2,3, \ldots\}$. It may also be noted that the language $\{a^n b^n: n=1,2,3, \ldots\}$ can be generated by the CFG, $S \rightarrow aSb|ab$

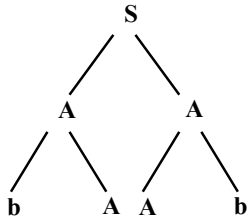
Example
Consider the following CFG
$S \rightarrow aXb|bXa$
$X \rightarrow aX|bX|\Lambda$
The above CFG generates the language of strings, defined over $\Sigma=\{a,b\}$, **beginning and ending in different letters**.

Trees
As in English language any sentence can be expressed by parse tree, so any word generated by the given CFG can also be expressed by the parse tree, *e.g.* consider the following CFG
$S \rightarrow AA$
$A \rightarrow AAA|bA|Ab|a$
Obviously, baab can be generated by the above CFG. To express the word baab as a parse tree, start with S. Replace S by the string AA, of nonterminals, drawing the downward lines from S to each character of this string as follows

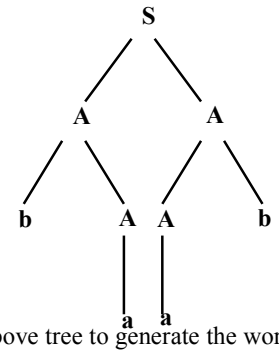
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Now let the left A be replaced by bA and the right one by Ab then the tree will be



Replacing both A's by a, the above tree will be



Thus the word baab is generated. The above tree to generate the word baab is called **Syntax tree or Generation tree or Derivation tree as well.**

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Theory of Automata

Lecture N0. 33
Reading Material

Introduction to Computer Theory Chapter 12

Summary

Example of trees, Polish Notation, examples, Ambiguous CFG, example

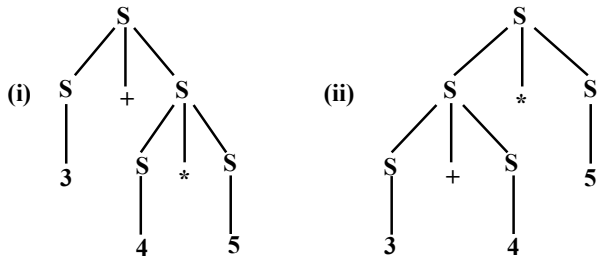
Example

Consider the following CFG

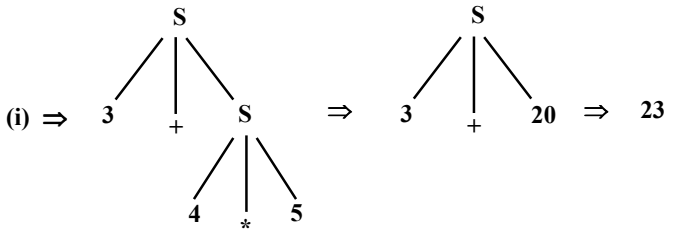
$S \rightarrow S+S|S*S|\underline{number}$

where S and number are non-terminals and the operators behave like terminals.

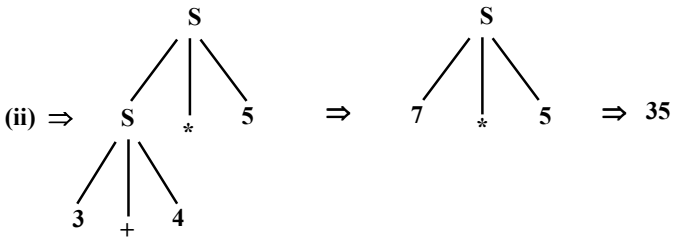
The above CFG creates ambiguity as the expression 3+4*5 has two possibilities (3+4)*5=35 and 3+(4*5)=23 which can be expressed by the following production trees



The expressions can be calculated starting from bottom to the top, replacing each nonterminal by the result of calculation *e.g.*



Similarly



The ambiguity that has been observed in this example can be removed with a change in the CFG as discussed in the following example

Example

$S \rightarrow (S+S)|(S*S)|\underline{number}$

where S and number are nonterminals, while (, *, +,) and the numbers are terminals.

Here it can be observed that

$S \Rightarrow (S+S)$
$\Rightarrow (S+(S*S))$
$\Rightarrow (3+(4*5)) = 23$

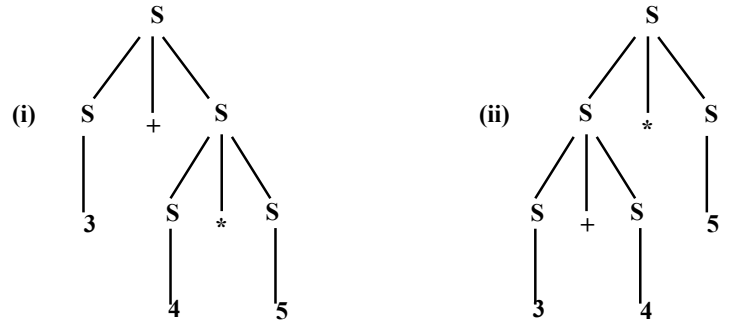
$S \Rightarrow (S*S)$
$\Rightarrow ((S+S)*S)$

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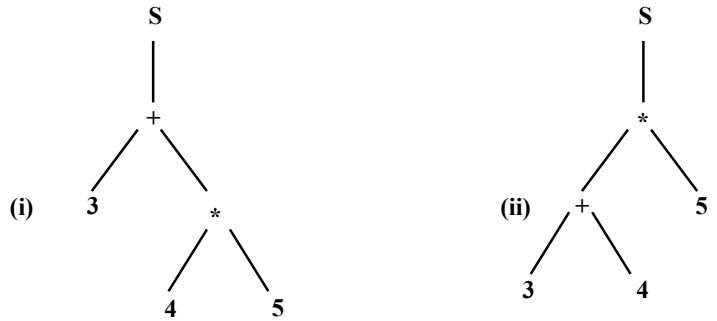
$\Rightarrow ((3+4)*5) = 35$

Polish Notation (o-o-o)

There is another notation for arithmetic expressions for the CFG, S→S+S|S*S|number. Consider the following derivation trees



The arithmetic expressions shown by the trees (i) and (ii) can be calculated from the following trees, respectively



Here most of the S's are eliminated.

The branches are connected directly with the operators. Moreover, the operators + and * are no longer terminals as these are to be replaced by numbers (results).

To write the arithmetic expression, it is required to traverse from the left side of S and going onward around the tree. The arithmetic expressions will be as under

(i) + 3 * 4 5 (ii) * +3 4 5

The above notation is called operator prefix notation.

To evaluate the strings of characters, the first substring (from the left) of the form operator-operand-operand (o-o-o) is found and is replaced by its calculation *e.g.*

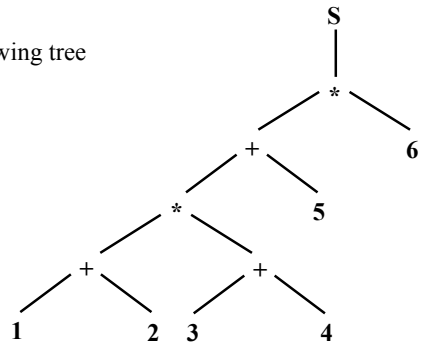
+3*4 5 = +3 20 = 23

*+3 4 5 = * 7 5 = 35

It may be noted that 4*5+3 is an infix arithmetic expression, while an arithmetic expression in (o-o-o) form is a prefix arithmetic expression.

Example

To calculate the arithmetic expression of the following tree



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It can be written as *+*+1 2+3 4 5 6
The above arithmetic expression in (o-o-o) form can be calculated as
++1 2+3 4 5 6 = *+*3+3 4 5 6
= *+*3 7 5 6 = *+21 5 6 = *26 6 = 156.

Note

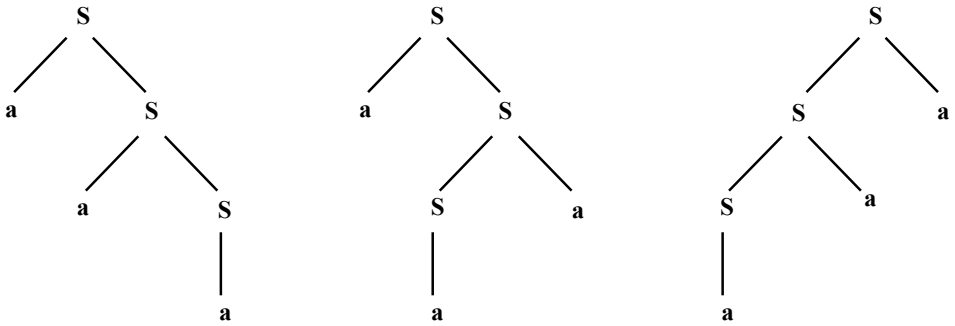
The previous prefix arithmetic expression can be converted into the following infix arithmetic expression as

++1 2+3 4 5 6
= *+*+1 2 (3+4) 5 6
= *+*(1+2) (3+4) 5 6
= *(((1+2)*(3+4)) + 5) 6
= (((1+2)*(3+4)) + 5)*6

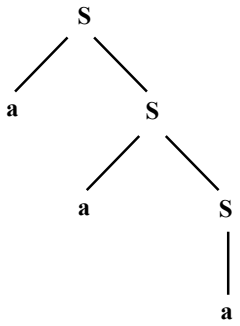
Ambiguous CFG

The CFG is said to be ambiguous if there exists atleast one word of it's language that can be generated by the different production trees.

Example: Consider the following CFG
S→aS|Sa|a
The word aaa can be generated by the following three different trees



Thus the above CFG is ambiguous, while the CFG, S→aS|a is not ambiguous as neither the word aaa nor any other word can be derived from more than one production trees. The derivation tree for aaa is as follows



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Theory of Automata

Lecture N0. 34
Reading Material

Introduction to Computer Theory Chapter 12,13

Summary
Example of Ambiguous Grammar, Example of Unambiguous Grammer (PALINDROME), Total Language tree with examples (Finite and infinite trees), Regular Grammar, FA to CFG, Semi word and Word, Theorem, Defining Regular Grammar, Method to build TG for Regular Grammar

Example
Consider the following CFG

$S \rightarrow aS \mid bS \mid aaS \mid \Lambda$

It can be observed that the word aaa can be derived from more than one production trees. Thus, the above CFG is ambiguous. This ambiguity can be removed by removing the production $S \rightarrow aaS$

Example
Consider the CFG of the language PALINDROME

$S \rightarrow aSa|bSb|a|b|\Lambda$

It may be noted that this CFG is unambiguous as all the words of the language PALINDROME can only be generated by a unique production tree.
It may be noted that if the production $S \rightarrow aaSaa$ is added to the given CFG, the CFG thus obtained will be no more unambiguous.

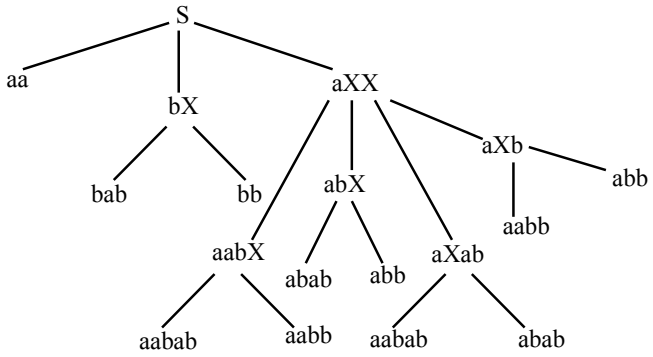
Total language tree
For a given CFG, a tree with the start symbol S as its root and whose nodes are working strings of terminals and non-terminals. The descendants of each node are all possible results of applying every production to the working string. This tree is called **total language tree**.

Example
Consider the following CFG

$S \rightarrow aa|bX|aXX$

$X \rightarrow ab|b$

then the total language tree for the given CFG may be



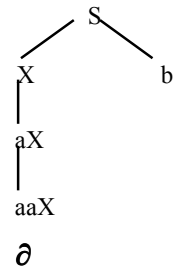
It may be observed from the above total language tree that dropping the repeated words, the language generated by the given CFG is {aa, bab, bb, aabab, aabb, abab, abb}

Example
Consider the following CFG

$S \rightarrow X|b$, $X \rightarrow aX$

then following will be the total language tree of the above CFG

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Note: It is to be noted that the only word in this language is b.

aaa …aX

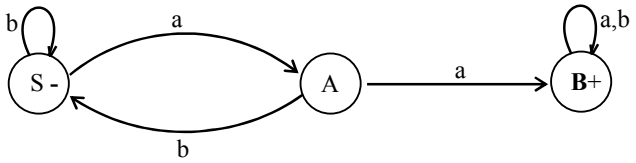
Regular Grammar

All regular languages can be generated by CFGs. Some nonregular languages can be generated by CFGs but not all possible languages can be generated by CFG, *e.g.* the CFG $S \rightarrow aSb|ab$ generates the language $\{a^n b^n : n=1,2,3, \ldots\}$, which is nonregular.

Note: It is to be noted that for every FA, there exists a CFG that generates the language accepted by this FA.

Example

Consider the language L expressed by **(a+b)*aa(a+b)*** *i.e.*the language of strings, defined over $\sum = \{a,b\}$, **containing aa.** To construct the CFG corresponding to L, consider the FA accepting L, as follows



CFG corresponding to the above FA may be

$S \rightarrow bS|aA$

$A \rightarrow aB|bS$

$B \rightarrow aB|bB|\Lambda$

It may be noted that the number of terminals in above CFG is equal to the number of states of corresponding FA where the nonterminal S corresponds to the initial state and each transition defines a production.

Semiword

A semiword is a string of terminals (may be none) concatenated with exactly one nonterminal on the right *i.e.* a semiword, in general, is of the following form

(terminal)(terminal)… (terminal)(nonterminal)

word

A word is a string of terminals. Λ is also a word.

Theorem

If every production in a CFG is one of the following forms

Nonterminal → semiword

Nonterminal → word

then the language generated by that CFG is regular.

Regular grammar

Definition

A CFG is said to be a **regular grammar** if it generates the regular language *i.e.* a CFG is said to be a **regular grammar** in which each production is one of the two forms

Nonterminal → semiword

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Nonterminal → word

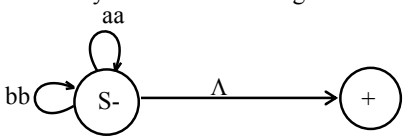
Examples
The CFG S → aaS|bbS|Λ is a regular grammar. It may be observed that the above CFG generates the language of strings expressed by the RE (aa+bb)*.
The CFG S → aA|bB, A → aS|a, B → bS|b is a regular grammar. It may be observed that the above CFG generates the language of strings expressed by RE $(aa+bb)^+$.

Following is a method of building TG corresponding to the regular grammar.

TG for Regular Grammar
For every regular grammar there exists a TG corresponding to the regular grammar.
Following is the method to build a TG from the given regular grammar
Define the states, of the required TG, equal in number to that of nonterminals of the given regular grammar. An additional state is also defined to be the final state. The initial state should correspond to the nonterminal S.
For every production of the given regular grammar, there are two possibilities for the transitions of the required TG

If the production is of the form nonterminal → semiword, then transition of the required TG would start from the state corresponding to the nonterminal on the left side of the production and would end in the state corresponding to the nonterminal on the right side of the production, labeled by string of terminals in semiword.
If the production is of the form nonterminal → word, then transition of the TG would start from the state corresponding to nonterminal on the left side of the production and would end on the final state of the TG, labeled by the word.

Example
Consider the following CFG
S → aaS|bbS| Λ
The TG accepting the language generated by the above CFG is given below



The corresponding RE may be (aa+bb)*.

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Theory of Automata

Lecture N0. 35
Reading Material

Introduction to Computer Theory Chapter 13

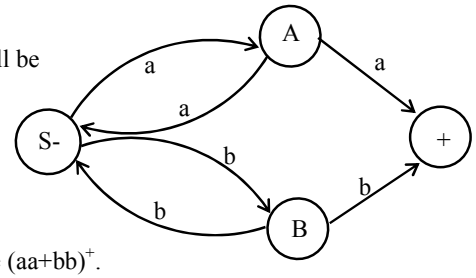
Summary

Examples of building TG's corresponding to the Regular Grammar, Null productions with examples, Nullable productions with examples, Unit production with example, Chomsky Normal Form (Definition)

Example
Consider the following CFG

$S \rightarrow aA|bB$
$A \rightarrow aS|a$
$B \rightarrow bS|b$

then the corresponding TG will be

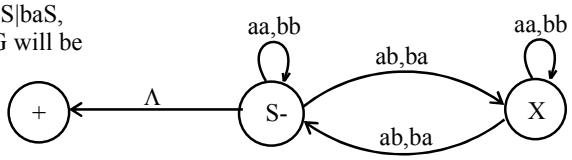


The corresponding RE may be $(aa+bb)^+$.

Example
Consider the following CFG

$S \rightarrow aaS|bbS|abX|baX|\Lambda$
$X \rightarrow aaX|bbX|abS|baS,$

then the corresponding TG will be



The corresponding language is EVEN-EVEN.

Null Production
Definition
The production of the form nonterminal $\rightarrow \Lambda$ is said to be ***null production***.

Example: Consider the CFG, $S \rightarrow aA|bB|\Lambda$, $A \rightarrow aa|\Lambda$, $B \rightarrow aS$
Here $S \rightarrow \Lambda$ and $A \rightarrow \Lambda$ are null productions.

Note
If a CFG has a null production, then it is possible to construct another CFG without null production accepting the same language with the exception of the word Λ *i.e.* if the language contains the word Λ then the new language cannot have the word Λ.

Following is a method to construct a CFG without null production for a given CFG

Method
Delete all the Null productions and add new productions *e.g.*
consider the productions of a certain CFG $X \rightarrow aNbNa$, $N \rightarrow \Lambda$, delete the production $N \rightarrow \Lambda$ and using the production $X \rightarrow aNbNa$, add the new productions $X \rightarrow aNba$, $X \rightarrow abNa$ and $X \rightarrow aba$
Thus the new CFG will contain the productions $X \rightarrow aNba|abNa|aba|aNbNa$
Note: It is to be noted that $X \rightarrow aNbNa$ will still be included in the new CFG.

Nullable Production

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Definition
A production is called ***nullable production*** if it is of the form $N \rightarrow \Lambda$
or
there is a derivation that starts at N and leads to Λ *i.e.* $N_1 \rightarrow N_2$, $N_2 \rightarrow N_3$, $N_3 \rightarrow N_4$, …, $N_n \rightarrow \Lambda$, where N, N_1, N_2, …, N_n are non terminals.

Example
Consider the following CFG

$S \rightarrow AA|bB$, $A \rightarrow aa|B$, $B \rightarrow aS \mid \Lambda$

Here $S \rightarrow AA$ and $A \rightarrow B$ are nullable productions, while $B \rightarrow \Lambda$ is null a production.
Following is an example describing the method to convert the given CFG containing null productions and nullable productions into the one without null productions

Example
Consider the following CFG

$S \rightarrow XaY|YY|aX|ZYX$
$X \rightarrow Za|bZ|ZZ|Yb$
$Y \rightarrow Ya|XY|\Lambda$
$Z \rightarrow aX|YYY$

It is to be noted that in the given CFG, the productions $S \rightarrow YY$, $X \rightarrow ZZ$, $Z \rightarrow YYY$ are Nullable productions, while $Y \rightarrow \Lambda$ is Null production.
Here the method of removing null productions, as discussed earlier, will be used along with replacing nonterminals corresponding to nullable productions like nonterminals for null productions are replaced.
Thus the required CFG will be

$S \rightarrow XaY|Xa|aY|a|YY|Y|aX|ZYX|YX|ZX|ZY|\ X|Z$
$X \rightarrow Za|a|bZ|b|ZZ|Z|Yb$
$Y \rightarrow Ya|a|XY|X|Y$
$Z \rightarrow aX|a|YYY|YY|Y$

Example
Consider the following CFG

$S \rightarrow XY$, $X \rightarrow Zb$, $Y \rightarrow bW$
$Z \rightarrow AB$, $W \rightarrow Z$, $A \rightarrow aA|bA|\Lambda$
$B \rightarrow Ba|Bb|\Lambda$.

Here $A \rightarrow \Lambda$ and $B \rightarrow \Lambda$ are null productions, while $Z \rightarrow AB$, $W \rightarrow Z$ are nullable productions. The new CFG after, applying the method, will be

$S \rightarrow XY$
$X \rightarrow Zb|b$
$Y \rightarrow bW|b$
$Z \rightarrow AB|A|B$
$W \rightarrow Z$
$A \rightarrow aA|a|bA|b$
$B \rightarrow Ba|a|Bb|b$

Note
While adding new productions all Nullable productions should be handled with care. All Nullable productions will be used to add new productions, but only the Null production will be deleted.

Unit production
The productions of the form nonterminal $\rightarrow$ one nonterminal, is called the ***unit production***.
Following is an example showing how ***to eliminate the unit productions from a given CFG.***

Example
Consider the following CFG

$S \rightarrow A|bb$
$A \rightarrow B|b$
$B \rightarrow S|a$

Separate the unit productions from the nonunit productions as shown below

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unit prods.	nonunit prods.
$S \rightarrow A$	$S \rightarrow bb$
$A \rightarrow B$	$A \rightarrow b$
$B \rightarrow S$	$B \rightarrow a$

$S \rightarrow A$ gives $S \rightarrow b$	(using $A \rightarrow b$)
$S \rightarrow A \rightarrow B$ gives $S \rightarrow a$	(using $B \rightarrow a$)
$A \rightarrow B$ gives $A \rightarrow a$	(using $B \rightarrow a$)
$A \rightarrow B \rightarrow S$ gives $A \rightarrow bb$	(using $S \rightarrow bb$)
$B \rightarrow S$ gives $B \rightarrow bb$	(using $S \rightarrow bb$)
$B \rightarrow S \rightarrow A$ gives $B \rightarrow b$	(using $A \rightarrow b$)

Thus the new CFG will be

$S \rightarrow a|b|bb$, $A \rightarrow a|b|bb$, $B \rightarrow a|b|bb$.

Which generates the finite language {a,b,bb}.

Chomsky Normal Form

If a CFG has only productions of the form

nonterminal → string of two nonterminals

or

nonterminal → one terminal

then the CFG is said to be in Chomsky Normal Form (CNF).

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Theory of Automata

Lecture N0. 36
Reading Material

Introduction to Computer Theory Chapter 13,14

Summary
Chomsky Normal Form, Theorem regarding CNF, examples of converting CFG to be in CNF, Example of an FA corresponding to Regular CFG, Left most and Right most derivations, New format of FAs.

Chomsky Normal Form (CNF)
If a CFG has only productions of the form
nonterminal → string of two nonterminals
or
nonterminal → one terminal
then the CFG is said to be in Chomsky Normal Form (CNF).

Note
It is to be noted that any CFG can be converted to be in CNF, if the null productions and unit productions are removed. Also if a CFG contains nullable productions as well, then the corresponding new production are also to be added. Which leads the following theorem

Theorem
All NONNULL words of the CFL can be generated by the corresponding CFG which is in CNF *i.e.* the grammar in CNF will generate the same language except the null string.

Following is an example showing that a CFG in CNF generates all nonnull words of corresponding CFL.

Example
Consider the following CFG
S → aSa|bSb|a|b|aa|bb
To convert the above CFG to be in CNF, introduce the new productions as
A → a, B → b, then the new CFG will be
S → ASA|BSB|AA|BB|a|b
A → a
B → b
Introduce nonterminals R_1 and R_2 so that
S → AR_1|BR_2|AA|BB|a|b
R_1 → SA
R_2 → SB
A → a
B → b
which is in CNF.
It may be observed that the above CFG which is in CNF generates the NONNULLPALINDROME, which does not contain the null string.

Example
Consider the following CFG
S → ABAB
A → a|Λ
B → b|Λ
Here S → ABAB is nullable production and A → Λ, B → Λ are null productions. Removing the null productions
A → Λ and B → Λ, and introducing the new productions as
S → BAB|AAB|ABB|ABA|AA|AB|BA|BB|A|B

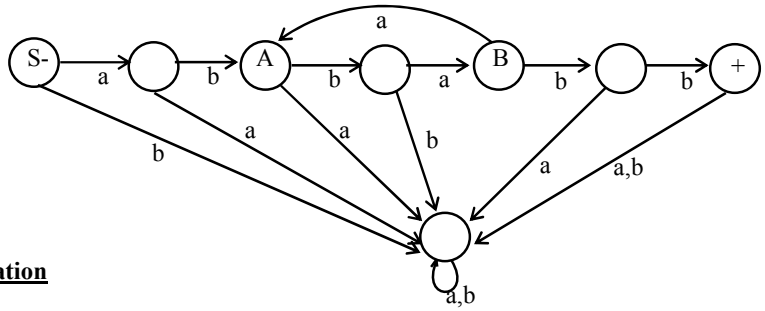
Now S → A|B are unit productions to be eliminated as shown below

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$S \rightarrow A$ gives $S \rightarrow a$ (using $A \rightarrow a$)
$S \rightarrow B$ gives $S \rightarrow b$ (using $B \rightarrow b$)
Thus the new resultant CFG takes the form
$S \rightarrow BAB|AAB|ABB|ABA|AA|AB|BA|BB|a|b$, $A \rightarrow a$, $B \rightarrow b$.
Introduce the nonterminal C where $C \rightarrow AB$, so that
$S \rightarrow BAB|AAB|ABB|ABA|AA|AB|BA|BB|a|b$

$S \rightarrow BC|AC|CB|CA|AA|C|BA|BB|a|b$
$A \rightarrow a$
$B \rightarrow b$
$C \rightarrow AB$
is the CFG in CNF.

Example
To construct an FA that accepts the grammar
$S \rightarrow abA$
$A \rightarrow baB$
$B \rightarrow aA|bb$
The language can be identified by the three words generated as follows
$S \Rightarrow abA$
$\Rightarrow abbaB$ (using $A \rightarrow baB$)
$\Rightarrow abba\ bb$ (using $B \rightarrow bb$)
$S \Rightarrow abA$
$\Rightarrow abbaB$ (using $A \rightarrow baB$)
$\Rightarrow abbaaA$ (using $B \rightarrow aA$)
$\Rightarrow abbaabaB$ (using $A \rightarrow baB$)
$\Rightarrow abbaababb$ (using $B \rightarrow bb$)
$S \Rightarrow abA$
$\Rightarrow abbaB$ (using $A \rightarrow baB$)
$\Rightarrow abbaaA$ (using $B \rightarrow aA$)
$\Rightarrow abbaabaB$ (using $A \rightarrow baB$)
$\Rightarrow abbaabaaA$ (using $B \rightarrow aA$)
$\Rightarrow abbaabaabaB$ (using $A \rightarrow baB$)
$\Rightarrow abbaabaababb$ (using $B \rightarrow bb$)
which shows that corresponding language has RE $abba(aba)^*bb$. Thus the FA accepting the given CFG may be



Left most derivation
Definition
The derivation of a word w, generated by a CFG, such that at each step, a production is applied to the left most nonterminal in the working string, is said to be ***left most derivation***.
It is to be noted that the nonterminal that occurs first from the left in the working string, is said to be ***left most nonterminal***.

Example
Consider the following CFG

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$S \rightarrow XY$
$X \rightarrow XX|a$
$Y \rightarrow YY|b$

then following are the two left most derivations of aaabb

$S \Rightarrow \underline{X}Y$
$\Rightarrow \underline{X}XY$
$\Rightarrow a\underline{X}Y$
$\Rightarrow a\underline{X}XY$
$\Rightarrow aa\underline{X}Y$
$\Rightarrow aaa\underline{Y}$
$\Rightarrow aaa\underline{Y}Y$
$\Rightarrow aaab\underline{Y}$
$= aaabb$

$S \Rightarrow \underline{X}Y$
$\Rightarrow \underline{X}XY$
$\Rightarrow \underline{X}XXY$
$\Rightarrow a\underline{X}XY$
$\Rightarrow aa\underline{X}Y$
$\Rightarrow aaa\underline{Y}$
$\Rightarrow aaa\underline{Y}Y$
$\Rightarrow aaab\underline{Y}$
$= aaabb$

Theorem

Any word that can be generated by a certain CFG has also a left most derivation.

It is to be noted that the above theorem can be stated for right most derivation as well.

Example

Consider the following CFG

$S \rightarrow YX$
$X \rightarrow XX|b$
$Y \rightarrow YY|a$

Following are the left most and right most derivations of abbbb

$S \Rightarrow \underline{Y}X$
$\Rightarrow a\underline{X}$
$\Rightarrow a\underline{X}X$
$\Rightarrow ab\underline{X}$
$\Rightarrow ab\underline{X}X$
$\Rightarrow abb\underline{X}$
$\Rightarrow abb\underline{X}X$
$\Rightarrow abbb\underline{X}$
$= abbbb$

$S \Rightarrow Y\underline{X}$
$\Rightarrow YX\underline{X}$
$\Rightarrow Y\underline{X}b$
$\Rightarrow YX\underline{X}b$
$\Rightarrow Y\underline{X}bb$
$\Rightarrow YX\underline{X}bb$
$\Rightarrow Y\underline{X}bbb$
$\Rightarrow \underline{Y}bbbb$
$= abbbb$

A new format for FAs

A class of machines (FAs) has been discussed accepting the regular language *i.e.* corresponding to a regular language there is a machine in this class, accepting that language and corresponding to a machine of this class there is a regular language accepted by this machine. It has also been discussed that there is a CFG corresponding to regular language and CFGs also define some nonregular languages, as well

There is a question whether there is a class of machines accepting the CFLs? The answer is yes. The new machines which are to be defined are more powerful and can be constructed with the help of FAs with new format.

To define the new format of an FA, some terms are defined in the next lecture.

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Theory of Automata

Lecture N0. 37
Reading Material

Introduction to Computer Theory Chapter 14

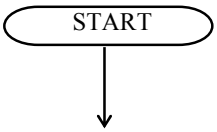
Summary
New format for FAs, input TAPE, START, ACCEPT , REJECT, READ states Examples of New Format of FA, PUSH Down STACK , PUSH and POP, Example of PDA

A new format for FAs
A class of machines (FAs) has been discussed accepting the regular language *i.e.* corresponding to a regular language there is a machine in this class, accepting that language and corresponding to a machine of this class there is a regular language accepted by this machine. It has also been discussed that there is a CFG corresponding to regular language and CFGs also define some nonregular languages, as well
There is a question whether there is a class of machines accepting the CFLs? The answer is yes. The new machines which are to be defined are more powerful and can be constructed with the help of FAs with new format.
To define the new format of an FA, the following are to be defined
Input TAPE
The part of an FA, where the input string is placed before it is run, is called the input TAPE.
The input TAPE is supposed to accommodate all possible strings. The input TAPE is partitioned with cells, so that each letter of the input string can be placed in each cell. The input string abbaa is shown in the following input TAPE.

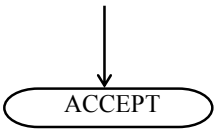
Cell i	Cell ii	Cell iii					
a	b	b	a	a	Δ	Δ	.

The character Δ indicates a blank in the TAPE. The input string is read from the TAPE starting from the cell (i). It is assumed that when first Δ is read, the rest of the TAPE is supposed to be blank.

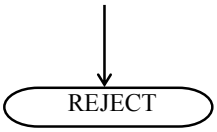
The START state
This state is like initial state of an FA and is represented by



An ACCEPT state
This state is like a final state of an FA and is expressed by



A REJECT state
This state is like dead-end non final state and is expressed by

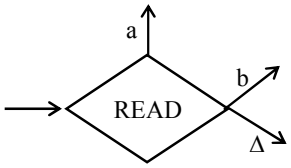


Note: It may be noted that the ACCEPT and REJECT states are called the halt states.

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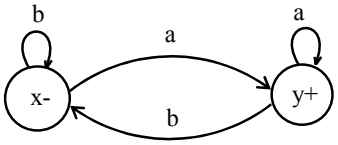
A READ state

This state is to read an input letter and lead to some other state. The READ state is expressed by

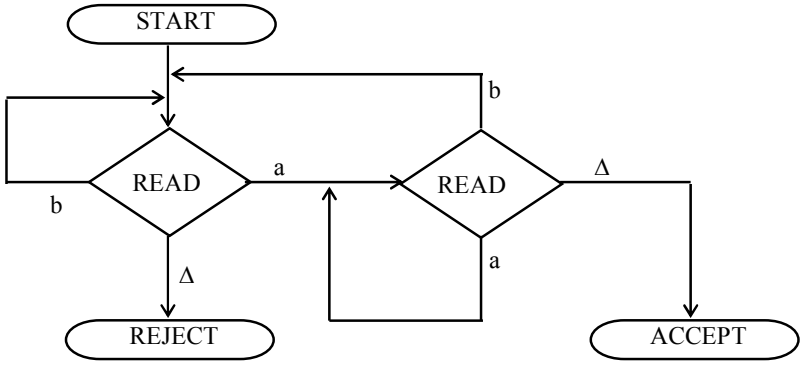


Example

Before some other states are defined consider the following example of an FA along with its new format



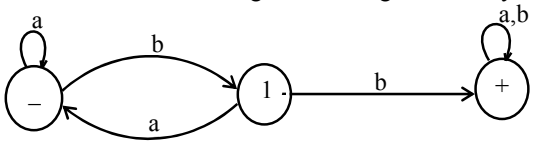
Obviously the above FA accepts the language of strings, expressed by $(a+b)^*a$. Following is the new format of the above FA



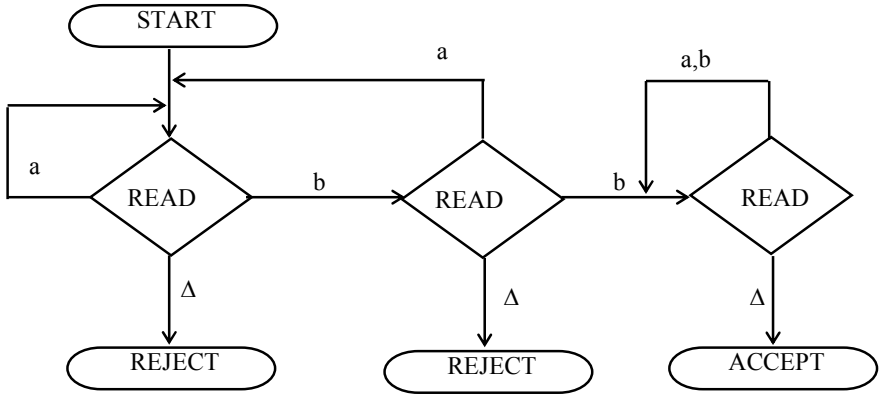
Note

The Δ edge should not be confused with **∧**-labeled edge. The Δ-edges start only from READ boxes and lead to halt states.

Example



The above FA accepts the language expressed by (a+b)*bb(a+b)*



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PUSHDOWN STACK or PUSHDOWN STORE

PUSHDOWN STACK is a place where the input letters can be placed until these letters are referred again. It can store as many letters as one can in a long column.

Initially the STACK is supposed to be empty *i.e.* each of its storage location contains a blank.

PUSH

A PUSH operator adds a new letter at the top of STACK, for *e.g.* if the letters a, b, c and d are pushed to the STACK that was initially blank, the STACK can be shown as

STACK

d
c
b
a
Δ
□

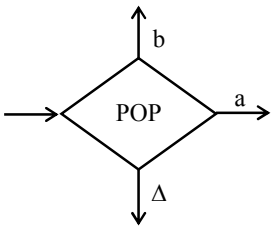
The PUSH state is expressed by



When a letter is pushed, it replaces the existing letter and pushes it one position below.

POP and STACK

POP is an operation that takes out a letter from the top of the STACK. The rest of the letters are moved one location up. POP state is expressed as

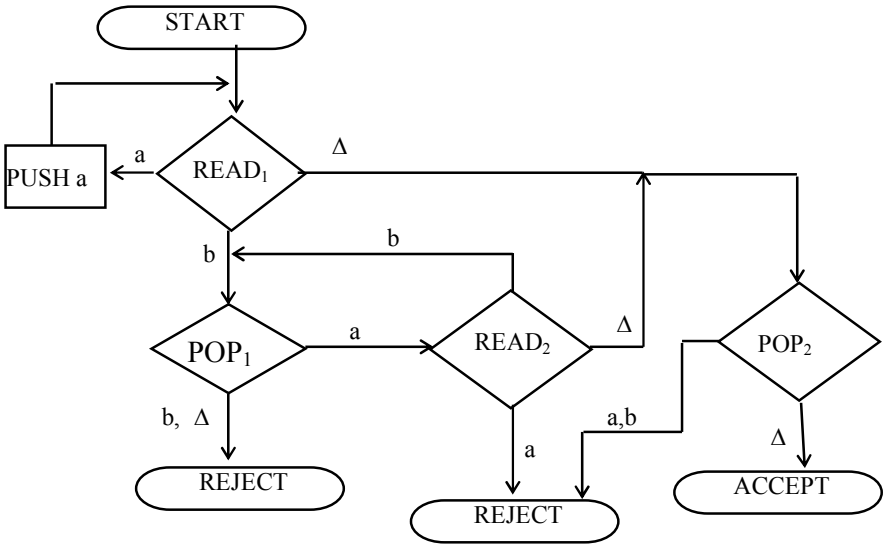


Note

It may be noted that popping an empty STACK is like reading an empty TAPE, *i.e.* popping a blank character Δ. It may also be noted that when the new format of an FA contains PUSH and POP states, it is called PUSHDOWN Automata or PDAs. It may be observed that if the PUSHDOWN STACK (the memory structure) is added to an FA then its language recognizing capabilities are increased considerably. Following is an example of PDA

Example

Consider the following PDA



The string aaabbb is to be run on this machine. Before the string is processed, the string is supposed to be placed on the TAPE and the STACK is supposed to be empty as shown below

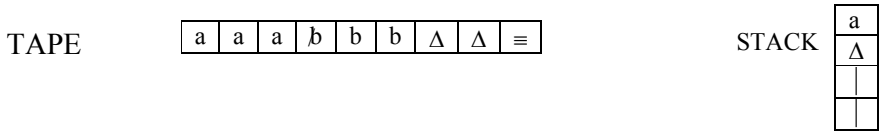
a	a	a	b	b	b	Δ	Δ	≡

STACK

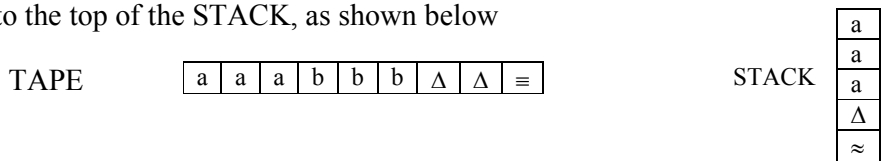
Δ
|
|
|

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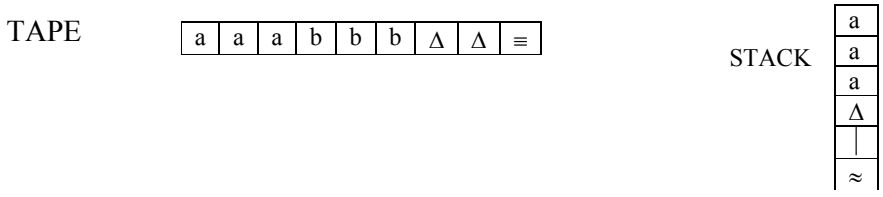
Reading first a from the TAPE we move from $READ_1$ State to PUSH a state, it causes the letter a deleted from the TAPE and added to the top of the STACK, as shown below



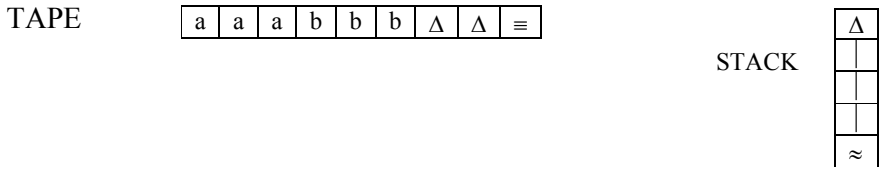
Reading next two a's successively, will delete further two a's from the TAPE and add these letters to the top of the STACK, as shown below



Then reading the next letter which is b from the TAPE will lead to the POP_1 state. The top letter at the STACK is a, which is popped out and $READ_2$ state is entered. Situation of TAPE and STACK is shown below



Reading the next two b's successively will delete two b's from the TAPE, will lead to the POP_1 state and these b's will be removed from the STACK as shown below



Now there is only blank character Δ left, to be read from the TAPE, which leads to POP_2 state. While the only blank characters is left in the STACK to be popped out and the ACCEPT state is entered, which shows that the string aaabbb is accepted by this PDA. It may be observed that the above PDA accepts the language

$\{a^n b^n: n = 0,1,2, \ldots\}$.

Since the null string is like a blank character, so to determine how the null string is accepted, it can be placed in the TAPE as shown below



Reading Δ at state $READ_1$ leads to POP_2 state and POP_2 state contains only Δ, hence it leads to ACCEPT state and the null string is accepted.

Note: The process of running the string aaabbb can also be expressed in the table given in the next lecture.

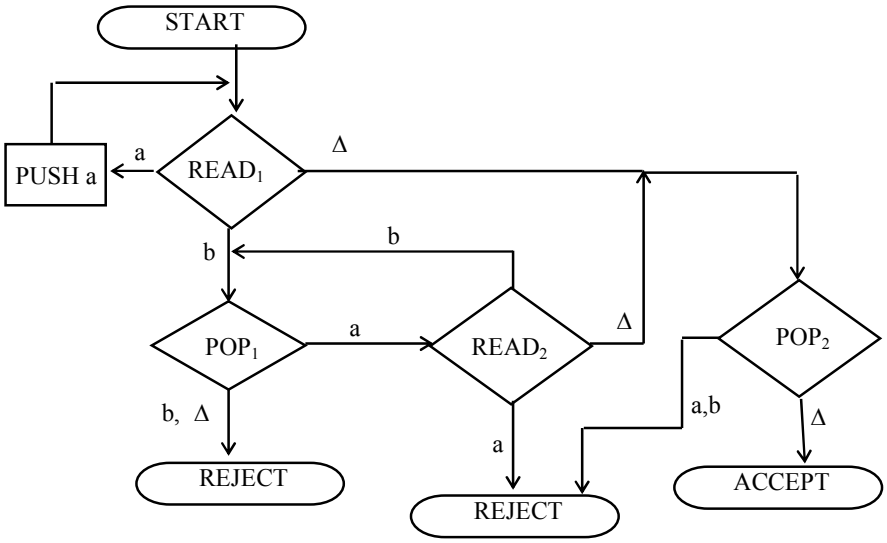
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Theory of Automata

Lecture N0. 38
Reading Material

Introduction to Computer Theory Chapter 14

Summary
Example of PDA with table for running a string, Equivalent PDA, PDA for EVEN EVEN Language. Non-Derterministic PDA, Example of Non-Derterministic PDA, Definition of PUSH DOWN Automata, Example of Non-Derterministic PDA.



Note
The process of running the string aaabbb can also be expressed in the following table

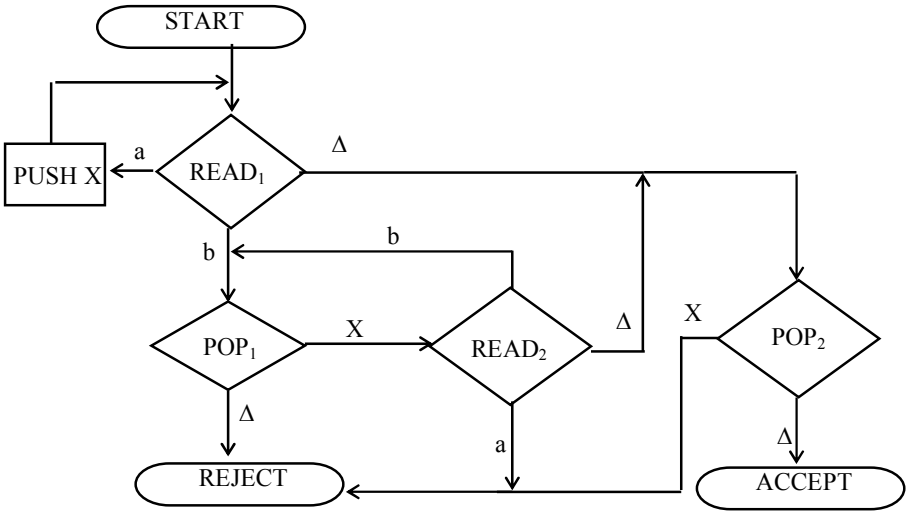
STATE	STACK	TAPE
START	Δ …	**aaabbb**Δ …
$READ_1$	Δ …	**a**aabbbΔ …
PUSH a	**a**Δ …	**a**aabbbΔ …
$READ_1$	**a**Δ …	**aa**abbbΔ …
PUSH a	**aa**Δ …	**aa**abbbΔ …
$READ_1$	**aa**Δ …	**aaa**bbbΔ …
PUSH a	**aaa**Δ …	**aaa**bbbΔ …
$READ_1$	**aaa**Δ …	**aaab**bbΔ …
POP_1	**aa**Δ …	**aaab**bbΔ …

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STATE	STACK	TAPE
$READ_1$	**aa**Δ …	**aaabbb**Δ …
POP_1	**a**Δ …	**aaabbb**Δ …
$READ_2$	**a**Δ …	**aaabbb**Δ …
POP_1	Δ …	**aaabbb**Δ …
$READ_2$	Δ …	**aaabbbΔ** …
POP_2	Δ …	**aaabbbΔ** …
ACCEPT	Δ …	**aaabbbΔ** …

It may be observed that the above PDA accepts the language $\{a^n b^n : n=0,1,2,3, \ldots\}$.

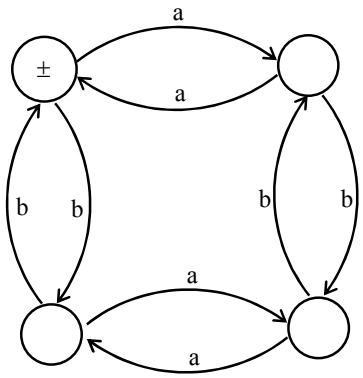
Note
It may be noted that the TAPE alphabet Σ and STACK alphabet Γ, may be different in general and hence the PDA equivalent to that accepting $\{a^n b^n: n=0,1,2,3\ldots\}$ discussed above may be



Following is an example of PDA corresponding to an FA

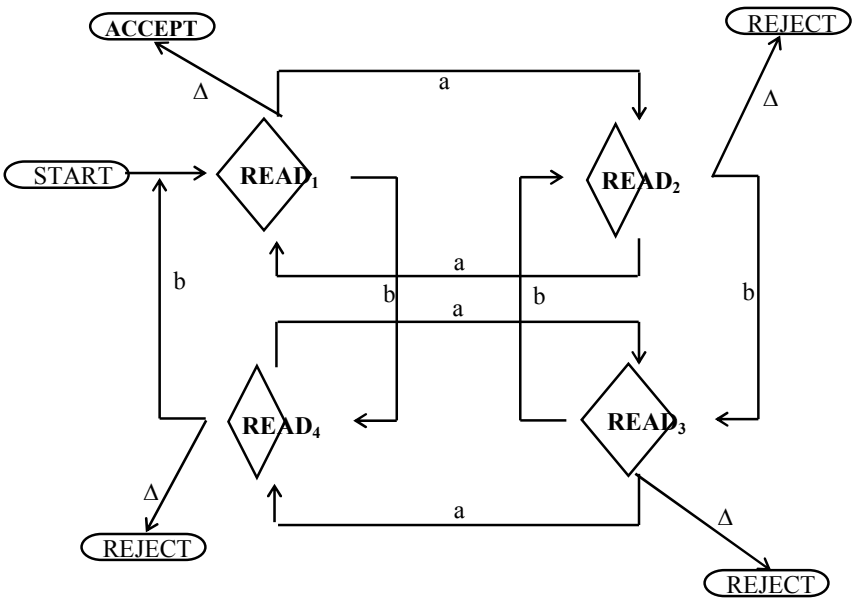
Example

Consider the following FA corresponding to the EVEN-EVEN language



The corresponding PDA will be

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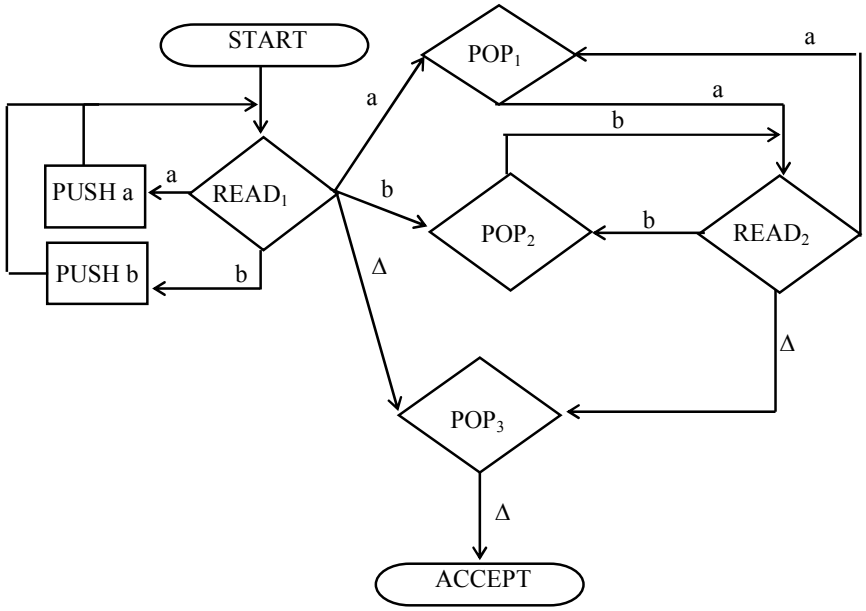
Nondeterministic PDA

Like TGs and NFAs, if in a PDA there are more than one outgoing edges at READ or POP states with one label, then it creates nondeterminism and the PDA is called nondeterministic PDA.

In nondeterministic PDA no edge is labeled by string of terminals or nonterminals, like that can be observed in TGs. Also if there is no edge for any letter to be read from the TAPE, the machine crashes and the string is rejected.

In nondeterministic PDA a string may trace more than one path. If there exists at least one path traced by a string leading to ACCEPT state, then the string is supposed to be accepted, otherwise rejected.

Following is an example of nondeterministic PDA



Here the nondeterminism can be observed at state $READ_1$. It can be observed that the above PDA accepts the language

EVENPALINDROME=$\{w \text{ reverse}(w): w \in \{a, b\}^*\}$

$=\{\Lambda, aa, bb, aaaa, abba, baab, bbbb, \ldots\}$

Now the definition of PDA including the possibility of nondeterminism may be given as follows

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PUSHDOWN AUTOMATON (PDA)[including the possibility of non determinism]

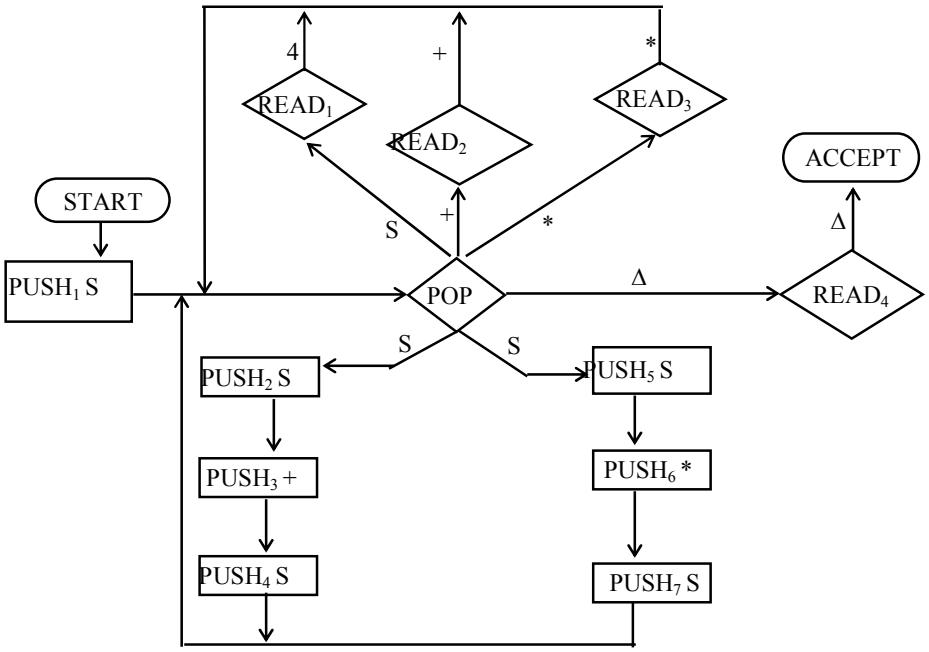
Pushdown Automaton (PDA), consists of the following

1. An alphabet Σ of input letters.
2. An input TAPE with infinite many locations in one direction. Initially the input string is placed in it starting from first cell, the remaining part of the TAPE is empty.
3. An alphabet Γ of STACK characters.
4. A pushdown STACK which is initially empty, with infinite many locations in one direction. Initially the STACK contains blanks.
5. One START state with only one out-edge and no in-edge.
6. Two halt states *i.e.* ACCEPT and REJECT states, with in-edges and no out-edges.
7. A PUSH state that introduces characters onto the top of the STACK.
8. A POP state that reads the top character of the STACK, (may contain more than one out-edges with same label).
9. A READ state that reads the next unused letter from the TAPE, (may contain more than one out-edges with same label).

Example: Consider the CFG

$S \rightarrow S+S|S*S|4$

Following is the PDA accepting the corresponding CFL



The string 4 + 4 * 4 traces the path shown in the following table

STATE	STACK	TAPE
START	Δ	**4+4*4**
$PUSH_1$ S	**S**	**4+4*4**
POP	Δ	**4+4*4**
$PUSH_2$ S	**S**	**4+4*4**
$PUSH_3$ +	**+S**	**4+4*4**
$PUSH_4$ S	**S+S**	**4+4*4**
POP	**+S**	**4+4*4**
$READ_1$	**+S**	**+4*4**
POP	**S**	**+4*4**

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STATE	STACK	TAPE
$READ_2$	**S**	**4*4**
POP	Δ	**4*4**
$PUSH_5$ S	**S**	**4*4**
$PUSH_6$ *	***S**	**4*4**
$PUSH_7$ S	**S*S**	**4*4**
POP	***S**	**4*4**
$READ_1$	***S**	***4**
POP	**S**	***4**
$READ_3$	**S**	**4**
POP	Δ	**4**
$READ_1$	Δ	Δ
POP	Δ	Δ
$READ_4$	Δ	Δ
ACCEPT	Δ	Δ

Note

It may be noted that the letters are deleted from the TAPE instead of underlined.

It may also be noted that the choice of path at POP state can be determined by the left most deviation of the string belonging to the CFL.

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Theory of Automata

Lecture N0. 39
Reading Material

Introduction to Computer Theory Chapter 15

Summary
PDA corresponding to CFG, Examples of PDA corresponding to CFG

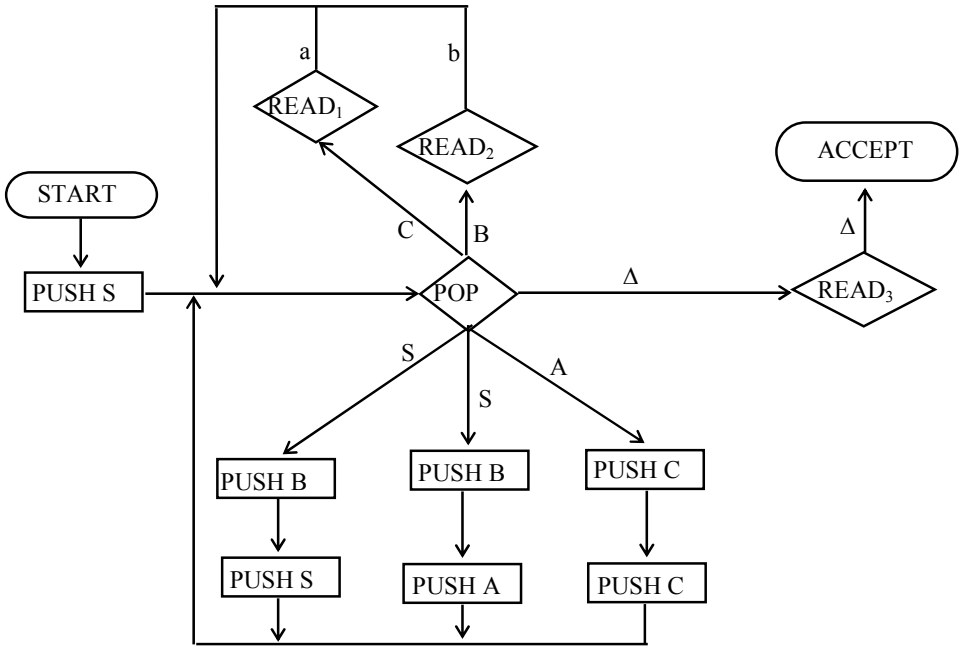
PDA corresponding to CFG
Theorem
Corresponding to any CFG there exists a PDA accepting the language generated by the CFG.

Since an algorithm has already been discussed to convert the CFG in CNF, so the PDA can be constructed corresponding to the CFG. As the CFG in CNF generates all the nonnull words of the corresponding CFL, so accepting the null string (if it is contained in the CFL), can be managed separately.

Example
Consider the following CFG which is in CNF and does not generate the null string

$S \rightarrow SB|AB$
$A \rightarrow CC$
$B \rightarrow b$
$C \rightarrow a$

The corresponding PDA will be



Here the STACK alphabet $\Gamma = \{S, A, B, C\}$, where the TAPE alphabet $\Sigma = \{a, b\}$

Note: It may be noted that when the POP state is entered either a nonterminal is replaced by two nonterminals at the top of the STACK accommodating a production, or a nonterminal is popped out from the top of the stack and a READ state is entered to read a specified letter from the TAPE or else the machine crashes.
The choice of path taken at POP state to accommodate the word belonging to the CFL can be determined by the left most derivation of the word. Consider the word aab with its left most derivation, as follows

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Working-String Generation	Production	Used
$S \Rightarrow AB$	$S \rightarrow AB$	step 1
$\Rightarrow CCB$	$A \rightarrow CC$	step 2
$\Rightarrow aCB$	$C \rightarrow a$	step 3
$\Rightarrow aaB$	$C \rightarrow a$	step 4
$\Rightarrow aab$	$B \rightarrow b$	step 5

First of all the START state is entered

STACK	TAPE
Δ…	aabΔ…

The PUSH S state is entered

STACK	TAPE
S	aabΔ…

The POP state is entered and to accommodate the production $S \rightarrow AB$, PUSH B and PUSH A states are entered.

STACK	TAPE
AB	aabΔ…

Then the POP state is entered and to accommodate the production $A \rightarrow CC$, PUSH C, PUSH C states are entered

STACK	TAPE
CCB	aab

The POP state is entered and to accommodate the production $C \rightarrow a$, $READ_1$ is entered and the letter a is read from the TAPE.

STACK	TAPE
CB	aab

The POP state is entered and to accommodate the production $C \rightarrow a$, $READ_1$ state is entered and the letter a is read from the TAPE

STACK	TAPE
B	aab

The POP state is entered and to accommodate the production $B \rightarrow b$, $READ_2$ state is entered and the letter b is read from the TAPE

STACK	TAPE
Δ	aab

The Δ shown in the STACK indicates that there are no nonterminals in the working string and Δ is read from the STACK which leads to $READ_3$ state where the Δ is read from the TAPE and the ACCEPT state is entered which shows that the word aab is accepted by the PDA.
Following is the table showing all the observations discussed above, for the word aab

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Left most derivation	STATE	STACK	TAPE
	START	Δ	aab
S	PUSH S	S	aab
	POP	Δ	aab
	PUSH B	B	aab
⇒AB	PUSH A	AB	aab
	POP	B	aab
	PUSH C	CB	aab
⇒CCB	PUSH C	CCB	aab
	POP	CB	aab
⇒aCB	$READ_1$	CB	aab
	POP	B	aab
⇒aaB	$READ_1$	B	aab
	POP	Δ	aab
⇒aab	$READ_2$	Δ	aab
	POP	Δ	aab
	$READ_3$	Δ	aab
	ACCEPT	Δ	aab

Following is an example of building the PDA corresponding to the given CFG

Example

Consider the following CFG

$S \rightarrow XY$

$X \rightarrow aX \mid bX \mid a$

$Y \rightarrow Ya \mid Yb \mid a$

First of all, converting the CFG to be in CNF, introduce the nonterminals A and B as

$A \rightarrow a$

$B \rightarrow b$

The following CFG is in CNF

$S \rightarrow XY$

$X \rightarrow AX \mid BX \mid a$

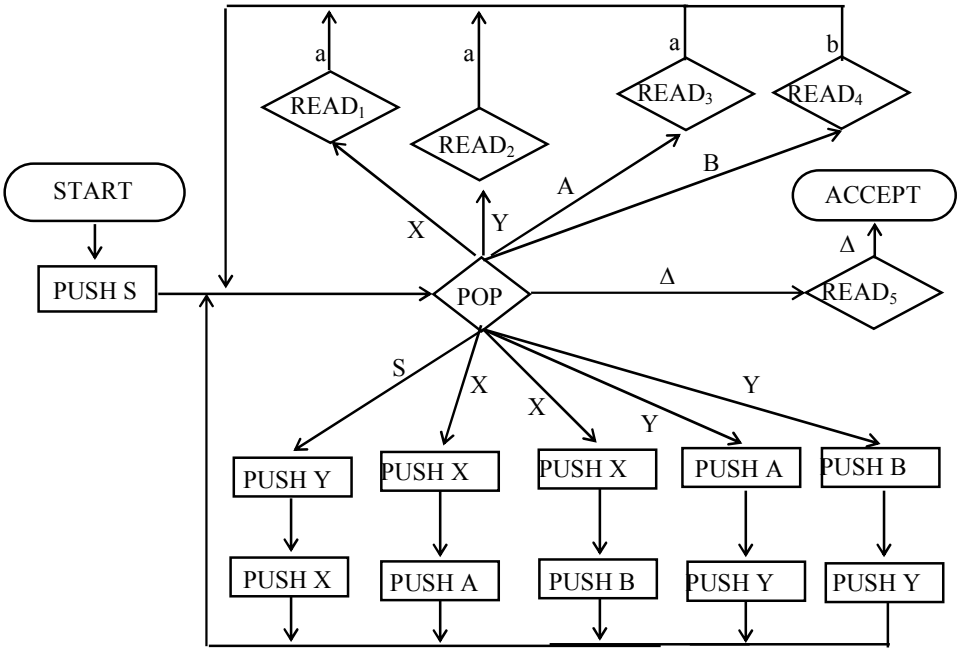
$Y \rightarrow YA \mid YB \mid a$

$A \rightarrow a$

$B \rightarrow b$

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The PDA corresponding to the above CFG will be



The word aaab can be generated as

Working-String Generation	Production	Used
$S \Rightarrow XY$	$S \rightarrow XY$	step 1
$\Rightarrow AXY$	$X \rightarrow AX$	step 2
$\Rightarrow aXY$	$A \rightarrow a$	step 3
$\Rightarrow aaY$	$X \rightarrow a$	step 4
$\Rightarrow aaYB$	$Y \rightarrow YB$	step 5
$\Rightarrow aaaB$	$Y \rightarrow a$	step 6
$\Rightarrow aaab$	$B \rightarrow b$	step 7

STACK	TAPE	STACK	TAPE
(START) Δ	aaab	(POP) Y	aaab
(PUSH S) S	aaab	($READ_1$) Y	aaab
(POP) Δ	aaab	(POP) Δ	aabb
(PUSH Y) Y	aaab	(PUSH B) B	aabb
(PUSH X) XY	aaab	(PUSH Y) YB	aabb
(POP) Y	aaab	(POP) B	aaab
(PUSH X) XY	aaab	($READ_2$) B	aaab
(PUSH A) AXY	aaab	(POP) Δ	aaab
(POP) XY	aaab	($READ_4$) Δ	aaab
($READ_3$) XY	aaab	(POP) Δ	Δ

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Theory of Automata

Lecture N0. 40
Reading Material

Introduction to Computer Theory Chapter 15

Summary
Recap of example of PDA corresponding to CFG, CFG corresponding to PDA. Theorem, HERE state, Definition of Conversion form, different situations of PDA to be converted into conversion form

Example
Consider the following CFG

$S \rightarrow XY$
$X \rightarrow aX \mid bX \mid a$
$Y \rightarrow Ya \mid Yb \mid a$

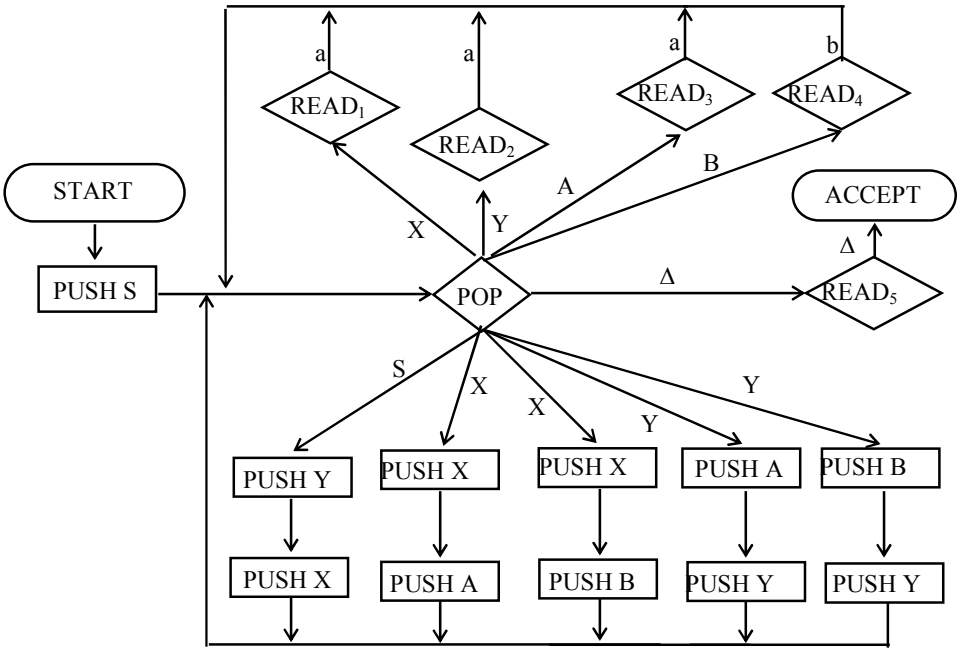
First of all, converting the CFG to be in CNF, introduce the nonterminals A and B as

$A \rightarrow a$
$B \rightarrow b$

The following CFG is in CNF

$S \rightarrow XY$
$X \rightarrow AX \mid BX \mid a$
$Y \rightarrow YA \mid YB \mid a$
$A \rightarrow a$
$B \rightarrow b$

The PDA corresponding to the above CFG will be

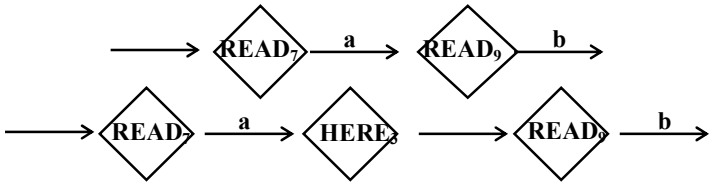


Theorem
Given a PDA that accepts the language L, there exists a CFG that generates exactly L.

Before the CFG corresponding to the given PDA is determined, the PDA is converted into the **standard form** which is called the **conversion form**.
Before the PDA is converted into conversion form a new state HERE is defined which is placed in the middle of any edge.
Like READ and POP states, HERE states are also numbered *e.g.*

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becomes



Conversion form of PDA

Definition

A PDA is in conversion form if it fulfills the following conditions:

There is only one ACCEPT state.

There are no REJECT states.

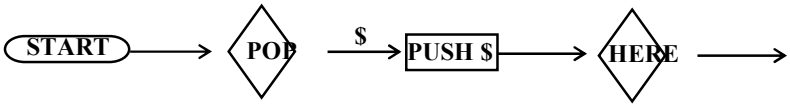
Every READ or HERE is followed immediately by a POP *i.e.* every edge leading out of any READ or HERE state goes directly into a POP state.

No two POPs exist in a row on the same path without a READ or HERE between them whether or not there are any intervening PUSH states (*i.e.* the POP states must be separated by READs or HEREs).

All branching, deterministic or nondeterministic occurs at READ or HERE states, none at POP states and every edge has only one label.

Even before we get to START, a "bottom of STACK" symbol $ is placed on the STACK. If this symbol is ever popped in the processing it must be replaced immediately. The STACK is never popped beneath this symbol. Right before entering ACCEPT this symbol is popped out and left.

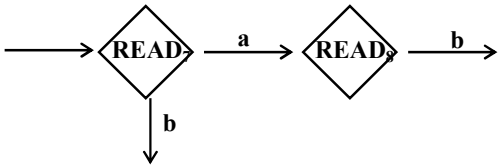
The PDA must begin with the sequence



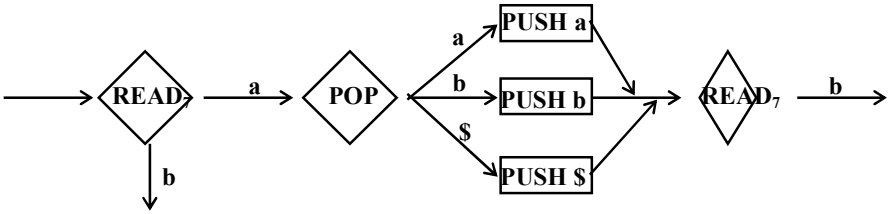
The entire input string must be read before the machine can accept the word.

Different situations of a PDA to be converted into **conversion form** are discussed as follows

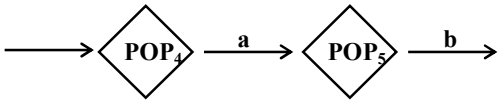
To satisfy condition 3,



becomes



To satisfy condition 4,

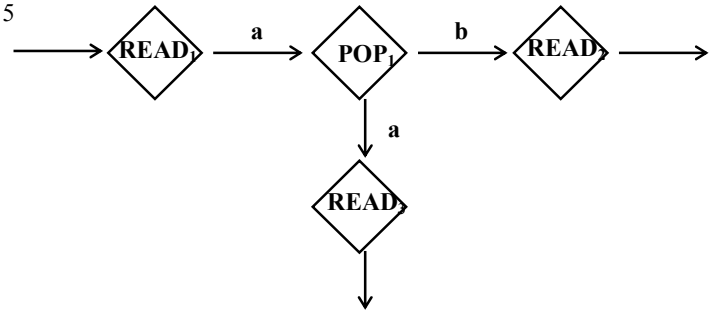


becomes

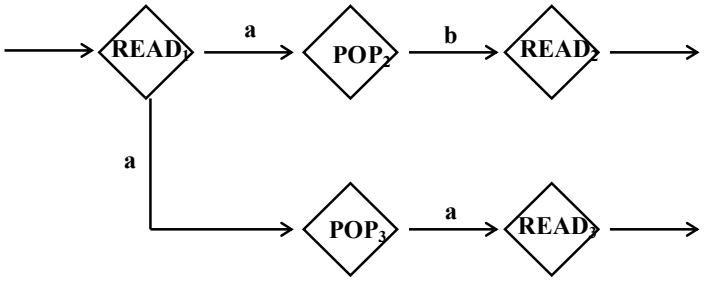


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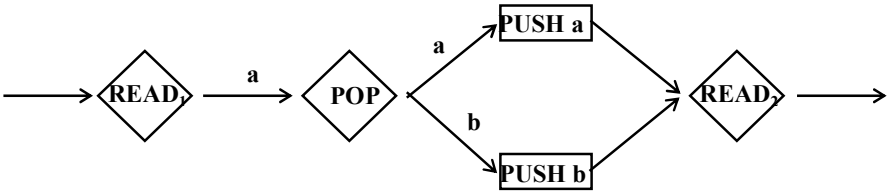
To satisfy condition 5



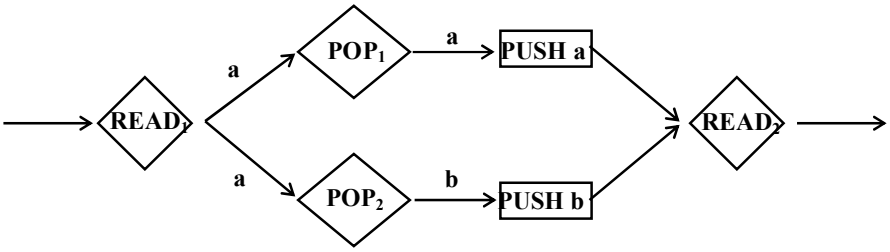
becomes



To satisfy condition 5



becomes



To satisfy condition 6, it is supposed that the STACK is initially in the position shown below

STACK

$
Δ

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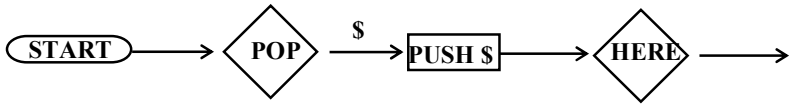
Theory of Automata

Lecture N0. 41
Reading Material

Introduction to Computer Theory Chapter 15

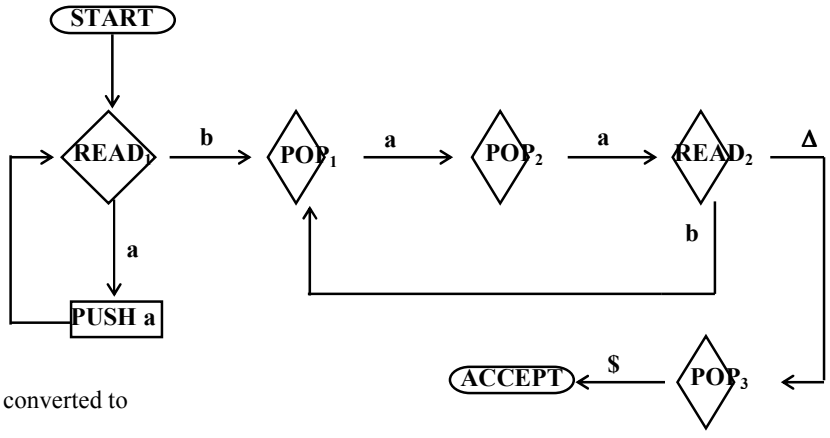
Summary
Recap of PDA in conversion form, example of PDA in conversion form, joints of the machine, new pictorial representation of PDA in conversion form, summary table, row sequence, row language.

Conversion form of PDA
Definition
A PDA is in conversion form if it fulfills the following conditions:
There is only one ACCEPT state.
There are no REJECT states.
Every READ or HERE is followed immediately by a POP *i.e.* every edge leading out of any READ or HERE state goes directly into a POP state.
No two POPs exist in a row on the same path without a READ or HERE between them whether or not there are any intervening PUSH states (*i.e.* the POP states must be separated by READs or HEREs).
All branching, deterministic or nondeterministic occurs at READ or HERE states, none at POP states and every edge has only one label.
Even before we get to START, a "bottom of STACK" symbol $ is placed on the STACK. If this symbol is ever popped in the processing it must be replaced immediately. The STACK is never popped beneath this symbol.
Right before entering ACCEPT this symbol is popped out and left.
The PDA must begin with the sequence



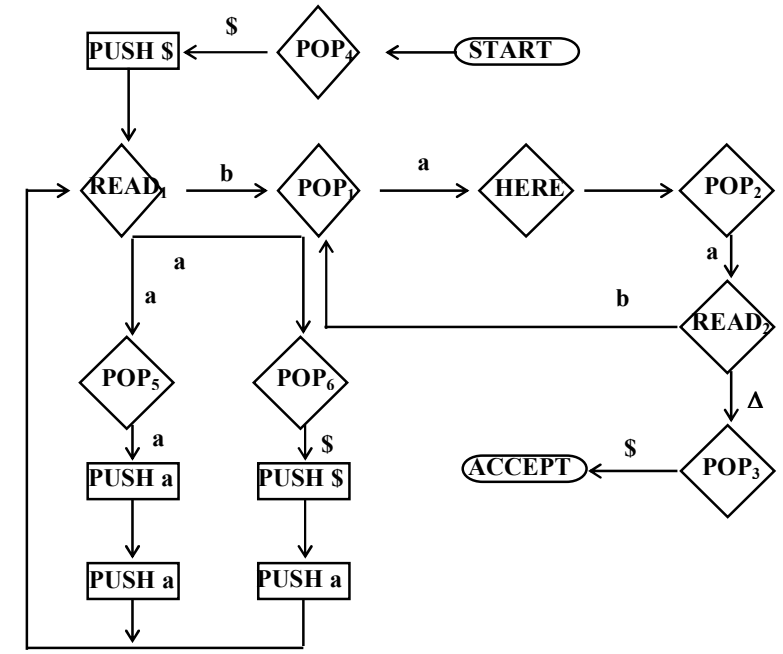
The entire input string must be read before the machine can accept the word.

Example
Consider the following PDA accepting the language $\{a^{2n}b^n : n = 1,2,3, \ldots\}$



Which may be converted to

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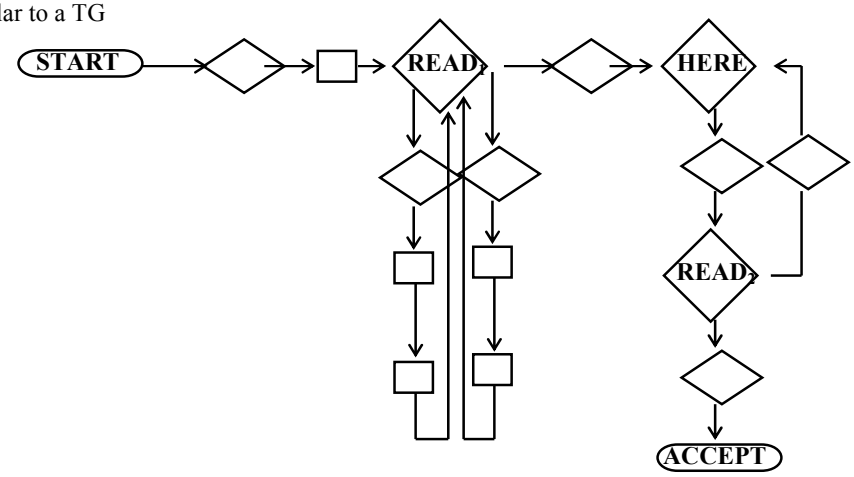


The above PDA accepts exactly the same language

Note
It may be noted that any PDA which is in **conversion form** can be considered to be the collection of path segments, where each path segment is of the following form

FROM	TO	READ	POP	PUSH
START or READ or HERE	READ or HERE or ACCEPT	ONE or no input letter	Exactly one STACK character	Any string onto the STACK

START, READ, HERE and ACCEPT states are called the **joints of the machine.** Between two consecutive joints on a path exactly one character is popped and any number of characters can be pushed.
The PDA which is in the **conversion form** can be supposed to be the set of joints with path segments in between, similar to a TG



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The above entire machine can be described as a list of all joint-to-joint path segments, called summary table. The PDA converted to the **conversion form** has the following summary table

FROM Where	TO Where	READ What	POP What	PUSH What	ROW Number
START	$READ_1$	Λ	$	$	1
$READ_1$	$READ_1$	a	$	a$	2
$READ_1$	$READ_1$	a	a	aa	3
$READ_1$	HERE	b	a	--	4
HERE	$READ_2$	Λ	a	--	5
$READ_2$	HERE	b	a	--	6
$READ_2$	AT	Δ	$	--	7

Consider the word aaaabb. This word is accepted by the above PDA through the following path START–POP_4–PUSH$–$READ_1$–$POP_6$–PUSH $–PUSH a–$READ_1$–$POP_5$–PUSH a–PUSH a– $READ_1$–POP_5–PUSH a–PUSH a–$READ_1$–POP_5–PUSH a–PUSH a–$READ_1$–POP_1– HERE–POP_2–$READ_2$–POP_1–HERE–POP_2–$READ_2$–POP_3–ACCEPT.
The above path can also be expressed by the following path in terms of sequence of rows
Row_1 –Row_2 –Row_3 –Row_3 –Row_3 –Row_4 –Row_5 –Row_6 –Row_5 –Row_7
It can be observed that the above path is not only **joint-to-joint consistent** but **STACK consistent** as well.
It may be noted that in FAs, paths correspond to strings of letters, while in PDAs, paths correspond to strings of rows from the summary table.

Note
It may be noted that since the HERE state reads nothing from the TAPE, therefore Λ is kept in the **READ what** column.
It may also be noted that the summary table contains all the information of the PDA which is in the pictorial representation. Every path through the PDA is a sequence of rows of the summary table. However, not every sequence of rows from the summary table represents a viable path, *i.e.* every sequence of rows may not be **STACK consistent**.
It is very important to determine which sequences of rows do correspond to possible paths through the PDA, because the paths are directly related to the language accepted, *e.g.* Row_4 cannot be immediately followed by Row_6 because Row_4 leaves in HERE, while Row_6 begins in $Read_2$. Some information must be kept about the STACK before rows are concatenated.
To represent a path, a sequence of rows must be **joint-consistent** (the rows meet up end to end) and **STACK-consistent** (when a row pops a character it should be there at the top of the STACK).
The next target is to define **row language** whose alphabet is $\Sigma = \{Row_1, Row_2, \ldots, Row_7\}$ *i.e.* the alphabet consists of the letters which are the names of the rows in the summary table.

Note
It may be noted that the words of the row language trace **joint-to-joint** and **STACK consistent** paths, which shows that all the words of this language begin with Row_1 and end in Row_7. Consider the following row sequence Row_5 Row_5 Row_3 Row_6
This is string of 4 letters, but not word of the row language because
It does not represent a path starting from START and ending in ACCEPT state.
It is not joint consistent.
It is not STACK consistent.
Before the CFG that generates the language accepted by the given PDA, is determined, the CFG that generates the **row language** is to be determined. For this purpose new nonterminals are to be introduced that contain the information needed to ensure **joint** and **STACK consistency.**
It is not needed to maintain any information about what characters are read from the TAPE.

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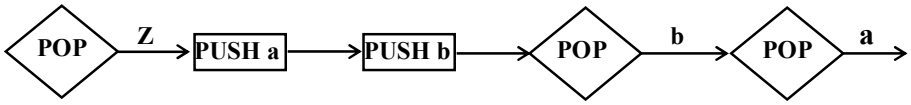
Theory of Automata

Lecture N0. 42
Reading Material

Introduction to Computer Theory Chapter 15

Summary
Row language, nonterminals defined from summary table, productions defined by rows, rules for defining productions, all possible productions of CFG for row language of the example under consideration, CFG corresponding to the given PDA

Note
As has already been discussed that the Row language is the language whose alphabet
$\Sigma = \{Row_1, Row_2, \ldots, Row_7\}$, for the example under consideration, so to determine the CFG of Row language, the nonterminals of this CFG are introduced in the form **Net(X, Y, Z)**
where X and Y are joints and Z is any STACK character. Following is an example of Net(X, Y, Z)



If the above is the path segment between two joints then, the net STACK effect is same as POP Z.
For a given PDA, some sets of all possible sentences Net(X, Y, Z) are true, while other are false. For this purpose every row of the summary table is examined whether the net effect of popping is exactly one letter.
Consider the Row_4 of the summary table developed for the PDA of the language $\{a^{2n}b^n\}$

FROM Where	TO Where	READ What	POP What	PUSH What	ROW Number
$READ_1$	HERE	b	a	--	4

The nonterminal corresponding to the above row may be written as Net ($READ_1$, HERE, a) *i.e.* Row_4 is a single Net row.
Consider the following row from an arbitrary summary table

FROM Where	TO Where	READ What	POP What	PUSH What	ROW Number
$READ_9$	$READ_3$	b	b	abb	11

which shows that Row_{11} is not Net style sentence because the trip from $READ_9$ to $READ_3$ does not pop one letter form the STACK, while it adds two letters to the STACK. However Row_{11} can be concatenated with some other Net style sentences *e.g.* $Row_{11}Net(READ_3, READ_7, a)Net(READ_7, READ_1, b)Net(READ_1, READ_8, b)$
Which gives the nonterminal
$Net(READ_9, READ_8, b)$, now the whole process can be written as
$Net(READ_9, READ_8, b) \rightarrow Row_{11}Net(READ_3, READ_7, a)\ Net(READ_7, READ_1, b)Net(READ_1, READ_8, b)$
Which will be a production in the CFG of the corresponding **row language**.
In general to create productions from rows of summary table, consider the following row in certain summary table

FROM Where	TO Where	READ What	POP What	PUSH What	ROW Number
$READ_x$	$READ_y$	u	w	$\mathbf{m_1m_2\ldots m_n}$	i

then for any sequence of joint states $S_1, S_2, \ldots S_n$, the production in the row language can be included as

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$Net(READ_x, S_n, w) \rightarrow Row_i Net(READ_y, S_1, m_1) \ldots Net(S_{n-1}, S_n, m_n)$
It may be noted that in CFG, in general, replacing a nonterminal with string of some other nonterminals does not always lead to a word in the corresponding CFL *e.g.* $S \rightarrow X|Y$, $X \rightarrow ab$, $Y \rightarrow aYY$
Here $Y \rightarrow aYY$ does not lead to any word of the language.
Following are the three rules of defining all possible productions of CFG of the row language
The trip starting from START state and ending in ACCEPT state with the NET style
Net(START, ACCEPT, $) gives the production of the form S → Net(START, ACCEPT, $)
From the summary table the row of the following form

FROM Where	TO Where	READ What	POP What	PUSH What	ROW Number
X	Y	anything	z	--	i

Defines the productions of the form $Net(X,Y,z) \rightarrow Row_i$
For each row that pushes string of characters on to the STACK of the form

FROM Where	TO Where	READ What	POP What	PUSH What	ROW Number
$READ_x$	$READ_y$	u	w	$\mathbf{m_1 m_2 \ldots m_n}$	i

then for any sequence of joint states $S_1, S_2, \ldots S_n$, the production in the row language can be included as
$Net(READ_X, S_n, w) \rightarrow Row_i Net(READ_Y, S_1, m_1) \ldots Net(S_{n-1}, S_n, m_n)$
It may be noted that this rule introduces new productions. It does not mean that each production of the form Nonterminal → string of nonterminals, helps in defining some word of the language.

Note
Considering the example of PDA accepting the language $\{a^{2n}b^n : n=1, 2, 3, \ldots\}$, using rule1, rule2 and rule3 the possible productions for the CFG of the row language are
$S \rightarrow Net(START, ACCEPT, \$)$
$Net(READ_1, HERE, a) \rightarrow Row_4$
$Net(HERE, READ_2, a) \rightarrow Row_5$
$Net(READ_2, HERE, a) \rightarrow Row_6$
$Net(READ_2, ACCEPT, \$) \rightarrow Row_7$
$Net(START, READ_1, \$) \rightarrow Row_1 Net(READ_1, READ_1, \$)$
$Net(START, READ_2, \$) \rightarrow Row_1 Net(READ_1, READ_2, \$)$
$Net(START, HERE, \$) \rightarrow Row_1 Net(READ_1, HERE, \$)$
$Net(START, ACCEPT, \$) \rightarrow Row_1 Net(READ_1, ACCEPT, \$)$
$Net(READ_1, READ_1, \$) \rightarrow Row_2 Net(READ_1, READ_1, a) Net(READ_1, READ_1, \$)$
$Net(READ_1, READ_1, \$) \rightarrow Row_2 Net(READ_1, READ_2, a) Net(READ_2, READ_1, \$)$
$Net(READ_1, READ_1, \$) \rightarrow Row_2 Net(READ_1, HERE, a) Net(HERE, READ_1, \$)$
$Net(READ_1, READ_2, \$) \rightarrow Row_2 Net(READ_1, READ_1, a) Net(READ_1, READ_2, \$)$
$Net(READ_1, READ_2, \$) \rightarrow Row_2 Net(READ_1, READ_2, a) Net(READ_2, READ_2, \$)$
$Net(READ_1, READ_2, \$) \rightarrow Row_2 Net(READ_1, HERE, a) Net(HERE, READ_2, \$)$
$Net(READ_1, HERE, \$) \rightarrow Row_2 Net(READ_1, READ_1, a) Net(READ_1, HERE, \$)$
$Net(READ_1, HERE, \$) \rightarrow Row_2 Net(READ_1, READ_2, a) Net(READ_2, HERE, \$)$
$Net(READ_1, HERE, \$) \rightarrow Row_2 Net(READ_1, HERE, a) Net(HERE, HERE, \$)$
$Net(READ_1, ACCEPT, \$) \rightarrow Row_2 Net(READ_1, READ_1, a) Net(READ_1, ACCEPT, \$)$
$Net(READ_1, ACCEPT, \$) \rightarrow Row_2 Net(READ_1, READ_2, a) Net(READ_2, ACCEPT, \$)$
$Net(READ_1, ACCEPT, \$) \rightarrow Row_2 Net(READ_1, HERE, a) Net(HERE, ACCEPT, \$)$
$Net(READ_1, READ_1, a) \rightarrow Row_3 Net(READ_1, READ_1, a) Net(READ_1, READ_1, a)$
$Net(READ_1, READ_1, a) \rightarrow Row_3 Net(READ_1, READ_2, a) Net(READ_2, READ_1, a)$
$Net(READ_1, READ_1, a) \rightarrow Row_3 Net(READ_1, HERE, a) Net(HERE, READ_1, a)$
$Net(READ_1, READ_2, a) \rightarrow Row_3 Net(READ_1, READ_1, a) Net(READ_1, READ_2, a)$
$Net(READ_1, READ_2, a) \rightarrow Row_3 Net(READ_1, READ_2, a) Net(READ_2, READ_2, a)$
$Net(READ_1, READ_2, a) \rightarrow Row_3 Net(READ_1, HERE, a) Net(HERE, READ_2, a)$

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$Net(READ_1, HERE, a) \rightarrow Row_3Net(READ_1, READ_1, a)Net(READ_1, HERE, a)$
$Net(READ_1, HERE, a) \rightarrow Row_3Net(READ_1, READ_2, a)Net(READ_2, HERE, a)$
$Net(READ_1, HERE, a) \rightarrow Row_3Net(READ_1, HERE, a)Net(HERE, HERE, a)$
$Net(READ_1, ACCEPT, a) \rightarrow Row_3Net(READ_1, READ_1, a)Net(READ_1, ACCEPT, a)$
$Net(READ_1, ACCEPT, a) \rightarrow Row_3Net(READ_1, READ_2, a)Net(READ_2, ACCEPT, a)$
$Net(READ_1, ACCEPT, a) \rightarrow Row_3Net(READ_1, HERE, a)Net(HERE, ACCEPT, a)$

Following is a left most derivation of a word of row language

$S \Rightarrow Net(START, ACCEPT, \$)$ … using 1
$\Rightarrow Row_1Net(READ_1, ACCEPT, \$)$ … using 9
$\Rightarrow Row_1Row_2Net(RD_1, RD_2, a)Net(RD_2, AT, \$)$ … using 20
$\Rightarrow Row_1Row_2Row_3Net(RD_1, HERE, a)Net(RD_2, HERE, a)Net(RD_2, AT, \$)$ … using 27
$\Rightarrow Row_1Row_2Row_3Row_4Net(HERE, RD_2, a)Net(RD_2, ACCEPT, \$)$ … using 2
$\Rightarrow Row_1Row_2Row_3Row_4Row_5Net(HERE, ACCEPT, \$)$ … using 3
$\Rightarrow Row_1Row_2Row_3Row_4Row_5Row_7$ … using 5

Which is the shortest word in the whole row language.

It can be observed that each left most derivation generates the sequence of rows of the summary table, which are both **joint-** and **STACK- consistent**.

Note: So far the rules have been defined to create all possible productions for the CFG of the row language. Since in each row in the summary table, the READ column contains Λ and Δ in addition to the letters of the alphabet of the language accepted by the PDA, so each word of the row language generates the word of the language accepted by the given PDA.

Thus the following rule 4 helps in completing the CFG corresponding to the given PDA

Each row of the summary table defines a production of the form $Row_i \rightarrow a$ where in Row_i the READ column consists of letter a.

Application of rule 4 to the summary table for the PDA accepting $\{a^{2n}b^n : n=1,2,3,\ldots\}$ under consideration adds the following productions

$Row_1 \rightarrow \Lambda$
$Row_2 \rightarrow a$
$Row_3 \rightarrow a$
$Row_4 \rightarrow b$
$Row_5 \rightarrow \Lambda$
$Row_6 \rightarrow b$
$Row_7 \rightarrow \Delta$

Which shows that the word $Row_1Row_2Row_3Row_4Row_5Row_7$ of the row language is converted to $\Lambda aab\Lambda\Delta = abb$

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Theory of Automata

Lecture N0. 43
Reading Material

Introduction to Computer Theory Chapter 16

Summary
Non-Context-Free-languages, Live Production, Dead Production, Theorem, self- embedded nonterminal, Pumping lemma for CFLs, Examples

Non-Context-Free language
There arises a question, whether all languages are CFL? The answer is no.
Languages which are not Context-Free, are called Non-CFL.

To prove the claim that all languages are not Context-Free, the study of machines of word production from the grammar is needed

Live production: A production of the form nonterminal $\rightarrow$ string of two nonterminals is called a live production.

Dead production: A production of the form nonterminal $\rightarrow$ terminal is called a dead production.

It may be noted that every CFG in CNF has only these types of productions.

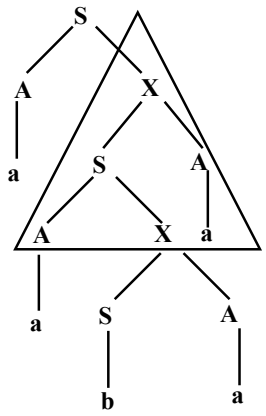
Theorem
If a CFG is in CNF and if there is restriction to use the live production at most once each, then only the finite many words can be generated.
It may be noted that every time a live production is applied during the derivation of a word it increases the number of nonterminals by one.
Similarly applying dead production decreases the nonterminals by one. Which shows that to generate a word, one more dead production are applied than the live productions *e.g.*

$S \Rightarrow XY$
$\Rightarrow aY$
$\Rightarrow aa$

Here one live and two dead productions are used.
In general, if a CFG in CNF has p live and q dead productions then all words generated without repeating any live production have at most (p+1) letters.

Theorem
If a CFG is in CNF with p live and q dead productions and if w is word generated by the CFG, having more than 2^p letters then any derivation tree for w has a nonterminal z which is used twice, where the second z is the descended from the first z.

It can be observed from the above theorem that generation tree of word w has more than p rows.
Self-embedded nonterminal
A nonterminal is said to be self-embedded, if in a given derivation of a word, it ever occurs as a tree descendant of itself, as shown in figure aside



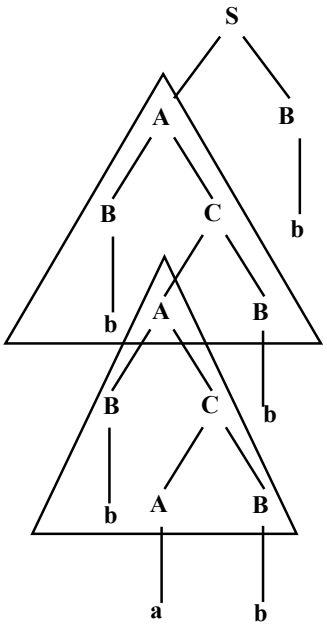
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Here the nonterminal X is self-embedded.

Note
Consider the following CFG in CNF

$S \rightarrow AB$
$A \rightarrow BC$
$C \rightarrow AB$
$A \rightarrow a$
$B \rightarrow b$

and the derivation tree of the word bbabbb



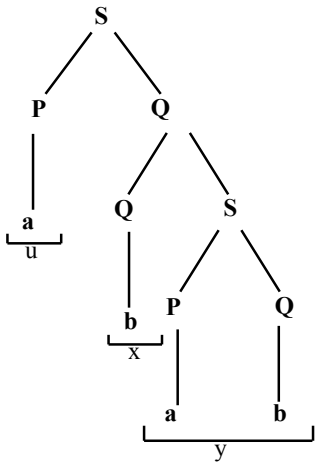
Note
The part of tree enclosed in upper triangle is identical to that enclosed in lower triangle, there is still another option of replacing A by the same sequence of production shown in lower triangle.
The above fact provides the following pumping lemma for the CFLs.

Pumping lemma for CFLs
Theorem
If G is any CFG in CNF with p live productions, then every word w of length more than 2^p can be partitioned into five substrings as w = uvxyz, where x is not null string and v and y are not both null string.
Then all the words of the form uv^nxy^nz, n = 1,2,3,… can also be generated by G.

Example
Consider the following CFG which is in CNF

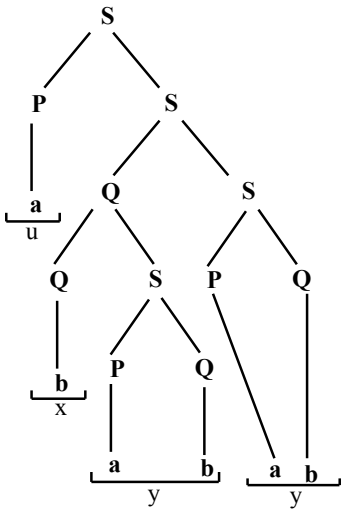
$S \rightarrow PQ$
$Q \rightarrow QS|b$
$P \rightarrow a$

and a word abab generated by the above CFG with the following derivation tree



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Then w can be broken up as w = uvxyz where u = a, v = Λ, x = b, y = ab, z = Λ
Repeating the triangle from the second Q just as it descends from the first Q, the corresponding tree may be expressed as follows



Which shows that uvvxyyz=aΛΛbababΛ=ababab belongs to the language generated by the given CFG.
So, it can be generalized that words of the form uv^nxy^nz, n=1,2,3,… belong to the language generated by the given CFG.

Note
It may be noted that the pumping lemma is satisfied by all CFLs and the languages which don't hold this pumping lemma, can't be Context Free languages. Such languages are non-CFLs.

Example
Consider the language
$L=\{a^nb^nc^n : n=1,2,3,\ldots\}$, let the language L be Context Free language and let the word $w=a^{200}b^{200}c^{200}$ of length more than 2^p, where p is the number of live productions of its CFG in CNF.

Note
It can be observed that no matter what choices are made for the substrings u,v,x,y and z, uv^2xy^2z can't belong to L, as all the words in $a^nb^nc^n$ have

Only one substring ab
Only one substring bc
No substring ac
No substring ba
No substring ca
No substring cb
For any n=1,2,3,…

The above observations shows that if v or y is not single letter or Λ, then uv^2xy^2z may contain either two or more substrings ab or bc or one or more substrings ac or ba or ca or cb *i.e.* these strings may be in the number more than the number they are supposed to be.
Moreover, if v and y are either single letter or Λ, then one or two of letters a,b,c will be increased, where as the other letter will not be increased in uv^2xy^2z, which shows uv^2xy^2z does not belong to L.
Thus pumping lemma is not satisfied. Hence L is non CFL.

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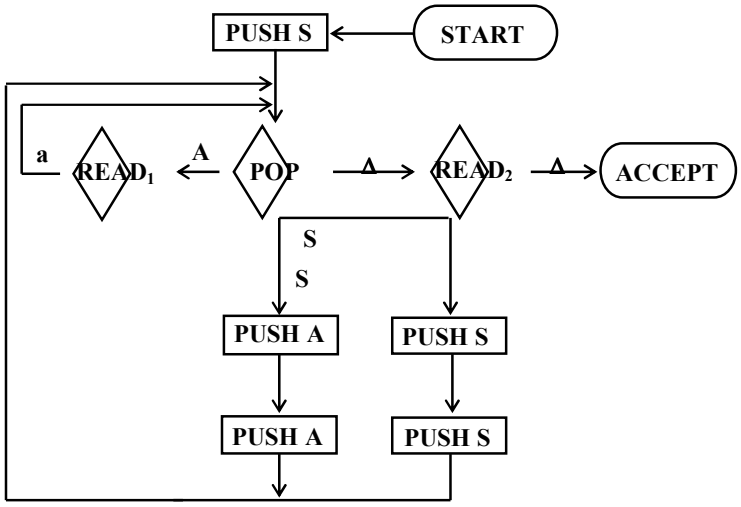
It may be noted that the pumping lemma discussed for infinite regular language L, **the word w can be decomposed into three parts w=xyz, such that all words of the form xy^nz, n=1,2,3,…, belong to L.**
Similarly, the pumping lemma discussed for CFLs can also stated as
If w is a large enough word in a CF: then, w can be decomposed into w=uvxyz such that all words of the form uv^nxy^nz belong to L
It may be noted that proof of pumping lemma for regular languages needed that path of word w to be so large enough so that it contains a circuit and circuit can be looped as many times as one can. The proof of the pumping lemma for CFLs needs the derivation for w to be so large that it contains a sequence of productions that can be repeated as many times as one can.
Moreover, the pumping lemma for regular languages does not hold for non regular language as that language does not contain both xyz and xyyz.
Similarly pumping lemma for CFLs does not hold for non-CFL as that language does not contain both uvxyz and uvvxyyz.
There is another difference between the pumping lemma for regular languages and that for CFLs that first one acts on machines while other acts on algebraic structures *i.e.* grammar.
To achieve full power the pumping lemma for regular languages has modified by pumping lemma version II.
Similarly, full power for pumping lemma for CFLs is achieved by stating the following theorem

Theorem
If L is a CFL in CNF with p live productions then any word W in L of length more than 2^p can be decomposed as w=uvxyz *s.t.* length(vxy) $\leq 2^p$, length(x) > 0, length(v)+length(y) > 0
then the words of the form uv^nxy^nz : n=1,2,3,… belong to L.

Example
Consider the language
L= $\{a^nb^ma^nb^m$:m,n=1,2,3,…$\}$
$=\{$abab,aabaab, abbabb, aabbaabb, aaabaaab,… $\}$
The first version of pumping lemma for CFLs may be satisfied by L, but to apply the second version of pumping lemma to L, let L be generated by CFG which is in CNF and has p live productions.
Consider the word decomposing w into uvxyz where length(vxy) < 2^p which shows that v and y can't be single letters separated by clumps of other letter because the separator letter is longer than the length of whole substring vxy, which shows that uvvxyyz is not contained in L. Thus pumping lemma is not satisfied and L is non CFL.

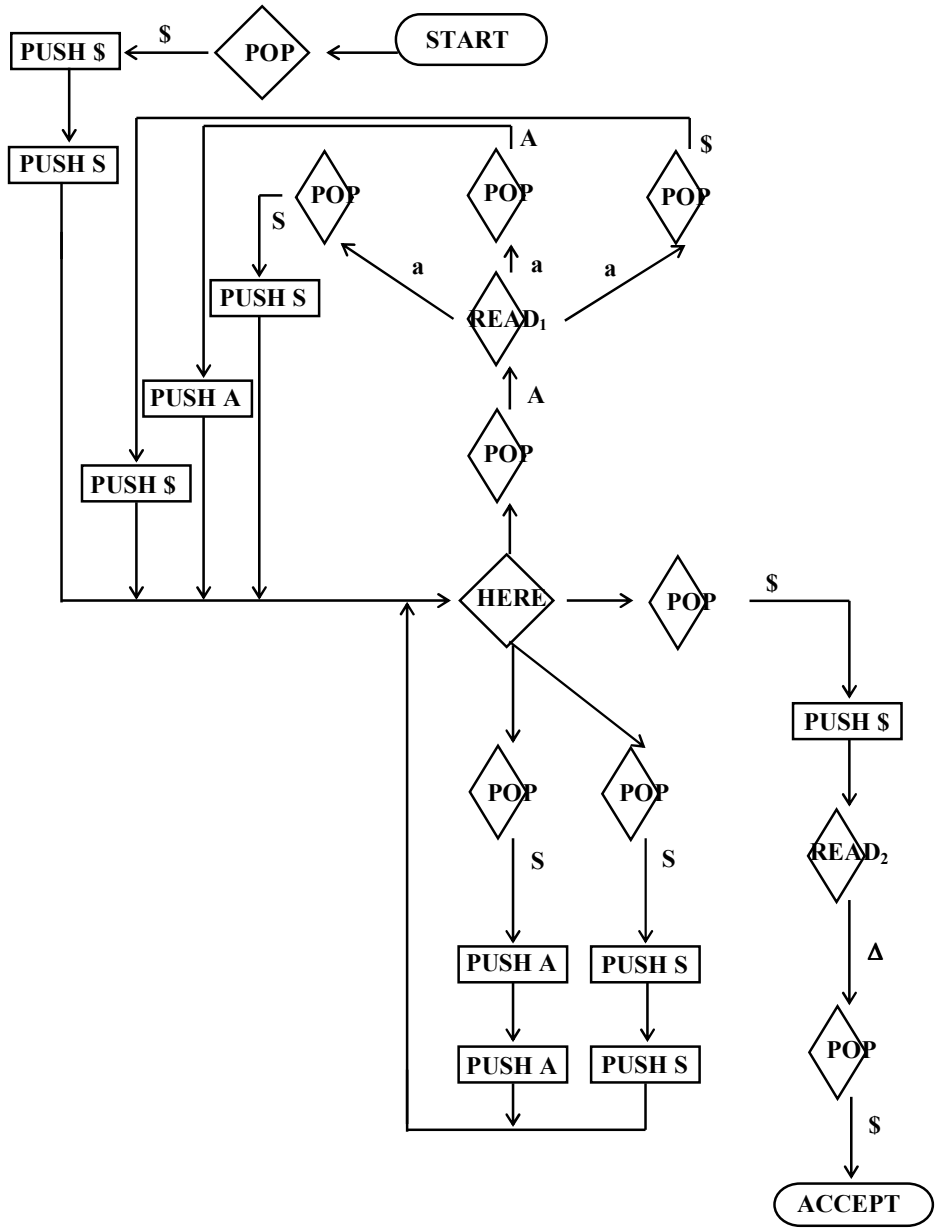
Example

Consider the language EVENA *i.e.*
EVENA=$(aa)^n=a^{2n}=\{$aa, aaaa, aaaaaa, …$\}$
The grammar for this language must be
S → SS|aa and its CNF will be
S → SS|AA, A → a, the PDA for this grammar will be as under



Its corresponding **conversion form** will be

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The summary table corresponding to the above PDA in **conversion form** can be expressed as

FROM	TO	READ	POP	PUSH	ROW
START	HERE	Λ	$	S$	1
HERE	HERE	Λ	S	SS	2
HERE	HERE	Λ	S	AA	3
HERE	$READ_1$	Λ	A	--	4
$READ_1$	HERE	a	S	S	5
$RAED_1$	HERE	a	$	$	6
$READ_1$	HERE	a	A	A	7
HERE	$READ_2$	Λ	$	$	8
$READ_2$	ACCEPT	Δ	$	--	9

Following are the productions defined from the summary table

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$S \rightarrow Net(START, ACCEPT, \$)$
$Net(HERE, READ_1, A) \rightarrow Row_4$
$Net(READ_2, ACCEPT, \$) \rightarrow Row_9$
$Net(START, X, \$) \rightarrow Row_1\ Net(HERE, Y, S)Net(Y, X, \$)$
$Net(HERE, X, S) \rightarrow Row_2\ Net(HERE, Y, S)Net(Y, X, S)$
$Net(START, X, S) \rightarrow Row_3\ Net(HERE, Y, A)Net(Y, X, A)$
$Net(READ_1, X, S) \rightarrow Row_5\ Net(HERE, X, S)$ gives four productions
$Net(READ_1, X, \$) \rightarrow Row_6\ Net(HERE, X, \$)$ gives four productions
$Net(READ_1, X, A) \rightarrow Row_7\ Net(HERE, X, A)$ gives four productions
$Net(HERE, ACCEPT, \$) \rightarrow Row_8\ Net(READ_2, ACCEPT, \$)$

Where X and Y are the corresponding joints

In addition to 44 productions following 9 productions complete the required CFG

$Row_1 \rightarrow \Lambda$
$Row_2 \rightarrow \Lambda$
$Row_3 \rightarrow \Lambda$
$Row_4 \rightarrow \Lambda$
$Row_5 \rightarrow a$
$Row_6 \rightarrow a$
$Row_7 \rightarrow a$
$Row_8 \rightarrow \Lambda$
$Row_9 \rightarrow \Lambda$

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Theory of Automata

Lecture N0. 44
Reading Material

Introduction to Computer Theory Chapter 18

Summary
Decidability, whether a CFG generates certain string (emptiness), examples, whether a nonterminal is used in the derivation of some word (uselessness), examples, whether a CFL is finite (finiteness), example, whether the given string is generated by the given CFG (membership), example, parsing techniques, top down parsing, example

Decidablity
Following are the decidable problems w.r.t. CFG
Whether or not the given CFG generates any word? Problem of emptiness of CFL.
Whether or not the given CFG generates the finite language? Problem of finiteness.
Whether or not the given string w can be generated by the given CFG? Problem of membership.

Following are algorithms showing that the answers to the above three questions are yes.

Algorithm 1 (Emptiness)
If the given CFG contains a production of the form $S \rightarrow \Lambda$, then obviously the corresponding CFL is not empty.
If the CFG contains the production of the form $S \rightarrow t$, where t is a terminal or string of terminal then t is a word of the corresponding CFL and CFL is not empty.
If the CFG contains no such production then
For each nonterminal N with $N \rightarrow t$, pick one production for N (if there are more than one) and replace N by t in the right side of each production wherever it lies. Remove all such productions from the CFG. Doing so the CFG will be changed, it will generate atleast one word of the old CFL.

Repeat the process until either it eliminates S or no new nonterminal is eliminated.
If S has been eliminated then CFG generates some words otherwise not.

Example

$S \rightarrow AB$, $A \rightarrow BSB$, $B \rightarrow CC$
$C \rightarrow SS$
$A \rightarrow a|b$
$C \rightarrow b|bb$

Step (1). Picking $A \rightarrow a$, $C \rightarrow b$, it can be written as

$S \rightarrow aB$
$A \rightarrow BSB$
$A \rightarrow bb$
$B \rightarrow aaS$
$B \rightarrow bb$
$C \rightarrow SS$

Step (1). Picking $B \rightarrow bb$ and $A \rightarrow bb$, it can be written as

$S \rightarrow abb$
$A \rightarrow bbSbb$
$B \rightarrow aaS$
$C \rightarrow SS$

Since $S \rightarrow abb$ has been obtained so, abb is a word in the corresponding CFL.

To determine whether the nonterminal X is ever used in the derivation of word from the given CFG, following algorithm is used

Algorithm 2 (Uselessness)
Find all unproductive nonterminals (the nonterminal is unproductive if it cannot produce a string of terminals).
Eliminate all productions involving unproductive nonterminals.

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Paint all X's blue.
If any nonterminal is in the left side of the production with any blue nonterminal in the right side, paint that nonterminal blue and paint that nonterminal blue at all occurrences of it throughout the grammar.
Repeat step 4 until no new nonterminal is painted.
If S is blue then X is useful member of CFG, otherwise not.

Example
Consider the following CFG

S→Aba | bAZ | b
A→Xb | bZa
B→bAA
X→aZa|aaa
Z→ZAbA

To determine whether X is ever used to generate some words, unproductive nonterminals are determined. Z is unproductive nonterminal, so eliminating the productions involving Z.

S→Aba|b
A→Xb
B→bAA
X→aaa

X is blue, so A is blue. Thus B and S are also blue. Since S is blue so X can be used to generate certain word from the given CFG.
Note: It may be noted that a nonterminal is called useless if it cannot be used in a production of some word.
Following algorithm is used to determine whether the given CFG generate the finite language

Algorithm 3 (Finiteness)
Determine all useless nonterminals and eliminate all productions involving these nonterminals.
For each of the remaining nonterminals, determine whether they are self-embedded (using the following steps).
Stop if a self-embedded nonterminal is discovered.
To test whether X is self-embedded
Change all X's on the left side of the productions into a Greek letter Ψ and keep all X's on the right side as such.
Paint all X's blue.
If Y is any nonterminal on the left side of the production with X in the right side, then paint Y blue.
Repeat step (c) until no new nonterminal is painted.
If Ψ is painted, then the X is self-embedded, otherwise not.
If any nonterminal, left in the grammar, after step 1, is self-embedded then the language generated is infinite, otherwise finite.

Example
Consider the CFG
S→ABa|bAZ|b
A→Xb|bZa
B→bAA
X→aZa|bA|aaa
Z→ZAbA
Here the nonterminal Z is useless, while all other are used in the derivation of some word. So eliminating the productions involving Z
S→ABa|b
A→Xb
B→bAA
X→bA|aaa
Starting with nonterminal X. Replacing X on left side of the production by Ψ
S→ABa|b
A→Xb
B→bAA
Ψ →bA|aaa
X is blue so A is blue and so Ψ is blue. Since A is blue, so B is blue and so S is blue. Since Ψ is blue so X is self-embedded and hence the CFG generates the infinite language.
To determine whether a string is generated by the given CFG, following algorithm is used

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Algorithm 4 (The CYK algorithm)

This algorithm was invented by John Cocke and later was published by Tandao Kasami and Daniel H. Younger.

Convert the given CFG in CNF.

Let the string x under consideration has the form $x=x_1x_2x_3\ldots x_n$ where all x_is may not be different. List all the nonterminals in the given CFG, say, S, N_1,N_2, …

List the nonterminals that generates single letter substrings of x *i.e.*

Substring	All producing nonterminals
x_1	N…
x_2	N…
x_3	∂
∂	∂
x_n	N…

List the nonterminals that generates substrings of length 2 *i.e.*

Substring	All producing nonterminals
$x_1\ x_2$	N…
$x_2\ x_3$	N…
$x_3\ x_4$	∂
∂	∂
$x_{n-1}\ x_n$	N…

Similarly, list of nonterminals generating substring of x of length 3

Substring	All producing nonterminals
$x_1\ x_2\ x_3$	N…
$x_2\ x_3\ x_4$	N…
$x_3\ x_4\ x_5$	∂
∂	∂
$x_{n-2}\ x_{n-1}\ x_n$	N…

Continuing the process, the nonterminals that generate $x_1x_2x_3\ldots x_n$ can be determined as

Substring	All producing nonterminals
$x_1\ x_2\ x_3\ldots x_n$	N…

If S is among the set of all producing nonterminals, then x can be generated by the CFG, otherwise not.

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Example
Consider the following CFG in CNF
S→AA
A→AA
A→a
Let x = aaa. To determine whether x can be generated from the given CFG let $x = x_1x_2x_3$ where $x_1 = x_2 = x_3 = a$
According to CYK algorithm, the list of nonterminals producing single letter double letter substrings of x and the string x itself, can be determined as follows

Substring	All producing nonterminals
$x_1 = a$	A
$x_2 = a$	A
$x_3 = a$	A
$x_1 x_2$	S, A
$x_2 x_3$	S, A
$x = x_1 x_2 x_3$	S, A

Since S is in the list of producing nonterminals, so aaa can be generated by the given CFG.

Parsing Techniques
Recall the CFG for arithmetic expression
S→S+S|S*S|number
It was observed that the word 3+4*5 created ambiguity by considering its value either 23 or 35. To remove this ambiguity, the CFG was modified to
S→(S+S)|(S*S)|number
There arises a question that whether a new CFG can be defined without having parentheses with operator hierarchy (*i.e.* * before +)? The answer is yes. Following is the required PLUS-TIMES grammar
S→E, E→T+E|T, T→F*T|F, F→(E)|i
Where i stands for any identifier *i.e.* number or of storage location name (variable). Following is the derivation of i+i*i

S ⇒E
⇒T+E
⇒F+E
⇒i+E
⇒i+T
⇒i+F*T
⇒i+i*T
⇒i+i*F
⇒i+i*i

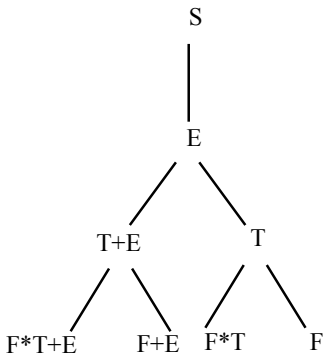
Parsing of word
Definition
The process of finding the derivation of word generated by particular grammar is called **parsing**.
There are different parsing techniques, containing the following three
Top down parsing.
Bottom up parsing.
Parsing technique for particular grammar of arithmetic expression.

Top down parsing
Following is an example showing top down parsing technique

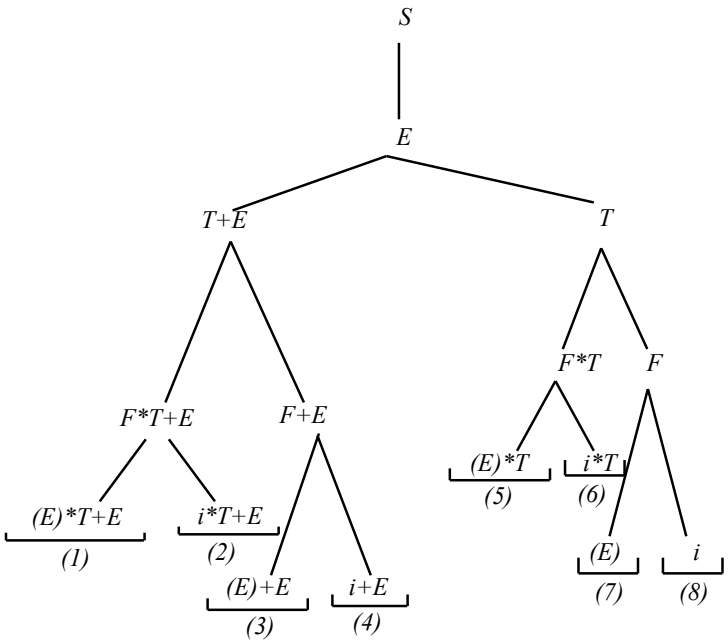
Example
Consider PLUS-TIMES grammar and a word i+i*i.

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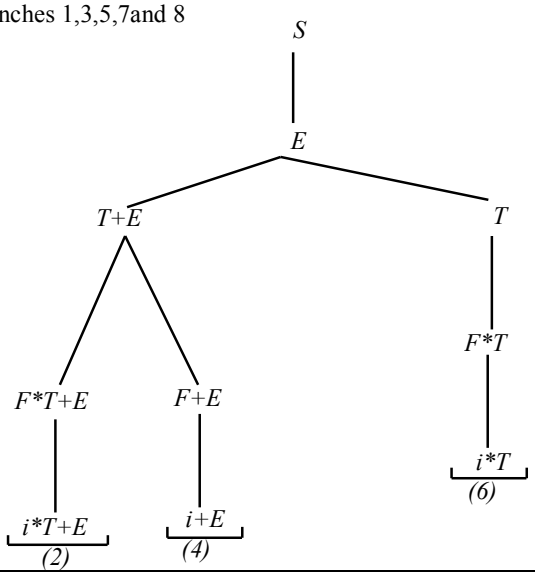
As can be observed from the name of top down parsing, the parsing starts from the nonterminal S and the structure similar to that of total language tree is developed. The branches of the tree are extended till the required word is found as a branch.
Some unwanted branches (the branches that don't lead to the required word) are dropped. For the word i+i*i , the total language tree can be started as



Which can further be extended to

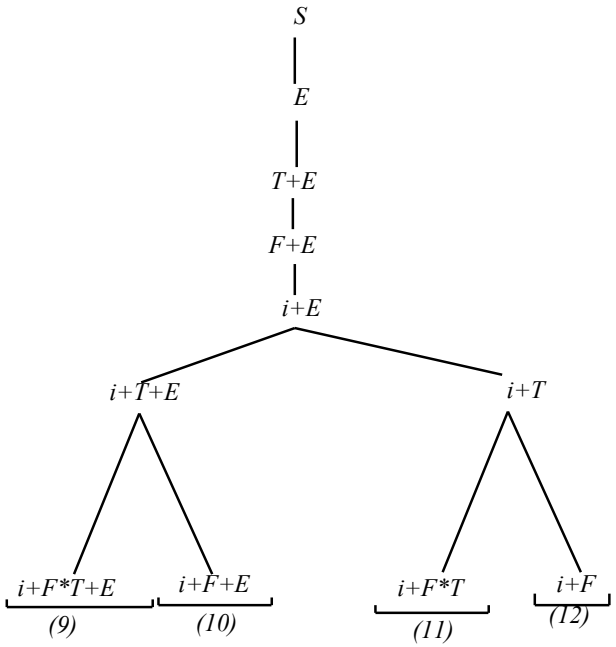


Dropping the unwanted branches 1,3,5,7and 8

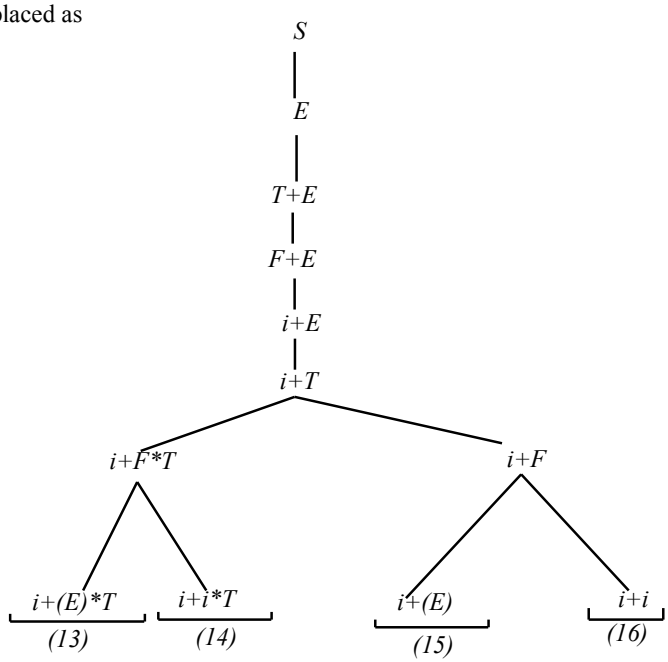


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Since first two letters in branches 2 and 6 are not that in i+i*i , so 2 and 6 can be dropped and using left most derivation the nonterminal T is replaced as

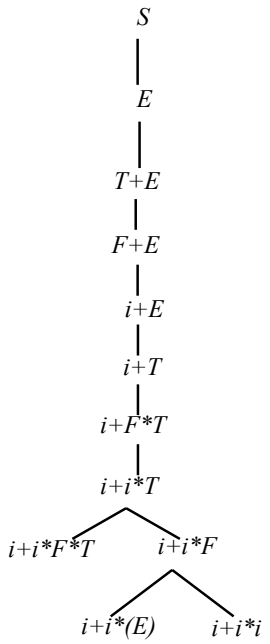


since (9) gives more than five letters and (10) contains two + so (9) and (10) are dropped and left most nonterminal F is replaced as



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(13), (15) and (16) are again unwanted, so it can be written as



The above tree confirms the required derivation

S ⇒E
⇒T+E
⇒F+E
⇒i+E
⇒i+T
⇒i+F*T
⇒i+i*T
⇒i+i*F
⇒i+i*i

Note

It can be noted that Bottom Up Parsing can be determined similar to that of Top Down Parsing with the change that in this case, the process is started with the given string and the tree is extended till S is obtained.

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Theory of Automata

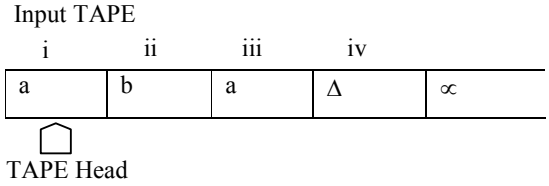
Lecture N0. 45
Reading Material

Introduction to Computer Theory Chapter 19

Summary
Turing machine, examples, DELETE subprogram, example, INSERT subprogram, example.

Turing machine
The mathematical models (FAs, TGs, PDAs) that have been discussed so far can decide whether a string is accepted or not by them *i.e.* these models are language identifiers. However, there are still some languages which can't be accepted by them *e.g.* there does not exist any FA or TG or PDA accepting any non-CFLs. Alan Mathison Turing developed the machines called Turing machines, which accept some non-CFLs as well, in addition to CFLs.

Definition
A Turing machine (TM) consists of the following
An alphabet ∑ of input letters.
An input TAPE partitioned into cells, having infinite many locations in one direction. The input string is placed on the TAPE starting its first letter on the cell i, the rest of the TAPE is initially filled with blanks (Δ's).



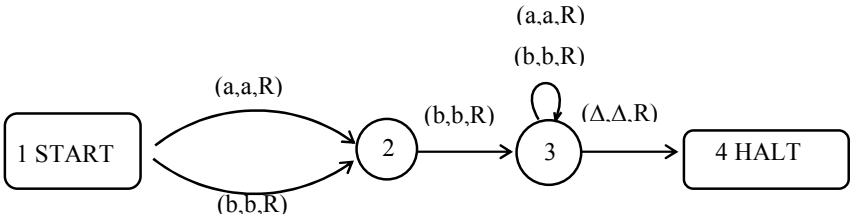
A tape Head can read the contents of cell on the TAPE in one step. It can replace the character at any cell and can reposition itself to the next cell to the right or to the left of that it has just read. Initially the TAPE Head is at the cell i. The TAPE Head can't move to the left of cell i. the location of the TAPE Head is denoted by .

An alphabet Γ of characters that can be printed on the TAPE by the TAPE Head. Γ may include the letters of ∑. Even the TAPE Head can print blank Δ, which means to erase some character from the TAPE.
Finite set of states containing exactly one START state and some (may be none) HALT states that cause execution to terminate when the HALT states are entered.
A **program** which is the set of rules, which show that which state is to be entered when a letter is read form the TAPE and what character is to be printed. This program is shown by the states connected by directed edges labeled by triplet (letter, letter, direction). It may be noted that the first letter is the character the TAPE Head reads from the cell to which it is pointing. The second letter is what the TAPE Head prints the cell before it leaves. The direction tells the TAPE Head whether to move one cell to the right, R, or one cell to the left, L.

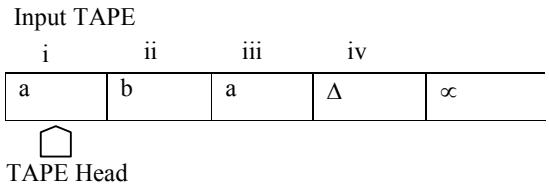
Note
It may be noted that there may not be any outgoing edge at certain state for certain letter to be read from the TAPE, which creates nondeterminism in Turing machines. It may also be noted that at certain state, there can't be more than one out going edges for certain letter to be read from the TAPE. The machine crashes if there is not path for a letter to be read from the TAPE and the corresponding string is supposed to be rejected.
To terminate execution of certain input string successfully, a HALT state must be entered and the corresponding string is supposed to be accepted by the TM. The machine also crashes when the TAPE Head is instructed to move one cell to the left of cell i.
Following is an example of TM

Example
Consider the following Turing machine

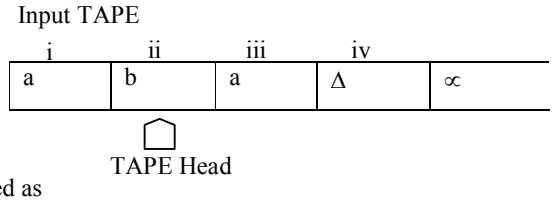
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Let the input string aba be run over this TM



Starting from the START state, reading a form the TAPE and according to the TM program, a will be printed *i.e.* a will be replaced by a and the TAPE Head will be moved one cell to the right.
Which can be seen as



This process can be expressed as

$$\underline{a}ba \longrightarrow a\underline{b}a$$

At state 2 reading b, state 3 is entered and the letter b is replaced by b, *i.e.*

$$\underline{a}ba \longrightarrow a\underline{b}a \longrightarrow ab\underline{a}$$

At state 3 reading a, will keep the state of the TM unchanged. Lastly, the blank Δ is read and Δ is replaced by Δ and the HALT state is entered. Which can be expressed as

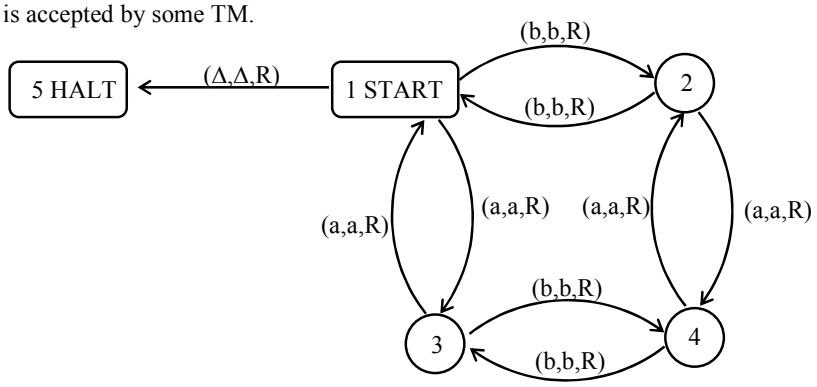
$$\underline{a}ba \longrightarrow a\underline{b}a \longrightarrow ab\underline{a} \longrightarrow aba\underline{\Delta} \longrightarrow \text{HALT}$$

Which shows that the string aba is accepted by this machine. It can be observed, from the program of the TM, that the machine accepts the language expressed by (a+b)b(a+b)*.

Theorem

Every regular language is accepted by some TM.

Example

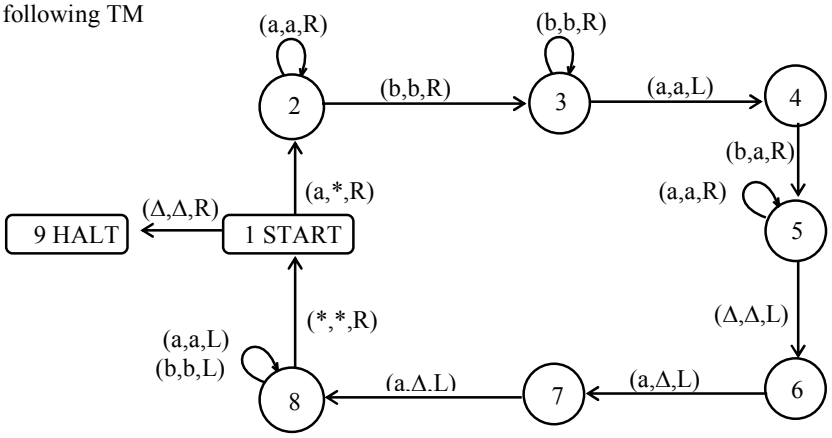


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Consider the EVEN-EVEN language. Following is a TM accepting the EVEN-EVEN language.

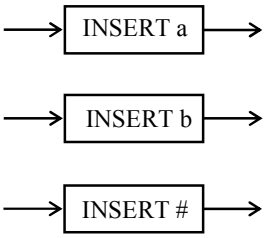
It may be noted that the above diagram is similar to that of FA corresponding to EVEN-EVEN language. Following is another example

Example
Consider the following TM



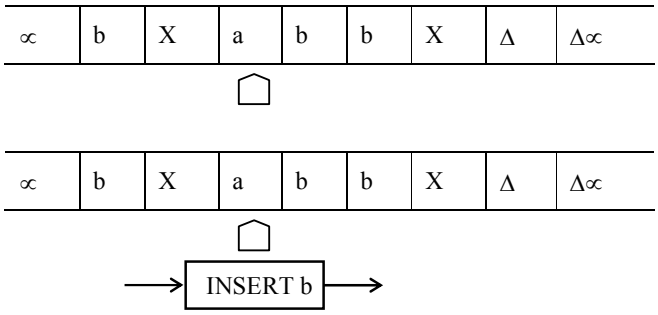
The string aaabbbaaa can be observed to be accepted by the above TM. It can also be observed that the above TM accepts the non-CFL $\{a^n b^n a^n\}$.

INSERT subprogram
Sometimes, a character is required to be inserted on the TAPE exactly at the spot where the TAPE Head is pointing, so that the character occupies the required cell and the other characters on the TAPE are moved one cell right. The characters to the left of the pointed cell are also required to remain as such.
In the situation stated above, the part of TM program that executes the process of insertion does not affect the function that the TM is performing. The subprogram of insertion is independent and can be incorporated at any time with any TM program specifying what character to be inserted at what location. The subprogram of insertion can be expressed as



The above diagrams show that the characters a,b and # are to be inserted, respectively. Following is an example showing how does the subprogram INSERT perform its function

Example
If the letter b is inserted at the cell where the TAPE Head is pointing as shown below



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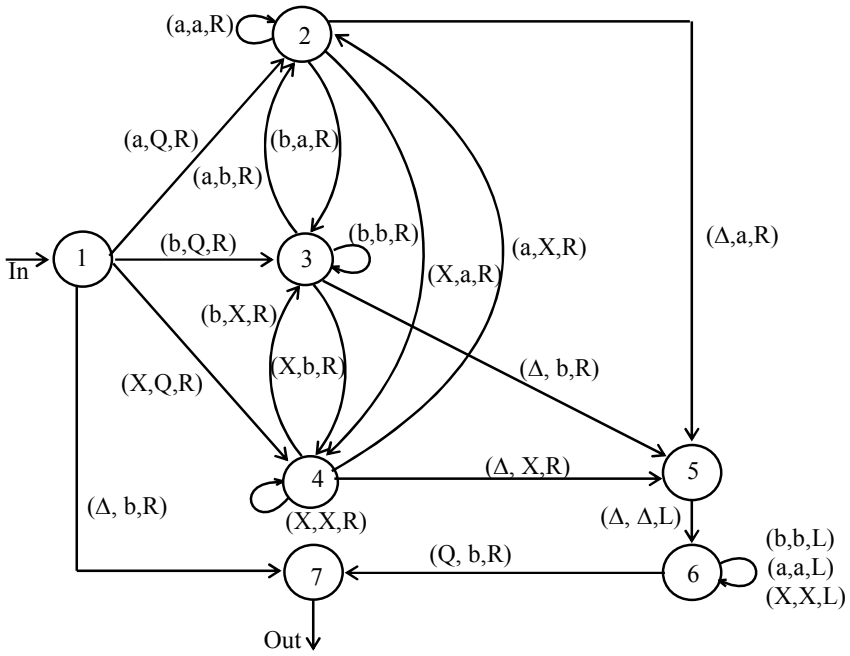
The function of subprogram INSERT b can be observed from the following diagram

∝	b	X	b	a	b	b	X	Δ	Δ∝

Following is the INSERT subprogram

The subprogram INSERT

Keeping in view the same example of inserting b at specified location, to determine the required subprogram, first Q will be inserted as marker at the required location, so that the TAPE Head must be able to locate the proper cell to the right of the insertion cell. The whole subprogram INSERT is given as



It is supposed that machine is at state 1, when b is to be inserted. All three possibilities of reading a, b or X are considered by introducing the states 2,3 and 4 respectively. These states remember what letter displaced during the insertion of Q.

Consider the same location where b is to be inserted

∝	b	X	a	b	b	X	Δ	Δ∝

After reading a from the TAPE, the program replaces a by Q and the TAPE Head will be moved one step right. Here the state 2 is entered. Reading b at state 2, b will be replaced by a and state 3 will be entered. At state 3, b is read which is not replaced by any character and the state 3 will not be left.

At state 3, the next letter to be read is X, which will be replaced by b and the state 4 will be entered. At state 4, Δ will be read, which will be replaced by X and state 5 will be entered. At state 5, Δ will be read and without any change state 6 will be entered, while TAPE Head will be moved one step left. The state 6 makes no change whatever (except Q) is read at that state. However at each step, the TAPE Head is moved one step left. Finally, Q is read which is replaced by b and the TAPE Head is moved to one step right.

Hence, the required situation of the TAPE can be shown as

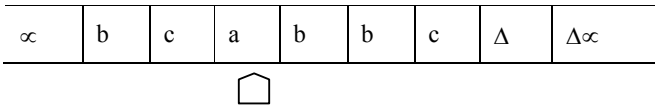
∝	b	X	a	b	b	X	Δ	Δ∝

DELETE subprogram

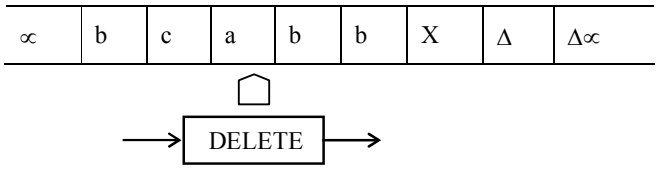
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Sometimes, a character is required to be DELETED on the TAPE exactly at the spot where the TAPE Head is pointing, so that the other characters on the right of the TAPE Head are moved one cell left. The characters to the left of the pointed cell are also required to remain as such.
In the situation stated above, the part of TM program that executes the process of deletion does not affect the function that the TM is performing. The subprogram of deletion is independent and can be incorporated at any time with any TM program specifying what character to be deleted at what location. The subprogram of deletion can be expressed as

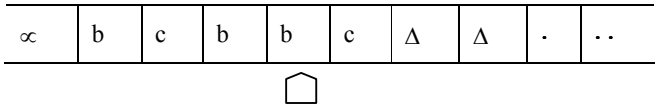
Example
If the letter a is to be deleted from the string bcabbc, shown below



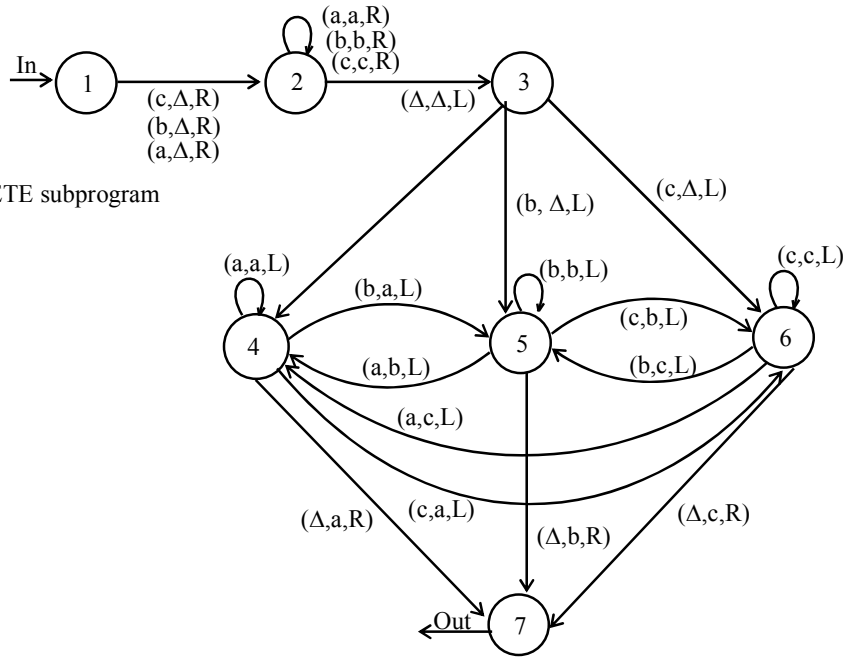
then, it is expressed as



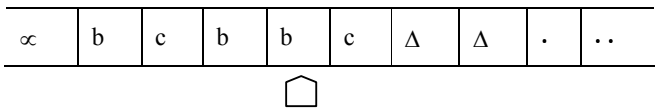
The function of subprogram DELETE can be observed from the following diagram



Following is the DELETE subprogram



The process of deletion of letter a from the string bcabbc can easily be checked, giving the TAPE situation as shown below



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CS402-Theory of Automata

(Solved MCS's)

LECTURE FROM

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ALL answers are verified if found any mistake then Correct ACCORDINGLY

1. If Σ = {aa, bb} , then Σ* will not contain
- **aaabbb**
- aabbbb
- aabbaa
- bbaabbbb
2. "One language can have _____TG"s".
- Only one
- Only two
- **More than one**
- Only three
3. According to 1st part of the Kleene"s theorem, If a language can be accepted by an FA then it can be accepted by a _____as well.
- FA
- CFG
- GTG
- **TG**
4. Even-palindrome is a _____language.
- **Non-regular**
- Regular
- Regular but infinite
- Regular but finite
5. If L is a regular language then, Lc is also a _____language.
- **Regular (Page 66)**
- Non-regular
- Regular but finite

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- None of the given

6. Pumping lemma is generally used to prove that:

- A given language is infinite
- **A given language is not regular**
- Whether two given regular expressions of a regular language are equivalent or not
- None of these

7. the FA has N states, then test the words of length less than N. If no word is accepted by this FA, then it will_____word/words.

- accept all
- **accept no (Page 85)**
- accept some
- reject no

8. In CFG, the symbols that can't be replaced by anything are called_____.

- **Terminal (Page 87)**
- Non-Terminal
- Production
- All of given

9. Which of the following is a regular language?

- **String of odd number of zeroes**
- Set of all palindromes made up of 0"s and 1"s
- String of 0"s whose length is a prime number
- All of these

10. Which of the following pairs of regular expressions are equivalent?

- 1(001)* and (10)*10
- **x(xx)* and (x)*x**
- X + and X*
- X + and X* X +

11. An alphabet of Σ is valid if

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- No letter of Σ appears in middle of any other letter
- No letter of Σ appears at end of any other letter
- **No letter of Σ appears at start of any other letter**
- No letter of Σ appears at end or middle of any other letter

12. Which of the following statement is true

- **The length of the output string is greater than length of input string in moore machine.**
- The length of the output string is greater than length of input string in mealy machine.
- The length of the output string is equal to length of input string in moore machine.
- The length of the output string is less than length of input string in mealy machine.

13. If a CFG has only productions of the form nonterminal → string of two nonterminals or nonterminal → one terminal then the CFG is said to be in _____

- **Chomsky Normal Form**
- Ambiguous Form
- Left Aligned Form
- Right Aligned Form

14. We can also represent an FA using different states e.g Accept state; Reject state, Read state etc. The_____state behaves as final state of an FA

- **Accept (Page 105)**
- Pop
- Push
- Reject

15. where the input string is placed before it is run, is called _____

- Date tape
- **Input Tape (Page 105)**
- Output Tape

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- Magnetic tape

16. An FSM can be considered as TM

- Of finite tape length, rewinding capability and unidirectional tape movement
- Of finite tape length, without rewinding capability and bidirectional tape movement
- Of finite tape length, rewinding capability and bidirectional tape movement
- **Of finite tape length, without rewinding capability and unidirectional tape movement**

17. The process of finding the derivation of the word generated by particular grammar is called _____

- Processing
- **Parsing (Page 136)**
- Programming
- Planning

18. The first rule of converting the given "CFG in CNF", is _____

- CNK algorithm
- **CYK algorithm (Page 135) Algorithm 4 (The CYK algorithm)**
- CKY algorithm
- KYC algorithm

19. Alphabet Σ = {a, bc, cc} has number of letters

- One
- Two
- **Three**
- Four

20. We cannot write regular expressions for all_____.

- FA"s
- TG"s
- NFA"s
- **CFG's (Page 97)**

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21. For every Context Free Grammar (CFG), we can make the corresponding_____.

- FA
- TG
- **PDA**
- Regular Grammar

22. Pumping Lemma II says that length(x) + length(y) should be _____.

- **Less than number of states (Page 75)**
- Equal to number of states
- Greater than number of states
- Greater than or equal to number of states

23. Chomsky normal form (CYK) algorithm was proposed by _____.

- **John cock (Page 135)**
- James Cock
- Daniel I.A.
- John Weiss

24. The language of Palindromes defined over an alphabet set {a, b} can be recognized by_____.

- FA
- NFA
- TG
- **PDA**

25. Which of the following is the first phase of compiler on the basis of functionality?

- Parser
- Lexical analyzer
- **Scanner**
- Interpreter

26. (Σ* - L) represent the _____of a language L.

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- **Complement (Page 66)**
- Kleene‟s closure
- Union
- intersection

27. If we have two transition graphs then their union will be expressed by

- **taking a common start state and joining them by two null transitions (Page 65)**
- just connecting both start states by null transitions
- connecting final state of first TG to the initial state of second TG
- connecting the final state of first TG to the final state of second TG

28. _____and_____are removed in order to make a CFG in Chomsky Normal Form(CNF).

- Null, nullable productions
- Nullable , unit productions
- **Null, unit productions (Page 102)**
- String of length 0, null

29. If L1 and L2 are expressed by regular languages then L1 + L2 is also a_____Language.

- **Regular (Page 10)**
- Ir-regular
- PDA
- Hybrid

30. Which of the following is a regular Context Free Grammar:

- **S → abS| baS | ^ ab(ab+ba)*ba + ba(ab+ba)*ab**
- S → aSb| baS | ^
- S → abS| bSa | ^
- S → aSb| Sa | ^

31. A read state can have _____outgoing edge/ edges.

- 1
- 2

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- 3
- **Any number of (Page 111)**

32. Who did not invent the Turing machine?

- Alan Turing
- **A. M. Turing (Page 140)**
- Turing
- None of these

33. Which statement is true?

- **The tape of turing machine is infinite. (Page 140)**
- The tape of turing machine is finite.
- The tape of turing machine is infinite when the language is regular
- The tape of turing machine is finite when the language is nonregular.

34. Every regular expression can be expressed as CFG but every CFG cannot be expressed as a regular expression. This statement is:

- Depends on the language
- None of the given options
- **True (Page 97)**
- False

35. Consider the language L of strings, defined over Σ = {a,b}, ending in a

- **There are finite many classes generated by L, so L is regular (Page 76)**
- There are infinite many classes generated by L, so L is regular
- There are finite many classes generated by L, so L is non-regular
- There are infinite many classes generated by L, so L is non-regular

36. The word „formal“ in formal languages means

► The symbols used have well defined meaning

► They are unnecessary, in reality

► **Only the form of the string of symbols is significant**

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► None of these

37. Let A = {0, 1}. The number of possible strings of length „n‟ that can be formed by the elements of the set A is

► n!

► n2

► nm

► 2n

38. Choose the correct statement.

► A Mealy machine generates no language as such

► A Moore machine generates no language as such

► A Mealy machine has no terminal state

► All of these

39. TM is more powerful than FSM because

► The tape movement is confined to one direction

► It has no finite state control

► It has the capability to remember arbitrary long sequences of input symbols

► None of these

40. Like TG, a PDA can also be non-deterministic

- **True (Page 111)**
- False

41. The language of all words (made up of a‟s and b‟s) with at least two a‟s can not be described by the regular expression.

- a(a+b)*a(a+b)*(a+b)*ab*

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- (a+b)* ab* a(a+b)*
- b*ab* a(a+b)*
- **none of these**

42. If L is a regular language then, Lc is also a _____language.

- **Regular (Page 66) rep**
- Non-regular
- Regular but finite
- None of the given

43. In CFG, the symbols that can‟t be replaced by anything are called__

- **Terminal (Page 87) rep**
- Non-Terminal
- Production
- All of given

44. Which of the following is NOT a regular language?

- String of 0‟s whose length is a perfect squere
- Set of all palindromes made up of 0‟s and 1‟s
- String of 0‟s whose length is a prime number
- **All of the given options**

45. Choose the incorrect (FALSE) statement.

- A Mealy machine generates no language as such
- A Mealy machine has no terminal state
- **For a given input string, length of the output string generated by a Moore machine is not more than the length of the output string generated by that of a Mealy machine**
- All of these

46. Choose the incorrect statement:

- (a+b)*aa(a+b)* generates Regular language.
- **A language consisting of all strings over ∑={a,b} having equal number of a's and b's is a regular language**

- Every language that can be expressed by FA can also be expressed by RE
- None of these

47. Left hand side of a production in CFG consists of:

- One terminal
- More than one terminal
- **One non-terminal (Page 87)**
- Terminals and non-terminals

48. PDA is only used to represent a regular language.

- True
- **False**

49. A production of the form non-terminal □ string of two non-terminal is called a live Production.

- **True (Page 127)**
- False

50. We can find a CFG corresponding to a DFA.

- **True (Page 97)**
- False

51. START, READ, HERE and ACCEPTS are conversions of the machine

- **True (Page 122)**
- False

52. A CFG is said to be ambiguous if there exists at least one word of its language that can be generated by different production trees

- **True (Page 95)**
- False

53. Syntax tree or Generation tree or Derivation tree are same tree

- **True (Page 92)**
- False

54. The symbols that cannot be replaced by anything are called terminals

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- **True (Page 87) repeat**
- False

55. The production of the form non-terminal □ one non-terminal is called unit production

- **True (Page 100)**
- False

56. DFA and PDA are equal in power.

- True
- **False (Page 105)**

57. A production of the form non-terminal □ non-terminal is called a dead Production.

- True
- **False (Page 127)**

58. Semi-word is a string having some terminals and one non-terminal at the right of string.

- **True (Page 97)**
- False

59. Two FAs are equivalent if they have same no. of states.

- **True (Page 15)**
- False

60. There exist exactly two different derivations in an ambiguous CFG for a word.

- **True (Page 93)**
- False

61. Regular languages are closed under Union, Concatenation and Kleene star.

- **True (Page 10)**
- False

62. CFG may also represent a regular language.

- **True (Page 97)**

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- False

63. PDA is stronger than FA.

- **True (Page 105)**
- False

64. A Total Language Tree has

- All languages over Σ
- **All strings over Σ (Page 96)**
- All words of all languages over Σ
- All words of one language over Σ

65. What Turing Machine does not have?

- Stack
- Tape
- Head
- **Word**

66. CFG given S bS|Sb|aa represents language b*aa

- aab*
- b*aab*
- **b*(aa)*b***

67. The values of input (say a & b) does not remain same in one cycle due to

- NAND gate
- Click plus
- OR gate
- **NOT gate**

68. Set of all palindromes over {a,b}is regular

- True
- **False (Page 74)**

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69. In CFG, the symbols that cannot be replaced by anything are called

- **Terminals (Page 87) rep**
- Non terminals
- Productions
- None of the given options

70. a^n b^n generates the ..... language

- regular
- non regular
- **EQUAL and non regular (Page 71)**
- EQUAL and regular

71. The grammatical rules which involves meaning of words are called:

- **Semantic (Page 87)**
- Sytactics
- Alphabets
- None of the given options

72. If an FA has N state then it must accept the word of length

- **N-1**
- N+1
- N+2N

73. Two languages are said to belong to same class if they end in the same state when they run over an FA, that state

- Must be final state
- **May be final state or not (Page 75)**
- May be start or not
- None of the given options

74. In pref(Q in R) Q is …… to (than) R

- Equal
- **Not Equal (Page 79)**
- Greater

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- Smaller

75. According to Myhill Nerode theorem, if L generates finite no. of classes then L is.....

- Finite
- Infinite
- **Regular (Page 76)**
- Non Regular

76. If the intersection of two regular languages is regular then the complement of the intersection of these two languages is also regular

- **True (Page 68)**
- False

77. In pumping lemma theorem (x y^n z) the range of n is

- **n=1,2,3,4 ..... (Page 74)**
- n=0,1,2,3,4....
- n=-3,-2,-1,0,1,2,3,4.....
- n=-3,-2,-1,1,2,3,4.....

78. The complement of a regular language is also a regular

- **True repeat**
- False

79. For a non regular language there exist …… FA

- One
- At least one
- At most one
- **No (Page 71)**

80. The strings or words which do not belong to a language is called..... of that language

- Intersection
- Union
- **Complement (Page 66)**
- Quotient

81. A non regular language can be represented by

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- RE
- FA
- TG
- **None of the given options (Page 71)**

82. For language L defined over {a, b},then L partitions {a, b}* into …… classes

- Infinite
- Finite
- **Distinct (Page 75)**
- Non distinct

83. If an FA accept a word then there must exist a path from

- **Initial to final state (Page 81)**
- Initial to each state
- Initial to each state but not to final state
- Initial to final state by traversing each state

84. Which of the following statement is true about NFA with Null String?

- Infinite states
- Infinite set of letters
- Infinite set of transitions
- **Transition of null string is allowed at any stage (Page 71)**

85. FA corresponding to an NFA can be built by introducing an empty state for a letter having

- **no transition at certain state (Page 43)**
- one transition at certain state
- two transition at certain state
- more than two transitions at certain state

86. Let FA3 be an FA corresponding to FA1FA2, then the initial state of FA3 must correspond to the initial state of

- **FA1 only (Page 35)**
- FA2 only

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- FA1 or FA2
- FA1 and FA2

87. (a* + b*)* = (a + b)* this expression is _____

- True
- **False**

88. If S = {ab, bb}, then S* will not contain

- Abbbab
- **Bbba**
- ababbb
- bbbbab

89. What do automata mean?

- Something done manually
- **Something done automatically**
- What is false about the term alphabet?
- It is a finite set of symbols.

90. Consider the following production (of a CFG): S->XYZ Here _____is left most nonterminal in working string. Note: S, X, Y and Z are all nonterminals

- S
- **X**
- Y
- Z

91. A PDA is called nondeterministic PDA if _____

- **There are more than one outgoing edges at READ or POP states with one label (Page 111)**
- There are more than one PUSH states
- There are mroe than one POP states
- All of the given options

92. A PDA consists of the following:

- An alphabet (Sigma) of input letters.
- An input TAPE with infinite many locations in one direction

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- One START state with only one out-edge and no in-edge
- **All of the given options (Page 105)**

93. The CFG S --> aSa | bSb | a | b | ^ represents the language

- EVEN-EVEN
- **PALINDROM (Page 91)**
- EQUAL
- ODD-ODD

94. Halt states are

- Start and Accept
- **Accept and Reject (Page 105)**
- Start and Reject
- Read and Reject

95. Choice of path can be determined by left most derivation of the string belonging to CFL at..... state

- **Accept (Page 104)**
- Reject
- Push
- POP

96. The unit and null productions can be deleted from a CFG

- **True (Page 99-100)**
- False

97. Identify the TRUE statement about following CFG: S -> SB|AB A -> CC B -> b C -> a

- The given CFG has 8 Nonterminals
- The given CFG has 8 Terminals
- **The given CFG is in CNF (Page 101)**
- The given CFG is not in CNF

98. The structure given below is called_____S -> aA|bB A -> aS|a B -> bS|b

- RE
- TG

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- **CFG (Page 87)**
- PDA

99. Which of the following states is not part of PDA

- START
- ACCEPT
- **WRITE (Page 107)**
- REJECT

100. The production of the form: nonterminal --> one nonterminal is called the _____

- **Unit production (Page 100)**
- NULL production
- Terminal production
- Non Terminal production

101. A_____is the one for which every input string has a unique path through the machine.

- **Deterministic PDA (Page 111)**
- nondeterministic PDA
- PUSHDOWN store
- Input Tape

102. In the null production N --> ^ , N is a

- Terminal
- **Non terminal (Page 99)**
- Word
- None of the given options

103. The major problem in the earliest computers was

- To store the contents in the registers
- **To display mathematical formulae (Page 87)**
- To load the contents from the registers
- To calculate the mathematical formula

104. In polish notation, (o-o-o) is the abbreviation of……?

- Operand - Operator – Operand

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- Operand - Operand- Operator
- **Operator -Operand – Operand (Page 94)**
- Operand -Operand – Operand

105. The CFG is said to be ambiguous if there exist at least one word of its language that can be generated by the..... production trees

- One
- Two
- **More than one (Page 95)**
- At most one

106. The input string is placed, before it runs, in

- Stack
- Memory
- **Tape (Page 105)**
- Ram

107. The production S --> SS | a | b | ^ can be expressed by RE

- (a+b)+
- (a+b)
- **(a+b)* (Page 88)**
- (ab)*

108. The locations into which we put the input letters on "Input Tap" are called _____

- Words
- alphabets
- **cells (Page 105)**
- elements

109. "CFG" stands for _____

- Context Free Graph
- **Context Free Grammer (Page 87)**
- Context Finite Graph
- Context Finite Grammer

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110. In a CFG the nonterminal that occurs first from the left in the working string, is said to be _____

- Least Significant nonterminal
- Most Significant nonterminal
- **Left most nonterminal (Page 103)**
- Left most derivate

111. The unit production is

- Terminal --> Terminal
- Terminal --> Non Terminal
- Non terminal --> Terminal
- **Non terminal --> Non Terminal (Page 100)**

112. A_____operator adds a new letter at the top of STACK

- **PUSH (Page 107)**
- POP
- READ
- APPEND

113. PDA stands for _____

- Push and Drop Automaton
- Pop and Drop Automaton
- **Push Down Automaton (Page 112)**
- None of given options

114. The production of the form: Nonterminal-> ^ is said to be _____ production

- **NULL (Page 99)**
- UNIT
- Chomsky form production
- None of the given options

115. In a STACK:

- The element PUSHed first is POPed first
- **The element PUSHed first is POPed in the last (Page 107 concept)**
- The element PUSHed in last is POPed in last

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- None of given options

116. For a given input, it provides the compliment of Boolean AND output.

- **NAND box (NOT AND) (Page 63)**
- DELAY box
- OR box
- AND box

117. It delays the transmission of signal along the wire by one step (clock pulse).

- NAND box (NOT AND)
- **DELAY box (Page 63)**
- OR box
- AND box

118. Any language that can not be expressed by a RE is said to be regular language.

- True
- **False**

119. The current in the wire is indicated by 1 and 0 indicates the absence of the current.

- **True (Page 63)**
- False

120. For the given input, AND box provides the Boolean AND output.

- **True (Page 63)**
- False

121. Let L be a language defined over an alphabet Σ, then the language of strings, defined over Σ, not belonging to L, is called Complement of the language L, denoted by Lc or L‟.

- **True (Page 66)**
- False

122. To describe the complement of a language, it is very important to describe the----------of that language over which the language is defined.

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- **Alphabet (Page 66)**
- Regular Expression
- String
- Word

123. For a certain language L, the complement of Lc is the given language L i.e. (Lc)c = Lc

- True
- **False (Page 66)**

124. If L is a regular language then, --------- is also a regular language.

- Lm
- Ls
- Lx
- **Lc (Page 66)**

125. Converting each of the final states of F to non-final states and old non-final states of F to final states, FA thus obtained will reject every string belonging to L and will accept every string, defined over Σ, not belonging to L. is called

- Transition Graph of L
- Regular expression of L
- **Complement of L (Page 66)**
- Finite Automata of L

126. If L1 and L2 are two regular languages, then L1 U L2 is not a regular.

- True
- **False (Page 65)**

127. If L1 and L2 are regular languages, then these can be expressed by the corresponding FAs.

- **True (Page 68)**
- False

128. The language that can be expressed by any regular expression is called a Non regular language.

- True

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- **False (Page 71)**

129. The languages----------are the examples of non regular languages.

- **PALINDROME and PRIME (Page 71)**
- PALINDROME and EVEN-EVEN
- EVEN-EVEN and PRIME
- FACTORIAL and SQURE

130. Let L be any infinite regular language, defined over an alphabet Σ then there exist three strings x, y and z belonging to Σ* such that all the strings of the form xy z n for n=1,2,3, … are the words in L. called.

- Complement of L
- **Pumping Lemma (Page 72)**
- Kleene‟s theorem
- None in given

131. Languages are proved to be regular or non regular using pumping lemma.

- **True (Page 74)**
- False

132. is obviously infinite language.

- EQUAL-EQUAL
- EVEN-EVEN
- **PALINDROME (Page 75)**
- FACTORIAL

133. If, two strings x and y, defined over Σ, are run over an FA accepting the language L, then x and y are said to belong to the same class if they end in the same state, no matter that state is final or not.

- **True (Page 75)**
- False

134. Myhill Nerode theorem is consisting of the followings,

- L partitions Σ* into distinct classes.
- If L is regular then, L generates finite number of classes.
- If L generates finite number of classes then L is regular.

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- **All of above (Page 75)**

135. The language Q is said to be quotient of two regular languages P and R, denoted by--- if PQ=R.

- R=Q/P
- **Q=R/P (Page 78)**
- Q=P/R
- P=R/Q

136. If two languages R and Q are given, then the prefixes of Q in R denoted by Pref(Q in R).

- **True (Page 78)**
- False

137. Let Q = {aa, abaaabb, bbaaaaa, bbbbbbbbbb} and R = {b, bbbb, bbbaaa, bbbaaaaa} Pref (Q in R) is equal to,

- **{b,bbba,bbbaaa} (Page 78)**
- {b,bba,bbaaa}
- {ab,bba,bbbaa}
- {b,bba,bbba}

138. If R is regular language and Q is any language (regular/ non regular), then Pref (Q in R) is ---------.

- Non-regular
- Equal
- **Regular (Page 79)**
- Infinite

139. _____states are called the halt states.

- **ACCEPT and REJECT (Page 105)**
- ACCEPT and READ
- ACCEPT AND START
- ACCEPT AND WRITE

140. The part of an FA, where the input string is placed before it is run, is called _____

- State
- Transition

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- **Input Tape (Page 105)**
- Output Tape

141. In new format of an FA (discussed in lecture 37), This state is like dead-end non final state

- ACCEPT
- **REJECT (Page 105)**
- STATR
- READ

142. Between the two consecutive joints on a path:

- One character can be pushed and one character can be popped
- **Any no. of characters can be pushed and one character can be popped (Page 122)**
- One character can be pushed and any no. of characters can be popped
- Any no. of characters can be pushed and any no. of characters can be popped

143. The PDA is called non-deterministic PDA when there are more than one out going edges from…… state

- START or READ
- POP or REJECT
- **READ or POP (Page 111)**
- PUSH or POP

144. Identify the TRUE statement:

- A PDA is non-deterministic, if there are more than one READ states in PDA
- A PDA is never non-deterministic
- **Like TG, A PDA can also be non-deterministic (Page 111)**
- A PDA is non-deterministic, if there are more than one REJECT states in PDA

145. There is a problem in deciding whether a state of FA should be marked or not when the language Q is infinite.

- **True (Page 79)**
- False

146. If an effectively solvable problem has answered in yes or no, then this solution is called ---------

- **Decision procedure (Page 80)**
- Decision method
- Decision problem
- Decision making

147. The following problem(s) ---------- is/are called decidable problem(s).

- The two regular expressions define the same language
- The two FAs are equivalent
- **Both a and b (Page 80)**
- None of given

148. To examine whether a certain FA accepts any words, it is required to seek the paths from ------- state.

- Final to initial
- Final to final
- **Initial to final (Page 81)**
- Initial to initial

149. The high level language is converted into assembly language codes by a program called compiler.

- **TRUE (Page 87)**
- FALSE

150. Grammatical rules which involve the meaning of words are called _____

- **Semantics (Page 87)**
- Syntactic
- Both a and b
- None of given

151. Grammatical rules which do not involve the meaning of words are called -_____

- Semantics
- **Syntactic (Page 87)**

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- Both a and b
- None of given

152. The symbols that must be replaced by other things are called _____

- Productions
- Terminals
- **Non-terminals (Page 87)**
- None of given

153. The grammatical rules are often called_____

- **Productions (Page 87)**
- Terminals
- Non-terminals
- None of given

154. The terminals are designated by _____letters, while the non-terminals are designated by_____letters.

- Capital, bold
- **Small, capital (Page 87)**
- Capital, small
- Small, bold

155. The language generated by _____is called Context Free Language (CFL).

- FA
- TG
- **CFG (Page 87)**
- TGT

156. S → aXb|bXa X → aX|bX|Λ The given CFG generates the language in English _____

- **Beginning and ending in different letters (Page 91)**
- Beginning and ending in same letter
- Having even-even language
- None of given

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157. The CFG is not said to be ambiguous if there exists atleast one word of its language that can be generated by the different production trees,

- TRUE
- **FALSE (Page 95)**

158. The language generated by that CFG is regular if _____

- No terminal → semi
- word No terminal → word
- **Both a and b (Page 97)**
- None of given

159. The production of the form no terminal → Λ is said to be null production.

- **TRUE (Page 99)**
- FALSE

160. CNF is stands for

- Context Normal Form
- Complete Normal Form
- **Chomsky Normal Form (Page 102)**
- Compared Null Form

161. Kleene‟s theorem states

- All representations of a regular language are equivalent.
- All representations of a context free language are equivalent.
- All representations of a recursive language are equivalent
- **Finite Automata are less powerful than Pushdown Automata. (Page 105)**

162. Null production is a

- Word
- String
- Terminal
- **All of the given options**

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163. In nondeterministic PDA a string is supposed to be accepted, if there exists at least one path traced by the string, leading to _____ state.

- **ACCEPT (Page 111)**
- REJECT
- START
- READ

164. The CFG which generates the regular language is called:

- Regular expression
- Finite Automata
- **Regular grammar (Page 97)**
- None of the given options

165. If a CFG has a null production, then it is possible to construct another CFG accepting the same language without null production

- TRUE
- **FALSE (Page 99)**

166. In large FA with thousands of states and millions of directed edges, without an effective procedure it is_____to find a path from initial to final state.

- Always easy
- **Impossible (Page 81)**
- may be good
- always impossible

167. If there is no final state of two FAs then their_____also have no___state

- initial, union
- final, union
- **union,final (Page 83)**
- union, initial

168. Set of all palindromes over {a,b} is:

- Regular
- Regular and finite

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- Regular and infinite
- **Non-regular (Page 71)**

169. In the context of Myhill Nerode theorem, for even-even language sigma star can be partitioned into_____number of classes.

- 3
- **4 (Page 77)**
- 5
- 6

170. The product of two regular languages is_____.

- **Regular (Page 78)**
- Infinite
- non-regular
- closure of a regular language

171. incase of Myhill Nerode theorem, if a language L partitions sigma star into distinct classes and L is also regular then L generates_____number of classes.

- Infinite
- specified
- **finite (Page 75)**
- odd

172. While determining regular expression for a given FA, it is _____to write its regular expression.

- Always possible easily
- **Sometime impossible (Page 80)**
- always impossible
- None of the given options

173. If (L1 ∩ L2c) ∪ (L1c ∩ L2) is regular language that accepts the words which are in L1 but not in L2 or else in L2 but not in L1 . The corresponding FA cannot accept any word which is in_____L1 and L2.

- Not both
- **Both (Page 80)**

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- At least in one
- None of the given options

174. A problem that has decision procedure is called _____ problem.

- Regular language
- un-decidable
- Infinite
- **Decidable (Page 80)**

175. The product of two regular languages is_____.

- **Regular (Page 78)**
- Infinite
- non-regular
- closure of a regular language

176. In new format of an FA (discussed in lecture 37):……state is like a final state of an FA

- START
- ACCEPT (Handouts Page # 119)
- REJECT
- READ

177. In conversion form of PDA there is no ..... state

- PUSH
- ACCEPT
- REJECT (Handouts Page # 119)
- READ

178. Given a PDA that accepts the language L

- There does not exist any CFG that generates exactly L
- that PDA will also accept Language L' (complement of L)
- There exists a CFG that generates exactly L (Handouts Page # 118)
- None of given options

179. In a CFG the non-terminal that occurs first from the left in the working string. is said to be

- Least Significant nonterminal

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- Left most nonterminal (Handouts Page # 103)
- Most Significant nonterminal
- Left most derivate

180. The structure given below is called

S -> aAlbB

A -> aSla

B -> bSIb

- RE
- PDA
- CFG (It is form of CFG)
- TG

181. An FA has N states then it must accept the word of length

- 2N
- N
- N-1
- N-1

182. To examine whether a certain FA accepts any words. it is required to seek the paths ..... state.

- from initial to final (Handouts Page # 81)
- from initial-to-initial back
- from final to initial
- from final to back final

183. In nondeterministic PDA. a string is supposed to be accepted if there exists at least one path traced by the string. leading to state.

- START
- REJECT
- READ
- ACCEPT (Handouts Page # 111)

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184. If a CFG has a null production, then it is

- Called Null CFG
- Not possible to construct another CFG without null production accepting the same language with the exception of the word
- Called Chmosky Normal Form (CNF)
- Possible to construct another CFG without null production accepting the same language with the exception of the word

185. There is at least one production in CFG that has one……. on its left side.

- Non terminal (Handouts Page # 87)
- Null production
- Terminal
- Unit production

186. In large FA with thousands of states and millions of directed edges, without an effective procedure it is……to find a path from initial to final state.

- Impossible (Handouts Page # 81)
- Always easy
- always impossible
- may be good

187. By removing null and unit productions _____ _.

- CNF can be converted into FA
- CNF can be converted into CFG
- CFG can be converted into CNF (Handouts Page # 102)
- CNF can be converted into Turing machine

188. A……is the one for which every input string has a unique path through the machine

- deterministic PDA
- Input Tape
- nondeterministic PDA
- PUSHDOWN store

189. PDA stands for

- Push Down Automaton (Handouts Page # 112)

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- Pop and Drop Automaton
- Push and Drop Automaton
- Push Deterministic Automaton

190. A PDA is called nondeterministic PDA if

- there are more than one POP states
- there are more than one PUSH states
- there are more than one outgoing edges at READ or POP states with one label
- every READ state is followed by a HERE state.

191. Which of the following cannot be represented by a regular expression?

- String of 0's with an odd length
- Language of even-even
- Language of odd-odd
- String of 0s with a prime length (Because Prime is not regular Langue)

192. In conversion form of PDA. there is……..accept state(s).

- At most one
- At least one
- More than One
- Exactly one (Handouts Page # 119)

193. If there is no final state of two FAs, then their…….also have no .....state

- union, initial
- union. Final (Handouts Page # 83)
- final, union
- initial, union

194. The tree which produces all the strings of a language is called

- Derivation tree
- Total language tree (Handouts Page # 96)
- Non ambiguous tree
- Ambiguous tree

195. In new format of an FA (discussed in lecture 37),..... state is like dead-end non final state.

- READ
- REJECT (Handouts Page # 105)
- START
- ACCEPT

196. To write the expression from the tree, it is required to traverse from

- Top to bottom of the tree
- Right side of the tree
- Left side of the tree (Handouts Page # 94)
- Bottom to top of the tree

197. A PDA consists of the following:

- An alphabet (Sigma) of input letters.
- An input TAPE with infinite many locations in one direction
- One START state with only one out-edge and no in-edge
- All the given options (Handouts Page 105)

198. If R is regular language and Q is any language (regular/ non-regular), then Pref…….in …….. is regular.

- R R
- R.Q
- Q.R (Handouts Page # 79)
- Q.Q

199. ……is an operation that takes out a letter from the top of the STACK.

- WRITE
- APPEND
- PUSH
- POP (Handouts Page # 107)

200. Before the CFG corresponding to the given PDA is determined, the PDA is converted into the standard form which is called.

- Finite Automaton
- Conversion form (Handouts Page # 118)
- None of given options
- Chomsky Normal Form (CNF)

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201. The part of an FA, where the input string is placed before it is run, is called

- Transition
- Input Tape (Handouts Page # 105)
- Output Tape
- State

202. A problem is said to be ……if there exists an algorithm that provides the solution in……. number of steps.

- Effectively unsolvable, infinite
- Effectively solvable, infinite
- Effectively unsolvable, finite
- Effectively solvable, finite (Handouts Page # 80)

203. …… states are called the halt states.

- ACCEPT AND START
- ACCEPT and READ
- ACCEPT and REJECT (Handouts Page # 105)
- ACCEPT AND WRITE

204. The grammatical rules which involve meaning of words are called

- Semantics (Handouts Page # 87)
- Syntactics
- strings
- alphabets

205. The PDA is called non-deterministic PDA when there are more than one out going edges from state

- READ or POP (Handouts Page # 111)
- START or READ
- POP or
- PUSH or POP

206. Which of the following states is not part of PDA?

- REJECT
- ACCEPT
- START
- WRITE (All other are parts of PDA)

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207. The major problem in the earliest computers was

- To calculate the mathematical formula
- To display mathematical formulas (Handouts Page # 87)
- To store the contents in the registers
- To load the contents from the registers

208. The operators like (` +) in the parse tree are considered as

- Terminals (Handouts Page # 93)
- productions
- non-terminals
- intermediates

209. If L1 and L2 are two regular languages, then they……expressed by FAs.

- cannot be
- May be
- may or may not be
- can be (Handouts Page # 68)

210. Before running the input string on PDA it is first placed on

- Stack
- Ram
- Memory
- Tape (Handouts Page # 107)

211. Which is the correct option

- The element PUSHED in last is POPED in last
- The element PUSHED first is POPED in the last (LIFO Method, from Book)
- The element PUSHED first is POPED first
- None of given options

212. Null production is a

- String
- Word (Handouts Page # 97)

- Terminal
- All the above

213. A/an……operator adds a new letter at the top of STACK

- Push (Handouts Page # 107)
- Append
- Read
- Pop

214. In conversion form of PDA, no two…….states exist in a row without state

- POP. READ (Handouts Page # 119)
- POP. REJECT
- PUSH. START
- PUSH READ

215. Given a PDA that accepts the language L

- that PDA will also accept Language L' (complement of Ll
- There exists a CFG that generates exactly L (Handouts Page # 118)
- None of given options
- There does not exist any CFG that generates exactly

216. In large FA with thousands of states and millions of directed edges, without an effective procedure it is ……. to find a path from initial to final state.

- Always easy
- always impossible
- may be good
- Impossible (Handouts Page # 81)

217. The CFG there generates the regular language is called

- Regular expression
- finite automata
- regular grammars (Handouts Page # 97)
- now regular grammars

218. Consider the following CFG: (Note: ^ means NULL)

S-→Xa

X→aX|bX|^

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- Above give a CFG can be represented by RE
- a*b*
- a (a + b) *a
- (a + b) *a
- A*b*a

S production will give us Xa. As the X is nonterminal and we must only change X and the terminal a will be on the last of the R.E.

Now we will change X production.

X production will give us as many a as we want. or if we use the second production which will give us as many b as we want. And last production will give us ^ (lemda)

So the answer would be a*b*a

Last a is the terminal which we got from the very first production.

219. For a machine with N number of states, the total number of strings to be tested, defined over an alphabet of m letters is

- mN +mN+1 + mN+2 +… +m2N-1 (Handouts Page # 86)
- mN
- Nm
- Nm +mN+1 + mN+2 +… +

220. Consider the CFG given below.

A→B|b

B→a

Which of the following is a unit production?

- S→bb
- A→b
- A→B
- B→a

221. The CFG is said to be ambiguous if there exist at least one word of its language that can be generated by the.....production trees

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- One
- Two
- More than one (Handouts Page # 95)
- At most one

222. If Q = {xx, xyxxxy} and R = { xyxyxyxxyy, xyxyyyxx} then Pref {Q in R} = _____

- Xx
- Xyxyxy
- Xyxyyy (Solved by my self 100% sure)
- Xxy

223. The unit production is

- Terminal --> Terminal
- Terminal --> Non Terminal
- Non terminal --> Terminal
- Non terminal --> Non-Terminal (Hand out Page # 100)

224. Which of the following statement is FALSE?

- For every PDA, there always exists a regular expression (Not sure)
- Every CFG cannot be expressed as Regular Expression
- Every Regular Expression be expressed by a CFG.
- For a PDA, there exists a CFG that represent the same language

225. The CFG S → aSb|ab|^

- Palindrome
- Prime
- Equal
- Even

The production will give us ab and non-terminal S inside the a and b. If we change the S into next production of ab, we will get abab but instead of using 2nd production, if we use the last which will give us only ^ (Lemda), thus we will get ^ab.

So, we will get Equal language (same number of a and same number of b)

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226. Before the PDA is converted into conversion form, a new state ---------- is defined which is placed in the middle of any edge.

- HERE (Hand out Page # 118)
- STOP
- START
- REJECT

227. A PDA is in conversion form if it fulfills the following conditions:

- There is only one ACCEPT state. (Hand out Page # 119)
- There are one REJECT state.
- There are more than one ACCEPT states.
- There is only one Accept state.

228. Identify the false statement about the following CFG
S→SB|AB
A→CC
B→
C→a

- CFG has 8 Non terminals
- all the given option (All are false as There are 4 terminals , It is in CNF and it does not generate any null string)
- CFG is not in CNF
- CFG generate null string

229. This CFG there generates to the regular language is called

- Regular grammar (Hand out Page # 97)
- nonregular grammar
- finite automata
- regular expression

230. The derivation of the word W generated by CFG such that at each step a production is applied to the leftmost nonterminal in the working string is said to be

- Left most terminal

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- right most terminal
- left most derivation (Hand out Page # 103)
- right most derivation

231. If the FA has N states, then test these words of length less than N. If no word is accepted by this FA, then it will word/words

- Accept no (Hand out Page # 85)
- Accept some
- Reject No
- Accept All

232. In a CFG then non terminals are denoted by

- small letters
- numbers
- capital letters (Hand out Page # 87)
- small letters and numbers

233. "CFG"stands for_____

- Context finite graph
- contacts finite grammar
- contact free graph
- Context free grammar (Hand out Page # 87)

234. Consider the following production (of a CFG) S→XYZ
Here _____is left most non terminals in working string
note XY and Z are all known terminals

- X (X is on the most left side)
- S
- Z
- Y

235. Consider the following CFG
S→ a|Xb|aYa
X→ Y|^ (Note: ^ means NULL)
Y→ b|X
which nonterminal is/are not nullable

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- Y
- S, X and Y
- S
- X (X is Null Production and not a null able)

236. In new format of an FA (discussed in lecture 37):……state is like a final state of an FA

- START
- ACCEPT (Handouts Page # 119)
- REJECT
- READ

237. In conversion form of PDA there is no ..... state

- PUSH
- ACCEPT
- REJECT (Handouts Page # 119)
- READ

238. Given a PDA that accepts the language L

- There does not exist any CFG that generates exactly L
- that PDA will also accept Language L' (complement of L)
- There exists a CFG that generates exactly L (Handouts Page # 118)
- None of given options

239. In a CFG the non-terminal that occurs first from the left in the working string. is said to be

- Least Significant nonterminal
- Left most nonterminal (Handouts Page # 103)
- Most Significant nonterminal
- Left most derivate

240. The structure given below is called

S -> aAlbB

A -> aSla

B -> bSIb

- RE
- PDA
- CFG (It is form of CFG)
- TG

241. An FA has N states then it must accept the word of length

- 2N
- N
- N-1
- N-1

242. To examine whether a certain FA accepts any words. it is required to seek the paths ..... state.

- from initial to final (Handouts Page # 81)
- from initial-to-initial back
- from final to initial
- from final to back final

243. In nondeterministic PDA. a string is supposed to be accepted if there exists at least one path traced by the string. leading to.....state.

- START
- REJECT
- READ
- ACCEPT (Handouts Page # 111)

244. If a CFG has a null production, then it is

- Called Null CFG
- Not possible to construct another CFG without null production accepting the same language with the exception of the word
- Called Chmosky Normal Form (CNF)
- Possible to construct another CFG without null production accepting the same language with the exception of the word

245. There is at least one production in CFG that has one……. on its left side.

- Non terminal (Handouts Page # 87)
- Null production

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- Terminal
- Unit production

246. In large FA with thousands of states and millions of directed edges, without an effective procedure it is……to find a path from initial to final state.

- Impossible (Handouts Page # 81)
- Always easy
- always impossible
- may be good

247. By removing null and unit productions _____ _.

- CNF can be converted into FA
- CNF can be converted into CFG
- CFG can be converted into CNF (Handouts Page # 102)
- CNF can be converted into Turing machine

248. A……is the one for which every input string has a unique path through the machine

- deterministic PDA
- Input Tape
- nondeterministic PDA
- PUSHDOWN store

249. PDA stands for

- Push Down Automaton (Handouts Page # 112)
- Pop and Drop Automaton
- Push and Drop Automaton
- Push Deterministic Automaton

250. A PDA is called nondeterministic PDA if

- there are more than one POP states
- there are more than one PUSH states
- there are more than one outgoing edges at READ or POP states with one label
- every READ state is followed by a HERE state.

251. Which of the following cannot be represented by a regular expression?

- String of 0's with an odd length
- Language of even-even
- Language of odd-odd
- String of 0s with a prime length (Because Prime is not regular Langue)

252. In conversion form of PDA. there is……..accept state(s).

- At most one
- At least one
- More than One
- Exactly one (Handouts Page # 119)

253. If there is no final state of two FAs, then their…….also have no .....state

- union, initial
- union. Final (Handouts Page # 83)
- final, union
- initial, union

254. The tree which produces all the strings of a language is called

- Derivation tree
- Total language tree (Handouts Page # 96)
- Non ambiguous tree
- Ambiguous tree

255. In new format of an FA (discussed in lecture 37),..... state is like dead-end non final state.

- READ
- REJECT (Handouts Page # 105)
- START
- ACCEPT

256. To write the expression from the tree, it is required to traverse from

- Top to bottom of the tree
- Right side of the tree
- Left side of the tree (Handouts Page # 94)
- Bottom to top of the tree

257. A PDA consists of the following:

- An alphabet (Sigma) of input letters.
- An input TAPE with infinite many locations in one direction
- One START state with only one out-edge and no in-edge
- All the given options (Handouts Page 105)

258. If R is regular language and Q is any language (regular/ non-regular), then Pref…….in …….. is regular.

- R R
- R.Q
- Q.R (Handouts Page # 79)
- Q.Q

259. ……is an operation that takes out a letter from the top of the STACK.

- WRITE
- APPEND
- PUSH
- POP (Handouts Page # 107)

260. Before the CFG corresponding to the given PDA is determined, the PDA is converted into the standard form which is called.

- Finite Automaton
- Conversion form (Handouts Page # 118)
- None of given options
- Chomsky Normal Form (CNF)

261. The part of an FA, where the input string is placed before it is run, is called

- Transition
- Input Tape (Handouts Page # 105)
- Output Tape
- State

262. A problem is said to be ……if there exists an algorithm that provides the solution in……. number of steps.

- Effectively unsolvable, infinite
- Effectively solvable, infinite
- Effectively unsolvable, finite
- Effectively solvable, finite (Handouts Page # 80)

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263. ……. states are called the halt states.

- ACCEPT AND START
- ACCEPT and READ
- ACCEPT and REJECT (Handouts Page # 105)
- ACCEPT AND WRITE

264. The grammatical rules which involve meaning of words are called

- Semantics (Handouts Page # 87)
- Syntactics
- strings
- alphabets

265. The PDA is called non-deterministic PDA when there are more than one out going edges from state

- READ or POP (Handouts Page # 111)
- START or READ
- POP or
- PUSH or POP

266. Which of the following states is not part of PDA?

- REJECT
- ACCEPT
- START
- WRITE (All other are parts of PDA)

267. The major problem in the earliest computers was

- To calculate the mathematical formula
- To display mathematical formulas (Handouts Page # 87)
- To store the contents in the registers
- To load the contents from the registers

268. The operators like (` +) in the parse tree are considered as

- Terminals (Handouts Page # 93)
- productions
- non-terminals

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- intermediates

269. If L1 and L2 are two regular languages, then they……expressed by FAs.

- cannot be
- May be
- may or may not be
- can be (Handouts Page # 68)

270. Before running the input string on PDA it is first placed on

- Stack
- Ram
- Memory
- Tape (Handouts Page # 107)

271. Which is the correct option

- The element PUSHED in last is POPED in last
- The element PUSHED first is POPED in the last (LIFO Method, from Book)
- The element PUSHED first is POPED first
- None of given options

272. Null production is a

- String
- Word (Handouts Page # 97)
- Terminal
- All the above

273. A/an……operator adds a new letter at the top of STACK

- Push (Handouts Page # 107)
- Append
- Read
- Pop

274. In conversion form of PDA, no two……states exist in a row without state

- POP. READ (Handouts Page # 119)
- POP. REJECT
- PUSH. START
- PUSH READ

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275. Given a PDA that accepts the language L

- that PDA will also accept Language L' (complement of Ll
- There exists a CFG that generates exactly L (Handouts Page # 118)
- None of given options
- There does not exist any CFG that generates exactly

276. In large FA with thousands of states and millions of directed edges, without an effective procedure it is .....to find a path from initial to final state.

- Always easy
- always impossible
- may be good
- Impossible (Handouts Page # 81)

277. The CFG there generates the regular language is called

- Regular expression
- finite automata
- regular grammars (Handouts Page # 97)
- now regular grammars

278. Consider the following CFG: (Note: ^ means NULL)

S-→Xa

X→aX|bX|^

- Above give a CFG can be represented by RE
- a*b*
- a (a + b) *a
- (a + b) *a
- A*b*a

279. For a machine with N number of states, the total number of strings to be tested, defined over an alphabet of m letters is

- mN +mN+1 + mN+2 +… +m2N-1 (Handouts Page # 86)
- mN
- Nm
- Nm +mN+1 + mN+2 +… +

280. Consider the CFG given below.

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A→B|b

B→a

Which of the following is a unit production?

- S→bb
- A→b
- A→B
- B→a

281. The CFG is said to be ambiguous if there exist at least one word of its language that can be generated by the ……. production trees

- One
- Two
- More than one (Handouts Page # 95)
- At most one

282. If Q = {xx, xyxxxy} and R = { xyxyxyxxyy, xyxyyyxx} then Pref {Q in R} = _____

- Xx
- Xyxyxy
- Xyxyyy (Solved by my self 100% sure)
- Xxy

283. The unit production is

- Terminal --> Terminal
- Terminal --> Non Terminal
- Non terminal --> Terminal
- Non terminal --> Non-Terminal (Hand out Page # 100)

284. Which of the following statement is FALSE?

- For every PDA, there always exists a regular expression (Not sure)
- Every CFG cannot be expressed as Regular Expression
- Every Regular Expression be expressed by a CFG.
- For a PDA, there exists a CFG that represent the same language

285. The CFG S → aSb|ab|^

- Palindrome

- Prime
- Equal
- Even

286. Before the PDA is converted into conversion form, a new state ---------- is defined which is placed in the middle of any edge.

- HERE (Hand out Page # 118)
- STOP
- START
- REJECT

287. A PDA is in conversion form if it fulfills the following conditions:

- There is only one ACCEPT state. (Hand out Page # 119)
- There are one REJECT state.
- There are more than one ACCEPT states.
- There is only one Accept state.

288. Identify the false statement about the following CFG

S→SB|AB

A→CC

B→

C→a

- CFG has 8 Non terminals
- all the given option (All are false as There are 4 terminals , It is in CNF and it does not generate any null string)
- CFG is not in CNF
- CFG generate null string

289. This CFG there generates to the regular language is called

- Regular grammar (Hand out Page # 97)
- nonregular grammar
- finite automata
- regular expression

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290. The derivation of the word W generated by CFG such that at each step a production is applied to the leftmost nonterminal in the working string is said to be

- Left most terminal
- right most terminal
- left most derivation (Hand out Page # 103)
- right most derivation

291. If the FA has N states, then test these words of length less than N. If no word is accepted by this FA, then it will word/words

- Accept no (Hand out Page # 85)
- Accept some
- Reject No
- Accept All

292. In a CFG then non terminals are denoted by

- small letters
- numbers
- capital letters (Hand out Page # 87)
- small letters and numbers

293. "CFG"stands for_____

- Context finite graph
- contacts finite grammar
- contact free graph
- Context free grammar (Hand out Page # 87)

294. Consider the following production (of a CFG) S→XYZ Here _____is left most non terminals in working string note XY and Z are all known terminals

- X (X is on the most left side)
- S
- Z
- Y

295. Consider the following CFG

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S→ a|Xb|aYa

X→ Y|^ (Note: ^ means NULL)

Y→ b|X

which nonterminal is/are not nullable

- Y
- S, X and Y
- S
- X (X is Null Production and not a nullable

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CS-402

Final Term

Quiz-3,4

by Attiq Kundi

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CS402

Up Dated on 30-01-2023

If the intersection of two regular languages is regular then the complement of the intersection of these two languages is _____.

Regular

The language of all strings partition ∑* into _____ class(es).

Two

The language of all strings not beginning with 'b' partitions ∑* into _____distinct classes.

Two

If Q = {xx, xyxxxy }, and R = {xyxyxyxxyy, xyxyyyxx} then Pref(Q in R) = _____

xyxyyy

A language ending with ‘b’ partitions ∑* into _____distinct classes.

Three

If R is regular language and Q is any language (regular/ non-regular), then Pref(_____in _____) is regular.

Q,R

The reverse of the string sbfsbb over { sb, f, b }

Bsbfsb

The basic approach of Myhill Nerode theorem is similar to the concept of:

concatenation of FAs

If there is no final state of two FAs then their _____ also have no _____ state

union, final

If an FA has N states then it must accept the word of length

N

For a machine with N number of sta., the total number of strings to be tested, defined over an alphabet of m letters, is

mN +mN+1+ mN,-2 ,m2N-1

Which of the following is not a true theorem?

Pseudo theorem

The language "PRIME" is an example of language.

non regular

The product of two regular languages is

Regular

One language can have _____ CFG(s).

More than one

Which of the following is a non-regular language?

Prime

If the FA has N states. then test the words of length less than N. If no word is accepted by this FA. then it will _____ word/words.

accept no

In large FA with thousands of states and millions of directed edges, without an effective procedure it is _____ to find a path from initial to final state.

Impossible

A problem that has decision procedure is called problem.

Decidable problem

Which one of the following languages is a non regular language?

Palindrome

Using Myhill Nerode theorem we partition sigma star into distinct

Classes

In pumping lemma theorem (x y"n z) the range of n is

n=1. 2, 3, 4.....

The values of input (say a & b) do not remain same in one cycle due to

NOT gate

The operators like (*" +) in the parse tree are considered as

Terminals

Even-Even language partitions ∑* into _____ distinct classes.

Four

The strings or words which do not belong to a language are called _____ of that language.

Complement

The production S —> SS lalbl^ can be expressed by Regular expression

(a+b)+

if L1 and L2 are two regular languages. then they expressed by FAs.

can be

The grammatical rules which involve meaning of words are called

Semantics

Set of all palindromes over (a,b) is:

Regular and finite

The language of all strings not beginning with partitions “b" into distinct classes.

Two

The CFG is said to be ambiguous if there exist at least one word of its language that can be generated by production trees.

More than one

The CFG S —> aSb|abl^ is used to express the language

Palindrome

A non regular language can be represented by

None of the given options

In large FA with thousands of states and millions of directed edges, without an effective procedure it is _____ to find a path from initial to final state.

Impossible

In polish notation, (o-o-o) is the abbreviation of _____.

Operator - Operand – Operand

If an FA has N states then it must accept the word of length

N

Using Myhill Nerode theorem we partition sigma star into distinct _____.

classes

If a language is regular it must generate _____ number of distinct classes.

finite

If L1 and L2 are regular languages then which statement is NOT true?

L1/L2 is always regular

If the FA has N states, then test the words of length less than N. If no word is accepted by this FA, then it will _____ word/words.

accept no

A problem that has decision procedure is called _____ problem.

decidable

If L1 and L2 are two regular languages, then they _____ expressed by FAs.

can be

A language that can be expressed by RE, is said to be a _____ language.

regular

The values of input (say a & b) do not remain same in one cycle due to

NOT gate

Prime is a _____ language.

non-regular

If an effectively solvable problem has answer in YES or NO, then the solution is called _____.

decision procedure

To write the expression from the tree, it is required to traverse from _____.

Left side of the tree

If there is no final state of two FAs then their _____ also have no _____ state

union,final

In CFG, symbols that cannot be replaced by anything are called _____.

terminals

Finite **Automaton (FA) must have _____ number of states while a language has _____ words.**

finite, infinite

The language "PRIME" is an example of _____ language.

non regular

Using Myhill Nerode theorem we partition sigma star into distinct _____.

classes

Even-Even language partitions ∑* into _____ distinct classes.

four

What will be the 9's complement of the number 872?

127

In pref(Q in R), Q is _____ to/than R.

Not equal

There is at least one production in CFG that has one _____ on its left side.

Non terminal

In pumping lemma theorem (x y^n z) the range of n is

n=1, 2, 3, 4…….

A language ending with 'b' partitions ∑* into _____distinct classes.

three

The operators like (* , +) in the parse tree are considered as _____.

terminals

For a machine with N number of states, the total number of strings to be tested, defined over an alphabet of m letters, is _____.

mN +mN+1+ mN+2 +… +m2N-1

In a CFG, the non-terminals are denoted by _____.

Capital letters

If an FA accepts a word then there must exist a path from _____.

Incase of Myhill Nerode theorem, if a language L partitions sigma star into distinct classes and L is also regular then L generates _____ number of classes.

finite

Which of the following is pumped to generate further strings in the definition of Pumping Lemma?

y

The complement of a regular language is also _____.

regular

An FA has same initial and _____ state, then it means that it has no _____ state.
Select correct option:

final, initial

The product of two regular languages is _____.

regular

According to Myhill Nerode theorem, if L generates finite no. of classes then L is.....
Select correct option:

Regular

Two languages are said to belong to same class if they end in the same state when they run over an FA, that state

May be final state or not

In pref(Q in R) Q is …… to (than) R

Not equal

For language L defined over {a, b},then L partitions {a, b}* into …… classes
Distinct

Which of the following is not a true theorem?
Pseudo theorem

If a regular expression contains * then it _____ define an _____ language.

may, infinite

a^n b^n generates the …… language

non regular

To examine whether a certain FA accepts any words, it is required to seek the paths _____ state.

from initial to final

If r1 and r2 are regular expressions then which of the following is not regular expression

r1 – r2

Kleene star closure can be defined

Over any set of string

Which of following string(s) belongs to the language of the regular expression (aa*b)*?

aabaab

According to theory of automata there are _____ types of languages
Select correct option:

Two

If S = {aa, bb}, then S* will not contain

aaabbb

Every non deterministic Finite Automata can be converted into

Regular Expression
Deterministic Finite Automata
Trasition Graph
All of the given options

The states in which there is no way to leave after entry are called

Davey John Lockers

Dead States

Waste Baskets

All of the given options

(a* + b*)* = (a + b)* this expression is _____

True

What is false about the PALINDROME LANGUAGE?

Every word is reverse of itself.

It is an infinite language.

FA can be build for it.

None of the given optio

FA is also called

DFA

Kleene star closure can be defined

Over any set of string

[(a + b)(a + b)]*, given RE cannot generate the string _____

bbbbbb

In an FA, when there is no path starting from initial state and ending in final state then that FA

does not accept any string

While finding RE corresponding to TG, we connect the new start state to the old start state by the transition labeled by

null string

According to theory of automata there are _____ types of languages

Two

While finding RE corresponding to TG, If TG has more than one final state then Introduce the new final state

The states in which there is no way to leave after entry are called

Davey John Lockers
Dead States
Waste Baskets
All of the given options

What is false about the term alphabet?
It can be an empty set.

Which of the following is used to delay the transmission of signal along the wire by one step (clock pulse)?

Delay box

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The basic approach of Myhill Nerode theorem is similar to the concept of:

Select the correct option

- (●) concatenation of FAs
- ( ) distinguishable and indistinguishable strings
- ( ) closure of FAs
- ( ) union of FAs

Click to Save Answer & Move to Next Question

5:34 PM
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vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=d090b867-bffe-4ed8-b1b0-bdab31937ea5

The language of all strings not beginning with 'b' partitions Σ^* into _____distinct classes.

Select the correct option

- ( ) five
- ( ) four
- (●) two
- ( ) three

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The language L defined over Σ, not belonging to L, is called _____ of the language L.

Select the correct option

- closure
- complement
- intersection
- union

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vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=cbecc20e-5d91-4ceb-99be-67b01584d5a8

What will be the 9's complement of the number 872?

Select the correct option

- 271
- 721
- 172
- 127

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A problem that has decision procedure is called _____ problem.

Select the correct option

- (●) decidable
- ( ) Regular language
- ( ) un-decidable
- ( ) Infinite

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The values of input (say a & b) do not remain same in one cycle due to

Select the correct option

- ( ) Clock pulse
- ( ) OR gate
- ( ) NAND gate
- (●) NOT gate

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vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=3ba484bc-5387-403f-b6f0-3303eacf7f29

In large FA with thousands of states and millions of directed edges, without an effective procedure it is _____ to find a path from initial to final state.

Select the correct option

- (●) Impossible
- ( ) Always easy
- ( ) always impossible
- ( ) may be good

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vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=1a92b484-88ec-417f-8d13-b3dd5ac23407

A non regular language can be represented by

Select the correct option

- ( ) TG
- ( ) RE
- ( ) None of the given options
- (●) FA

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If Q = {xx, xyxxxy }, and R = {xyxyxyxxyy, xyxyyyxx} then Pref(Q in R) = _____

Select the correct option

- [ ] xyxyxy
- [ ] xxy
- [ ] xx
- [x] xyxyyy

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vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=90d6c354-715e-4e2f-8cde-6764ab7659f4

If the intersection of two regular languages is regular then the complement of the intersection of these two languages is _____.

Select the correct option

- [ ] irregular but finite
- [ ] regular
- [x] irregular
- [ ] irregular but infinite

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a^n b^n generates the ..... language

Select the correct option

- non regular
- EQUAL and non regular
- regular
- EQUAL and regular

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Prime is a _____ language.

Select the correct option

- finite
- non-regular
- regular
- both context free and regular

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Set of all palindromes over {a,b} is:

Select the correct option

- Regular and infinite
- (●) Regular and finite
- Non-regular
- Regular

Click to Save Answer & Move to Next Question

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The language "PRIME" is an example of _____ language.

Select the correct option

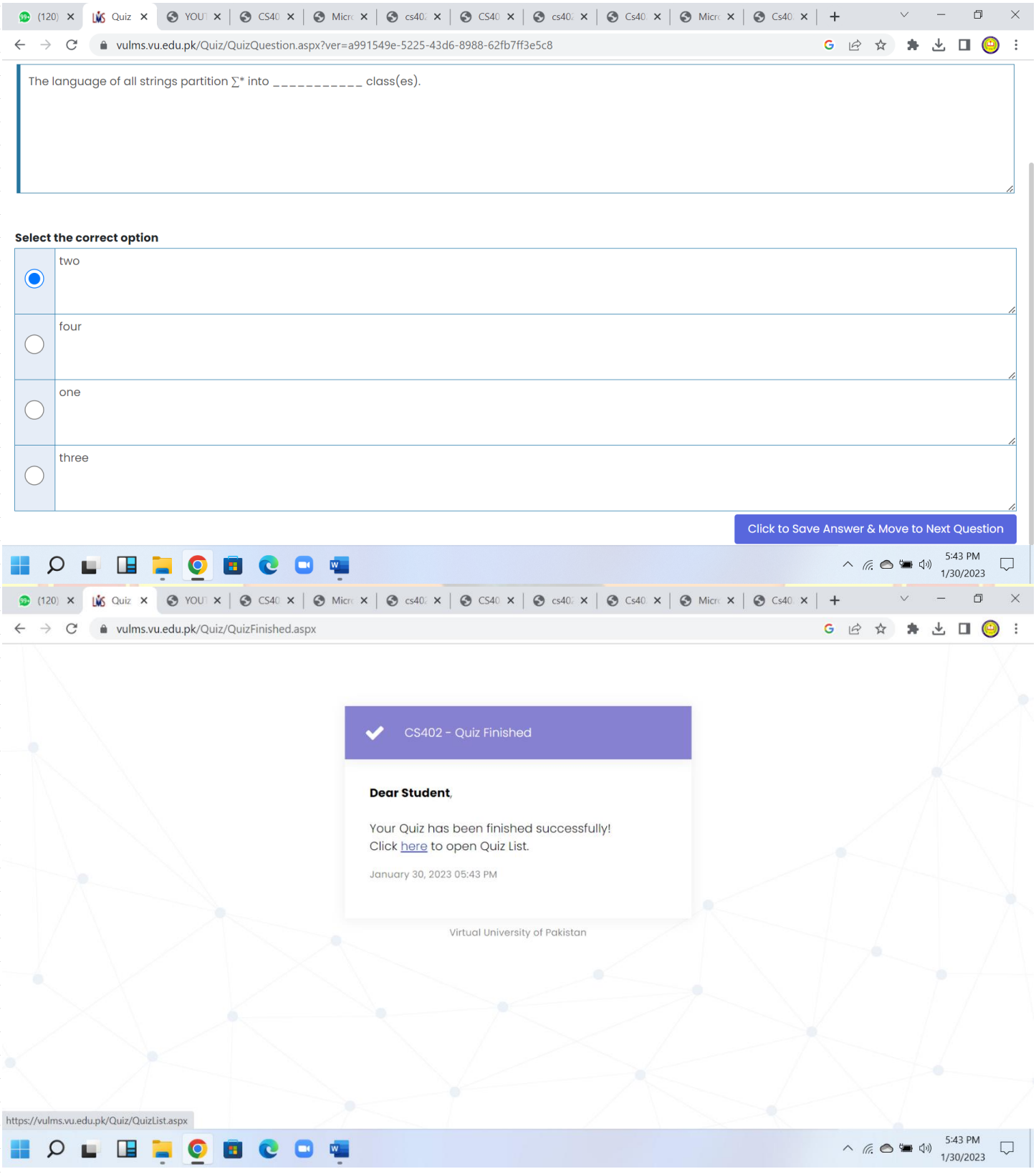
- regular but finite
- non regular but finite
- regular
- (●) non regular

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CS-402

Final Term

Quiz-3,4

by Attiq Kundi

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CS402

Up Dated on 30-01-2023

If the intersection of two regular languages is regular then the complement of the intersection of these two languages is _____.

Regular

The language of all strings partition ∑* into _____ class(es).

Two

The language of all strings not beginning with 'b' partitions ∑* into _____distinct classes.

Two

If Q = {xx, xyxxxy }, and R = {xyxyxyxxyy, xyxyyyxx} then Pref(Q in R) = _____

xyxyyy

A language ending with ‘b’ partitions ∑* into _____distinct classes.

Three

If R is regular language and Q is any language (regular/ non-regular), then Pref(_____in _____) is regular.

Q,R

The reverse of the string sbfsbb over { sb, f, b }

Bsbfsb

The basic approach of Myhill Nerode theorem is similar to the concept of:

concatenation of FAs

If there is no final state of two FAs then their _____ also have no _____ state

union, final

If an FA has N states then it must accept the word of length

N

For a machine with N number of sta., the total number of strings to be tested, defined over an alphabet of m letters, is

mN +mN+1+ mN,-2 ,m2N-1

Which of the following is not a true theorem?

Pseudo theorem

The language "PRIME" is an example of language.

non regular

The product of two regular languages is

Regular

One language can have _____ CFG(s).

More than one

Which of the following is a non-regular language?

Prime

If the FA has N states. then test the words of length less than N. If no word is accepted by this FA. then it will _____ word/words.

accept no

In large FA with thousands of states and millions of directed edges, without an effective procedure it is _____ to find a path from initial to final state.

Impossible

A problem that has decision procedure is called problem.

Decidable problem

Which one of the following languages is a non regular language?

Palindrome

Using Myhill Nerode theorem we partition sigma star into distinct

Classes

In pumping lemma theorem (x y"n z) the range of n is

n=1. 2, 3, 4.....

The values of input (say a & b) do not remain same in one cycle due to

NOT gate

The operators like (*" +) in the parse tree are considered as

Terminals

Even-Even language partitions ∑* into _____ distinct classes.

Four

The strings or words which do not belong to a language are called _____ of that language.

Complement

The production S —> SS lalbl^ can be expressed by Regular expression

(a+b)+

if L1 and L2 are two regular languages. then they expressed by FAs.

can be

The grammatical rules which involve meaning of words are called

Semantics

Set of all palindromes over (a,b) is:

Regular and finite

The language of all strings not beginning with partitions “b" into distinct classes.

Two

The CFG is said to be ambiguous if there exist at least one word of its language that can be generated by production trees.

More than one

The CFG S —> aSb|abl^ is used to express the language

Palindrome

A non regular language can be represented by

None of the given options

In large FA with thousands of states and millions of directed edges, without an effective procedure it is _____ to find a path from initial to final state.

Impossible

In polish notation, (o-o-o) is the abbreviation of _____.

Operator - Operand – Operand

If an FA has N states then it must accept the word of length

N

Using Myhill Nerode theorem we partition sigma star into distinct _____.

classes

If a language is regular it must generate _____ number of distinct classes.

finite

If L1 and L2 are regular languages then which statement is NOT true?

L1/L2 is always regular

If the FA has N states, then test the words of length less than N. If no word is accepted by this FA, then it will _____ word/words.

accept no

A problem that has decision procedure is called _____ problem.

decidable

If L1 and L2 are two regular languages, then they _____ expressed by FAs.

can be

A language that can be expressed by RE, is said to be a _____ language.

regular

The values of input (say a & b) do not remain same in one cycle due to

NOT gate

Prime is a _____ language.

non-regular

If an effectively solvable problem has answer in YES or NO, then the solution is called _____.

decision procedure

To write the expression from the tree, it is required to traverse from _____.

Left side of the tree

If there is no final state of two FAs then their _____ also have no _____ state

union,final

In CFG, symbols that cannot be replaced by anything are called _____.

terminals

Finite **Automaton (FA) must have _____ number of states while a language has _____ words.**

finite, infinite

The language "PRIME" is an example of _____ language.

non regular

Using Myhill Nerode theorem we partition sigma star into distinct _____.

classes

Even-Even language partitions ∑* into _____ distinct classes.

four

What will be the 9's complement of the number 872?

127

In pref(Q in R), Q is _____ to/than R.

Not equal

There is at least one production in CFG that has one _____ on its left side.

Non terminal

In pumping lemma theorem (x y^n z) the range of n is

n=1, 2, 3, 4…….

A language ending with 'b' partitions ∑* into _____distinct classes.

three

The operators like (* , +) in the parse tree are considered as _____.

terminals

For a machine with N number of states, the total number of strings to be tested, defined over an alphabet of m letters, is _____.

mN +mN+1+ mN+2 +… +m2N-1

In a CFG, the non-terminals are denoted by _____.

Capital letters

If an FA accepts a word then there must exist a path from _____.

Incase of Myhill Nerode theorem, if a language L partitions sigma star into distinct classes and L is also regular then L generates _____ number of classes.

finite

Which of the following is pumped to generate further strings in the definition of Pumping Lemma?

y

The complement of a regular language is also _____.

regular

An FA has same initial and _____ state, then it means that it has no _____ state.
Select correct option:

final, initial

The product of two regular languages is _____.

regular

According to Myhill Nerode theorem, if L generates finite no. of classes then L is.....
Select correct option:

Regular

Two languages are said to belong to same class if they end in the same state when they run over an FA, that state

May be final state or not

In pref(Q in R) Q is …… to (than) R

Not equal

For language L defined over {a, b},then L partitions {a, b}* into …… classes
Distinct

Which of the following is not a true theorem?
Pseudo theorem

If a regular expression contains * then it _____ define an _____ language.

may, infinite

a^n b^n generates the …… language

non regular

To examine whether a certain FA accepts any words, it is required to seek the paths _____ state.

from initial to final

If r1 and r2 are regular expressions then which of the following is not regular expression

r1 – r2

Kleene star closure can be defined

Over any set of string

Which of following string(s) belongs to the language of the regular expression (aa*b)*?

aabaab

According to theory of automata there are _____ types of languages
Select correct option:

Two

If S = {aa, bb}, then S* will not contain

aaabbb

Every non deterministic Finite Automata can be converted into

Regular Expression
Deterministic Finite Automata
Trasition Graph
All of the given options

The states in which there is no way to leave after entry are called

Davey John Lockers

Dead States

Waste Baskets

All of the given options

(a* + b*)* = (a + b)* this expression is _____

True

What is false about the PALINDROME LANGUAGE?

Every word is reverse of itself.

It is an infinite language.

FA can be build for it.

None of the given optio

FA is also called

DFA

Kleene star closure can be defined

Over any set of string

[(a + b)(a + b)]*, given RE cannot generate the string _____

bbbbbb

In an FA, when there is no path starting from initial state and ending in final state then that FA

does not accept any string

While finding RE corresponding to TG, we connect the new start state to the old start state by the transition labeled by

null string

According to theory of automata there are _____ types of languages

Two

While finding RE corresponding to TG, If TG has more than one final state then Introduce the new final state

The states in which there is no way to leave after entry are called

Davey John Lockers
Dead States
Waste Baskets
All of the given options

What is false about the term alphabet?
It can be an empty set.

Which of the following is used to delay the transmission of signal along the wire by one step (clock pulse)?

Delay box

FINALTERM EXAMINATION
Spring 2010
CS402- Theory of Automata (Session - 1)

Question No: 1 (Marks: 1) - Please choose one
If r1 = (aa + bb) and r2 = (a + b) then the language (aa + bb)(a + b) will be generated by

- **(r1)(r2) (Page 10)**
- (r1 + r2)
- (r2)(r1)
- (r1)*

Question No: 2 (Marks: 1) - Please choose one
"One language can be expressed by more than one FA". This statement is _____

- **True (Page 14)**
- False
- Some times true & sometimes false
- None of these

Question No: 3 (Marks: 1) - Please choose one
Who did not invent the Turing machine?

- Alan Turing
- **A. M. Turing (Page 140)**
- Turing
- None of these

Question No: 4 (Marks: 1) - Please choose one
Which statement is true?

- **The tape of turing machine is infinite. (Page 140)**
- The tape of turing machine is finite.
- The tape of turing machine is infinite when the language is regular
- The tape of turing machine is finite when the language is nonregular.

دنیا میں سب سے مشکل کام اپنی اصلاح اور سب سے آسان کام دوسروں پر نکتہ چینی کرنا ہے

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Question No: 5 (Marks: 1) - Please choose one

A regular language:

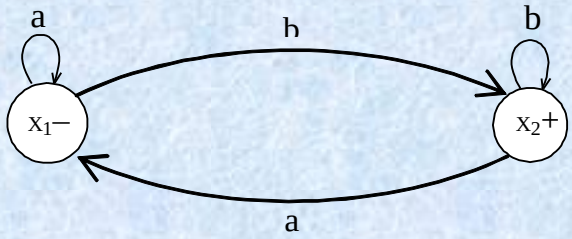
- ► **Must be finite (Page 11)**
- ► Must be infinite
- ► Can be finite or infinite
- ► Must be finite and cannot be infinite

Question No: 6 (Marks: 1) - Please choose one

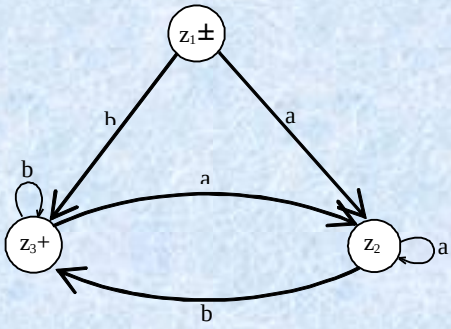
Every regular expression can be expressed as CFG but every CFG cannot be expressed as a regular expression. This statement is:

- ► Depends on the language
- ► None of the given options
- ► **True (Page 97)**
- ► False

Question No: 7 (Marks: 1) - Please choose one



Above given FA corresponds RE r. then FA corresponding to r* will be



This statement is

- ► **True (Page 38)**
- ► False
- ► Depends on language
- ► None of these

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

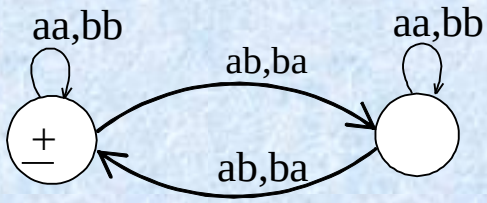
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Question No: 8 (Marks: 1) - Please choose one

Consider the language L of strings, defined over Σ = {a,b}, ending in a

- **There are finite many classes generated by L, so L is regular (Page 76)**
- There are infinite many classes generated by L, so L is regular
- There are finite many classes generated by L, so L is non-regular
- There are infinite many classes generated by L, so L is non-regular

Question No: 9 (Marks: 1) - Please choose one



Above given TG has _____ RE.

- (aa+aa+(ab+ab)(aa+ab)*(ab+ba))*
- **(aa+bb+(ab+ba)(aa+bb)*(ab+ba))* (Page 22)**
- (aa+bb+(ab+ba)(aa+bb)(ab+ba))*
- None of these

Question No: 10 (Marks: 1) - Please choose one

The word 'formal' in formal languages means

- The symbols used have well defined meaning
- They are unnecessary, in reality
- **Only the form of the string of symbols is significant** Click here for detail
- None of these

Question No: 11 (Marks: 1) - Please choose one

Let A = {0, 1}. The number of possible strings of length 'n' that can be formed by the elements of the set A is

- n!
- n^2
- n^m
- **2^n**

Question No: 12 (Marks: 1) - Please choose one

Choose the correct statement.

- A Mealy machine generates no language as such
- A Moore machine generates no language as such
- A Mealy machine has no terminal state
- **All of these** click here for detail

اللہ کا خوف سب سے بڑی دانائی ہے

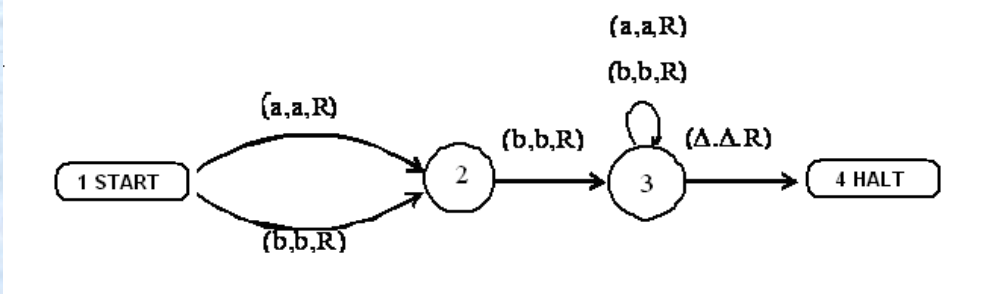
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Question No: 13 (Marks: 1) - Please choose one
TM is more powerful than FSM because
- ► The tape movement is confined to one direction
- ► It has no finite state control
- ► **It has the capability to remember arbitrary long sequences of input symbols** Click here for detail
- ► None of these

Question No: 14 (Marks: 1) - Please choose one
If L1 and L2 are expressed by regular expressions r1 and r2, respectively then the language expressed by r1 + r2 will be _____
- ► **Regular (Page 10)**
- ► Ir-regular
- ► Can't be decided
- ► Another Language which is not listed here

Question No: 15 (Marks: 1) - Please choose one
Like TG, a PDA can also be non-deterministic
- ► **True (Page 111)**
- ► False

Question No: 16 (Marks: 1) - Please choose one



The above machine is a/anTG _____
- ► Finite Automata
- ► **Turing machine (Page 141)**
- ► FA
- ► TG

Question No: 17 (Marks: 1) - Please choose one
The language of all words (made up of a's and b's) with at least two a's can not be described by the regular expression.
- ► a(a+b)*a(a+b)*(a+b)*ab*
- ► (a+b)* ab* a(a+b)*
- ► b*ab* a(a+b)*
- ► **none of these**

a^nb^n {where n>0} is the language will at least one a and b and cannot be described by RE.

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Question No: 18 (Marks: 1) - Please choose one
In FA, if one enters in a specific state but there is no way to leave it, then that specific state is called
- ► Dead State
- ► Waste Basket
- ► Davey John Locker
- ► **All of these (Page 17)**

Question No: 19 (Marks: 1) - Please choose one
If L is a regular language then, L^c is also a _____ language.
- ► **Regular (Page 66)**
- ► Non-regular
- ► Regular but finite
- ► None of the given

Question No: 20 (Marks: 1) - Please choose one
In CFG, the symbols that can't be replaced by anything are called___
- ► **Terminal (Page 87)**
- ► Non-Terminal
- ► Production
- ► All of given

Question No: 21 (Marks: 1) - Please choose one
Which of the following is NOT a regular language?
- ► String of 0's whose length is a perfect squere
- ► Set of all palindromes made up of 0's and 1's
- ► String of 0's whose length is a prime number
- ► **All of the given options** Click here for detail

Question No: 22 (Marks: 1) - Please choose one
Choose the incorrect (FALSE) statement.
- ► A Mealy machine generates no language as such
- ► A Mealy machine has no terminal state
- ► **For a given input string, length of the output string generated by a Moore machine is not more than the length of the output string generated by that of a Mealy machine** click here for detail
- ► All of these

Question No: 23 (Marks: 1) - Please choose one
Pumping lemma is generally used to prove that:
- ► A given language is infinite
- ► **A given language is not regular** Click here for detail
- ► Whether two given regular expressions of a regular language are equivalent or not
- ► None of these

دنیا کی سب سے بڑی فتح نفس پر قابو رکھنا ہے

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Question No: 24 (Marks: 1) - Please choose one
Which of the following is a regular language?
- ► **String of odd number of zeroes** Click here for detail
- ► Set of all palindromes made up of 0's and 1's
- ► String of 0's whose length is a prime number
- ► All of these

Question No: 25 (Marks: 1) - Please choose one
Choose the incorrect statement:
- ► (a+b)*aa(a+b)* generates Regular language.
- ► **A language consisting of all strings over ∑={a,b} having equal number of a's and b's is a regular language**
- ► Every language that can be expressed by FA can also be expressed by RE
- ► None of these

Question No: 26 (Marks: 1) - Please choose one
Left hand side of a production in CFG consists of:
- ► One terminal
- ► More than one terminal
- ► **One non-terminal (Page 87)**
- ► Terminals and non-terminals

FINALTERM EXAMINATION
SPRING 2007

Question No: 1 (Marks: 1) - Please choose one
PDA is only used to represent a regular language.
- ► True
- ► **False** Click here for detail

Question No: 2 (Marks: 1) - Please choose one
If L is a regular language then LC is also a regular language.
- ► **True (Page 66)**
- ► False

ایماندار کو غصہ دیر سے آتا ہے اور جلدی دور ہو جاتا ہے

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Question No: 3 (Marks: 1) - Please choose one
A production of the form non-terminal ⇾ string of two non-terminal is called a live Production.
► **True (Page 127)**
► False

Question No: 4 (Marks: 1) - Please choose one
We can find a CFG corresponding to a DFA.
► **True (Page 97)**
► False

Question No: 5 (Marks: 1) - Please choose one
START, READ, HERE and ACCEPTS are conversions of the machine
► **True (Page 122)**
► False

Question No: 6 (Marks: 1) - Please choose one
A CFG is said to be ambiguous if there exists at least one word of its language that can be generated by different production trees
► **True (Page 95)**
► False

Question No: 7 (Marks: 1) - Please choose one
Syntax tree or Generation tree or Derivation tree are same tree
► **True (Page 92)**
► False

Question No: 8 (Marks: 1) - Please choose one
The symbols that cannot be replaced by anything are called terminals
► **True (Page 87)**
► False

Question No: 9 (Marks: 1) - Please choose one
The production of the form non-terminal ⇾ one non-terminal is called unit production
► **True (Page 100)**
► False

Question No: 10 (Marks: 1) - Please choose one
DFA and PDA are equal in power.
► True
► **False (Page 105)**

زندگی میں کامیابی کا یہی راز ہے کہ پریشانیوں سے پریشان مت بنو

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FINALTERM EXAMINATION
Spring 2006
CS402- Theory of Automata

Question No. 1
A production of the form non-terminal ⇾ non-terminal is called a dead Production.
True
False (Page 127)

Question No. 2
Semi-word is a string having some terminals and one non-terminal at the right of string.
True (Page 97)
False

Question No. 3
Two FAs are equivalent if they have same no. of states.
True (Page 15)
False

Question No. 4
There exist exactly two different derivations in an ambiguous CFG for a word.
True (Page 93)
False

Question No. 6
Regular languages are closed under Union, Concatenation and Kleene star.
True (Page 10)
False

Question No. 7
CFG may also represent a regular language.
True (Page 97)
False

Question No. 9 Marks : 1
PDA is stronger than FA.
True (Page 105)
False

جھوٹ انسان اور ایمان دونوں کا دشمن ہے

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FINALTERM EXAMINATION
Spring 2005
CS402- Theory of Automata

Question No. 1
A Total Language Tree has
All languages over Σ
All strings over Σ (Page 96)
All words of all languages over Σ
All words of one language over Σ

Question No. 2
What Turing Machine does not have?
Stack
Tape
Head
Word
Turing machine has stack but insertion and deletion can be done from both sides. Tape and head to.

Question No. 3
CFG given S ⇗ bS|Sb|aa represents language
b*aa
aab*
b*aab*
b*(aa)*b*

Question No. 4
A Language that is finite but not regular
Λ
(a+b)*
Φ (not sure)
All strings of a's in Σ = {a,.b}

عقل مند کہتا ہے میں کچھ نہیں جانتا جبکہ بے وقوف کہتا ہے کہ میں سب کچھ جانتا ہوں

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CS402 – Quiz No.3

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
The values of input (say a & b) does not remain same in one cycle due to
NAND gate
Click plus
OR gate
NOT gate

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
Set of all palindromes over {a,b}is regular

True
False **(Page 74)**

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
In CFG, the symbols that cannot be replaced by anything are called

Terminals **(Page 87)**
Non terminals
Productions
None of the given options

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
a^n b^n generates the …… language

regular
non regular
EQUAL and non regular **(Page 71)**
EQUAL and regular

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
The grammatical rules which involves meaning of words are called:

Semantic **(Page 87)**
Sytactics
Alphabets
None of he given options

خود کو تمھیں سے بڑھ کر کوئی اچھا مشورہ نہیں دے سکتا

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Question # 6 of 10 (Total Marks: 1) **Select correct option:**
The reverse of the string sbfsbb over { sb, f, b }

bbsfbs
bsbfsb
sbbfsb
bsfbsb

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
If an FA has N state then it must accept the word of length

N-1
N+1
N+2N

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
Two languages are said to belong to same class if they end in the same state when they run over an FA, that state

Must be final state
May be final state or not (Page 75)
May be start or not
None of the given options

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
In pref(Q in R) Q is …… to (than) R

Equal
Not Equal (Page 79)
Greater
Smaller

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
According to Myhill Nerode theorem, if L generates finite no. of classes then L is.....

Finite
Infinite
Regular (Page 76)
Non Regular

جو شخص ناکامیوں سے ڈر کر بھاگتا ہے کامیابی اُس سے ڈر کر بھاگتی ہے

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Question # 1 of 10 (Total Marks: 1) **Select correct option:**
If the intersection of two regular languages is regular then the complement of the intersection of these two languages is also regular

True (Page 68)
False

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
In pumping lemma theorem (x y^n z) the range of n is

n=1,2,3,4..... (Page 74)
n=0,1,2,3,4....
n=-3,-2,-1,0,1,2,3,4.....
n=-3,-2,-1,1,2,3,4.....

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
The complement of a regular language is also a regular

True rep
False

CS402 – Quiz No.3

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
For a non regular language there exist …… FA

One
At least one
At most one
No (Page 71)

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
The strings or words which do not belong to a language is called…… of that language

Intersection
Union
Complement (Page 66)
Quotient

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Question # 3 of 10 (Total Marks: 1) **Select correct option:**
A non regular language can be represented by
RE
FA
TG
None of the given options (Page 71)

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
For language L defined over {a, b},then L partitions {a, b}* into …… classes

Infinite
Finite
Distinct (Page 75)
Non distinct

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
If an FA accept a word then there must exist a path from

Initial to final state (Page 81)
Initial to each state
Initial to each state but not to final state
Initial to final state by traversing each state

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
Does the empty string match the regular expression |y+a|?

Yes
No (Page 3)

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
If an FA already accepts the language expressed by the closure of certain RE, then the given FA is the required FA.

True (Page 37)
False

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
Which of the following statement is true about NFA with Null String?

Infinite states
Infinite set of letters
Infinite set of transitions
Transition of null string is allowed at any stage (Page 71)

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Question # 9 of 10 (Total Marks: 1) **Select correct option:**
If R is a regular language and L is some language, and L U R is a regular language, then L must be a regular language.

True (page 10)
False

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
FA corresponding to an NFA can be built by introducing an empty state for a letter having

no transition at certain state (Page 43)
one transition at certain state
two transition at certain state
more than two transitions at certain state

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
Let FA3 be an FA corresponding to FA1FA2, then the initial state of FA3 must correspond to the initial state of

FA1 only (Page 35)
FA2 only
FA1 or FA2
FA1 and FA2

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
(a* + b*)* = (a + b)* this expression is _____
True
False

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
If S = { x }, then S* will be
{x,xx,xxx,xxxx,…}
{^ ,x,xx,xxx,xxxx,…} (Page 10)

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
The states in which there is no way to leave after entry are called
Davey John Lockers
Dead States
Waste Baskets
All of the given options (Page 17)

جو لوگوں کے سامنے فخر کرتا ہے وہ لوگوں کی نظروں سے گر جاتا ہے

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Question # 5 of 10 (Total Marks: 1) **Select correct option:**
If S = {ab, bb}, then S* will not contain
Abbbab
Bbba
ababbb
bbbbab

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
According to theory of automata there are _____ types of languages

1
2 (Page 3)
3
4

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
What do automata mean?
Something done manually
Something done automatically (Page 3)

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
What is false about the term alphabet?
It is a finite set of symbols.
It is usually denoted by Greek letter sigma
It can be an empty set. (Page 3)
Strings are made up of its elements

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
Formal is also known as _____
Syntactic language (page 3)
Semantic language
Informal language
Nsone of these

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
Following are types of languages

Formal Languages (Syntactic languages)
Informal Languages (Semantic languages)
Both (Page 3)
None of above

عقل مند اپنے عیب خود دیکھتا ہے اور بیوقوفوں کے عیب دنیا دیکھتی ہے

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CS402 – Quiz No.4

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
Consider the following production (of a CFG): S->XYZ Here _____ is left most nonterminal in working string. Note: S, X, Y and Z are all nonterminals
S
X
Y
Z

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
A PDA is called nondeterministic PDA if _____

There are more than one outgoing edges at READ or POP states with one label (Page 111)
There are more than one PUSH states
There are mroe than one POP states
All of the given options

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
A PDA consists of the following:

An alphabet (Sigma) of input letters.
An input TAPE with infinite many locations in one direction
One START state with only one out-edge and no in-edge
All of the given options (Page 105)

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
The CFG S --> aSa | bSb | a | b | ^ represents the language

EVEN-EVEN
PALINDROM (Page 91)
EQUAL
ODD-ODD

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
Halt states are

Start and Accept
Accept and Reject (Page 105)
Start and Reject
Read and Reject

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Question # 6 of 10 (Total Marks: 1) **Select correct option:**
Choice of path can be determined by left most derivation of the string belonging to CFL at…… state

Accept (Page 104)
Reject
Push
POP

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
The unit and null productions can be deleted from a CFG

True (Page 99-100)
False

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
Identify the TRUE statement about following CFG:
S -> SB|AB
A -> CC
B -> b
C -> a

The given CFG has 8 Nonterminals
The given CFG has 8 Terminals
The given CFG is in CNF (Page 101)
The given CFG is not in CNF

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
The structure given below is called _____ S -> aA|bB A -> aS|a B -> bS|b

RE
TG
CFG (Page 87)
PDA

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
Which of the following states is not part of PDA

START
ACCEPT
WRITE (Page 107)
REJECT

بد صورت چہرہ بد صورت دماغ سے بہتر ہے

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CS402 – Quiz No.4

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
The production of the form: nonterminal --> one nonterminal is called the _____

Unit production (Page 100)
NULL production
Terminal production
Non Terminal production

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
A _____ is the one for which every input string has a unique path through the machine.

Deterministic PDA (Page 111)
nondeterministic PDA
PUSHDOWN store
Input Tape

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
In the null production N --> ^ , N is a

Terminal
Non terminal (Page 99)
Word
None of the given options

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
The major problem in the earliest computers was

To store the contents in the registers
To display mathematical formulae (Page 87)
To load the contents from the registers
To calculate the mathematical formula

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
In polish notation, (o-o-o) is the abbreviation of……?

Operand - Operator – Operand
Operand - Operand- Operator
Operator -Operand – Operand (Page 94)
Operand -Operand – Operand

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Question # 6 of 10 (Total Marks: 1) **Select correct option:**
The CFG is said to be ambiguous if there exist at least one word of its language that can be generated by the …… production trees

One
Two
More than one (Page 95)
At most one

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
The input string is placed, before it runs, in

Stack
Memory
Tape (Page 105)
Ram

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
The production S --> SS | a | b | ^ can be expressed by RE

(a+b)+
(a+b)
(a+b)* (Page 88)
(ab)*

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
The locations into which we put the input letters on "Input Tap" are called _____

words
alphabets
cells (Page 105)
elements

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
"CFG" stands for _____

Context Free Graph
Context Free Grammer (Page 87)
Context Finite Graph
Context Finite Grammer

عقل مند آدمی اس وقت تک نہیں بولتا جب تک خاموشی نہیں ہو جاتی

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Question # 1 of 10 (Total Marks: 1) **Select correct option:**
In a CFG the nonterminal that occurs first from the left in the working string, is said to be _____

Least Significant nonterminal
Most Significant nonterminal
Left most nonterminal (Page 103)
Left most derivate

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
The unit production is

Terminal --> Terminal
Terminal --> Non Terminal
Non terminal --> Terminal
Non terminal --> Non Terminal (Page 100)

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
A _____ operator adds a new letter at the top of STACK

PUSH (Page 107)
POP
READ
APPEND

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
PDA stands for _____

Push and Drop Automaton
Pop and Drop Automaton
Push Down Automaton (Page 112)
None of given options

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
The production of the form: Nonterminal-> ^ is said to be _____ production

NULL (Page 99)
UNIT
Chomsky form production
None of the given options

انسان دکھ نہیں دیتے بلکہ انسانوں سے وابستہ امیدیں دکھ دیتی ہیں

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Question # 6 of 10 (Total Marks: 1) **Select correct option:**
If a CFG has a null production, then it is _____

Posiible to construct another CFG without null production accepting the same language with the exception of the word ^ (Page 99)
Not possible to construct another CFG without null production accepting the same language with the exception of the word ^
Called NULL CFG
Called Chmosky Normal Form (CNF)

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
In a STACK:

The element PUSHed first is POPed first
The element PUSHed first is POPed in the last (Page 107 concept)
The element PUSHed in last is POPed in last
None of given options

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
Kleene star closure can be defined
Over any set of string (Page 7)
Over specific type of string

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
While finding RE corresponding to TG, we connect the new start state to the old start state by the transition labeled by
A
B
null string (Page 26)
None of the given options

خوبصورتی علم وادب سے ہوتی ہے لباس و حسن سے نہیں

بہترین تجربہ وہ ہے جس سے نصیحت حاصل ہو

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Some More Quizzes

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
For a given input, it provides the compliment of Boolean AND output.

NAND box (NOT AND) **(Page 63)**
DELAY box
OR box
AND box

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
It delays the transmission of signal along the wire by one step (clock pulse).

NAND box (NOT AND)
DELAY box **(Page 63)**
OR box
AND box

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
For the given input, it provides the Boolean OR output

NAND box (NOT AND)
DELAY box
OR box **(Page 63)**
AND box

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
For the given input, AND box provides the Boolean AND output.
True **(Page 63)**
False

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
The current in the wire is indicated by 1 and 0 indicates the absence of the current.
True **(Page 63)**
False

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
Any language that can not be expressed by a RE is said to be regular language.
True
False **(Page 71)**

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Question # 7 of 10 (Total Marks: 1) **Select correct option:**
If L1 and L2 are regular languages_____ is/are also regular language(s).

L1 + L2
L1L2
L1*
All of above (Page 10)

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
Let L be a language defined over an alphabet Σ, then the language of strings, defined over Σ, not belonging to L, is called Complement of the language L, denoted by Lc or L'.
True (Page 66)
False

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
To describe the complement of a language, it is very important to describe the ---------- of that language over which the language is defined.

Alphabet (Page 66)
Regular Expression
String
Word

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
For a certain language L, the complement of Lc is the given language L *i.e.* (Lc)c = Lc
True
False (Page 66)

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
If L is a regular language then, --------- is also a regular language.
Lm
Ls
Lx
Lc (Page 66)

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
Converting each of the final states of F to non-final states and old non-final states of F to final states, FA thus obtained will reject every string belonging to L and will accept every string, defined over Σ, not belonging to L. is called
Transition Graph of L
Regular expression of L
Complement of L (Page 66)
Finite Automata of L

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Question # 3 of 10 (Total Marks: 1) **Select correct option:**
If L1 and L2 are two regular languages, then L1 U L2 is not a regular.
True
False (Page 65)

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
De-Morgan's law for sets is expressed by,
$(L_1^c \cap L_2^c)^c = L_1^c \cap L_2^c$
$(L_1^c \cap L_2^c)^c = L_1^c \cap L_2^c$
$(L_1^c \cap L_2^c)^c = L_1 \cap L_2$
$(L_1^c \cap L_2^c)^c = L_1 \cup L_2$ **CORRECT (page 68)**

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
If L1 and L2 are regular languages, then these can be expressed by the corresponding FAs.
True (Page 68)
False

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
L= language of words containing even number of a's. Regular Expression is

(a+b)*aa(a+b)*
(b+ab*a)* (Page 68)
a+bb*aab*a
(a+b)*ab(a+b)*

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
The regular expression defining the language $L_1 \cap L_2$ can be obtained, converting and reducing the previous ---------- into a ---------- as after eliminating states.

GTG, TG
FA, GTG (Page 69)
FA, TG
TG, RE

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
The language that can be expressed by any regular expression is called a Non regular language.
True
False (Page 71)

تم اچھا کرو زمانہ تم کو برا سمجھے یہ اس سے بہتر ہے کہ تم برا کرو اور زمانہ تم کو اچھا سمجھے

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Question # 9 of 10 (Total Marks: 1) **Select correct option:**
The languages ---------- are the examples of non regular languages.

PALINDROME and PRIME (Page 71)
PALINDROME and EVEN-EVEN
EVEN-EVEN and PRIME
FACTORIAL and SQURE

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
Let L be any infinite regular language, defined over an alphabet Σ then there exist three strings x, y and z belonging to Σ* such that all the strings of the form xy^nz for n=1,2,3, … are the words in L. called.

Complement of L
Pumping Lemma (Page 72)
Kleene's theorem
None in given

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
Languages are proved to be regular or non regular using pumping lemma.
True (Page 74)
False

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
---------- is obviously infinite language.
EQUAL-EQUAL
EVEN-EVEN
PALINDROME (Page 75)
FACTORIAL

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
If, two strings x and y, defined over Σ, are run over an FA accepting the language L, then x and y are said to belong to the same class if they end in the same state, no matter that state is final or not.
True (Page 75)
False

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
Myhill Nerode theorem is consisting of the followings,

L partitions Σ* into distinct classes.
If L is regular then, L generates finite number of classes.
If L generates finite number of classes then L is regular.
All of above (Page 75)

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Question # 5 of 10 (Total Marks: 1) **Select correct option:**
The language Q is said to be quotient of two regular languages P and R, denoted by--- if PQ=R.
R=Q/P
Q=R/P (Page 78)
Q=P/R
P=R/Q

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
If two languages R and Q are given, then the prefixes of Q in R denoted by Pref(Q in R).
True (Page 78)
False

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
Let Q = {aa, abaaabb, bbaaaaa, bbbbbbbbbb} and R = {b, bbbb, bbbaaa, bbbaaaaa} Pref (Q in R) is equal to,

{b,bbba,bbbaaa} (Page 78)
{b,bba,bbaaa}
{ab,bba,bbbaa}
{b,bba,bbba}

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
If R is regular language and Q is any language (regular/ non regular), then Pref (Q in R) is ---------.

Non-regular
Equal
Regular (Page 79)
Infinite

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
_____ states are called the halt states.
ACCEPT and REJECT (Page 105)
ACCEPT and READ
ACCEPT AND START
ACCEPT AND WRITE

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
The part of an FA, where the input string is placed before it is run, is called _____

State
Transition
Input Tape (Page 105)
Output Tape

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Question # 1 of 10 (Total Marks: 1) **Select correct option:**
In new format of an FA (discussed in lecture 37), This state is like dead-end non final state

ACCEPT
REJECT (Page 105)
STATR
READ

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
Between the two consecutive joints on a path

One character can be pushed and one character can be popped
Any no. of characters can be pushed and one character can be popped (Page 122)
One character can be pushed and any no. of characters can be popped
Any no. of characters can be pushed and any no. of characters can be popped

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
The PDA is called non-deterministic PDA when there are more than one out going edges from…… state

START or READ
POP or REJECT
READ or POP (Page 111)
PUSH or POP

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
Identify the TRUE statement:
A PDA is non-deterministic, if there are more than one READ states in PDA
A PDA is never non-deterministic
Like TG, A PDA can also be non-deterministic (Page 111)
A PDA is non-deterministic, if there are more than one REJECT states in PDA

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
There is a problem in deciding whether a state of FA should be marked or not when the language Q is infinite.
True (Page 79)
False

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
If an effectively solvable problem has answered in yes or no, then this solution is called ---------

Decision procedure (Page 80)
Decision method
Decision problem
Decision making

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Question # 7 of 10 (Total Marks: 1) **Select correct option:**
The following problem(s) ---------- is/are called decidable problem(s).

The two regular expressions define the same language
The two FAs are equivalent
Both a and b (Page 80)
None of given

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
To examine whether a certain FA accepts any words, it is required to seek the paths from ------- state.

Final to initial
Final to final
Initial to final (Page 81)
Initial to initial

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
The high level language is converted into assembly language codes by a program called compiler.

TRUE (Page 87)
FALSE

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
Grammatical rules which involve the meaning of words are called ----------
Semantics (Page 87)
Syntactic
Both a and b
None of given

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
Grammatical rules which do not involve the meaning of words are called ----------
Semantics
Syntactic (Page 87)
Both a and b
None of given

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
The symbols that must be replaced by other things are called _____

Productions
Terminals
Non-terminals (Page 87)
None of given

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Question # 3 of 10 (Total Marks: 1) **Select correct option:**

The grammatical rules are often called_____

Productions (Page 87)

Terminals

Non-terminals

None of given

Question # 4 of 10 (Total Marks: 1) **Select correct option:**

The terminals are designated by _____ letters, while the non-terminals are designated by _____ letters.

Capital, bold

Small, capital (Page 87)

Capital, small

Small, bold

Question # 5 of 10 (Total Marks: 1) **Select correct option:**

The language generated by _____ is called Context Free Language (CFL).

FA

TG

CFG (Page 87)

TGT

Question # 6 of 10 (Total Marks: 1) **Select correct option:**

$\Sigma = \{a,b\}$ Productions $S \rightarrow XaaX$

$X \rightarrow aX$

$X \rightarrow bX$

$X \rightarrow \Lambda$

This grammar defines the language expressed by_____

(a+b)*aa(a+b)* (Page 89)

(a+b)*a(a+b)*a

(a+b)*aa(a+b)*aa

(a+b)*aba+b)*

Question # 7 of 10 (Total Marks: 1) **Select correct option:**

$S \rightarrow aXb|b$

$XaX \rightarrow aX|bX|\Lambda$

The given CFG generates the language in English _____

Beginning and ending in different letters (Page 91)

Beginning and ending in same letter

Having even-even language

None of given

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Question # 8 of 10 (Total Marks: 1) **Select correct option:**
The CFG is not said to be ambiguous if there exists atleast one word of its language that can be generated by the different production trees,

TRUE
FALSE (Page 95)

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
The language generated by that CFG is regular if _____

No terminal → semi word
No terminal → word
Both a and b (Page 97)
None of given

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
The production of the form no terminal → Λ is said to be null production.

TRUE (Page 99)
FALSE

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
CNF is stands for

Context Normal Form
Complete Normal Form
Chomsky Normal Form (Page 102)
Compared Null Form

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
Proof(Kleene's Theorem Part II)
If a TG has more than one start states, then

Introduce the new start state (Page 26)
Eliminate the old start state
Replace the old start state with final state
Replace the old final state with new start state

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
Which of the following regular expression represents same language?
a. (a+ab)*
b. (ba+a)*
c. a*(aa*b)*
d. (a*b*)* (a+b)*a(a+b)*b(a+b)*+ (a+b)*b(a+b)*a(a+b)*.

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{ x}*, { x}+, {a+b}*

Select correct option:
a and b (correct)
a and c
c and d

Question # 4 of 10 (Total Marks: 1) Select correct option:
Let FA3 be an FA corresponding to FA1+FA2, then the initial state of FA3 must correspond to the initial state of

FA1 only
FA2 only
FA1 or FA2 (Page 32)
FA1 and FA2

Question # 5 of 10 (Total Marks: 1) Select correct option:
Which of the following statement is NOT true about TG?

There exists exactly one path for certain string (Page 19)
There may exist more than one paths for certain string
There may exist no path for certain string
There may be no final state

Question # 6 of 10 (Total Marks: 1) Select correct option:
Kleene's theorem states

All representations of a regular language are equivalent.
All representations of a context free language are equivalent.
All representations of a recursive language are equivalent
Finite Automata are less powerful than Pushdown Automata. (Page 105)

Question # 7 of 10 (Total Marks: 1) Select correct option:
A language accepted by an FA is also accepted by

TG only
GTG only
RE only
All of the given (Page 25)

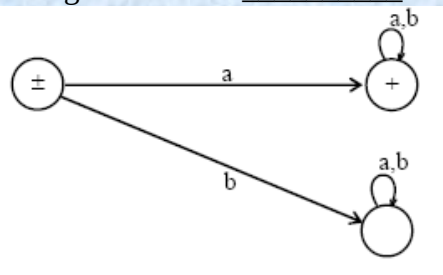
انسان کے لئے بری صحبت سے بڑھ کر بری کوئی چیز نہیں

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FINALTERM EXAMINATION
Fall 2012
CS402- Theory of Automata

Question No: 1 (Marks: 1) - Please choose one
If Σ = {aa, bb} , then Σ* will not contain
►**aaabbb**
►aabbbb
►aabbaa
►bbaabbbb

Question No: 2 (Marks: 1) - Please choose one
Below given FA has _____ RE.



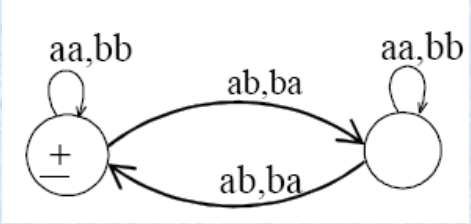
►**a(a+b)*** **(Page 14)**
► (a(a+b)*)*
►a(a+b)*a
►a(a+b)*a + b(a+b)*b

Question No: 3 (Marks: 1) - Please choose one
"One language can have _____ TG's".
►Only one
►Only two
►**More than one**
►Only three

دنیا میں سب سے مشکل کام اپنی اصلاح اور سب سے آسان کام دوسروں پر نکتہ چینی کرنا ہے

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Question No: 4 (Marks: 1) - Please choose one



Above given TG represents the language i.e.

►**EVEN-EVEN (Page 22)**
►PALINDROME
►FACTORIAL
►None of these

Question No: 5 (Marks: 1) - Please choose one

According to 1st part of the Kleene's theorem, If a language can be accepted by an FA then it can be accepted by a _____ as well.

►FA
►CFG
►GTG
►**TG (Page 25)**

Question No: 6 (Marks: 1) - Please choose one

Even-palindrome is a _____ language.

►**Non-regular** click here for detail
►Regular
►Regular but infinite
►Regular but finite

Question No: 7 (Marks: 1) - Please choose one

If L is a regular language then, L^c is also a _____ language.

► **Regular (Page 66)**
► Non-regular
► Regular but finite
► None of the given

Question No: 8 (Marks: 1) - Please choose one

Pumping lemma is generally used to prove that:

► A given language is infinite
► **A given language is not regular** Click here for detail
► Whether two given regular expressions of a regular language are equivalent or not
► None of these

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

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Question No: 9 (Marks: 1) - Please choose one
If the FA has N states, then test the words of length less than N. If no word is accepted by this FA, then it will _____ word/words.

►accept all
►**accept no (Page 85)**
►accept some
►reject no

Question No: 10 (Marks: 1) - Please choose one
In CFG, the symbols that can't be replaced by anything are called___.

► **Terminal (Page 87)**
► Non-Terminal
► Production
► All of given

Question No: 11 (Marks: 1) - Please choose one
Which of the following is a regular language?

► **String of odd number of zeroes** Click here for detail
► Set of all palindromes made up of 0's and 1's
► String of 0's whose length is a prime number
► All of these

Question No: 12 (Marks: 1) - Please choose one
Which of the following pairs of regular expressions are equivalent?

►1(001)* and (10)*10
►**x(xx)* and (x)*x**
► X^+ and X*
► X^+ and X* X^+

Question No: 13 (Marks: 1) - Please choose one
An alphabet of Σ is valid if

►No letter of Σ appears in middle of any other letter
►No letter of Σ appears at end of any other letter
►**No letter of Σ appears at start of any other letter (Page 4)**
►No letter of Σ appears at end or middle of any other letter



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Question No: 14 (Marks: 1) - Please choose one
Which of the following statement is true

►The length of the output string is greater than length of input string in moore machine. Click here for detail
►The length of the output string is greater than length of input string in mealy machine.
►The length of the output string is equal to length of input string in moore machine.
►The length of the output string is less than length of input string in mealy machine.

Question No: 15 (Marks: 1) - Please choose one
If a CFG has only productions of the form
nonterminal → string of two nonterminals
or
nonterminal → one terminal
then the CFG is said to be in _____

►**Chomsky Normal Form (Page 101)**
►Ambiguous Form
►Left Aligned Form
►Right Aligned Form

Question No: 16 (Marks: 1) - Please choose one
We can also represent an FA using different states e.g Accept state; Reject state, Read state etc.
The _____ state behaves as final state of an FA
►**Accept (Page 105)**
►Pop
►Push
►Reject

Question No: 17 (Marks: 1) - Please choose one
where the input string is placed before it is run, is called _____
►Date tape
►**Input Tape (Page 105)**
►Output Tape
►Magnetic tape

Question No: 18 (Marks: 1) - Please choose one
An FSM can be considered as TM

►Of finite tape length, rewinding capability and unidirectional tape movement
►Of finite tape length, without rewinding capability and bidirectional tape movement
►Of finite tape length, rewinding capability and bidirectional tape movement
►**Of finite tape length, without rewinding capability and unidirectional tape movement** click here for detail

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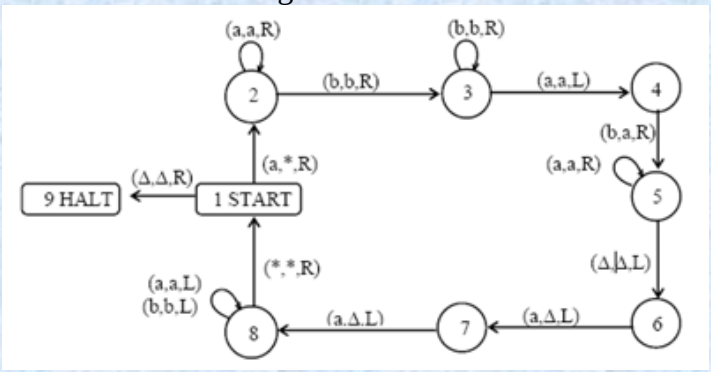
Question No: 19 (Marks: 1) - Please choose one
The process of finding the derivation of the word generated by particular grammar is called _____

►Processing
►**Parsing (Page 136)**
►Programming
►Planning

Question No: 20 (Marks: 1) - Please choose one
The first rule of converting the given "CFG in CNF", is _____

►CNK algorithm
►**CYK algorithm (Page 135) Algorithm 4 (The CYK algorithm)**
►CKY algorithm
►KYC algorithm

Question No: 21 (Marks: 1) - Please choose one
Consider the following TM



►Above TM accepts the non-CFL {a b c}
►**Above TM accepts the non-CFL** $\{a^n\ b^n\ a^n\}$ **(Page 142)**
►Above TM accepts the non-CFL $\{a^n\ b^{n+2}\ a^n\}$

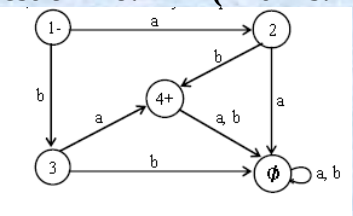
Question No: 22 (Marks: 1) - Please choose one
Alphabet Σ = {a, bc, cc} has number of letters

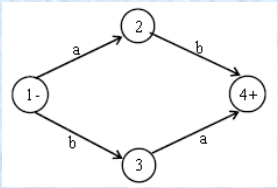
►One
►Two
►**Three**
►Four

دنیا کی سب سے بڑی فتح نفس پر قابو رکھنا ہے

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Question No: 23 (Marks: 1) - Please choose one
If r1 is a regular expression then r1* is a _____
►GTG
►NFA
►FA
►**RE (Page 9)**

Question No: 24 (Marks: 1) - Please choose one
(a)


(b)


► **(a) is FA, (b) is NFA (Page 43)**
► (a) is NFA, (b) is FA
► (a) is TG, (b) is FA
► (a) is TG, (b) is GTG

Question No: 25 (Marks: 1) - Please choose one
We cannot write regular expressions for all _____.

►FA's
►TG's
►NFA's
►**CFG's (Page 97)**

Question No: 26 (Marks: 1) - Please choose one
For every Context Free Grammar (CFG), we can make the corresponding _____.

►FA
►TG
►**PDA** click here for detail
►Regular Grammar

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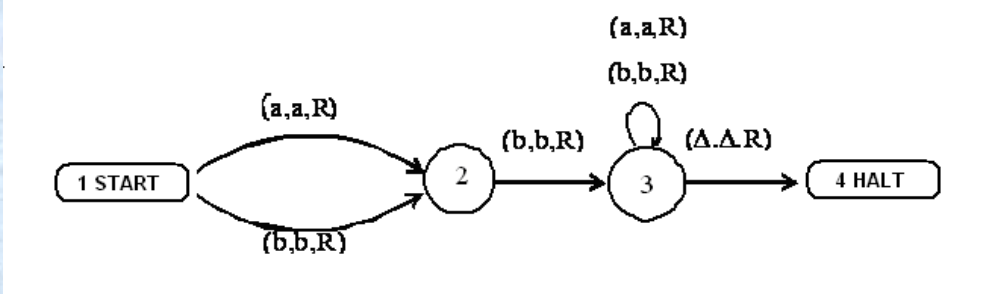
Question No: 27 (Marks: 1) - Please choose one
Pumping Lemma II says that length(x) + length(y) should be _____.

►**Less than number of states (Page 75)**
►Equal to number of states
►Greater than number of states
►Greater than or equal to number of states

Question No: 28 (Marks: 1) - Please choose one
Chomsky normal form (CYK) algorithm was proposed by _____.

►**John cock (Page 135)**
►James Cock
►Daniel I.A.
►John Weiss

Question No: 29 (Marks: 1) - Please choose one



The above machine is a/anTG _____

► Finite Automata
► **Turing machine (Page 141)**
► FA
► TG

Question No: 30 (Marks: 1) - Please choose one
The language of Palindromes defined over an alphabet set {a, b} can be recognized by _____.

►FA
►NFA
►TG
►**PDA (Page 91)**
Hint: - as it is non-regular so its CFG and PDA are possible.

ایماندار کو غصہ دیر سے آتا ہے اور جلدی دور ہو جاتا ہے

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Question No: 31 (Marks: 1) - Please choose one
Which of the following statement(s) is/are true or false?
(1)The Turing Machine is similar to a finite automation but with an unlimited and unrestricted memory.
(2) A Turing machine much more accurate model of a general purpose computer.

►**Statement 1 is true** Click here for detail
►**Statement 2 is true** Click here for detail
►Both statements (1 & 2) are false
►Statements 2 is false

Question No: 32 (Marks: 1) - Please choose one
Which of the following is the first phase of compiler on the basis of functionality?

►Parser
►Lexical analyzer
►**Scanner** Click here for detail
►Interpreter
Hint: - The first phase of a compiler is called lexical analysis (and is also known as a lexical scanner).

Question No: 33 (Marks: 1) - Please choose one
(Σ* - L) represent the _____ of a language L.

►**Complement (Page 66)**
►Kleene's closure
►Union
►intersection

Question No: 34 (Marks: 1) - Please choose one
If we have two transition graphs then their union will be expressed by
►**taking a common start state and joining them by two null transitions (Page 65)**
►just connecting both start states by null transitions
►connecting final state of first TG to the initial state of second TG
►connecting the final state of first TG to the final state of second TG

Question No: 35 (Marks: 1) - Please choose one
_____ and _____ are removed in order to make a CFG in Chomsky Normal Form(CNF).

►Null, nullable productions
►Nullable , unit productions
►**Null, unit productions (Page 102)**
►String of length 0, null

زندگی میں کامیابی کا یہی راز ہے کہ پریشانیوں سے پریشان مت بنو

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Question No: 36 (Marks: 1) - Please choose one
If L1 and L2 are expressed by regular languages then L1 + L2 is also a_____ Language.

► **Regular (Page 10)**
► Ir-regular
► PDA
► Hybrid

Question No: 37 (Marks: 1) - Please choose one
Which of the following is a regular Context Free Grammar:

►**S → abS| baS | ^** **ab(ab+ba)*ba + ba(ab+ba)*ab**
►S → aSb| baS | ^
►S → abS| bSa | ^
►S → aSb| Sa | ^
Hint :- remaining represents palindromes language which is non-regular

Question No: 38 (Marks: 1) - Please choose one
A read state can have _____ outgoing edge/ edges.

►1
►2
►3
►**Any number of (Page 111)**

Question No: 39 (Marks: 1) - Please choose one
Finite Automation (FA) and Nondeterministic Finite Automation (NFA) are equivalent if

►**FA and NFA accept the same language (Page 43)** Also click here for detail
►FA shape is same like an NFA
►FA accept the null string also
►NFA accept the null string also

Question No: 40 (Marks: 1) - Please choose one
_____ is always Deterministic.

►**Finite Automation (Page 25)**
►Transition Graph
►Generalize Transition Graph
►Non-deterministic finite automation

جھوٹ انسان اور ایمان دونوں کا دشمن ہے

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FINALTERM EXAMINATION
Spring 2010
CS402- Theory of Automata (Session - 1)

Question No: 1 (Marks: 1) - Please choose one
If r1 = (aa + bb) and r2 = (a + b) then the language (aa + bb)(a + b) will be generated by

- **(r1)(r2) (Page 10)**
- (r1 + r2)
- (r2)(r1)
- (r1)*

Question No: 2 (Marks: 1) - Please choose one
"One language can be expressed by more than one FA". This statement is _____

- **True (Page 14)**
- False
- Some times true & sometimes false
- None of these

Question No: 3 (Marks: 1) - Please choose one
Who did not invent the Turing machine?

- Alan Turing
- **A. M. Turing (Page 140)**
- Turing
- None of these

Question No: 4 (Marks: 1) - Please choose one
Which statement is true?

- **The tape of turing machine is infinite. (Page 140)**
- The tape of turing machine is finite.
- The tape of turing machine is infinite when the language is regular
- The tape of turing machine is finite when the language is nonregular.

Question No: 5 (Marks: 1) - Please choose one
A regular language:

- **Must be finite (Page 11)**
- Must be infinite
- Can be finite or infinite
- Must be finite and cannot be infinite

عقل مند کہتا ہے میں کچھ نہیں جانتا جبکہ بے وقوف کہتا ہے کہ میں سب کچھ جانتا ہوں

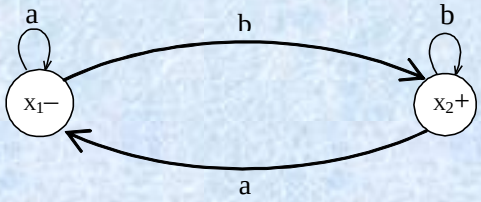
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Question No: 6 (Marks: 1) - Please choose one

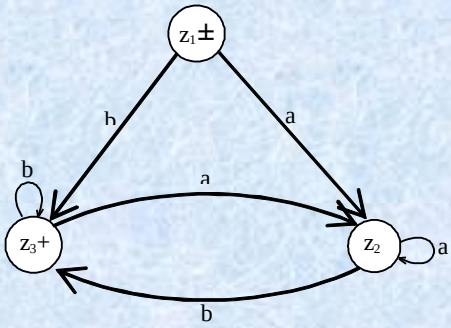
Every regular expression can be expressed as CFG but every CFG cannot be expressed as a regular expression. This statement is:

- ► Depends on the language
- ► None of the given options
- ► **True (Page 97)**
- ► False

Question No: 7 (Marks: 1) - Please choose one



Above given FA corresponds RE r. then FA corresponding to r* will be



This statement is

- ► **True (Page 38)**
- ► False
- ► Depends on language
- ► None of these

Question No: 8 (Marks: 1) - Please choose one

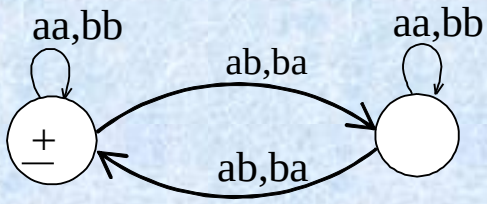
Consider the language L of strings, defined over Σ = {a,b}, ending in a

- ► **There are finite many classes generated by L, so L is regular (Page 76)**
- ► There are infinite many classes generated by L, so L is regular
- ► There are finite many classes generated by L, so L is non-regular
- ► There are infinite many classes generated by L, so L is non-regular

خود کو تمھیں سے بڑھ کر کوئی اچھا مشورہ نہیں دے سکتا

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Question No: 9 (Marks: 1) - Please choose one



Above given TG has _____ RE.

- ► (aa+aa+(ab+ab)(aa+ab)*(ab+ba))*
- ► **(aa+bb+(ab+ba)(aa+bb)*(ab+ba))*** **(Page 22)**
- ► (aa+bb+(ab+ba)(aa+bb)(ab+ba))*
- ► None of these

Question No: 10 (Marks: 1) - Please choose one
The word 'formal' in formal languages means

- ► The symbols used have well defined meaning
- ► They are unnecessary, in reality
- ► **Only the form of the string of symbols is significant** Click here for detail
- ► None of these

Question No: 11 (Marks: 1) - Please choose one
Let A = {0, 1}. The number of possible strings of length 'n' that can be formed by the elements of the set A is

- ► n!
- ► n^2
- ► n^m
- ► **2^n**

Question No: 12 (Marks: 1) - Please choose one
Choose the correct statement.

- ► A Mealy machine generates no language as such
- ► A Moore machine generates no language as such
- ► A Mealy machine has no terminal state
- ► **All of these** click here for detail

Question No: 13 (Marks: 1) - Please choose one
TM is more powerful than FSM because

- ► The tape movement is confined to one direction
- ► It has no finite state control
- ► **It has the capability to remember arbitrary long sequences of input symbols** Click here for detail
- ► None of these

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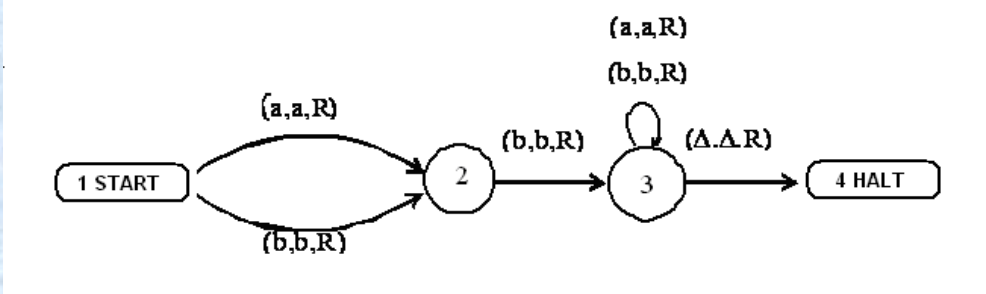
Question No: 14 (Marks: 1) - Please choose one
If L1 and L2 are expressed by regular expressions r1 and r2, respectively then the language expressed by r1 + r2 will be _____

- **Regular (Page 10)**
- Ir-regular
- Can't be decided
- Another Language which is not listed here

Question No: 15 (Marks: 1) - Please choose one
Like TG, a PDA can also be non-deterministic

- **True (Page 111)**
- False

Question No: 16 (Marks: 1) - Please choose one



The above machine is a/an _____

- Finite Automata
- **Turing machine (Page 141) rep**
- FA
- TG

Question No: 17 (Marks: 1) - Please choose one
The language of all words (made up of a's and b's) with at least two a's can not be described by the regular expression.

- a(a+b)*a(a+b)*(a+b)*ab*
- (a+b)* ab* a(a+b)*
- b*ab* a(a+b)*
- **none of these**

a^nb^n {where n>0} is the language will at least one a and b and cannot be described by RE.

Question No: 18 (Marks: 1) - Please choose one
In FA, if one enters in a specific state but there is no way to leave it, then that specific state is called

- Dead State
- Waste Basket
- Davey John Locker
- **All of these (Page 17)**

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Question No: 19 (Marks: 1) - Please choose one
If L is a regular language then, L^c is also a _____ language.

► **Regular (Page 66) rep**
► Non-regular
► Regular but finite
► None of the given

Question No: 20 (Marks: 1) - Please choose one
In CFG, the symbols that can't be replaced by anything are called___

► **Terminal (Page 87) rep**
► Non-Terminal
► Production
► All of given

Question No: 21 (Marks: 1) - Please choose one
Which of the following is NOT a regular language?

► String of 0's whose length is a perfect squere
► Set of all palindromes made up of 0's and 1's
► String of 0's whose length is a prime number
► **All of the given options** Click here for detail

Question No: 22 (Marks: 1) - Please choose one
Choose the incorrect (FALSE) statement.

► A Mealy machine generates no language as such
► A Mealy machine has no terminal state
► **For a given input string, length of the output string generated by a Moore machine is not more than the length of the output string generated by that of a Mealy machine** click here for detail
► All of these

Question No: 23 (Marks: 1) - Please choose one
Pumping lemma is generally used to prove that:

► A given language is infinite
► **A given language is not regular** Click here for detail **rep**
► Whether two given regular expressions of a regular language are equivalent or not
► None of these

Question No: 24 (Marks: 1) - Please choose one
Which of the following is a regular language?

► **String of odd number of zeroes** Click here for detail **rep**
► Set of all palindromes made up of 0's and 1's
► String of 0's whose length is a prime number
► All of these

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Question No: 25 (Marks: 1) - Please choose one
Choose the incorrect statement:
► (a+b)*aa(a+b)* generates Regular language.
► **A language consisting of all strings over ∑={a,b} having equal number of a's and b's is a regular language**
► Every language that can be expressed by FA can also be expressed by RE
► None of these

Question No: 26 (Marks: 1) - Please choose one
Left hand side of a production in CFG consists of:
► One terminal
► More than one terminal
► **One non-terminal (Page 87)**
► Terminals and non-terminals

FINALTERM EXAMINATION
SPRING 2007

Question No: 1 (Marks: 1) - Please choose one
PDA is only used to represent a regular language.
► True
► **False** Click here for detail

Question No: 2 (Marks: 1) - Please choose one
If L is a regular language then LC is also a regular language.
► **True (Page 66) rep**
► False

Question No: 3 (Marks: 1) - Please choose one
A production of the form non-terminal ⇾ string of two non-terminal is called a live Production.
► **True (Page 127)**
► False

Question No: 4 (Marks: 1) - Please choose one
We can find a CFG corresponding to a DFA.
► **True (Page 97)**
► False

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Question No: 5 (Marks: 1) - Please choose one
START, READ, HERE and ACCEPTS are conversions of the machine
► **True (Page 122)**
► False

Question No: 6 (Marks: 1) - Please choose one
A CFG is said to be ambiguous if there exists at least one word of its language that can be generated by different production trees
► **True (Page 95)**
► False

Question No: 7 (Marks: 1) - Please choose one
Syntax tree or Generation tree or Derivation tree are same tree
► **True (Page 92)**
► False

Question No: 8 (Marks: 1) - Please choose one
The symbols that cannot be replaced by anything are called terminals
► **True (Page 87) rep**
► False

Question No: 9 (Marks: 1) - Please choose one
The production of the form non-terminal ⇾ one non-terminal is called unit production
► **True (Page 100)**
► False

Question No: 10 (Marks: 1) - Please choose one
DFA and PDA are equal in power.
► True
► **False (Page 105)**

FINALTERM EXAMINATION
Spring 2006
CS402- Theory of Automata

Question No. 1
A production of the form non-terminal ⇾ non-terminal is called a dead Production.
True
False (Page 127)

جو شخص ناکامیوں سے ڈر کر بھاگتا ہے کامیابی اُس سے ڈر کر بھاگتی ہے

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Question No. 2
Semi-word is a string having some terminals and one non-terminal at the right of string.
True (Page 97)
False

Question No. 3
Two FAs are equivalent if they have same no. of states.
True (Page 15)
False

Question No. 4
There exist exactly two different derivations in an ambiguous CFG for a word.
True (Page 93)
False

Question No. 6
Regular languages are closed under Union, Concatenation and Kleene star.
True (Page 10)
False

Question No. 7
CFG may also represent a regular language.
True (Page 97)
False

Question No. 9 Marks : 1
PDA is stronger than FA.
True (Page 105)
False

FINALTERM EXAMINATION
Spring 2005
CS402- Theory of Automata

Question No. 1
A Total Language Tree has
All languages over Σ
All strings over Σ (Page 96)
All words of all languages over Σ
All words of one language over Σ

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Question No. 2
What Turing Machine does not have?
Stack
Tape
Head
Word
Turing machine has stack but insertion and deletion can be done from both sides. Tape and head to.

Question No. 3
CFG given S ⇗ bS|Sb|aa represents language
b*aa
aab*
b*aab*
b*(aa)*b*

Question No. 4
A Language that is finite but not regular
Λ
(a+b)*
Φ (not sure)
All strings of a's in Σ = {a,.b}

جو لوگوں کے سامنے فخر کرتا ہے وہ لوگوں کی نظروں سے گر جاتا ہے

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CS402 – Quiz No.3

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
The values of input (say a & b) does not remain same in one cycle due to
NAND gate
Click plus
OR gate
NOT gate

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
Set of all palindromes over {a,b}is regular

True
False (Page 74)

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
In CFG, the symbols that cannot be replaced by anything are called

Terminals (Page 87) rep
Non terminals
Productions
None of the given options

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
a^n b^n generates the …… language

regular
non regular
EQUAL and non regular (Page 71)
EQUAL and regular

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
The grammatical rules which involves meaning of words are called:

Semantic (Page 87)
Sytactics
Alphabets
None of the given options

عقل مند اپنے عیب خود دیکھتا ہے اور بیوقوفوں کے عیب دنیا دیکھتی ہے

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Question # 6 of 10 (Total Marks: 1) **Select correct option:**
The reverse of the string sbfsbb over { sb, f, b }

bbsfbs
bsbfsb
sbbfsb
bsfbsb

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
If an FA has N state then it must accept the word of length

N-1
N+1
N+2N

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
Two languages are said to belong to same class if they end in the same state when they run over an FA, that state

Must be final state
May be final state or not (Page 75)
May be start or not
None of the given options

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
In pref(Q in R) Q is …… to (than) R

Equal
Not Equal (Page 79)
Greater
Smaller

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
According to Myhill Nerode theorem, if L generates finite no. of classes then L is.....

Finite
Infinite
Regular (Page 76)
Non Regular

عقل مند آدمی اس وقت تک نہیں بولتا جب تک خاموشی نہیں ہو جاتی

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Question # 1 of 10 (Total Marks: 1) **Select correct option:**
If the intersection of two regular languages is regular then the complement of the intersection of these two languages is also regular

True (Page 68)
False

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
In pumping lemma theorem (x y^n z) the range of n is

n=1,2,3,4..... (Page 74)
n=0,1,2,3,4....
n=-3,-2,-1,0,1,2,3,4.....
n=-3,-2,-1,1,2,3,4.....

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
The complement of a regular language is also a regular

True rep
False

CS402 – Quiz No.3

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
For a non regular language there exist …… FA

One
At least one
At most one
No (Page 71)

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
The strings or words which do not belong to a language is called……. of that language

Intersection
Union
Complement (Page 66)
Quotient

انسان دکھ نہیں دیتے بلکہ انسانوں سے وابستہ امیدیں دکھ دیتی ہیں

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Question # 3 of 10 (Total Marks: 1) **Select correct option:**
A non regular language can be represented by
RE
FA
TG
None of the given options (Page 71)

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
For language L defined over {a, b},then L partitions {a, b}* into …… classes

Infinite
Finite
Distinct (Page 75)
Non distinct

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
If an FA accept a word then there must exist a path from

Initial to final state (Page 81)
Initial to each state
Initial to each state but not to final state
Initial to final state by traversing each state

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
Does the empty string match the regular expression |y+a|?

Yes
No (Page 3)

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
If an FA already accepts the language expressed by the closure of certain RE, then the given FA is the required FA.

True (Page 37)
False

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
Which of the following statement is true about NFA with Null String?

Infinite states
Infinite set of letters
Infinite set of transitions
Transition of null string is allowed at any stage (Page 71)

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Question # 9 of 10 (Total Marks: 1) **Select correct option:**
If R is a regular language and L is some language, and L U R is a regular language, then L must be a regular language.

True (page 10)
False

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
FA corresponding to an NFA can be built by introducing an empty state for a letter having

no transition at certain state (Page 43)
one transition at certain state
two transition at certain state
more than two transitions at certain state

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
Let FA3 be an FA corresponding to FA1FA2, then the initial state of FA3 must correspond to the initial state of

FA1 only (Page 35)
FA2 only
FA1 or FA2
FA1 and FA2

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
(a* + b*)* = (a + b)* this expression is _____
True
False

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
If S = { x }, then S* will be
{x,xx,xxx,xxxx,…}
{^ ,x,xx,xxx,xxxx,…} (Page 10)

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
The states in which there is no way to leave after entry are called
Davey John Lockers
Dead States
Waste Baskets
All of the given options (Page 17)

خوبصورتی علم وادب سے ہوتی ہے لباس و حسن سے نہیں

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Question # 5 of 10 (Total Marks: 1) **Select correct option:**
If S = {ab, bb}, then S* will not contain
Abbbab
Bbba
ababbb
bbbbab

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
According to theory of automata there are _____ types of languages

1
2 (Page 3)
3
4

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
What do automata mean?
Something done manually
Something done automatically (Page 3)

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
What is false about the term alphabet?
It is a finite set of symbols.
It is usually denoted by Greek letter sigma
It can be an empty set. (Page 3)
Strings are made up of its elements

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
Formal is also known as _____
Syntactic language (page 3)
Semantic language
Informal language
Nsone of these

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
Following are types of languages

Formal Languages (Syntactic languages)
Informal Languages (Semantic languages)
Both (Page 3)
None of above

بہترین تجربہ وہ ہے جس سے نصیحت حاصل ہو

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CS402 – Quiz No.4

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
Consider the following production (of a CFG): S->XYZ Here _____ is left most nonterminal in working string. Note: S, X, Y and Z are all nonterminals
S
X
Y
Z

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
A PDA is called nondeterministic PDA if _____

There are more than one outgoing edges at READ or POP states with one label (Page 111)
There are more than one PUSH states
There are mroe than one POP states
All of the given options

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
A PDA consists of the following:

An alphabet (Sigma) of input letters.
An input TAPE with infinite many locations in one direction
One START state with only one out-edge and no in-edge
All of the given options (Page 105)

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
The CFG S --> aSa | bSb | a | b | ^ represents the language

EVEN-EVEN
PALINDROM (Page 91)
EQUAL
ODD-ODD

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
Halt states are

Start and Accept
Accept and Reject (Page 105)
Start and Reject
Read and Reject

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Question # 6 of 10 (Total Marks: 1) **Select correct option:**
Choice of path can be determined by left most derivation of the string belonging to CFL at…… state

Accept (Page 104)
Reject
Push
POP

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
The unit and null productions can be deleted from a CFG

True (Page 99-100)
False

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
Identify the TRUE statement about following CFG:
S -> SB|AB
A -> CC
B -> b
C -> a

The given CFG has 8 Nonterminals
The given CFG has 8 Terminals
The given CFG is in CNF (Page 101)
The given CFG is not in CNF

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
The structure given below is called _____ S -> aA|bB A -> aS|a B -> bS|b

RE
TG
CFG (Page 87)
PDA

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
Which of the following states is not part of PDA

START
ACCEPT
WRITE (Page 107)
REJECT

تم اچھا کرو زمانہ تم کو برا سمجھے یہ اس سے بہتر ہے کہ تم برا کرو اور زمانہ تم کو اچھا سمجھے

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CS402 – Quiz No.4

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
The production of the form: nonterminal --> one nonterminal is called the _____

Unit production (Page 100)
NULL production
Terminal production
Non Terminal production

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
A _____ is the one for which every input string has a unique path through the machine.

Deterministic PDA (Page 111)
nondeterministic PDA
PUSHDOWN store
Input Tape

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
In the null production N --> ^ , N is a

Terminal
Non terminal (Page 99)
Word
None of the given options

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
The major problem in the earliest computers was

To store the contents in the registers
To display mathematical formulae (Page 87)
To load the contents from the registers
To calculate the mathematical formula

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
In polish notation, (o-o-o) is the abbreviation of……?

Operand - Operator – Operand
Operand - Operand- Operator
Operator -Operand – Operand (Page 94)
Operand -Operand – Operand

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Question # 6 of 10 (Total Marks: 1) **Select correct option:**
The CFG is said to be ambiguous if there exist at least one word of its language that can be generated by the …… production trees

One
Two
More than one (Page 95)
At most one

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
The input string is placed, before it runs, in

Stack
Memory
Tape (Page 105)
Ram

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
The production S --> SS | a | b | ^ can be expressed by RE

(a+b)+
(a+b)
(a+b)* (Page 88)
(ab)*

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
The locations into which we put the input letters on "Input Tap" are called _____

words
alphabets
cells (Page 105)
elements

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
"CFG" stands for _____

Context Free Graph
Context Free Grammer (Page 87)
Context Finite Graph
Context Finite Grammer

بدصورت چہرہ بدصورت دماغ سے بہتر ہے

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Question # 1 of 10 (Total Marks: 1) **Select correct option:**
In a CFG the nonterminal that occurs first from the left in the working string, is said to be _____

Least Significant nonterminal
Most Significant nonterminal
Left most nonterminal (Page 103)
Left most derivate

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
The unit production is

Terminal --> Terminal
Terminal --> Non Terminal
Non terminal --> Terminal
Non terminal --> Non Terminal (Page 100)

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
A _____ operator adds a new letter at the top of STACK

PUSH (Page 107)
POP
READ
APPEND

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
PDA stands for _____

Push and Drop Automaton
Pop and Drop Automaton
Push Down Automaton (Page 112)
None of given options

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
The production of the form: Nonterminal-> ^ is said to be _____ production

NULL (Page 99)
UNIT
Chomsky form production
None of the given options

عقل مند آدمی اس وقت تک نہیں بولتا جب تک خاموشی نہیں ہو جاتی

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Question # 6 of 10 (Total Marks: 1) **Select correct option:**
If a CFG has a null production, then it is _____

Posiible to construct another CFG without null production accepting the same language with the exception of the word ^ (Page 99)
Not possible to construct another CFG without null production accepting the same language with the exception of the word ^
Called NULL CFG
Called Chmosky Normal Form (CNF)

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
In a STACK:

The element PUSHed first is POPed first
The element PUSHed first is POPed in the last (Page 107 concept)
The element PUSHed in last is POPed in last
None of given options

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
Kleene star closure can be defined
Over any set of string (Page 7)
Over specific type of string

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
While finding RE corresponding to TG, we connect the new start state to the old start state by the transition labeled by
A
B
null string (Page 26)
None of the given options

انسان کے لئے بری صحبت سے بڑھ کر بری کوئی چیز نہیں

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Some More Quizzes

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
For a given input, it provides the compliment of Boolean AND output.

NAND box (NOT AND) **(Page 63)**
DELAY box
OR box
AND box

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
It delays the transmission of signal along the wire by one step (clock pulse).

NAND box (NOT AND)
DELAY box **(Page 63)**
OR box
AND box

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
For the given input, it provides the Boolean OR output

NAND box (NOT AND)
DELAY box
OR box **(Page 63)**
AND box

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
For the given input, AND box provides the Boolean AND output.
True **(Page 63)**
False

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
The current in the wire is indicated by 1 and 0 indicates the absence of the current.
True **(Page 63)**
False

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
Any language that can not be expressed by a RE is said to be regular language.
True
False **(Page 71)**

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Question # 7 of 10 (Total Marks: 1) **Select correct option:**
If L1 and L2 are regular languages_____ is/are also regular language(s).

L1 + L2
L1L2
L1*
All of above (Page 10)

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
Let L be a language defined over an alphabet Σ, then the language of strings, defined over Σ, not belonging to L, is called Complement of the language L, denoted by Lc or L'.
True (Page 66)
False

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
To describe the complement of a language, it is very important to describe the ---------- of that language over which the language is defined.

Alphabet (Page 66)
Regular Expression
String
Word

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
For a certain language L, the complement of Lc is the given language L *i.e.* (Lc)c = Lc
True
False (Page 66)

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
If L is a regular language then, --------- is also a regular language.
Lm
Ls
Lx
Lc (Page 66)

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
Converting each of the final states of F to non-final states and old non-final states of F to final states, FA thus obtained will reject every string belonging to L and will accept every string, defined over Σ, not belonging to L. is called
Transition Graph of L
Regular expression of L
Complement of L (Page 66)
Finite Automata of L

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Question # 3 of 10 (Total Marks: 1) **Select correct option:**
If L1 and L2 are two regular languages, then L1 U L2 is not a regular.
True
False (Page 65)

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
De-Morgan's law for sets is expressed by,
$(L_1^c \cap L_2^c)^c = L_1^c \cap L_2^c$
$(L_1^c \cap L_2^c)^c = L_1^c \cap L_2^c$
$(L_1^c \cap L_2^c)^c = L_1 \cap L_2$
$(L_1^c \cap L_2^c)^c = L_1 \cup L_2$ **CORRECT (page 68)**

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
If L1 and L2 are regular languages, then these can be expressed by the corresponding FAs.
True (Page 68)
False

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
L= language of words containing even number of a's. Regular Expression is

(a+b)*aa(a+b)*
(b+ab*a)* (Page 68)
a+bb*aab*a
(a+b)*ab(a+b)*

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
The regular expression defining the language $L_1 \cap L_2$ can be obtained, converting and reducing the previous ---------- into a ---------- as after eliminating states.

GTG, TG
FA, GTG (Page 69)
FA, TG
TG, RE

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
The language that can be expressed by any regular expression is called a Non regular language.
True
False (Page 71)

خاموشی غصے کا بہترین علاج ہے

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Question # 9 of 10 (Total Marks: 1) **Select correct option:**
The languages ---------- are the examples of non regular languages.

PALINDROME and PRIME (Page 71)
PALINDROME and EVEN-EVEN
EVEN-EVEN and PRIME
FACTORIAL and SQURE

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
Let L be any infinite regular language, defined over an alphabet Σ then there exist three strings x, y and z belonging to Σ* such that all the strings of the form xy^nz for n=1,2,3, … are the words in L. called.

Complement of L
Pumping Lemma (Page 72)
Kleene's theorem
None in given

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
Languages are proved to be regular or non regular using pumping lemma.
True (Page 74)
False

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
---------- is obviously infinite language.
EQUAL-EQUAL
EVEN-EVEN
PALINDROME (Page 75)
FACTORIAL

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
If, two strings x and y, defined over Σ, are run over an FA accepting the language L, then x and y are said to belong to the same class if they end in the same state, no matter that state is final or not.
True (Page 75)
False

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
Myhill Nerode theorem is consisting of the followings,

L partitions Σ* into distinct classes.
If L is regular then, L generates finite number of classes.
If L generates finite number of classes then L is regular.
All of above (Page 75)

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Question # 5 of 10 (Total Marks: 1) **Select correct option:**
The language Q is said to be quotient of two regular languages P and R, denoted by--- if PQ=R.
R=Q/P
Q=R/P (Page 78)
Q=P/R
P=R/Q

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
If two languages R and Q are given, then the prefixes of Q in R denoted by Pref(Q in R).
True (Page 78)
False

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
Let Q = {aa, abaaabb, bbaaaaa, bbbbbbbbbb} and R = {b, bbbb, bbbaaa, bbbaaaaa} Pref (Q in R) is equal to,

{b,bbba,bbbaaa} (Page 78)
{b,bba,bbaaa}
{ab,bba,bbbaa}
{b,bba,bbba}

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
If R is regular language and Q is any language (regular/ non regular), then Pref (Q in R) is ---------.

Non-regular
Equal
Regular (Page 79)
Infinite

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
_____ states are called the halt states.
ACCEPT and REJECT (Page 105)
ACCEPT and READ
ACCEPT AND START
ACCEPT AND WRITE

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
The part of an FA, where the input string is placed before it is run, is called _____

State
Transition
Input Tape (Page 105)
Output Tape

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Question # 1 of 10 (Total Marks: 1) **Select correct option:**
In new format of an FA (discussed in lecture 37), This state is like dead-end non final state

ACCEPT
REJECT (Page 105)
STATR
READ

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
Between the two consecutive joints on a path

One character can be pushed and one character can be popped
Any no. of characters can be pushed and one character can be popped (Page 122)
One character can be pushed and any no. of characters can be popped
Any no. of characters can be pushed and any no. of characters can be popped

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
The PDA is called non-deterministic PDA when there are more than one out going edges from…… state

START or READ
POP or REJECT
READ or POP (Page 111)
PUSH or POP

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
Identify the TRUE statement:
A PDA is non-deterministic, if there are more than one READ states in PDA
A PDA is never non-deterministic
Like TG, A PDA can also be non-deterministic (Page 111)
A PDA is non-deterministic, if there are more than one REJECT states in PDA

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
There is a problem in deciding whether a state of FA should be marked or not when the language Q is infinite.
True (Page 79)
False

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
If an effectively solvable problem has answered in yes or no, then this solution is called ---------

Decision procedure (Page 80)
Decision method
Decision problem
Decision making

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Question # 7 of 10 (Total Marks: 1) **Select correct option:**
The following problem(s) ---------- is/are called decidable problem(s).

The two regular expressions define the same language
The two FAs are equivalent
Both a and b (Page 80)
None of given

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
To examine whether a certain FA accepts any words, it is required to seek the paths from ------- state.

Final to initial
Final to final
Initial to final (Page 81)
Initial to initial

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
The high level language is converted into assembly language codes by a program called compiler.

TRUE (Page 87)
FALSE

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
Grammatical rules which involve the meaning of words are called ----------
Semantics (Page 87)
Syntactic
Both a and b
None of given

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
Grammatical rules which do not involve the meaning of words are called ----------
Semantics
Syntactic (Page 87)
Both a and b
None of given

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
The symbols that must be replaced by other things are called _____

Productions
Terminals
Non-terminals (Page 87)
None of given

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Question # 3 of 10 (Total Marks: 1) **Select correct option:**
The grammatical rules are often called_____

Productions (Page 87)
Terminals
Non-terminals
None of given

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
The terminals are designated by _____ letters, while the non-terminals are designated by _____ letters.
Capital, bold
Small, capital (Page 87)
Capital, small
Small, bold

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
The language generated by _____ is called Context Free Language (CFL).
FA
TG
CFG (Page 87)
TGT

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
$\Sigma = \{a,b\}$ Productions $S \rightarrow XaaX$
$X \rightarrow aX$
$X \rightarrow bX$
$X \rightarrow \Lambda$
This grammar defines the language expressed by_____

(a+b)*aa(a+b)* (Page 89)
(a+b)*a(a+b)*a
(a+b)*aa(a+b)*aa
(a+b)*aba+b)*

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
$S \rightarrow aXb|bXa$
$X \rightarrow aX|bX|\Lambda$
The given CFG generates the language in English _____

Beginning and ending in different letters (Page 91)
Beginning and ending in same letter
Having even-even language
None of given

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Question # 8 of 10 (Total Marks: 1) **Select correct option:**
The CFG is not said to be ambiguous if there exists atleast one word of its language that can be generated by the different production trees,

TRUE
FALSE (Page 95)

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
The language generated by that CFG is regular if _____

No terminal → semi word
No terminal → word
Both a and b (Page 97)
None of given

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
The production of the form no terminal → Λ is said to be null production.

TRUE (Page 99)
FALSE

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
CNF is stands for

Context Normal Form
Complete Normal Form
Chomsky Normal Form (Page 102)
Compared Null Form

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
Proof(Kleene's Theorem Part II)
If a TG has more than one start states, then

Introduce the new start state (Page 26)
Eliminate the old start state
Replace the old start state with final state
Replace the old final state with new start state

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
Which of the following regular expression represents same language?
a. (a+ab)*
b. (ba+a)*
c. a*(aa*b)*
d. (a*b*)* (a+b)*a(a+b)*b(a+b)*+ (a+b)*b(a+b)*a(a+b)*.

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{ x}*, { x}+, {a+b}*

Select correct option:
a and b (correct)
a and c
c and d

Question # 4 of 10 (Total Marks: 1) Select correct option:
Let FA3 be an FA corresponding to FA1+FA2, then the initial state of FA3 must correspond to the initial state of

FA1 only
FA2 only
FA1 or FA2 (Page 32)
FA1 and FA2

Question # 5 of 10 (Total Marks: 1) Select correct option:
Which of the following statement is NOT true about TG?

There exists exactly one path for certain string (Page 19)
There may exist more than one paths for certain string
There may exist no path for certain string
There may be no final state

Question # 6 of 10 (Total Marks: 1) Select correct option:
Kleene's theorem states

All representations of a regular language are equivalent.
All representations of a context free language are equivalent.
All representations of a recursive language are equivalent
Finite Automata are less powerful than Pushdown Automata. (Page 105)

Question # 7 of 10 (Total Marks: 1) Select correct option:
A language accepted by an FA is also accepted by

TG only
GTG only
RE only
All of the given (Page 25)



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Quiz No. 4

Question # 1of 10 (Total Marks: 1) **Select correct option:**
Consider the Following CFG: (NOTE: ^ means NULL) S->Xa X->aX|bX|^
above given CFG can be represented by RE _____

a*b*
a*b*a
(a+b)*a
a(a+b)*a

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
Identify FALSE statement:

Every Regular Expression be expressed by CFG and every CFG can be expressed by a Regular Expression (Page 97)
Every regular expression can be expressed as CFG but every CFG cannot be expressed as a regular expression.
For a PDA, there exists a CFG, that represents the same language as represented by PDA.
None of the given options

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
Null production is a

Word
String
Terminal
All of the given options

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
In nondeterministic PDA a string is supposed to be accepted, if there exists at least one path traced by the string, leading to _____ state.

ACCEPT (Page 111)
REJECT
START
READ

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Question # 5 of 10 (Total Marks: 1) **Select correct option:**
The CFG which generates the regular language is called

Regular expression
Finite Automata
Regular grammar (Page 97)
None of the given options

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
If a CFG has a null production, then it is possible to construct another CFG accepting the same language without null production

TRUE (Page 99)
FALSE

CS402 – Quiz No.2 (15 Jun 2013)

Question # 1 of 10 (Total Marks: 1) Select correct option
In large FA with thousands of states and millions of directed edges, without an effective procedure it is _____ to find a path from initial to final state.
Always easy
Impossible (Page 81)
may be good
always impossible

Question # 2 of 10 (Total Marks: 1) Select correct option
If there is no final state of two FAs then their _____ also have no _____ state
initial, union
final, union
union,final (Page 83)
union, initial

Question # 3 of 10 (Total Marks: 1) Select correct option
Set of all palindromes over {a,b} is:
Regular
Regular and finite
Regular and infinite
Non-regular (Page 71)

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Question # 4 of 10 (Total Marks: 1) Select correct option
In the context of Myhill Nerode theorem, for even-even language sigma star can be partitioned into _____ number of classes.

3
4 (Page 77)
5
6

Question # 5 of 10 (Total Marks: 1) Select correct option
The product of two regular languages is _____.

Regular (Page 78)
infinite
non-regular
closure of a regular language

Question # 6 of 10 (Total Marks: 1) Select correct option
If the FA has N states, then test the words of length less than N. If no word is accepted by this FA, then it will _____ word/words.

accept all
accept no (Page 85) rep
accept some
reject no

Question # 7 of 10 (Total Marks: 1) Select correct option
An FA has same initial and _____ state, then it means that it has no _____ state.

initial, final
final, initial
initial, initial
none of the given options

Question # 8 of 10 (Total Marks: 1) Select correct option
A problem that has decision procedure is called _____ problem.

Regular language
un-decidable
Infinite
Decidable (Page 80)

اطمینان قلب چاہتے ہو تو حسد سے دور رہو

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Question # 9 of 10 (Total Marks: 1) Select correct option
For a machine with N number of states, the total number of strings to be tested, defined over an alphabet of m letters, is_____.
Select correct option:

Nm +Nm+1+ N m+2 +… + N2m-1
m^N + m^N+1 + m^N + 2 +… +m2N-1 (Page 86)
Nm
mN

Question # 10 of 10 (Total Marks: 1) Select correct option
If (L1 ∩ L2c) ∪ (L1c ∩ L2) is regular language that accepts the words which are in L1 but not in L2 or else in L2 but not in L1 . The corresponding FA cannot accept any word which is in _____ L1 and L2.
Not both
Both (Page 80)
At least in one
None of the given options

Question # 1 of 10 (Total Marks: 1) Select correct option
While determining regular expression for a given FA, it is _____ to write its regular expression.

Always possible easily
Sometime impossible (Page 80)
always impossible
None of the given options

Question # 2 of 10 (Total Marks: 1) Select correct option
Incase of Myhill Nerode theorem, if a language L partitions sigma star into distinct classes and L is also regular then L generates_____ number of classes.

infinite
specified
finite (Page 75)
odd

ہر چیز کی ایک پہچان ہوتی ہے اور عقلمند کی پہچان غور و فکر کرنا ہے اور غور و فکر کی پہچان خاموشی ہے

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CS402-Theory of Automata

(Solved MCS's)

LECTURE FROM

(23 to 45)

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ALL answers are verified if found any mistake then Correct ACCORDINGLY

1. If Σ = {aa, bb} , then Σ* will not contain
- **aaabbb**
- aabbbb
- aabbaa
- bbaabbbb
2. "One language can have _____TG"s".
- Only one
- Only two
- **More than one**
- Only three
3. According to 1st part of the Kleene"s theorem, If a language can be accepted by an FA then it can be accepted by a _____as well.
- FA
- CFG
- GTG
- **TG**
4. Even-palindrome is a _____language.
- **Non-regular**
- Regular
- Regular but infinite
- Regular but finite
5. If L is a regular language then, Lc is also a _____language.
- **Regular (Page 66)**
- Non-regular
- Regular but finite

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- None of the given

6. Pumping lemma is generally used to prove that:

- A given language is infinite
- **A given language is not regular**
- Whether two given regular expressions of a regular language are equivalent or not
- None of these

7. the FA has N states, then test the words of length less than N. If no word is accepted by this FA, then it will_____word/words.

- accept all
- **accept no (Page 85)**
- accept some
- reject no

8. In CFG, the symbols that can't be replaced by anything are called_____.

- **Terminal (Page 87)**
- Non-Terminal
- Production
- All of given

9. Which of the following is a regular language?

- **String of odd number of zeroes**
- Set of all palindromes made up of 0"s and 1"s
- String of 0"s whose length is a prime number
- All of these

10. Which of the following pairs of regular expressions are equivalent?

- 1(001)* and (10)*10
- **x(xx)* and (x)*x**
- X + and X*
- X + and X* X +

11. An alphabet of Σ is valid if

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- No letter of Σ appears in middle of any other letter
- No letter of Σ appears at end of any other letter
- **No letter of Σ appears at start of any other letter**
- No letter of Σ appears at end or middle of any other letter

12. Which of the following statement is true

- **The length of the output string is greater than length of input string in moore machine.**
- The length of the output string is greater than length of input string in mealy machine.
- The length of the output string is equal to length of input string in moore machine.
- The length of the output string is less than length of input string in mealy machine.

13. If a CFG has only productions of the form nonterminal → string of two nonterminals or nonterminal → one terminal then the CFG is said to be in _____

- **Chomsky Normal Form**
- Ambiguous Form
- Left Aligned Form
- Right Aligned Form

14. We can also represent an FA using different states e.g Accept state; Reject state, Read state etc. The_____state behaves as final state of an FA

- **Accept (Page 105)**
- Pop
- Push
- Reject

15. where the input string is placed before it is run, is called _____

- Date tape
- **Input Tape (Page 105)**
- Output Tape

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- Magnetic tape

16. An FSM can be considered as TM

- Of finite tape length, rewinding capability and unidirectional tape movement
- Of finite tape length, without rewinding capability and bidirectional tape movement
- Of finite tape length, rewinding capability and bidirectional tape movement
- **Of finite tape length, without rewinding capability and unidirectional tape movement**

17. The process of finding the derivation of the word generated by particular grammar is called _____

- Processing
- **Parsing (Page 136)**
- Programming
- Planning

18. The first rule of converting the given "CFG in CNF", is _____

- CNK algorithm
- **CYK algorithm (Page 135) Algorithm 4 (The CYK algorithm)**
- CKY algorithm
- KYC algorithm

19. Alphabet Σ = {a, bc, cc} has number of letters

- One
- Two
- **Three**
- Four

20. We cannot write regular expressions for all_____.

- FA"s
- TG"s
- NFA"s
- **CFG's (Page 97)**

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21. For every Context Free Grammar (CFG), we can make the corresponding_____.

- FA
- TG
- **PDA**
- Regular Grammar

22. Pumping Lemma II says that length(x) + length(y) should be _____.

- **Less than number of states (Page 75)**
- Equal to number of states
- Greater than number of states
- Greater than or equal to number of states

23. Chomsky normal form (CYK) algorithm was proposed by _____.

- **John cock (Page 135)**
- James Cock
- Daniel I.A.
- John Weiss

24. The language of Palindromes defined over an alphabet set {a, b} can be recognized by_____.

- FA
- NFA
- TG
- **PDA**

25. Which of the following is the first phase of compiler on the basis of functionality?

- Parser
- Lexical analyzer
- **Scanner**
- Interpreter

26. (Σ* - L) represent the _____of a language L.

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- **Complement (Page 66)**
- Kleene‟s closure
- Union
- intersection

27. If we have two transition graphs then their union will be expressed by

- **taking a common start state and joining them by two null transitions (Page 65)**
- just connecting both start states by null transitions
- connecting final state of first TG to the initial state of second TG
- connecting the final state of first TG to the final state of second TG

28. _____and_____are removed in order to make a CFG in Chomsky Normal Form(CNF).

- Null, nullable productions
- Nullable , unit productions
- **Null, unit productions (Page 102)**
- String of length 0, null

29. If L1 and L2 are expressed by regular languages then L1 + L2 is also a_____Language.

- **Regular (Page 10)**
- Ir-regular
- PDA
- Hybrid

30. Which of the following is a regular Context Free Grammar:

- **S → abS| baS | ^ ab(ab+ba)*ba + ba(ab+ba)*ab**
- S → aSb| baS | ^
- S → abS| bSa | ^
- S → aSb| Sa | ^

31. A read state can have _____outgoing edge/ edges.

- 1
- 2

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- 3
- **Any number of (Page 111)**

32. Who did not invent the Turing machine?

- Alan Turing
- **A. M. Turing (Page 140)**
- Turing
- None of these

33. Which statement is true?

- **The tape of turing machine is infinite. (Page 140)**
- The tape of turing machine is finite.
- The tape of turing machine is infinite when the language is regular
- The tape of turing machine is finite when the language is nonregular.

34. Every regular expression can be expressed as CFG but every CFG cannot be expressed as a regular expression. This statement is:

- Depends on the language
- None of the given options
- **True (Page 97)**
- False

35. Consider the language L of strings, defined over Σ = {a,b}, ending in a

- **There are finite many classes generated by L, so L is regular (Page 76)**
- There are infinite many classes generated by L, so L is regular
- There are finite many classes generated by L, so L is non-regular
- There are infinite many classes generated by L, so L is non-regular

36. The word „formal“ in formal languages means

► The symbols used have well defined meaning

► They are unnecessary, in reality

► **Only the form of the string of symbols is significant**

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► None of these

37. Let A = {0, 1}. The number of possible strings of length „n‟ that can be formed by the elements of the set A is

► n!

► n2

► nm

► 2n

38. Choose the correct statement.

► A Mealy machine generates no language as such

► A Moore machine generates no language as such

► A Mealy machine has no terminal state

► All of these

39. TM is more powerful than FSM because

► The tape movement is confined to one direction

► It has no finite state control

► It has the capability to remember arbitrary long sequences of input symbols

► None of these

40. Like TG, a PDA can also be non-deterministic

- **True (Page 111)**
- False

41. The language of all words (made up of a‟s and b‟s) with at least two a‟s can not be described by the regular expression.

- a(a+b)*a(a+b)*(a+b)*ab*

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- (a+b)* ab* a(a+b)*
- b*ab* a(a+b)*
- **none of these**

42. If L is a regular language then, Lc is also a _____language.

- **Regular (Page 66) rep**
- Non-regular
- Regular but finite
- None of the given

43. In CFG, the symbols that can‟t be replaced by anything are called__

- **Terminal (Page 87) rep**
- Non-Terminal
- Production
- All of given

44. Which of the following is NOT a regular language?

- String of 0‟s whose length is a perfect squere
- Set of all palindromes made up of 0‟s and 1‟s
- String of 0‟s whose length is a prime number
- **All of the given options**

45. Choose the incorrect (FALSE) statement.

- A Mealy machine generates no language as such
- A Mealy machine has no terminal state
- **For a given input string, length of the output string generated by a Moore machine is not more than the length of the output string generated by that of a Mealy machine**
- All of these

46. Choose the incorrect statement:

- (a+b)*aa(a+b)* generates Regular language.
- **A language consisting of all strings over ∑={a,b} having equal number of a's and b's is a regular language**

- Every language that can be expressed by FA can also be expressed by RE
- None of these

47. Left hand side of a production in CFG consists of:

- One terminal
- More than one terminal
- **One non-terminal (Page 87)**
- Terminals and non-terminals

48. PDA is only used to represent a regular language.

- True
- **False**

49. A production of the form non-terminal □ string of two non-terminal is called a live Production.

- **True (Page 127)**
- False

50. We can find a CFG corresponding to a DFA.

- **True (Page 97)**
- False

51. START, READ, HERE and ACCEPTS are conversions of the machine

- **True (Page 122)**
- False

52. A CFG is said to be ambiguous if there exists at least one word of its language that can be generated by different production trees

- **True (Page 95)**
- False

53. Syntax tree or Generation tree or Derivation tree are same tree

- **True (Page 92)**
- False

54. The symbols that cannot be replaced by anything are called terminals

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- **True (Page 87) repeat**
- False

55. The production of the form non-terminal □ one non-terminal is called unit production

- **True (Page 100)**
- False

56. DFA and PDA are equal in power.

- True
- **False (Page 105)**

57. A production of the form non-terminal □ non-terminal is called a dead Production.

- True
- **False (Page 127)**

58. Semi-word is a string having some terminals and one non-terminal at the right of string.

- **True (Page 97)**
- False

59. Two FAs are equivalent if they have same no. of states.

- **True (Page 15)**
- False

60. There exist exactly two different derivations in an ambiguous CFG for a word.

- **True (Page 93)**
- False

61. Regular languages are closed under Union, Concatenation and Kleene star.

- **True (Page 10)**
- False

62. CFG may also represent a regular language.

- **True (Page 97)**

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- False

63. PDA is stronger than FA.

- **True (Page 105)**
- False

64. A Total Language Tree has

- All languages over Σ
- **All strings over Σ (Page 96)**
- All words of all languages over Σ
- All words of one language over Σ

65. What Turing Machine does not have?

- Stack
- Tape
- Head
- **Word**

66. CFG given S bS|Sb|aa represents language b*aa

- aab*
- b*aab*
- **b*(aa)*b***

67. The values of input (say a & b) does not remain same in one cycle due to

- NAND gate
- Click plus
- OR gate
- **NOT gate**

68. Set of all palindromes over {a,b}is regular

- True
- **False (Page 74)**

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69. In CFG, the symbols that cannot be replaced by anything are called

- **Terminals (Page 87) rep**
- Non terminals
- Productions
- None of the given options

70. a^n b^n generates the ..... language

- regular
- non regular
- **EQUAL and non regular (Page 71)**
- EQUAL and regular

71. The grammatical rules which involves meaning of words are called:

- **Semantic (Page 87)**
- Sytactics
- Alphabets
- None of the given options

72. If an FA has N state then it must accept the word of length

- **N-1**
- N+1
- N+2N

73. Two languages are said to belong to same class if they end in the same state when they run over an FA, that state

- Must be final state
- **May be final state or not (Page 75)**
- May be start or not
- None of the given options

74. In pref(Q in R) Q is …… to (than) R

- Equal
- **Not Equal (Page 79)**
- Greater

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- Smaller

75. According to Myhill Nerode theorem, if L generates finite no. of classes then L is.....

- Finite
- Infinite
- **Regular (Page 76)**
- Non Regular

76. If the intersection of two regular languages is regular then the complement of the intersection of these two languages is also regular

- **True (Page 68)**
- False

77. In pumping lemma theorem (x y^n z) the range of n is

- **n=1,2,3,4 ..... (Page 74)**
- n=0,1,2,3,4....
- n=-3,-2,-1,0,1,2,3,4.....
- n=-3,-2,-1,1,2,3,4.....

78. The complement of a regular language is also a regular

- **True repeat**
- False

79. For a non regular language there exist …… FA

- One
- At least one
- At most one
- **No (Page 71)**

80. The strings or words which do not belong to a language is called..... of that language

- Intersection
- Union
- **Complement (Page 66)**
- Quotient

81. A non regular language can be represented by

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- RE
- FA
- TG
- **None of the given options (Page 71)**

82. For language L defined over {a, b},then L partitions {a, b}* into …… classes

- Infinite
- Finite
- **Distinct (Page 75)**
- Non distinct

83. If an FA accept a word then there must exist a path from

- **Initial to final state (Page 81)**
- Initial to each state
- Initial to each state but not to final state
- Initial to final state by traversing each state

84. Which of the following statement is true about NFA with Null String?

- Infinite states
- Infinite set of letters
- Infinite set of transitions
- **Transition of null string is allowed at any stage (Page 71)**

85. FA corresponding to an NFA can be built by introducing an empty state for a letter having

- **no transition at certain state (Page 43)**
- one transition at certain state
- two transition at certain state
- more than two transitions at certain state

86. Let FA3 be an FA corresponding to FA1FA2, then the initial state of FA3 must correspond to the initial state of

- **FA1 only (Page 35)**
- FA2 only

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- FA1 or FA2
- FA1 and FA2

87. (a* + b*)* = (a + b)* this expression is _____

- True
- **False**

88. If S = {ab, bb}, then S* will not contain

- Abbbab
- **Bbba**
- ababbb
- bbbbab

89. What do automata mean?

- Something done manually
- **Something done automatically**
- What is false about the term alphabet?
- It is a finite set of symbols.

90. Consider the following production (of a CFG): S->XYZ Here _____is left most nonterminal in working string. Note: S, X, Y and Z are all nonterminals

- S
- **X**
- Y
- Z

91. A PDA is called nondeterministic PDA if _____

- **There are more than one outgoing edges at READ or POP states with one label (Page 111)**
- There are more than one PUSH states
- There are mroe than one POP states
- All of the given options

92. A PDA consists of the following:

- An alphabet (Sigma) of input letters.
- An input TAPE with infinite many locations in one direction

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- One START state with only one out-edge and no in-edge
- **All of the given options (Page 105)**

93. The CFG S --> aSa | bSb | a | b | ^ represents the language

- EVEN-EVEN
- **PALINDROM (Page 91)**
- EQUAL
- ODD-ODD

94. Halt states are

- Start and Accept
- **Accept and Reject (Page 105)**
- Start and Reject
- Read and Reject

95. Choice of path can be determined by left most derivation of the string belonging to CFL at..... state

- **Accept (Page 104)**
- Reject
- Push
- POP

96. The unit and null productions can be deleted from a CFG

- **True (Page 99-100)**
- False

97. Identify the TRUE statement about following CFG: S -> SB|AB A -> CC B -> b C -> a

- The given CFG has 8 Nonterminals
- The given CFG has 8 Terminals
- **The given CFG is in CNF (Page 101)**
- The given CFG is not in CNF

98. The structure given below is called_____S -> aA|bB A -> aS|a B -> bS|b

- RE
- TG

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- **CFG (Page 87)**
- PDA

99. Which of the following states is not part of PDA

- START
- ACCEPT
- **WRITE (Page 107)**
- REJECT

100. The production of the form: nonterminal --> one nonterminal is called the _____

- **Unit production (Page 100)**
- NULL production
- Terminal production
- Non Terminal production

101. A_____is the one for which every input string has a unique path through the machine.

- **Deterministic PDA (Page 111)**
- nondeterministic PDA
- PUSHDOWN store
- Input Tape

102. In the null production N --> ^ , N is a

- Terminal
- **Non terminal (Page 99)**
- Word
- None of the given options

103. The major problem in the earliest computers was

- To store the contents in the registers
- **To display mathematical formulae (Page 87)**
- To load the contents from the registers
- To calculate the mathematical formula

104. In polish notation, (o-o-o) is the abbreviation of……?

- Operand - Operator – Operand

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- Operand - Operand- Operator
- **Operator -Operand – Operand (Page 94)**
- Operand -Operand – Operand

105. The CFG is said to be ambiguous if there exist at least one word of its language that can be generated by the..... production trees

- One
- Two
- **More than one (Page 95)**
- At most one

106. The input string is placed, before it runs, in

- Stack
- Memory
- **Tape (Page 105)**
- Ram

107. The production S --> SS | a | b | ^ can be expressed by RE

- (a+b)+
- (a+b)
- **(a+b)* (Page 88)**
- (ab)*

108. The locations into which we put the input letters on "Input Tap" are called _____

- Words
- alphabets
- **cells (Page 105)**
- elements

109. "CFG" stands for _____

- Context Free Graph
- **Context Free Grammer (Page 87)**
- Context Finite Graph
- Context Finite Grammer

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110. In a CFG the nonterminal that occurs first from the left in the working string, is said to be _____

- Least Significant nonterminal
- Most Significant nonterminal
- **Left most nonterminal (Page 103)**
- Left most derivate

111. The unit production is

- Terminal --> Terminal
- Terminal --> Non Terminal
- Non terminal --> Terminal
- **Non terminal --> Non Terminal (Page 100)**

112. A_____operator adds a new letter at the top of STACK

- **PUSH (Page 107)**
- POP
- READ
- APPEND

113. PDA stands for _____

- Push and Drop Automaton
- Pop and Drop Automaton
- **Push Down Automaton (Page 112)**
- None of given options

114. The production of the form: Nonterminal-> ^ is said to be _____ production

- **NULL (Page 99)**
- UNIT
- Chomsky form production
- None of the given options

115. In a STACK:

- The element PUSHed first is POPed first
- **The element PUSHed first is POPed in the last (Page 107 concept)**
- The element PUSHed in last is POPed in last

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- None of given options

116. For a given input, it provides the compliment of Boolean AND output.

- **NAND box (NOT AND) (Page 63)**
- DELAY box
- OR box
- AND box

117. It delays the transmission of signal along the wire by one step (clock pulse).

- NAND box (NOT AND)
- **DELAY box (Page 63)**
- OR box
- AND box

118. Any language that can not be expressed by a RE is said to be regular language.

- True
- **False**

119. The current in the wire is indicated by 1 and 0 indicates the absence of the current.

- **True (Page 63)**
- False

120. For the given input, AND box provides the Boolean AND output.

- **True (Page 63)**
- False

121. Let L be a language defined over an alphabet Σ, then the language of strings, defined over Σ, not belonging to L, is called Complement of the language L, denoted by Lc or L‟.

- **True (Page 66)**
- False

122. To describe the complement of a language, it is very important to describe the----------of that language over which the language is defined.

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- **Alphabet (Page 66)**
- Regular Expression
- String
- Word

123. For a certain language L, the complement of Lc is the given language L i.e. (Lc)c = Lc

- True
- **False (Page 66)**

124. If L is a regular language then, --------- is also a regular language.

- Lm
- Ls
- Lx
- **Lc (Page 66)**

125. Converting each of the final states of F to non-final states and old non-final states of F to final states, FA thus obtained will reject every string belonging to L and will accept every string, defined over Σ, not belonging to L. is called

- Transition Graph of L
- Regular expression of L
- **Complement of L (Page 66)**
- Finite Automata of L

126. If L1 and L2 are two regular languages, then L1 U L2 is not a regular.

- True
- **False (Page 65)**

127. If L1 and L2 are regular languages, then these can be expressed by the corresponding FAs.

- **True (Page 68)**
- False

128. The language that can be expressed by any regular expression is called a Non regular language.

- True

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- **False (Page 71)**

129. The languages----------are the examples of non regular languages.

- **PALINDROME and PRIME (Page 71)**
- PALINDROME and EVEN-EVEN
- EVEN-EVEN and PRIME
- FACTORIAL and SQURE

130. Let L be any infinite regular language, defined over an alphabet Σ then there exist three strings x, y and z belonging to Σ* such that all the strings of the form xy z n for n=1,2,3, … are the words in L. called.

- Complement of L
- **Pumping Lemma (Page 72)**
- Kleene‟s theorem
- None in given

131. Languages are proved to be regular or non regular using pumping lemma.

- **True (Page 74)**
- False

132. is obviously infinite language.

- EQUAL-EQUAL
- EVEN-EVEN
- **PALINDROME (Page 75)**
- FACTORIAL

133. If, two strings x and y, defined over Σ, are run over an FA accepting the language L, then x and y are said to belong to the same class if they end in the same state, no matter that state is final or not.

- **True (Page 75)**
- False

134. Myhill Nerode theorem is consisting of the followings,

- L partitions Σ* into distinct classes.
- If L is regular then, L generates finite number of classes.
- If L generates finite number of classes then L is regular.

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- **All of above (Page 75)**

135. The language Q is said to be quotient of two regular languages P and R, denoted by--- if PQ=R.

- R=Q/P
- **Q=R/P (Page 78)**
- Q=P/R
- P=R/Q

136. If two languages R and Q are given, then the prefixes of Q in R denoted by Pref(Q in R).

- **True (Page 78)**
- False

137. Let Q = {aa, abaaabb, bbaaaaa, bbbbbbbbbb} and R = {b, bbbb, bbbaaa, bbbaaaaa} Pref (Q in R) is equal to,

- **{b,bbba,bbbaaa} (Page 78)**
- {b,bba,bbaaa}
- {ab,bba,bbbaa}
- {b,bba,bbba}

138. If R is regular language and Q is any language (regular/ non regular), then Pref (Q in R) is ---------.

- Non-regular
- Equal
- **Regular (Page 79)**
- Infinite

139. _____states are called the halt states.

- **ACCEPT and REJECT (Page 105)**
- ACCEPT and READ
- ACCEPT AND START
- ACCEPT AND WRITE

140. The part of an FA, where the input string is placed before it is run, is called _____

- State
- Transition

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- **Input Tape (Page 105)**
- Output Tape

141. In new format of an FA (discussed in lecture 37), This state is like dead-end non final state

- ACCEPT
- **REJECT (Page 105)**
- STATR
- READ

142. Between the two consecutive joints on a path:

- One character can be pushed and one character can be popped
- **Any no. of characters can be pushed and one character can be popped (Page 122)**
- One character can be pushed and any no. of characters can be popped
- Any no. of characters can be pushed and any no. of characters can be popped

143. The PDA is called non-deterministic PDA when there are more than one out going edges from…… state

- START or READ
- POP or REJECT
- **READ or POP (Page 111)**
- PUSH or POP

144. Identify the TRUE statement:

- A PDA is non-deterministic, if there are more than one READ states in PDA
- A PDA is never non-deterministic
- **Like TG, A PDA can also be non-deterministic (Page 111)**
- A PDA is non-deterministic, if there are more than one REJECT states in PDA

145. There is a problem in deciding whether a state of FA should be marked or not when the language Q is infinite.

- **True (Page 79)**
- False

146. If an effectively solvable problem has answered in yes or no, then this solution is called ---------

- **Decision procedure (Page 80)**
- Decision method
- Decision problem
- Decision making

147. The following problem(s) ---------- is/are called decidable problem(s).

- The two regular expressions define the same language
- The two FAs are equivalent
- **Both a and b (Page 80)**
- None of given

148. To examine whether a certain FA accepts any words, it is required to seek the paths from ------- state.

- Final to initial
- Final to final
- **Initial to final (Page 81)**
- Initial to initial

149. The high level language is converted into assembly language codes by a program called compiler.

- **TRUE (Page 87)**
- FALSE

150. Grammatical rules which involve the meaning of words are called _____

- **Semantics (Page 87)**
- Syntactic
- Both a and b
- None of given

151. Grammatical rules which do not involve the meaning of words are called -_____

- Semantics
- **Syntactic (Page 87)**

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- Both a and b
- None of given

152. The symbols that must be replaced by other things are called _____

- Productions
- Terminals
- **Non-terminals (Page 87)**
- None of given

153. The grammatical rules are often called_____

- **Productions (Page 87)**
- Terminals
- Non-terminals
- None of given

154. The terminals are designated by _____letters, while the non-terminals are designated by_____letters.

- Capital, bold
- **Small, capital (Page 87)**
- Capital, small
- Small, bold

155. The language generated by _____is called Context Free Language (CFL).

- FA
- TG
- **CFG (Page 87)**
- TGT

156. S → aXb|bXa X → aX|bX|Λ The given CFG generates the language in English _____

- **Beginning and ending in different letters (Page 91)**
- Beginning and ending in same letter
- Having even-even language
- None of given

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157. The CFG is not said to be ambiguous if there exists atleast one word of its language that can be generated by the different production trees,

- TRUE
- **FALSE (Page 95)**

158. The language generated by that CFG is regular if _____

- No terminal → semi
- word No terminal → word
- **Both a and b (Page 97)**
- None of given

159. The production of the form no terminal → Λ is said to be null production.

- **TRUE (Page 99)**
- FALSE

160. CNF is stands for

- Context Normal Form
- Complete Normal Form
- **Chomsky Normal Form (Page 102)**
- Compared Null Form

161. Kleene‟s theorem states

- All representations of a regular language are equivalent.
- All representations of a context free language are equivalent.
- All representations of a recursive language are equivalent
- **Finite Automata are less powerful than Pushdown Automata. (Page 105)**

162. Null production is a

- Word
- String
- Terminal
- **All of the given options**

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163. In nondeterministic PDA a string is supposed to be accepted, if there exists at least one path traced by the string, leading to _____ state.

- **ACCEPT (Page 111)**
- REJECT
- START
- READ

164. The CFG which generates the regular language is called:

- Regular expression
- Finite Automata
- **Regular grammar (Page 97)**
- None of the given options

165. If a CFG has a null production, then it is possible to construct another CFG accepting the same language without null production

- TRUE
- **FALSE (Page 99)**

166. In large FA with thousands of states and millions of directed edges, without an effective procedure it is_____to find a path from initial to final state.

- Always easy
- **Impossible (Page 81)**
- may be good
- always impossible

167. If there is no final state of two FAs then their_____also have no___state

- initial, union
- final, union
- **union,final (Page 83)**
- union, initial

168. Set of all palindromes over {a,b} is:

- Regular
- Regular and finite

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- Regular and infinite
- **Non-regular (Page 71)**

169. In the context of Myhill Nerode theorem, for even-even language sigma star can be partitioned into_____number of classes.

- 3
- **4 (Page 77)**
- 5
- 6

170. The product of two regular languages is_____.

- **Regular (Page 78)**
- Infinite
- non-regular
- closure of a regular language

171. incase of Myhill Nerode theorem, if a language L partitions sigma star into distinct classes and L is also regular then L generates_____number of classes.

- Infinite
- specified
- **finite (Page 75)**
- odd

172. While determining regular expression for a given FA, it is _____to write its regular expression.

- Always possible easily
- **Sometime impossible (Page 80)**
- always impossible
- None of the given options

173. If (L1 ∩ L2c) ∪ (L1c ∩ L2) is regular language that accepts the words which are in L1 but not in L2 or else in L2 but not in L1 . The corresponding FA cannot accept any word which is in_____L1 and L2.

- Not both
- **Both (Page 80)**

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- At least in one
- None of the given options

174. A problem that has decision procedure is called _____ problem.

- Regular language
- un-decidable
- Infinite
- **Decidable (Page 80)**

175. The product of two regular languages is_____.

- **Regular (Page 78)**
- Infinite
- non-regular
- closure of a regular language

176. In new format of an FA (discussed in lecture 37):……state is like a final state of an FA

- START
- ACCEPT (Handouts Page # 119)
- REJECT
- READ

177. In conversion form of PDA there is no ..... state

- PUSH
- ACCEPT
- REJECT (Handouts Page # 119)
- READ

178. Given a PDA that accepts the language L

- There does not exist any CFG that generates exactly L
- that PDA will also accept Language L' (complement of L)
- There exists a CFG that generates exactly L (Handouts Page # 118)
- None of given options

179. In a CFG the non-terminal that occurs first from the left in the working string. is said to be

- Least Significant nonterminal

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- Left most nonterminal (Handouts Page # 103)
- Most Significant nonterminal
- Left most derivate

180. The structure given below is called

S -> aAlbB

A -> aSla

B -> bSIb

- RE
- PDA
- CFG (It is form of CFG)
- TG

181. An FA has N states then it must accept the word of length

- 2N
- N
- N-1
- N-1

182. To examine whether a certain FA accepts any words. it is required to seek the paths ..... state.

- from initial to final (Handouts Page # 81)
- from initial-to-initial back
- from final to initial
- from final to back final

183. In nondeterministic PDA. a string is supposed to be accepted if there exists at least one path traced by the string. leading to state.

- START
- REJECT
- READ
- ACCEPT (Handouts Page # 111)

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184. If a CFG has a null production, then it is

- Called Null CFG
- Not possible to construct another CFG without null production accepting the same language with the exception of the word
- Called Chmosky Normal Form (CNF)
- Possible to construct another CFG without null production accepting the same language with the exception of the word

185. There is at least one production in CFG that has one……. on its left side.

- Non terminal (Handouts Page # 87)
- Null production
- Terminal
- Unit production

186. In large FA with thousands of states and millions of directed edges, without an effective procedure it is……to find a path from initial to final state.

- Impossible (Handouts Page # 81)
- Always easy
- always impossible
- may be good

187. By removing null and unit productions _____ _.

- CNF can be converted into FA
- CNF can be converted into CFG
- CFG can be converted into CNF (Handouts Page # 102)
- CNF can be converted into Turing machine

188. A……is the one for which every input string has a unique path through the machine

- deterministic PDA
- Input Tape
- nondeterministic PDA
- PUSHDOWN store

189. PDA stands for

- Push Down Automaton (Handouts Page # 112)

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- Pop and Drop Automaton
- Push and Drop Automaton
- Push Deterministic Automaton

190. A PDA is called nondeterministic PDA if

- there are more than one POP states
- there are more than one PUSH states
- there are more than one outgoing edges at READ or POP states with one label
- every READ state is followed by a HERE state.

191. Which of the following cannot be represented by a regular expression?

- String of 0's with an odd length
- Language of even-even
- Language of odd-odd
- String of 0s with a prime length (Because Prime is not regular Langue)

192. In conversion form of PDA. there is……..accept state(s).

- At most one
- At least one
- More than One
- Exactly one (Handouts Page # 119)

193. If there is no final state of two FAs, then their…….also have no .....state

- union, initial
- union. Final (Handouts Page # 83)
- final, union
- initial, union

194. The tree which produces all the strings of a language is called

- Derivation tree
- Total language tree (Handouts Page # 96)
- Non ambiguous tree
- Ambiguous tree

195. In new format of an FA (discussed in lecture 37),..... state is like dead-end non final state.

- READ
- REJECT (Handouts Page # 105)
- START
- ACCEPT

196. To write the expression from the tree, it is required to traverse from

- Top to bottom of the tree
- Right side of the tree
- Left side of the tree (Handouts Page # 94)
- Bottom to top of the tree

197. A PDA consists of the following:

- An alphabet (Sigma) of input letters.
- An input TAPE with infinite many locations in one direction
- One START state with only one out-edge and no in-edge
- All the given options (Handouts Page 105)

198. If R is regular language and Q is any language (regular/ non-regular), then Pref…….in …….. is regular.

- R R
- R.Q
- Q.R (Handouts Page # 79)
- Q.Q

199. ……is an operation that takes out a letter from the top of the STACK.

- WRITE
- APPEND
- PUSH
- POP (Handouts Page # 107)

200. Before the CFG corresponding to the given PDA is determined, the PDA is converted into the standard form which is called.

- Finite Automaton
- Conversion form (Handouts Page # 118)
- None of given options
- Chomsky Normal Form (CNF)

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201. The part of an FA, where the input string is placed before it is run, is called

- Transition
- Input Tape (Handouts Page # 105)
- Output Tape
- State

202. A problem is said to be ……if there exists an algorithm that provides the solution in……. number of steps.

- Effectively unsolvable, infinite
- Effectively solvable, infinite
- Effectively unsolvable, finite
- Effectively solvable, finite (Handouts Page # 80)

203. …… states are called the halt states.

- ACCEPT AND START
- ACCEPT and READ
- ACCEPT and REJECT (Handouts Page # 105)
- ACCEPT AND WRITE

204. The grammatical rules which involve meaning of words are called

- Semantics (Handouts Page # 87)
- Syntactics
- strings
- alphabets

205. The PDA is called non-deterministic PDA when there are more than one out going edges from state

- READ or POP (Handouts Page # 111)
- START or READ
- POP or
- PUSH or POP

206. Which of the following states is not part of PDA?

- REJECT
- ACCEPT
- START
- WRITE (All other are parts of PDA)

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207. The major problem in the earliest computers was

- To calculate the mathematical formula
- To display mathematical formulas (Handouts Page # 87)
- To store the contents in the registers
- To load the contents from the registers

208. The operators like (` +) in the parse tree are considered as

- Terminals (Handouts Page # 93)
- productions
- non-terminals
- intermediates

209. If L1 and L2 are two regular languages, then they……expressed by FAs.

- cannot be
- May be
- may or may not be
- can be (Handouts Page # 68)

210. Before running the input string on PDA it is first placed on

- Stack
- Ram
- Memory
- Tape (Handouts Page # 107)

211. Which is the correct option

- The element PUSHED in last is POPED in last
- The element PUSHED first is POPED in the last (LIFO Method, from Book)
- The element PUSHED first is POPED first
- None of given options

212. Null production is a

- String
- Word (Handouts Page # 97)

- Terminal
- All the above

213. A/an……operator adds a new letter at the top of STACK

- Push (Handouts Page # 107)
- Append
- Read
- Pop

214. In conversion form of PDA, no two…….states exist in a row without state

- POP. READ (Handouts Page # 119)
- POP. REJECT
- PUSH. START
- PUSH READ

215. Given a PDA that accepts the language L

- that PDA will also accept Language L' (complement of Ll
- There exists a CFG that generates exactly L (Handouts Page # 118)
- None of given options
- There does not exist any CFG that generates exactly

216. In large FA with thousands of states and millions of directed edges, without an effective procedure it is ……. to find a path from initial to final state.

- Always easy
- always impossible
- may be good
- Impossible (Handouts Page # 81)

217. The CFG there generates the regular language is called

- Regular expression
- finite automata
- regular grammars (Handouts Page # 97)
- now regular grammars

218. Consider the following CFG: (Note: ^ means NULL)

S-→Xa

X→aX|bX|^

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- Above give a CFG can be represented by RE
- a*b*
- a (a + b) *a
- (a + b) *a
- A*b*a

S production will give us Xa. As the X is nonterminal and we must only change X and the terminal a will be on the last of the R.E.

Now we will change X production.

X production will give us as many a as we want. or if we use the second production which will give us as many b as we want. And last production will give us ^ (lemda)

So the answer would be a*b*a

Last a is the terminal which we got from the very first production.

219. For a machine with N number of states, the total number of strings to be tested, defined over an alphabet of m letters is

- mN +mN+1 + mN+2 +… +m2N-1 (Handouts Page # 86)
- mN
- Nm
- Nm +mN+1 + mN+2 +… +

220. Consider the CFG given below.

A→B|b

B→a

Which of the following is a unit production?

- S→bb
- A→b
- A→B
- B→a

221. The CFG is said to be ambiguous if there exist at least one word of its language that can be generated by the.....production trees

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- One
- Two
- More than one (Handouts Page # 95)
- At most one

222. If Q = {xx, xyxxxy} and R = { xyxyxyxxyy, xyxyyyxx} then Pref {Q in R} = _____

- Xx
- Xyxyxy
- Xyxyyy (Solved by my self 100% sure)
- Xxy

223. The unit production is

- Terminal --> Terminal
- Terminal --> Non Terminal
- Non terminal --> Terminal
- Non terminal --> Non-Terminal (Hand out Page # 100)

224. Which of the following statement is FALSE?

- For every PDA, there always exists a regular expression (Not sure)
- Every CFG cannot be expressed as Regular Expression
- Every Regular Expression be expressed by a CFG.
- For a PDA, there exists a CFG that represent the same language

225. The CFG S → aSb|ab|^

- Palindrome
- Prime
- Equal
- Even

The production will give us ab and non-terminal S inside the a and b. If we change the S into next production of ab, we will get abab but instead of using 2nd production, if we use the last which will give us only ^ (Lemda), thus we will get ^ab.

So, we will get Equal language (same number of a and same number of b)

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226. Before the PDA is converted into conversion form, a new state ---------- is defined which is placed in the middle of any edge.

- HERE (Hand out Page # 118)
- STOP
- START
- REJECT

227. A PDA is in conversion form if it fulfills the following conditions:

- There is only one ACCEPT state. (Hand out Page # 119)
- There are one REJECT state.
- There are more than one ACCEPT states.
- There is only one Accept state.

228. Identify the false statement about the following CFG
S→SB|AB
A→CC
B→
C→a

- CFG has 8 Non terminals
- all the given option (All are false as There are 4 terminals , It is in CNF and it does not generate any null string)
- CFG is not in CNF
- CFG generate null string

229. This CFG there generates to the regular language is called

- Regular grammar (Hand out Page # 97)
- nonregular grammar
- finite automata
- regular expression

230. The derivation of the word W generated by CFG such that at each step a production is applied to the leftmost nonterminal in the working string is said to be

- Left most terminal

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- right most terminal
- left most derivation (Hand out Page # 103)
- right most derivation

231. If the FA has N states, then test these words of length less than N. If no word is accepted by this FA, then it will word/words

- Accept no (Hand out Page # 85)
- Accept some
- Reject No
- Accept All

232. In a CFG then non terminals are denoted by

- small letters
- numbers
- capital letters (Hand out Page # 87)
- small letters and numbers

233. "CFG"stands for_____

- Context finite graph
- contacts finite grammar
- contact free graph
- Context free grammar (Hand out Page # 87)

234. Consider the following production (of a CFG) S→XYZ
Here _____is left most non terminals in working string
note XY and Z are all known terminals

- X (X is on the most left side)
- S
- Z
- Y

235. Consider the following CFG
S→ a|Xb|aYa
X→ Y|^ (Note: ^ means NULL)
Y→ b|X
which nonterminal is/are not nullable

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- Y
- S, X and Y
- S
- X (X is Null Production and not a null able)

236. In new format of an FA (discussed in lecture 37):……state is like a final state of an FA

- START
- ACCEPT (Handouts Page # 119)
- REJECT
- READ

237. In conversion form of PDA there is no ..... state

- PUSH
- ACCEPT
- REJECT (Handouts Page # 119)
- READ

238. Given a PDA that accepts the language L

- There does not exist any CFG that generates exactly L
- that PDA will also accept Language L' (complement of L)
- There exists a CFG that generates exactly L (Handouts Page # 118)
- None of given options

239. In a CFG the non-terminal that occurs first from the left in the working string. is said to be

- Least Significant nonterminal
- Left most nonterminal (Handouts Page # 103)
- Most Significant nonterminal
- Left most derivate

240. The structure given below is called

S -> aAlbB

A -> aSla

B -> bSIb

- RE
- PDA
- CFG (It is form of CFG)
- TG

241. An FA has N states then it must accept the word of length

- 2N
- N
- N-1
- N-1

242. To examine whether a certain FA accepts any words. it is required to seek the paths ..... state.

- from initial to final (Handouts Page # 81)
- from initial-to-initial back
- from final to initial
- from final to back final

243. In nondeterministic PDA. a string is supposed to be accepted if there exists at least one path traced by the string. leading to.....state.

- START
- REJECT
- READ
- ACCEPT (Handouts Page # 111)

244. If a CFG has a null production, then it is

- Called Null CFG
- Not possible to construct another CFG without null production accepting the same language with the exception of the word
- Called Chmosky Normal Form (CNF)
- Possible to construct another CFG without null production accepting the same language with the exception of the word

245. There is at least one production in CFG that has one……. on its left side.

- Non terminal (Handouts Page # 87)
- Null production

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- Terminal
- Unit production

246. In large FA with thousands of states and millions of directed edges, without an effective procedure it is……to find a path from initial to final state.

- Impossible (Handouts Page # 81)
- Always easy
- always impossible
- may be good

247. By removing null and unit productions _____ _.

- CNF can be converted into FA
- CNF can be converted into CFG
- CFG can be converted into CNF (Handouts Page # 102)
- CNF can be converted into Turing machine

248. A……is the one for which every input string has a unique path through the machine

- deterministic PDA
- Input Tape
- nondeterministic PDA
- PUSHDOWN store

249. PDA stands for

- Push Down Automaton (Handouts Page # 112)
- Pop and Drop Automaton
- Push and Drop Automaton
- Push Deterministic Automaton

250. A PDA is called nondeterministic PDA if

- there are more than one POP states
- there are more than one PUSH states
- there are more than one outgoing edges at READ or POP states with one label
- every READ state is followed by a HERE state.

251. Which of the following cannot be represented by a regular expression?

- String of 0's with an odd length
- Language of even-even
- Language of odd-odd
- String of 0s with a prime length (Because Prime is not regular Langue)

252. In conversion form of PDA. there is……..accept state(s).

- At most one
- At least one
- More than One
- Exactly one (Handouts Page # 119)

253. If there is no final state of two FAs, then their…….also have no .....state

- union, initial
- union. Final (Handouts Page # 83)
- final, union
- initial, union

254. The tree which produces all the strings of a language is called

- Derivation tree
- Total language tree (Handouts Page # 96)
- Non ambiguous tree
- Ambiguous tree

255. In new format of an FA (discussed in lecture 37),..... state is like dead-end non final state.

- READ
- REJECT (Handouts Page # 105)
- START
- ACCEPT

256. To write the expression from the tree, it is required to traverse from

- Top to bottom of the tree
- Right side of the tree
- Left side of the tree (Handouts Page # 94)
- Bottom to top of the tree

257. A PDA consists of the following:

- An alphabet (Sigma) of input letters.
- An input TAPE with infinite many locations in one direction
- One START state with only one out-edge and no in-edge
- All the given options (Handouts Page 105)

258. If R is regular language and Q is any language (regular/ non-regular), then Pref…….in …….. is regular.

- R R
- R.Q
- Q.R (Handouts Page # 79)
- Q.Q

259. ……is an operation that takes out a letter from the top of the STACK.

- WRITE
- APPEND
- PUSH
- POP (Handouts Page # 107)

260. Before the CFG corresponding to the given PDA is determined, the PDA is converted into the standard form which is called.

- Finite Automaton
- Conversion form (Handouts Page # 118)
- None of given options
- Chomsky Normal Form (CNF)

261. The part of an FA, where the input string is placed before it is run, is called

- Transition
- Input Tape (Handouts Page # 105)
- Output Tape
- State

262. A problem is said to be ……if there exists an algorithm that provides the solution in……. number of steps.

- Effectively unsolvable, infinite
- Effectively solvable, infinite
- Effectively unsolvable, finite
- Effectively solvable, finite (Handouts Page # 80)

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263. ……. states are called the halt states.

- ACCEPT AND START
- ACCEPT and READ
- ACCEPT and REJECT (Handouts Page # 105)
- ACCEPT AND WRITE

264. The grammatical rules which involve meaning of words are called

- Semantics (Handouts Page # 87)
- Syntactics
- strings
- alphabets

265. The PDA is called non-deterministic PDA when there are more than one out going edges from state

- READ or POP (Handouts Page # 111)
- START or READ
- POP or
- PUSH or POP

266. Which of the following states is not part of PDA?

- REJECT
- ACCEPT
- START
- WRITE (All other are parts of PDA)

267. The major problem in the earliest computers was

- To calculate the mathematical formula
- To display mathematical formulas (Handouts Page # 87)
- To store the contents in the registers
- To load the contents from the registers

268. The operators like (` +) in the parse tree are considered as

- Terminals (Handouts Page # 93)
- productions
- non-terminals

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- intermediates

269. If L1 and L2 are two regular languages, then they……expressed by FAs.

- cannot be
- May be
- may or may not be
- can be (Handouts Page # 68)

270. Before running the input string on PDA it is first placed on

- Stack
- Ram
- Memory
- Tape (Handouts Page # 107)

271. Which is the correct option

- The element PUSHED in last is POPED in last
- The element PUSHED first is POPED in the last (LIFO Method, from Book)
- The element PUSHED first is POPED first
- None of given options

272. Null production is a

- String
- Word (Handouts Page # 97)
- Terminal
- All the above

273. A/an……operator adds a new letter at the top of STACK

- Push (Handouts Page # 107)
- Append
- Read
- Pop

274. In conversion form of PDA, no two……states exist in a row without state

- POP. READ (Handouts Page # 119)
- POP. REJECT
- PUSH. START
- PUSH READ

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275. Given a PDA that accepts the language L

- that PDA will also accept Language L' (complement of Ll
- There exists a CFG that generates exactly L (Handouts Page # 118)
- None of given options
- There does not exist any CFG that generates exactly

276. In large FA with thousands of states and millions of directed edges, without an effective procedure it is .....to find a path from initial to final state.

- Always easy
- always impossible
- may be good
- Impossible (Handouts Page # 81)

277. The CFG there generates the regular language is called

- Regular expression
- finite automata
- regular grammars (Handouts Page # 97)
- now regular grammars

278. Consider the following CFG: (Note: ^ means NULL)

S-→Xa

X→aX|bX|^

- Above give a CFG can be represented by RE
- a*b*
- a (a + b) *a
- (a + b) *a
- A*b*a

279. For a machine with N number of states, the total number of strings to be tested, defined over an alphabet of m letters is

- mN +mN+1 + mN+2 +… +m2N-1 (Handouts Page # 86)
- mN
- Nm
- Nm +mN+1 + mN+2 +… +

280. Consider the CFG given below.

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A→B|b

B→a

Which of the following is a unit production?

- S→bb
- A→b
- A→B
- B→a

281. The CFG is said to be ambiguous if there exist at least one word of its language that can be generated by the ……. production trees

- One
- Two
- More than one (Handouts Page # 95)
- At most one

282. If Q = {xx, xyxxxy} and R = { xyxyxyxxyy, xyxyyyxx} then Pref {Q in R} = _____

- Xx
- Xyxyxy
- Xyxyyy (Solved by my self 100% sure)
- Xxy

283. The unit production is

- Terminal --> Terminal
- Terminal --> Non Terminal
- Non terminal --> Terminal
- Non terminal --> Non-Terminal (Hand out Page # 100)

284. Which of the following statement is FALSE?

- For every PDA, there always exists a regular expression (Not sure)
- Every CFG cannot be expressed as Regular Expression
- Every Regular Expression be expressed by a CFG.
- For a PDA, there exists a CFG that represent the same language

285. The CFG S → aSb|ab|^

- Palindrome

- Prime
- Equal
- Even

286. Before the PDA is converted into conversion form, a new state ---------- is defined which is placed in the middle of any edge.

- HERE (Hand out Page # 118)
- STOP
- START
- REJECT

287. A PDA is in conversion form if it fulfills the following conditions:

- There is only one ACCEPT state. (Hand out Page # 119)
- There are one REJECT state.
- There are more than one ACCEPT states.
- There is only one Accept state.

288. Identify the false statement about the following CFG

S→SB|AB

A→CC

B→

C→a

- CFG has 8 Non terminals
- all the given option (All are false as There are 4 terminals , It is in CNF and it does not generate any null string)
- CFG is not in CNF
- CFG generate null string

289. This CFG there generates to the regular language is called

- Regular grammar (Hand out Page # 97)
- nonregular grammar
- finite automata
- regular expression

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290. The derivation of the word W generated by CFG such that at each step a production is applied to the leftmost nonterminal in the working string is said to be

- Left most terminal
- right most terminal
- left most derivation (Hand out Page # 103)
- right most derivation

291. If the FA has N states, then test these words of length less than N. If no word is accepted by this FA, then it will word/words

- Accept no (Hand out Page # 85)
- Accept some
- Reject No
- Accept All

292. In a CFG then non terminals are denoted by

- small letters
- numbers
- capital letters (Hand out Page # 87)
- small letters and numbers

293. "CFG"stands for_____

- Context finite graph
- contacts finite grammar
- contact free graph
- Context free grammar (Hand out Page # 87)

294. Consider the following production (of a CFG) S→XYZ Here _____is left most non terminals in working string note XY and Z are all known terminals

- X (X is on the most left side)
- S
- Z
- Y

295. Consider the following CFG

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S→ a|Xb|aYa

X→ Y|^ (Note: ^ means NULL)

Y→ b|X

which nonterminal is/are not nullable

- Y
- S, X and Y
- S
- X (X is Null Production and not a nullable

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Q No 1.How Moore and Mealy machine works in Computer Memory what is their importance in Computing ?

Mealy & Moore Machines work in computing as incrementing machine & 1's complement machine etc. These operations as basic computer operations so these machines are very important.

Q No 2.What is sequential circuit ?

Sequential Circuit:

A sequential circuit contains a memory component.

The memory component provides a state input. A flip-flop is often used as a memory component.

The state variable indicates the states of the sequential machine, i.e. the status or stage or progress of the whole event.

The state of a sequential circuit is indicated by the output of a flip-flop. A single flip-flop can be used to indicate two states (q=0 and q=1). When there are more than two states, additional flip-flops are used. Given n flip-flops, a total of 2^n states can be represented.

In other words, a sequential machine can be put into a number of different states depending on the particular inputs given.

The output is a function of both the Present Inputs and the Present States.

In addition to the outputs, the circuit must also generate an update to the memory components so that the state of the machine can also be changed with respect to the new inputs.

The update is called the Next State Function and is also a function of the Present Inputs and the Present States.

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Both the output functions and the Next State Functions are combinational circuits. $Z=f(X,S^t)$

$S=g(X,S^t)$ The superscript t indicates the present time period while the superscript (t+1) indicates the next time period.

The characteristic of a sequential circuit is completely defined by a state transition diagram that enumerates all possible transitions for every possible input combination.

Q No 1. What is the concept of Pumping Lemma I and II and what is the difference between pumping Lemma 1 and pumping Lemma 2 ?

In fact PLI & PLII are same (A way to recognize Non Regular language). The only difference is that the conditions in pumping lemma II are more stricter than Pumping Lemma I some language that are difficult to proof Non Regular by Pumping Lemma I are proved Non Regular by pumping Lemma II easily.

Further mare in pumping lemma I we have to generate all words to of a language but in Pumping Lemma II we have to generate a single word to prove a language non regular.

Explanation:

Some languages like PALINDROME that are proved to be regular by first version

due to some of their symmetrical words when we pump these words they remain to be the parts of the language like bbabb

By pumping

lemma 1 Let y

= a

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Now repeating y three
times results in bbaaabb
That is also a valid word of PALINDROME so by pumping lemma I PALINDROME can not be proved non regular, so there was the need of pumping lemma version version 2.
Now consider for
the word bbabb
if we take N=2
Then by pumping y (let we take it b)
two times results in bbbbabb
That word is not in
PALINDROME. But if
we take N=3 and y = a
Then by pumping y two
times results in bbaaabb
That word is in PALINDROME. So be careful in taking total no of states of the FA and also the repeating factor (y) to prove an infinite language non regular you need to prove only one word that is not part of the language.

Q No 2. What is the significance of Pumping Lemma II ?

The significance of 2nd version of 'pumping lemma' is that there are some infinite non regular languages like PALINDROME we can built FA that can accept there certain words but if we increase the length of their words that FA don't accept these words so by pumping lemma

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version I it is very difficult to prove them non regular but with the second version we can prove that a language is Non regular even it's some words may be accepted by some FA's. See page 195 of the book for further example.

Q No 3.Moore and Mealy machine?

1. In order to run a string on a Mealy or Moore machine, you can take directions from transition table. Running string on Mealy or Moore machine is similar to running string on a FA. For example, if want to run abba on the machine, take start from initial state. Check what is the transition for a, what state it goes. After that check what is the path of b from that state and so on. In this way you will be able to run whole of the string. Note that there is no final state in Mealy or Moore machine. So there is no case of acceptance or rejection of string. You just have to determine what the output is. I hope that will clear your mind for further clarification please listens to your lecture carefully.
2. The string is taken for the testing purposes. You can take any sort of string and determine its output using machine.

Q No 1.What is the difference between semiword and word please also give an example regarding this?

Word:

A word is complete combinations of terminals only
e.g. abba or ab or a or null string.

Semiword:

A semiword is a string of terminals (may be none)

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concatenated with exactly one nonterminal on the right i.e. a semi word, in general, is of the following form

(terminal)(terminal)--- (terminal)(nonterminal)

For example

aaaaaaB , aabbaaaA , A.

What is the difference between derivation tree and total tree?

A Derivation tree is the one that shows how to derive any specific word of the language described by CFG but Total Language Tree shows all words of the

Q No 2.What does mean the LANGUAGE IS CLOSED?

When we say that a Language is closed it is always with respect to certain operation.

A simple example may be that the set of integers is closed under addition. It means when we take two numbers from set of integers say 3, 7 the result of their addition would also be in the set of integers. Similarly if the result of an operation on the words of a language results in the word of the same language we say that the language is closed under that operation.

Q No 3.What are the Productions?

Productions are the grammatical rules and regulations. These rules express the behavior of CFG. Using production in CFG terminals are converted into non-terminals and when all the terminals are converted using productions, a word

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is acquired.

Q No 4.What is the difference between concatenation and intersection of two FA's also what is the difference among Union of two FA's and addition of them?

In intersection of two FA's only those strings are accepted which are independently accepted by both FA's, while in concatenation of two FA's only those strings will be accepted in which first part of string is accepted by first FA and remaining part of string is accepted by the second FA.

While taking union of two FA's one can represent it using + sign. So (FA1 UFA2) and (FA + FA2) both are same. There is no difference between them.

FAQ's about

Lectures 36 to 40

Automata Theory

FAQ's about Lectures 36 to 40

Q No 1.What is the Difference between Nullable and Null production? How to make eliminate Nullable and for Null Productions from the CFG ?

The production of

the form

nonterminal ® L

is said to be ***null production***.

Example:

Consider the following CFG

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S ® aA|bB|L, A ® aa|L, B ® aS

Here S ® L and A ® L are null productions.

A production is called ***nullable production*** there is a derivation that starts at Non Terminal and leads to L

i.e.

S ----> *aA* | *bB* | *aa*

A -----> *C* | *bb*

C -----> L

Here A nullable Non Terminal due to Nullable production A > C as C leads to null.

Example:

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Consider the following CFG S ® XY, X ® Zb, Y ® bW Z ® AB, W ® Z, A ® aA|bA|L B ®Ba|Bb|L.

Here A ® L and B ® L are null productions, while Z ® AB, W ® Z are nullable productions.

Method:

Delete all the Null productions and add new productions *e.g.* Consider the following productions of a certain CFG X ® aNbNa, N ® L, delete the production N ® L and using the production X ® aNbNa, add the following new productions X ® aNba, X ® abNa and X ® aba

Thus the new CFG will contain the following productions X ® Nba|abNa|aba|aNbNa

Note: It is to be noted that X ® aNbNa will still be included in the new CFG.

Method:

Consider the following CFG S ® XY, X ® Zb, Y ® bW Z ® AB, W ® Z, A ® aA|bA|L B ®Ba|Bb|L.

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Here A ® L and B ® L are null productions, while Z ® AB, W ® Z are nullable productions. The new CFG after, applying the method, will be

S ® XY
X ®
Zb|b Y
®
bW|b
Z ®
AB|A|B
W ® Z
A ®
aA|a|bA|b
B
®Ba|a|Bb
|b

Note: While adding new productions all Nullable productions should be handled with care. All Nullable productions will be used to add new productions, but only the Null production will be deleted

Q No 2. Is it possible to make CFG for infix and postfix expression's using derivation tree ?

Derivation tree is only used to derive words of language that is described by a CFG. Yes, we can create CFG for languages infix expressions, postfix expressions.

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Q No 3 what is the uses of push down automata in computing ?

PDA is just an enhancement in FAs. i.e Memory is attached with machine that recognizes some language. FA is basic structure for most advanced electronic machines such as computer etc.

Q No 4 What is difference between PUSH DOWN STACK and PUSH DOWN STORE ?

No difference at all. Both terms are used to describe memory structure attached with FAs to store some characters in it.

Q No 5 How we can distinguish between "CFG" and "CNF" in the questions ?

Chomsky Normal Form (CNF)
If a CFG has only productions of the form
string of two nonterminals□nonterminal ---------
or
one terminal□Nonterminal ----------
Then the CFG is said to be in Chomsky Normal Form (CNF).
Thus if the given CFG is in the form specified above it will be called in CNF.

Q No 6.What is meant by the terms stack consistence and input tape consistence ?

Term Stack consistent means we can pop any character from the top of the stack only. PDA should not be able to pop any character other than that is present on the top of the stack.
Term Tape consistent means we can read only the first letter on the tape not any other letter of the

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tape after the first one.

Q No 7 What is the concept of unit production ?

The productions of the form

one Nonterminal□one Nonterminal --------

Are called unit

productions. For

example

A (Unit Produciton)□S --------

a | b□A--------

A. we can directly write□Here there is no need of Unit Production S S □ a | b

Q No 8 Why Context Free Grammars are called "Context Free? Context Free Grammars are called context free because the words of the languages of Context Free Grammars have words like

“aaabbb”(PALINDROME). In these words the value of letters (a, b) is the same on whatever position they appear. On the other hand in context sensitive grammars their value depend on the position they appear in the word a simple example may be as follows

Suppose we have a decimal number 141 in our language . When compiler reads it, it would be in the form of string. The compiler would calculate its decimal equivalent so that we can perform

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mathematical functions on it. In calculating its decimal value , weight of first "1" is different than the second "1" it means it is context sensitive (depends on in which position the "1" has appeared).
i.e.

$$1*10^2 + 4*10^1 + 1*10^0 = 14$$

(value of one is 100) (value of one is just one)
That is not the case with the words of Context Free Languages. (The value of "a" is always same in whatever position "a" appears).

Q No 9. What is Unit Production?

The production in which one non-terminal leads to only one non-terminal.

Q No 10.What is Left most Derivation in CFG?

It is a method of generation of strings from a CFG starting from left most letter of the string.

Q No 1.Give a example of converting a CFG to CNF?

Consider the CFG
given below S}
ABC
A} aa
| bB}
c
C}d

Its CNF

will be

S} DC

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D } AB
A}
EE | b
E}a
B}c
C}d

Q No 2.In the lecture 41 's example, we have converted PDA to conversion form and a word 'aaaabb' is derived from this conversion form PDA. What are the derivation steps.

The PDA converted to conversion form has some specific features that are important to understand first. These features are

The states named START, READ, HERE and ACCEPT are called joints of the machine.

With the help of the conversion form we have been able to achieve that POP state has only one path out of it and the path taking (multiple paths) decisions take place only on the READ state.

The word 'aaaabb' is generated as follows from the PDA START-

POP4-PUSH $

This step pops $ and then pushes it to ensure that stack contains $ at the beginning.

READ1-POP6-PUSH $-PUSH a

As first time after reading "a" there is $ at the top

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of stack so we will follow path segment READ1-
POP6-PUSH $-PUSH a
READ1-POP5-PUSH a-PUSH a
Now a is on the top of the stack so we will follow
READ1-POP5-PUSH a-PUSH a
READ1-POP5-PUSH a-PUSH a
Again following same
segment for a READ1-
POP5-PUSH a-PUSH

a
Again following same segment for a
READ1-POP1- HERE-POP2
As we read b on
input tape.
READ2-POP1-
HERE-POP2
As we read b on
input tape.
READ2-POP3-
ACCEPT.
As we read Δ from the input tape

Q No 3.How to differentiate between "wanted" and "unwanted branch" ? When we derive a word in Top down parsing beginning with the starting Non Terminal the branches of the tree that do not lead to our required word are left aside these branches are called unwanted branches.

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For example

for CFG S

---->AA

A->a | b

If we want to generate the word "aa" we will leave the branch generated by the production A >b.

Q No 4.What is the difference between intersection and union of a language?

Intersection of two languages will consist of all those words which are in both languages while union of two languages will consist of all those words which are present in at least one language.

Symbol for intersection is 、and for union is U.

Q No 5.What is the difference between Context free languages and regular languages?

Regular languages can be represented by FA's because we do not need any memory to recognize (accept or reject them on FA) them but there is another class of languages that can not be represented by FA's because these languages require that we have some memory (with the help of memory we can store letters of the string we are checking so that we can compare them with next coming letters in the string).

For example language anbn requires that we must store a's and then compare their count with next coming b's so that we can check whether a's are equal to b's or not.

Due to this reason we use Context Free Grammars to

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represent them because we can5t write RE's for them. So Context Free Languages represent a broader category this category also include regular languages as subcategory. It means that context free languages include regular languages as well as some other languages.

Q No 6 What is the difference between Moore and Mealey machines?

In Mealy Machine we read input string letters and generate output while moving along the paths from one state to another while in Moore machine we generate output on reaching the state so the output pattern of Moore machine contains one extra letter because we generated output for state q0 where we read nothing.

Q No 7.What does the following terms mean

- STACK Consistent
- Y-able Paths
- Working string
- Semi Word means

Stack consistence means that in the PDA converted in the conversion form, when we follow a path segment (which is formed by combining start, read or here state with next read, here or accept state on the path) along the PDA its pop state should have the path for the same letter that is present on the top of the stack at that stage. If this doesn't happen our PDA will crash because in conversion form of

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the PDA the pop state has only one letter path, so if we could not be able to find that letter on the top of the stack our PDA will crash (if will not find path where to go from that state)

Working string means the string present on the input tape.

Y-able Paths means that when we follow a certain sequence of rows from the row table to generate a path for a word form start state to accept state. The path (sequence of rows) should be stack as well as joint consistent it means that rows should end at the same read or here state (join consistency) and the rows should be able to pop the letter from the top that is indicated in the pop state of the row.

Semi word is the string of terminals it may be null string ending with a Non terminals on the right.

For example some
semi words are aaS
aab
bA
B

Is Automata Theory is a Programming Subject or theoretical?

Automata theory is the study of *abstract* computing devices, or "machines". This topic goes back to the days before digital computers and describes *what is possible to compute using an abstract machine*.

These ideas directly apply to creating compilers, programming languages, and designing applications.

They also provide a formal framework to analyze new types of computing devices, e.g. biocomputers or quantum computers

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What are practical Examples of the implications of Automata Theory and the formal Languages?

Grammars and languages are closely related to automata theory and are the basis of many important software components like:

- Compilers and interpreters
- Text editors and processors
- Text searching
- System verification

What are the Types of Automata?

- The Types of Automata Theory are
- Finite Automata

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- Regular Languages
- Linear-bounded Automata
- Context Sensitive Languages
- Push-Down Automata
- Context Free Languages
- Turing Machines
- Recursively innumerable languages

There are others as well like,

- Random Access Machines
- Parallel Random Access Machines
- Arrays of Automata

Question: How types of Automata Differ?

Answer:

They differ in the following areas

Complexity (or Simplicity)

Power

In the function that can be computed. In the languages that can be accepted.

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Question: What is the difference between the alphabet and an element of a set?

Answer: Alphabets is a set of letters nothing else but a set of strings (elements) can have more than one letters in one string.

Question: Difference between Palindrome and Reverse function?

Answer:

The language consisting of ⋯ and the strings s defined over ┆ such that Rev(s)=s.

It is to be denoted that the words of PALINDROME are called palindromes. Reverse (w) = w

Example: ┆={a,b},

PALINDROME={⋯, a, b, aa, bb, aaa, aba, bab, bbb, ...}

If a is a word in some language L, then reverse (a) is the same string of letters spelled backwards, called the reverse of a.

e.g

reverse (xxx)

= xxx reverse

(623) =

326

reverse (140) = 041

Question: Define Strings?

Answer: Concatenation of finite letters from the alphabet is called a string.

e.g If ┆={a,b} then a language L can be defined as

L = {a, abab, aaabb, abababababababababab,.}

it's mean all words with a's more or equal to b's

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Question: Define empty or null strings?

Answer: Concatenation of finite letters from the alphabet is called a string.

Sometimes a string with no symbol at all is used, denoted by (Small Greek letter Lambda) ⊢ or (Capital Greek letter Lambda) ⋯, is called an empty string or null string.

Question: Difference between string and word?

Answer: Any combination of letters of alphabet that follows rules of language is called a word.
A string is a finite sequence of symbols from an alphabet.

Question: There are as many palindromes of length 2n as there are of length 2n-1, please explain?

Answer: If we try to create palindromes then middle elements (2 in even palindromes & 1 in odd palindrome) does not cause any change in no. of palindromes
Defining the language PALINDROME, of length 2n and 2n-1 defined over S = {a,b}
e.g if we take n= 2 for 2n
Length (2n) = 4 and string can be written as
{aaaa, abba, baab, bbbb} And if we take n = 2 for 2n-1
Length (2n-1) = 3 and string can be written as
{aaa, aba, bab, bbb}

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Question: Define Kleene Star?

Answer: Given ┆, then the Kleene Star Closure of the alphabet ┆, denoted by ┆*, is the collection of all strings defined over ┆, including ¨.

It is to be noted that Kleene Star Closure can be defined over any set of strings. Examples

If ┆ = {x}
Then ┆* = {¨, x, xx, xxx, xxxx, ….} If ┆ = {0,1}
Then ┆* = {¨, 0, 1, 00, 01, 10, 11, ….}
If ┆ = {aaB, c}
Then ┆* = {¨, aaB, c, aaBaaB, aaBc, caaB, cc, ….} Note:

Languages generated by Kleene Star Closure of set of strings, are infinite languages. (By infinite language, it is supposed that the language contains infinite many words, each of finite length).

Question: Why do not we can write" ba" in the set of PALINDROME while it is reverse of "ab".

Answer: The language consisting of ¨ and the strings s defined over ┆ such that

Rev(s)=s.

It is to be denoted that the words of PALINDROME are called palindromes.

Exampl e For ┆ ={a,b},
PALINDROME={¨, a, b, aa, bb, aaa, aba, bab, bbb, ...}
All two length string cannot satisfied the palindrome. aa and bb in palindrome but ba and ab are not in palindrome.

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Question: What are the steps of Recursive Definition of Languages?
Answer: A recursive definition is characteristically a three-step process. First, we specify some basic objects in the set.

Second, we give rules for constructing more objects in the set from ones we already know.

Third, we declare that no objects except those constructed in this way are allowed in the set.

Question: Strings that ending in "a " and strings containing exactly one "a".
Answer:

Its means all string ending in a

e.g ¦= {a, b}

{a, aa, ba, aba,

baa,…….} Exactly a,

defined over ¦= {a, b}

{ab, ba, abb, bba,…}

Question: What is Lexical Analyzer?
Answer:

The first phase of the compiler is the lexical analyzer, also known as the scanner, which recognizes the basic language units, called tokens.

The exact characters in a token is called its lexeme.□

Tokens are classified by token types, e.g. identifiers, constant literals, strings, operators, punctuation marks, and key words.□

Different types of tokens may have their own semantic attributes (or values) which must be extracted and stored in the symbol table.□

The lexical analyzer may perform semantic actions to extract such values and insert

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them in the symbol table.□

Question: What is accepting string language?

Answer: The strings which follow rules for the language are accepted in language.

Let u and v be strings. Then uv denotes the string obtained by concatenating u with v, that is, uv is the string obtained by appending the sequence of symbols of v to that of u. For example if u = aab and v = bbab,□ then uv = aabbbab. Note that vu = bbabaab uv. We are going to use first few symbols of English alphabet such as a and b to denote symbols of an alphabet and those toward the end such as u and v for strings.

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Question: What is transition table?

Answer: A complete transition table contains one column for each character. To save space, table compression may be used. Only non-error entries are explicitly represented in the table, using hashing, indirection or linked structures. Tabular representation of a function that takes two arguments and returns a value

q0◊	q2	q0
*q1	q1	q1
q2	q2	q1

Rows correspond to states Columns correspond to inputs Entries correspond to next states. The start state is marked with an arrow The accepting states are marked with a star

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Question: What does it means by the transition?

Answer: Transition means which letter, after being read, is transfer from which place to which place. It is necessary to show transition of every letter from each and every state.

Question: What is Null?

Answer: ⋯ is a string having no letter in it. e.g (A box having no object in it).
Let us observe that if the alphabet has no letters, then its closure is the language with the null string as its only word, because ⋯ is always a word in a Kleen closure. Symbolically,
we write
if ┆ = {the empty set} then ┆* = {⋯},
This is not the same as if S = {⋯}
then S* = {⋯}
which is also true but for a different reason, that is ⋯⋯ = ⋯.
Example
If L is any language, then
L⋯ = ⋯L = L
If ⋯ string concatenates with any string S, it does not cause any change in the string S, if we use ⋯ in any string then it generates some result that is below here
⋯ for both side then string is ⋯aa⋯ = aa
b for left and ⋯ for right then string is baa⋯ = baa

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··· for left and b for right then string is ···aab = abb

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Question: What is the difference between Regular Languages and Non Regular Languages?

Answer: The language determined by a regular expression is regular otherwise non regular.

Question: What is NFA?

Answer: Nondeterminism plays a key role in the theory of computing. A nondeterministic finite state automaton is one in which the current state of the machine and the current input do not uniquely determine the next state. This just means that a number of subsequent states (zero or more) are possible next states of the automaton at every step of a computation.

Of course, nondeterminism is not realistic, because in real life, computers must be deterministic. Still, we can simulate nondeterminism with deterministic programs. Furthermore, as a mathematical tool for understanding computability, nondeterminism is invaluable.

As with deterministic finite state automata, a nondeterministic finite state automaton has five components.

a set of *states*

a finite *input alphabet* from which input strings can be constructed

a *transition function* that describes how the automaton changes states as it processes an input string

a single designated *starting state*

a set of *accepting states*

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The only difference lies in the transition function, which can now target subsets of the states of the automaton rather than a single next state for each state, input pair.

Question: What is a main difference between NFA and FA?

Answer: Finite Automata (FA)

A finite automaton with unique transitions from each state. There are no choices

There is only 1 letter of the alphabet per transition (the label on the edges in the graph is limited to 1)

- transitions are not allowed.

No implied trap states. That is, if the letter of alphabet has n letters in it, every state will have n edges coming out of it. If the letters are not part of a valid word, then the edges will go into a special state, called the trap states. Trap states are the NO states.

Nondeterministic Finite Automaton (NFA)

Has the freedom to do various different moves when in a state and seeing some input

This is modeled mathematically as

The ability to be in various states at once

Accepting a string whenever at least one of those states is accepting.

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Question: How to obtain 9's complement?

Answer: The (r – 1)'s Complement

Given a positive number N is base r with an integer part of n digits and a fraction part of m digits, the (r-1)'s complements of N is defined as

$r^n - r^{-m} - N$. Some numerical examples follow:

The 9's complement of $(52520)_{10}$ is $(10^5 - 1 - 52520) = 99999 - 52520 = 47479$.

No fraction part, so $10^{-m} = 10^0 = 1$.

The 9's complement of $(0.3267)_{10}$ is $(1 - 10^{-4} - 0.3267) = 0.9999 - 0.3267 =$

0.6732

No integer part, so $10^n = 10^0 = 1$.

The 9's complement of $(25.639)_{10}$ is $(10^2 - 10^{-3} - 25.639) = 99.9999 - 25.63967$

$= 74.360$

Question: What is DELAY box?

Answer: It is a component which held input for some time and then forwards it just a holder.

Question: What is the difference between pumping Lemma 1 and pumping Lemma 2?

Answer: Infact PLI & PLII are same (A way to recognize Non Regular language). The only difference is in PLII we take care about the substring x & y that length (x) + length (y) less than or equal no. of state of machine. This is because through PLI palindrome (that is Non Regular) is proved to be regular and through PLII this problem is fixed.

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Question: What is pumping lemma? And what is history?

Answer: A theorem to check validity (Regularity) of an infinite language should not be used with finite languages. Whenever an infinite is regular then there must be a loop (circuit) because without a loop means infinite no. of states that is not possible practically. (Machine can have finite states only)

Question: What is the difference between semi word and word?

Answer: A word is complete combinations of terminals only e.g. abba or ab or a or null string.

Semiword: A semiword is a string of terminals (may be none) concatenated with exactly one nonterminal on the right i.e. a semiword, in general, is of the following form

(terminal)(terminal)… (terminal)(nonterminal)

Question: What is the difference between derivation tree and total tree?

Answer: A Derivation tree is the one that shows how to derive any specific word of the language described by CFG but Total Language Tree shows all words of the

Language described by CFG on it.

Question: How to identify a production by it, ambiguity will be removed?

Answer: It is a matter of practice that one can know how to

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remove ambiguity from it, only practice makes you efficient enough to do it in less time.

Question: Difference between Nullable and Null production? How to make CFG for Nullable and for Null?

Answer: The production of the form
nonterminal ® L
is said to be ***null production***.

Example:
Consider the following CFG
S ® aA|bB|L, A ® aa|L, B ® aS
Here S ® L and A ® L are null productions.
A production is called ***nullable production*** if it is of the form N ® L
or
there is a derivation that starts at N and leads to L *i.e.*
N1 ® N2, N2 ® N3, N3 ® N4, …, Nn ® L,
where N, N1, N2, …, Nn are non terminals.

Example:
Consider the following CFG S ® XY, X ® Zb, Y ® bW
Z ® AB, W ® Z,
A ® aA|bA|L B ®Ba|Bb|L.
Here A ® L and B ® L are null productions, while Z ® AB, W ® Z are nullable productions.

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Method:

Delete all the Null productions and add new productions *e.g.*

Consider the following productions of a certain CFG X ® aNbNa, N ® L, delete the production N ® L and using the production X ® aNbNa, add the following new productions X ® aNba, X ® abNa and X ® aba

Thus the new CFG will contain the following productions X ® Nba|abNa|aba|aNbNa

Note: It is to be noted that X ® aNbNa will still be included in the new CFG.

Method:

Consider the following CFG S ® XY, X ® Zb, Y ® bW Z ® AB, W ® Z, A ® aA|bA|L B ®Ba|Bb|L.

Here A ® L and B ® L are null productions, while Z ® AB, W ® Z are nullable productions. The new CFG after, applying the method, will be

S ® XY X ® Zb|b Y ® bW|b Z ® AB|A|B W ® Z

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A ®

aA|a|bA|b

B

®Ba|a|Bb|

b

Note: While adding new productions all Nullable productions should be handled with care. All Nullable productions will be used to add new productions, but only the Null production will be deleted.

Question: Is it possible to make CFG for infix and postfix expression's using derivation tree?

Answer: Derivation tree is only used to derive words of language that is described by a CFG. Yes, we can create CFG for languages infix expressions, postfix expressions.

Question: What are the uses of push down automata in computing?

Answer: PDA is just an enhancement in FAs. i.e Memory is attached with machine that recognizes some language. FA is basic structure for most advanced electronic machines such as computer etc.

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Question: What is difference between PUSH DOWN STACK and PUSH DOWN STORE?

Answer: No difference at all. Both terms are used to describe memory structure attached with FAs to store some characters in it.

Question: How to accommodate NULL string if it is part of language during converting from CFG to CNF and building FA's?

Answer: When we convert CFG to CNF and Null is a part of language then null string is not part of language in CNF. This is the only change a language gets in CNF. When we draw an FA for a CFG, there is no change in language and simply draws FA that accepts the language of CFG.

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Question: How to accommodate NULL string if it is part of language during converting from CFG to CNF and in building PDA?

Answer: When we convert CFG to CNF and Null is a part of language then null string is not part of language in CNF. This is the only change a language gets in CNF. When we draw an PDA for a CFG, there is no change in language and simply draws PDA that accepts the language of CFG.

Question: What is Push down Automata?

Answer: Pushdown Automaton (PDA), consists of the following

An alphabet S of input letters.
An input TAPE with infinite many locations in one direction. Initially the input string is placed in it starting from first cell; the remaining part of the TAPE is empty.
An alphabet G of STACK characters.
A pushdown STACK which is initially empty, with infinite many locations in one direction. Initially the STACK contains blanks.
One START state with only one out-edge and no in-edge.
Two halt states i.e. ACCEPT and REJECT states, with in-edges and no out- edges.
A PUSH state that introduces characters onto the top of the STACK.
A POP state that reads the top character of the STACK (may contain more than one out-edges with same label).

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A READ state that reads the next unused letter from the TAPE (may contain more than one out-edges with same label).

Question: Why we study Automata?

Answer: Automata theory is the study of abstract computing devices, or "machines". This topic goes back to the days before digital computers and describes what is possible to compute using an abstract machine. These ideas directly apply to creating compilers, programming languages, and designing applications. They also provide a formal framework to analyze new types of computing devices, e.g. biocomputers or quantum computers. Finally, the course should help to turn you into mathematically mature computer scientists capable of precise and formal reasoning. More precisely, we'll focus primarily on the following topics. Don't worry about what all the terms mean yet, we'll cover the definitions as we go:

1. Finite state automata: Deterministic and non-deterministic finite state machines; regular expressions and languages. Techniques for identifying and describing regular languages; techniques for showing that a language is not regular. Properties of such languages.

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2. Context-free languages: Context-free grammars, parse trees, derivations and ambiguity. Relation to pushdown automata. Properties of such languages and techniques for showing that a language is not context-free.
3. Turing Machines: Basic definitions and relation to the notion of an algorithm or program. Power of Turing Machines.
4. Undecidability: Recursive and recursively enumerable languages. Universal Turing Machines. Limitations on our ability to compute; undecidable problems.
5. Computational Complexity: Decidable problems for which no sufficient algorithms are known. Polynomial time computability. The notion of NP-completeness and problem reductions. Examples of hard problems.

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Question: What is valid and invalid alphabets explain with example?

Answer: Example 1

If s=abc is a string defined over ¦= {a,b,c} then Rev(s) or sr = cba

¦= {a,b}

s=abbaa

Rev(s)=a

abba

When more then letter in the alphabet you have to be quite careful that don't reverse the symbols however you write the letter from right to left.

Example 2

¦= {B, aB,

bab, d}

s=BaBbabB

d

Rev(s)=dBb

abaBB

Example 3

¦= {ab, b,

aa} s=abbaa

Rev(s)=aaba

b

¦1= {B, aB, bab, d} is valid alphabet as there is no letter in ¦1 that lies in start of any other letter means all the tokens of any word (string) will be unique.

Whereas in ¦2= {B, Ba, bab, d} letter B lies in start of letter Ba.

This makes it difficult to decide which token to select

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at some point if B occurs in any string.

Question: Why we use Capital Letters for Languages.
Is it possible to combine two languages together like EVEN-EVEN & EQUAL, and so on?

Answer: We use capital letter for our convenient and yes you can combine two languages.

Question: What are the rules to form WORDS in languages developed by Automata?
Are strings not following any rule?

Answer: Rules are different for different languages.
e.g ┆={a,b}
The language L of strings of even length, defined over ┆ = {a,b}, can be written as Valid for even length L= {aa, bb, aabb, bbaa, baab, abba …..}
Invalid for even length L= {a, b, aaa, bbb, aba, bab, …….}
The language L of strings of odd length, defined over ┆ = {a,b}, can be written as Valid for odd length L1= {a, b, aaa, bbb, aba, bab, …….}
Invalid for odd length L= {aa, bb, aabb, bbaa, baab, abba …..} Strings cannot follow any rule.

Question: What are graphs of palindromes of length 2n and length 2n-1?

Answer: Palindromes of even length are always symmetric about the middle line and

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palindromes of odd length are always symmetric about middle letter.

If we try to create palindromes then middle elements (2 in even palindromes & 1 in odd palindrome) does not cause any change in no. of palindromes

Defining the language PALINDROME, of length 2n and 2n-1 defined over S = {a,b}

e.g if we take n= 2 for 2n

Length (2n) = 4 and string can be written as {aaaa, abba, baab, bbbb} And if we take n = 2 for 2n-1

Length (2n-1) = 3 and string can be written as {aaa, aba, bab, bbb}

Question: How can we write a RE for a given number of words?

Answer: In example let us consider a finite language L that contains all the strings of a's and b's of length three exactly:

L = {aaa aab aba abb baa bab bba bbb}

The first letter of each word in L is either an a or a b. The second letter of each word in L is either an a or a b. The third letter of each word in L is either an a or a b. So, we may write

L = language ((a+b)(a+b)(a+b)) Or

L = language $((a+b)^3)$

If we want to define the set of all seven-letter strings of a's and b's, we could write $(a+b)^7$. In general, if we want to refer to the set of all possible strings of a's and b's of any length whatsoever, we could write $(a+b)^*$

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This is the set of all possible strings of letters from the alphabet $\sum = \{a, b\}$ including the null string. This is a very important expression and we shall use it often.

Again, this expression represents a language. If we choose that * stands for 5, then

(a+
b)*
give
s

$(a+b)^5 = (a+b)(a+b)(a+b)(a+b)(a+b)$

We now have to make five more choices: either a or b for the first letter, either a or b for second letter, and so on.

21. Please explain that in some expression having more then one plus sign the resulting string is only one from them or it can be more then one? For example if there is an expression: a +b + c, with out any small bracket between them the resultant string is "a or b or c" OR it cans both "a or b and c","a and b or c" The resulting string is only one from them. Whenever we put + signs between n words that means one option (only) out of all available.

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Question: One Language generates many REs?

Answer: Sometimes we generate many Regular Expressions (RE) for one language these REs are called Equal RE.

Example:

Consider the following regular expressions r1= (a + b)* (aa + bb)

r2= (a + b)*aa + (a + b)*bb then

both regular expressions define the language of strings ending in aa or bb. This RE (aa+bb) is separated by +. Whenever RE is separated by +, two possibilities occur

1. Before + part
2. After + part

(a+b)*aa comes before + part, means RE generate all strings that end with aa. (a+b)*bb comes after + part this means RE generate all strings that end with bb. In short we can say r2 generates the language of strings ending either aa or bb this is equal to r1.

Question: Rules for determining RE for a given language defined on a set?

Answer: The following rules define the language associated with any regular expression: Rule 1: The language associated with the regular expression that is just a single letter is that one-letter word alone and the language associated with null is just {null}, a one-word language.

Rule 2: If r1 is a regular expression associated with the language L1 and r2 is regular expression associated with the language L2, then:

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The regular expression (r□1)(r2) is associated with the product L1L2 that is the language L1 times L2

language (r1r2) = L1L2

The regular expression r□1 + r2 is associated with the language formed by the union of the sets L1 and L2:

language (r1 + r2) = L1 + L2

The language associated with the regular expression (r□1)* is L1*, the Kleene closure of the set L1 as a set of words:

language(r1*) = L1*

Once again, this collection of rules proves recursively that there is some language associated with every regular expression. As we build up a regular expression from the rules, we simultaneously are building up the corresponding language.

Question: What is EVEN - EVEN LANGUAGE?

Answer: Even-Even means count of a's is even and cout of b's is also even.

Even + Even = Even (Proved)

So we can divide any string excluding ¨ which is also in Even-Even in substrings of length 2 each. It gives us following combinations

aa, bb makes no change in string status ab, ba create disorder in string status

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so (aa+bb+(ab+ba)(aa+bb)*(ab+ba))*.

Question: What is tokenizing string?

Answer: Tokenize a string means make its valid units. Question: What is whole star?

Answer: Whole star of any RE means all possible combinations of that RE including ¨.

Question: How to use + operator in Automata?

Answer: Plus Operation is same as Kleene Star Closure except that it does not generate ¨ (null string), automatically.

Question: How to know what is RE?

Answer: It is a matter of practice that you come to know what does an RE represent. You may start with simpler REs and later you will be able to recognize different REs gradually.

Question: Why we use null string in FA?

Answer: If null string is part of our language then we have to handle it in FA, its not compulsion.

Question: What is tree structure?

Answer: A tree is a connected undirected graph with no simple circuits. Since a tree cannot have a simple circuit, a tree cannot contain multiple edges or loops. Therefore any tree must be a simple graph.

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Question: What is difference between (a, b) & (a + b)?

Answer: (a, b) & (a + b) are same and both represents either a or b.

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Question: Difference between FA & TG?

Answer: Finite Automata (FA)

- A finite automaton with unique transitions from each state.
- There are no choices
- There is only 1 letter of the alphabet per transition (the label on the edges in the graph is limited to 1)
- ... transitions are not allowed.
- No implied trap states. That is, if the letter of alphabet has n letters in it, every state will have n edges

coming out of it. If the letters are not part of a valid word, then the edges will go into a special state, called the trap states. Trap states are the NO states. Transition Graphs (TG)

- TG are generalizations of FA.
- Can change state without an input.
- Can read more than one letter at a time.
- Can have a regular expression as a edge label.
- Can have more then one start state.

We are not bound in TG. We are only given a freedom or relation which we are not forced to use. We may or may not use it on will.

In fact we enjoy freedom of staying at multiple places at one time in TGs while reading any letter strings which provides us a facility that any one available path it leads us to final state, word is accepted.

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Question: Difference between $(a+b)^+$ and $(a+b)^*$?

Answer: $(a + b)^+$ means, we repeat RE a + b infinite (any) no. of times but minimum once, whereas $(a + b)^*$ means we may repeat a + b any no. of times even zero times. () are only used to clear (distinguish) the one RE from some other RE.

Question: What is length of string?

Answer: The length of a string indicates how many symbols are in that string.

For example, the string 0101 using the binary alphabet has a length of 4. The standard notation for a string w is to use |w|.
For example:
Length of string: |0101| is 4.
Length of string: |0010| = 4, |aa| = 2, | |= 0

Question: After looking the on diagram how can we say it is TG or is FA?

Answer: Every FA is also a TG but not every TG is FA.
In every FA, every state shows transition of all letters of given alphabet but in any TG it is not must. In TG, we may or may not show all letters transition according to requirement. We can also show transitions on reading any strings in TGs but it is not possible in FAs.

Question: Differentiate between FA, TG and GTG?

Answer: Every FA is also a TG but not every TG is FA.
In every FA, every state shows transition of all letters of given alphabet but in any TG it is not

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must. In TG, we may or may not show all letters transition according to requirement.

We can also show transitions on reading any strings in TGs but it is not possible in FAs. In GTG Directed edges connecting some pair of states labeled with regular expression. It may be noted that in GTG, the labels of transition edges are corresponding regular expressions. In TG we write strings and in GTG we are bound to write RE.

Question: Difference between even clumps and odd clumps?

Answer: Letters may be a's (for example) in even count at one place.

e.g.
aab =
valid baa
= valid
aba =
invalid
abab =
invalid

Question: Difference between containing and consisting?

Answer: Containing something (e.g. R1) means R1 is there (must), whatever other things are, we don't care. Consisting something (e.g. R1) means only R1 is there all other things should be rejected.

Question: Define the main formula of Regular expressions? Define the back ground of regular

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expression?

Answer: Regular expressions are a notation that you can think of similar to a programming language. In fact, regular expressions are quite fundamental in some programming languages like perl and applications like grep or lex.

Regular expressions are similar to NFA and end up describing the same things we can express with a finite automaton. However, regular expressions are declarative in what strings are accepted, while automata are machines that accept strings. We use Regular expressions for defining the languages.

Say that R is a regular expression if R is:

1. a for some a in the alphabet ¦, standing for the language {a}
2. ¨ , standing for the language {¨}
3. R1+R2 where R1 and R2 are regular expressions, and + signifies union
4. R1R2 where R1 and R2 are regular expressions and this signifies concatenation
5. R* where R is a regular expression and signifies closure
6. (R) where R is a regular expression, then a parenthesized R is also a regular expression

A set of strings from an alphabet. The set may be empty, finite or infinite.

The building blocks of regular languages are symbols, concatenation of symbols to make strings (words), set union of strings and Kleene closure (denoted as *, also called the Kleene star, it should be typed as a superscript but this is plain text.)

Informally, we use a syntax for regular expressions.

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Using sigma as the set {0, 1} (an alphabet of two symbols)

01110 is a string starting with the symbol 0 and then concatenating 1, then 1, then 1, and finally concatenating 0. No punctuation is used between symbols or strings that are concatenated.

(01+10) is a union of the two strings 01 and 10. The set {01, 10}

(00+11)* is the Kleene closure of the union of 0 concatenated with 0 and 1 concatenated with 1.

The Kleene closure (star) is defined as the concatenation of none, one, two, or any countable number strings it applies to.

Examples of Kleene star:

1* is the set of strings {¨, 1, 11, 111, 1111, 11111, etc. } This set is infinite.

(1100)* is the set of strings {¨, 1100, 11001100, 110011001100, etc. } (00+11)* is the set of strings {¨, 00, 11, 0000, 0011, 1100, 1111, 000000, 000011, 001100, etc. }

Note how the union (+ symbol) allows all possible choices of ordering when used with the Kleene star.

(0+1)* is all possible strings of zeros and ones, often written as ¦ * where ¦ = {0, 1}(0+1)*

(00+11) is all strings of zeros and ones that end with either 00 or 11. Note that concatenation does not have an operator symbol.

(w)+ is a shorthand for (w)(w)* w is any string or expression and the superscript plus, + , means one or more copies of w are in the set defined by this expression.

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Question: Are S* and S^+same?

Answer: No because S+ means same as Kleene Star Closure except that it does not generate ¨ (null string), automatically. So in the above example, there is no null string.

Concatenation of finite letters from the alphabet is called a string.

If we have S={a, bb, bab, abaab}, first we factorize the string like (a) (bb) (bab) (abaab) then concatenate to each other and make more string to its concatenation but the string abbabaabab is not in S* because the last member (ab) of the group does not belong to S, so abbabaabab is not in S.

Question: What is set?

Answer: Set is a collection of distinct objects.

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Question: How many Methods of defining the languages?

Answer: 1. You can describe a language in English like statement
2. You can define a language by putting all its words in a set
3. You can define a language in a mathematical way
4. You can define a language by Recursive Definition
5. You can define a language by Regular Expression
6. You can define a language by Finite Automata
7. You can define a language by Transition Graph
8. You can define a language by Context Free Grammar

Question: What is difference between Palindrome, Kleene star closure and plus operation?

Answer: The language consisting of ⋯ and the strings s defined over ┊ such that Rev(s)=s.

or

A string x is a palindrome if $x^R=x$.
It is to be denoted that the words of PALINDROME are called palindromes. Reverse (w) = w
Example: ┊={a,b},
PALINDROME={⋯, a, b, aa, bb, aaa, aba, bab, bbb, ...}
If a is a word in some language L, then reverse (a) is the same string of letters spelled backwards, called the reverse of a.
e.g
reverse (xxx)
= xxx reverse
(623) = 326

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reverse (140) = 041

Given ¦, then the Kleene Star Closure of the alphabet ¦, denoted by ¦*, is the collection of all strings defined over ¦, including ···.

Plus Operation is same as Kleene Star Closure except that it does not generate ··· (null string), automatically.

Question: What is inside language?

Answer: Any language contains words in it and certain rules to validate strings for that language.

Question: What is Equal RE?

Answer: Sometimes we generate many REs for one language these are called Equal RE.

Question: How can we make Finite Automaton from a language and a language from FA?

Answer: There is not any formal procedure to design FA for a language. This ability just improves with time and practice.

Question: Could we just use +, - symbols with x not with y in FA?

Answer: Yes, you can use only + or – for the place of x and y, but remember when you don't write + or –, So you should write start and final at the beginning or ending state Or you should indicated by an arrow and a final state by drawing box or another circle around its circle because if you don't write how can we indicate the start and final state.

Question: Explain the language L of string, defined over ∑= {0,1}, having double 0's and double 1's?

Answer: Language of strings with 00 or 11 means string that have substrings 00 or 11 in them at least.

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Minimum words which are included in this language are 00 and 11. This language does not accept null and it also does not accept 0 or 1.

Question: What are the difference between single 1 and 0 and double 1's and 0's?

Answer: 1. Consider the language L of strings, defined over

If ¦={0, 1}, having words with either 0's or 1's without null. The language L may be expressed by RE

$(0 + 1)^+$

When you make string by above RE you have all possible combination of 0's and 1's except null i.e {0, 1, 00, 01, 10, 11,… }. Minimum words which are included in this language are 0 and 1.

2. Consider the language L of strings, defined over

If ¦={0, 1}, having double 0's or double 1's, The language L may be expressed by the RE (0+1)* (00 + 11)

(0+1)*

Double 1's and 0's means clumps of letter which will always come together.

Minimum words which are included in this language are 00 and 11.

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Question: On what basis we select initial and final states?

Answer: It depends on the expression given to us.

Question: How we know that the given expression has how many states?

Answer: There is not any formal procedure to know the number of states. This ability just improves with time and practice.

Question: How will we develop the rules of transition?

Answer: Transition means which letter, after being read, is transfer from which place to which place. It is necessary to show transition of every letter from each and every state.

Question: Can we accept the strings going from final to initial?

Answer: It is to be noted that if any state start from the final state it does not accept any string. Even it does not accept the null string, because there is no path starting from initial state and ending in final state.

Question: What are the basic rules to build FA?

Answer: One and only rule is to build a Finite Automata (FA) should accept all words of the language and reject all the words which are not part of the language. Any FA that ensures these above things is the right FA for the language.
Note: One language can have many FA.

Question: What is Dead state?

Answer: The DEAD STATE is introduced to be able to

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make an automaton complete without altering its behavior.

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